PERSONAL FITNESS COACH PROGRAM STUDY GUIDE

Welcome to Your Program Guidebook

This guidebook brings all your knowledge together in one place, bridging the gap between theory and practice and helping you study smart and prepare effectively for each theoretical and practical session.

It is designed to guide you in *how to study, what to focus on,* and how to get the most out of your learning experience; so, you're not just showing up, but showing up prepared. Whether you're reviewing anatomy, learning how to cue an exercise, or preparing for your exams, this guide will keep everything clear, relevant, and within reach.

Each section is structured to help you:

- Understand what to expect in your sessions
- Know where to focus your attention
- Build confidence in your practical skills
- Be well-prepared for your assessments
- Link each session's content with the material on the platform and book

The guidebook is divided into sessions, and each session is broken down into lessons that include:

- Gamification
- Group activities
- Practical exercises
- Brainstorming prompts
- Space to take notes and reflect

It's built to support your growth in an *interactive, structured, and enjoyable* way as you build the mindset and skills of a future fitness professional.

Let's dive in and make every session count, starting with the way you prepare for it!

Below, you'll find the icons used throughout the guide to help you navigate each section with ease:



Group Exercise or Activity



Practice Time



Individual or Group brainstorming



Game Time

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Your Learning Map Starts Here!

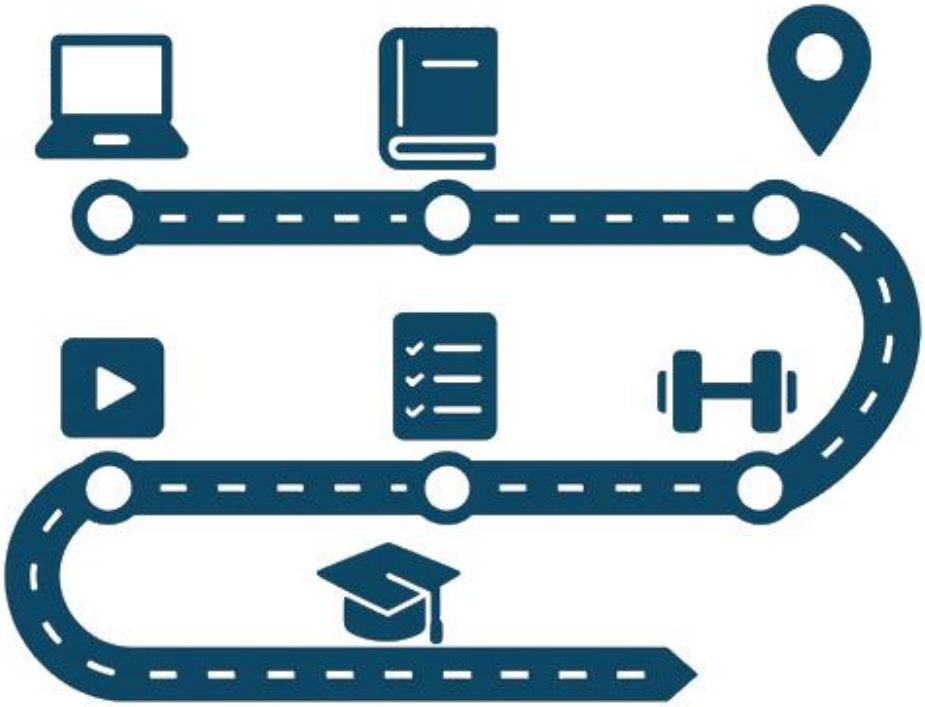
Think of the platform as your **navigation system**—it helps you stay on track, know where you're going, and arrive prepared for every session.

This is where you'll find all the resources that support your sessions and studies: from theoretical videos and reading materials to downloadable slides, to quizzes, assignments, sample exams. The guidebook will often point you back to the platform—so knowing your way around it will help you connect the dots, study smarter, and feel more confident walking into every session and into your exams.

Let's make sure you know where to find it all.

Ready?

Your future coach-self will thank you!

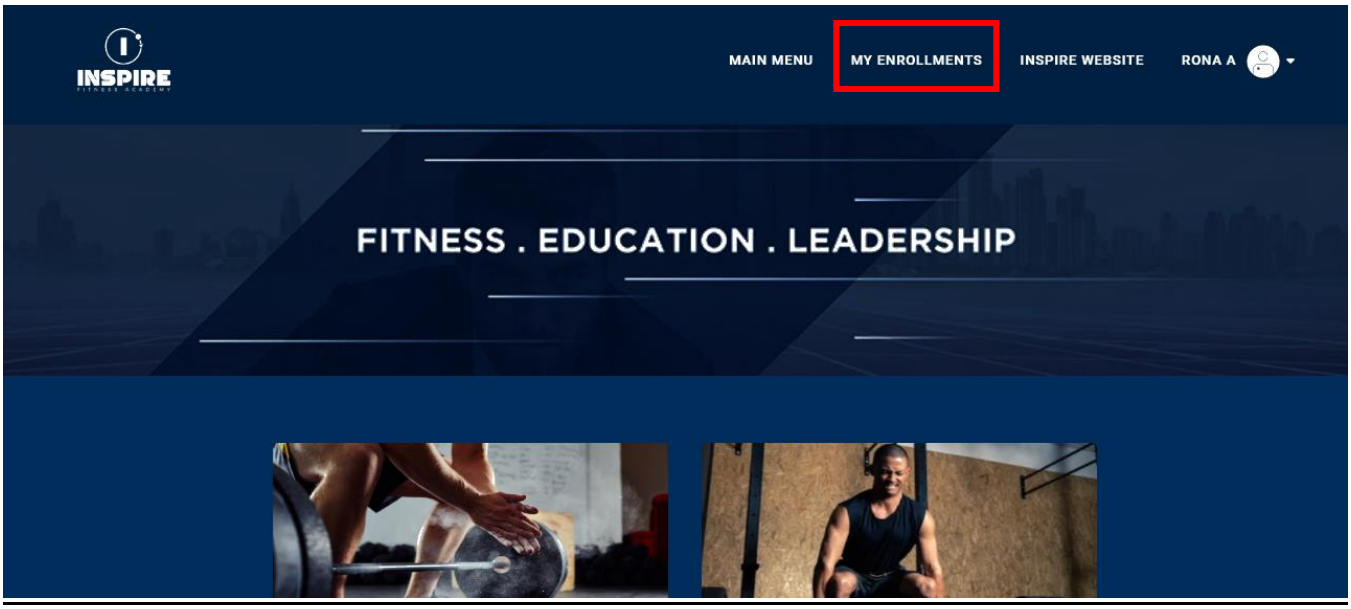


Learn from the Platform, Book and Guidebook with Commitment, Show up to the Sessions
Practice Consistently, Challenge yourself in the exams,
Earn the Certification that proves you have transformed into the Coach you are meant to be!

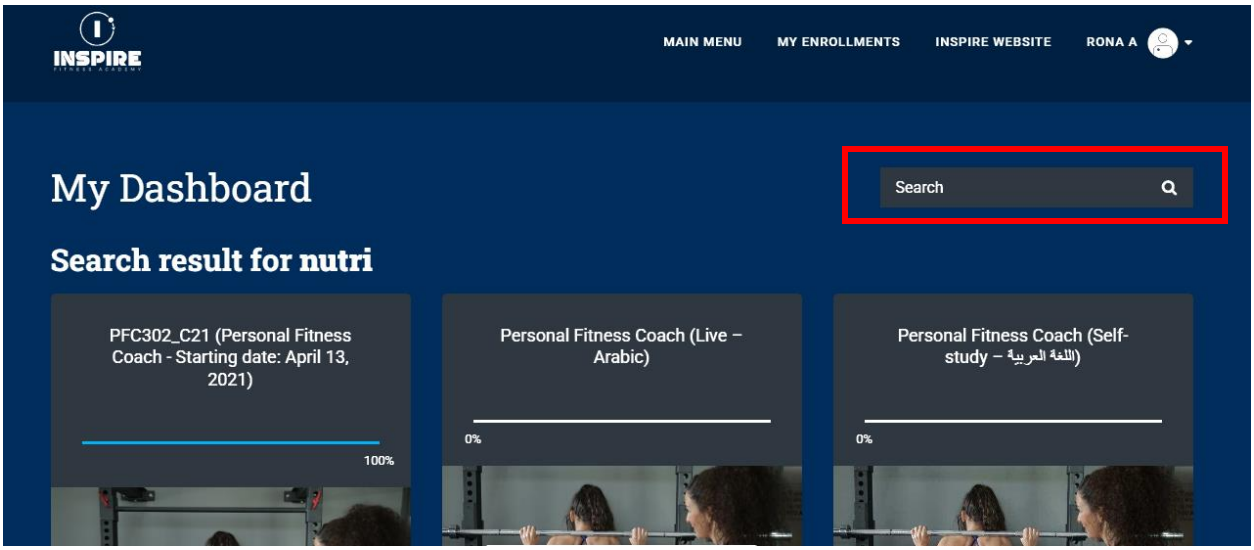
Platform Overview:

1- Sign-in using your username and password (check your email inbox)

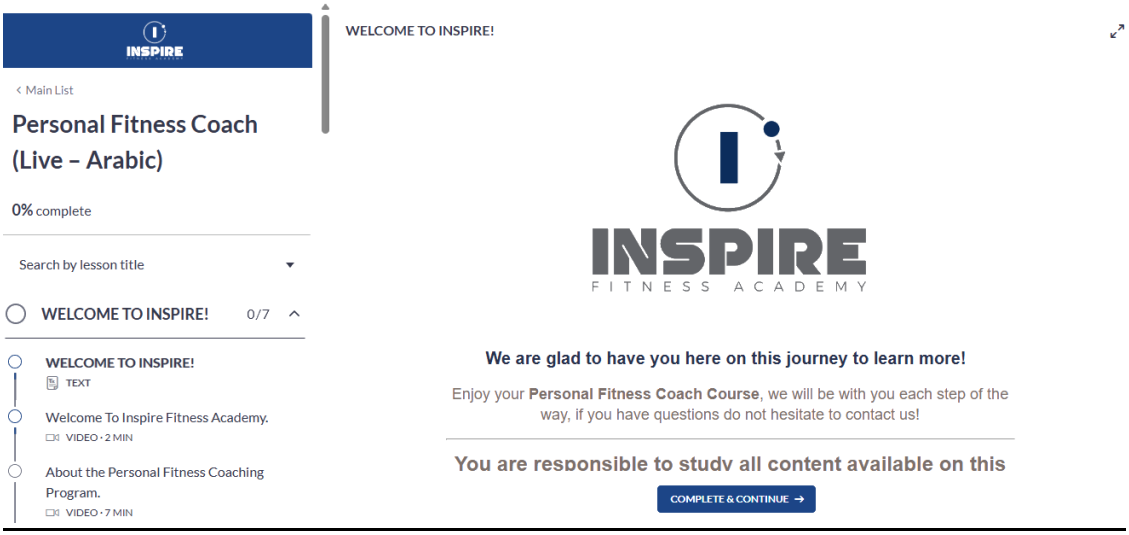
2- Click on “My Enrollments”



3- Search for “Personal Fitness Coach”



4- ENJOY!



INSPIRE'S MEMBERS BENEFIT PROGRAM



RULES & REGULATIONS

PARKING



FIRST AID KIT



ELEVATOR



FIRE EXTINGUISHERS



AED AT THE GYM



CELL PHONE USE



NO SMOKING



COACHING AND PRACTICING



Session 1: Introduction to Professional Personal Fitness Coaching

LESSON 1: BENEFITS OF PHYSICAL ACTIVITY



Objectives:

- State the common benefits of physical activity



State all benefits of physical activity you are aware of?

Instructor's feedback



Consolidation:

By the end of this lesson, I am aware that:

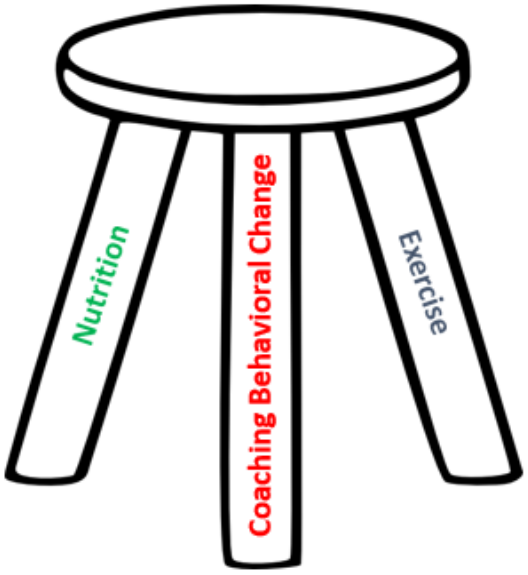
- Exercise has so many more benefits other than improving body shape

LESSON 2: BECOMING A QUALIFIED COACH

Objectives:

- Understand the difference between coach and trainer.
- Understand the importance of being a qualified coach.
- State the reasons why qualified coaches' matter.

- **COACHES VS TRAINERS:**



- **QUALIFIED VS CERTIFIED:**

Qualified coaches are expected to?

Qualified Coaches integrate Evidence, Ethics, and Accountability into Client Care.

- **Safety and Injury Risk Reduction:**

What are the steps that you would do as a coach when working with a client, to make sure that the exercises you will be asking them to do are safe?

- **Effectiveness and Client Outcomes:**

Beside designing a good training program, what else coaches should do to improve clients' results?

- **Ethics and Public Trust:**

Which unethical behaviors coaches might do that negatively affects client's trust?

Coaches as Partners in Public Health:

Coaches are members of a team of professionals; can you guess these professionals?

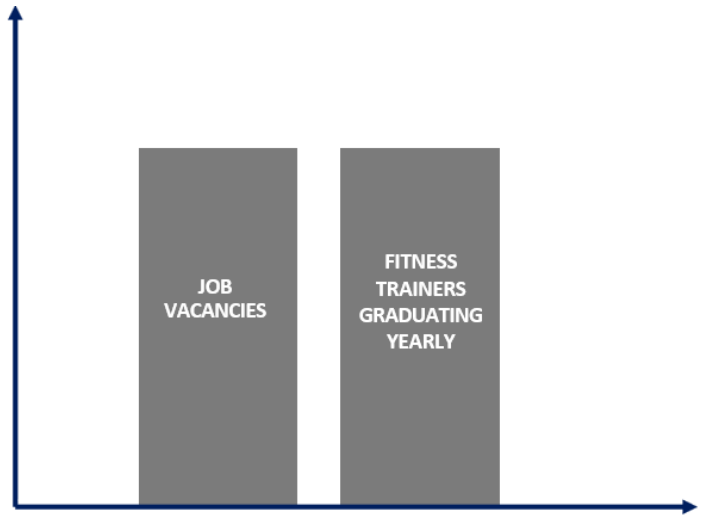


Consolidation:
By the end of this lesson, I am aware of:
- What it takes to become a qualified coach
- The importance of becoming a qualified coach

LESSON 3: FITNESS INDUSTRY BETWEEN TODAY AND TOMORROW

Objectives:
- Explore how to face the challenges within this technological growth

- **GLOBAL VS REGIONAL GROWTH:**



How can we react when we hear the market is saturated, and too many trainers are graduating yearly?

- **Digital Transformation, Technology and Data:**

Artificial Intelligence and coaching apps… should we fight them?

- **Integration with Healthcare:**

Why do many doctors still advise their clients to avoid lifting weights or training at the gym, and instead recommend simple walks or rest?

- **Future Landscape:**

Let's predict the future of this industry together!!!

Consolidation:

By the end of this lesson, I am aware of:

- How to deal with the fast technological growth in the field of fitness
- Better understanding of what the future holds for us

LESSON 4: SCOPE OF PRACTICE & CODE OF ETHICS

Objectives:
- Explore the scope of practice and code of ethics of personal fitness coaches

- **PERSONAL FITNESS CCOACHING GOES BEYOND TRAINING:**

GROUP ACTIVITY

Pair Activity: When discussing the role of Personal Fitness Coaches, **Label** each of the below words as **"Needed" - "Optional" - "Should Not Be Done"**

- Give orders to clients
- Client-centered
- Evidence-Based
- Accountability
- Support
- Guide clients to make the best decision
- Tell clients what they should do
- Empathy
- Believe in clients potential
- Set Training Goals
- Give and Receive Feedback
- Explain the training program to clients
- Assess clients
- Maintain Confidentiality

- **SCOPE OF PRATICE AND CODE OF ETHICS:**



Circle the MOST appropriate response for each scenario presented:

Client Scenario	Response A	Response B
1. Client experiences back pain while training	Help the client by improving strength and flexibility of his back and lower body muscles	Diagnosing the cause of the client's back pain then designing a training program that suits the diagnosis
2. Client wants to lose weight by trying the latest commercial diet	Helping the client to understand and implement the diet	Helping the client to make more healthful choices using the Dietary Guidelines and tools on my-plate or Harvard-plate…
3. Client is cleared for exercise following physical therapy for shoulder injury	Continuing with physical therapy exercises to treat the shoulder	Implementing exercises to improve shoulder stability and building on the work done in physical therapy
4. Client has tightness in one of her muscles	Teaching the client stretching exercises	Providing deep tissue massage to relieve tightness in the IT bands
5. Client has soreness following a weekend tennis tournament	Recommending use of over-the-counter anti-inflammatory medications	Discussing proper techniques for performing light, dynamic stretching activities for the affected areas
6. Client tells you they are depressed due to problems with their spouse	Listening to the client with empathy and maintaining their confidentiality	Listening to the client and providing them with recommendations for improving the situation
7. Client wants to lose weight	Inform your client that they will lose 5kg of fat in 5 days	Inform your client that we will work on improving our diet and exercise to reach our goal
8. Client is asking whether it is possible to pay the sessions fees late	Accept, since this client is funny and coach enjoys sessions with her	Accept only if coach can afford applying this option to all of his clients



Consolidation:
By the end of this lesson, I am aware of:
- Coach's role and responsibilities within their scope of practice and code of ethics.

LESSON 5: BECOMING A PROFESSIONAL

Objectives:
- Explore the way towards becoming a Pro

- **HOW CAN YOU SHOW PROFESSIONALISM IN YOUR ACTIONS AS A COACH?**

Take five minutes to write your personal "why." Why do you want to coach? What change do you hope to inspire in others? Keep this statement visible as a daily reminder of your purpose.



Consolidation:

By the end of this lesson, I am aware of:
- Coach's criteria to become a Professional one

LESSON 6: HOW TO PREPARE FOR YOUR EXAM



Objectives:
- Explore study methods to increase your chances to successfully complete your exams

- **ACTIVE VS PASSIVE LEARNING:**

Let's discuss some ideas you can benefit from to increase the chances of succeeding your exams

- **CONTINUING EDUCATION:**

What is the purpose of Continuing Education?

How can I collect continuing education credits / points to renew my certification?

Consolidation:

By the end of Session 1, I am aware of:

- Benefits of physical activity
- How to become a professional qualified fitness coach
- Scope of Practice as fitness coach
- How to deal with the recent technological outbreaks in the fitness industry
- How to prepare well for my exams
- Importance of continuing education

Post-Session 1 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 2.

Pre-Session 2 Preparation:

-Review all materials covered in Session 1

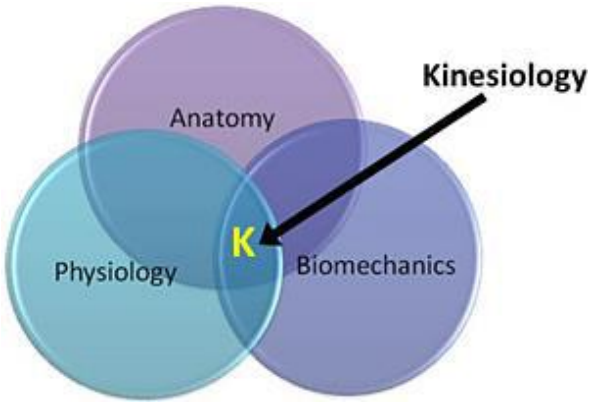
-Watch all Anatomy related videos of the Chapter titled: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System

Session 2: Functional Anatomy – Skeletal System

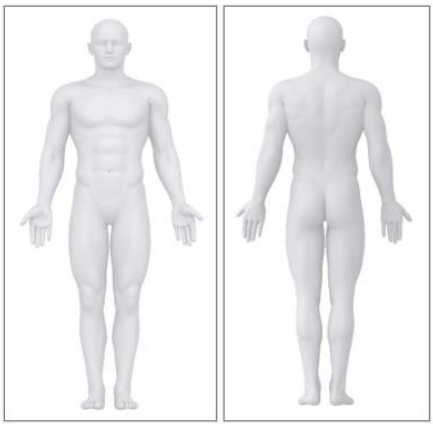
LESSON 1: TERMS & DEFINITIONS

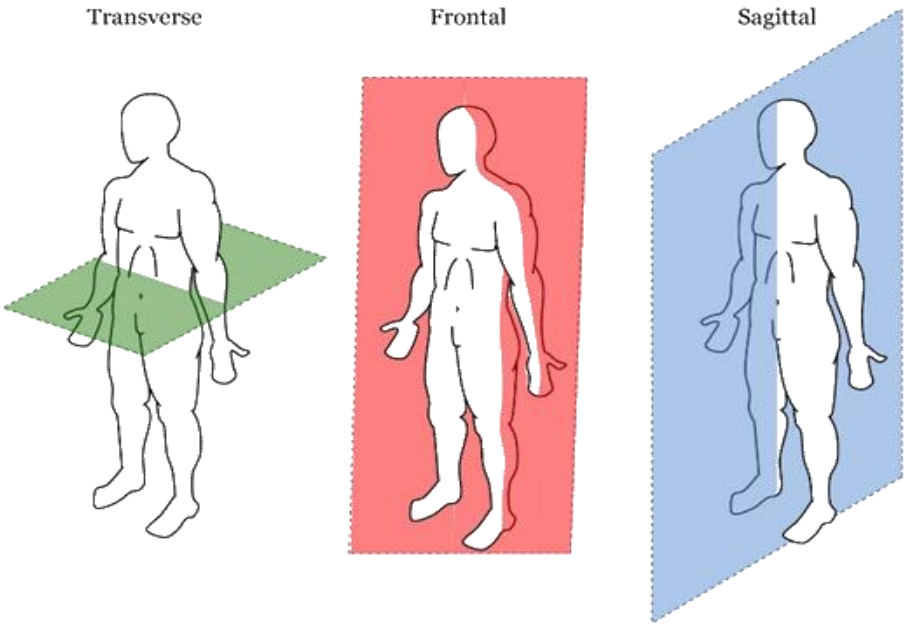
Objectives:
- Understand the importance of studying Anatomy.
- Understand the anatomical position and the planes of motion.
- State the common anatomical terms and definitions.

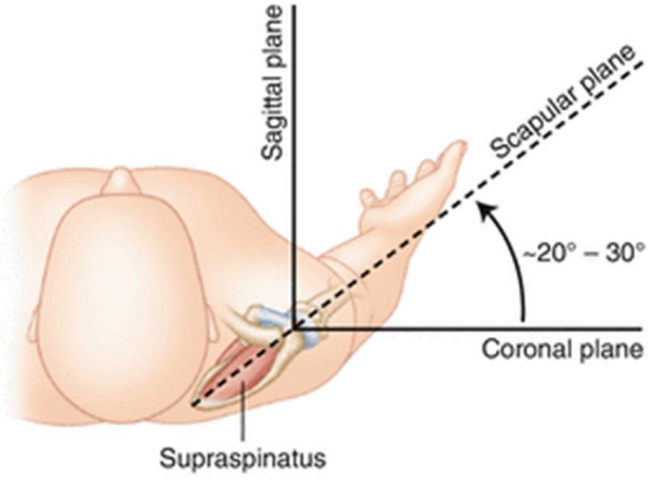
- **WHY STUDYING ANATOMY MATTERS?**



- **ANATOMICAL POSITION & PLANES OF MOTION:** *(Platform: Chapter title: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System – Video title: Anatomical Position & Directional Terms / Planes of Motion)*




Transverse
Frontal
Sagittal


Sagittal plane
Scapular plane
~20° – 30°
Coronal plane
Supraspinatus

- **TERMS AND DEFINITIONS:** *(Platform: Chapter title: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System – Video title Anatomical Position & Directional Terms)*

Term	Definition	Example
Anterior	Front of the body	Anterior Delt
Posterior	Back of the body	Posterior Delt
Superior	Toward the head	Knee is superior to the ankle
Inferior	Away from the head	Ankle is inferior to the knee
Medial	Toward the midline of the body	Medial/Internal Rotation
Lateral	Away from the midline of the body	Lateral/External Rotation
Dorsal	Top of the foot	Dorsiflexion
Plantar	Bottom of the foot	Plantar Flexion
Supine	Lying face up	Lying on the spine
Prone	Lying face down	Prone Leg Curl
Proximal	Extremity close to the midline of the body	Top of the humerus
Distal	Extremity far from the midline of the body	Bottom of the humerus
Superficial	Close to the surface	Rectus Abdominis
Deep	Below the surface	Transverse Abdominis
Arm	Region from the shoulder to the elbow	
Forearm	Region from the elbow to the wrist	
Thigh	Region from the hip to the knee	
Leg (Lower Leg)	Region from the knee to the ankle	
Ipsilateral	On the same side	Side Plank
Contralateral	On the opposite side	Resisted Side Bending
Unilateral	On one side	Single Leg Squat
Bilateral	On both sides	Squat
Palmar	Anterior ventral surface of the hand	
Hyper	Above normal	Hyper lordosis - Hyperglycemia
Hypo	Below normal	Hypoglycemia

- **<u>Terminology Charades:</u>**

In teams, take 5 minutes to go over the terms together; then each member of the team will draw a card with a term and act it out silently using gestures / body movements for their teammates to figure out and earn a point. Let's see which team wins 💪
OH! You only have 20 seconds to guess the correct term…

Consolidation:
By the end of this lesson, I am aware of:
- The importance of Anatomy in motion analysis
- Anatomical Position
- Planes of Motion and their importance
- Anatomical Terms & Definitions

LESSON 2: ANATOMY of THE SKELETAL SYSTEM

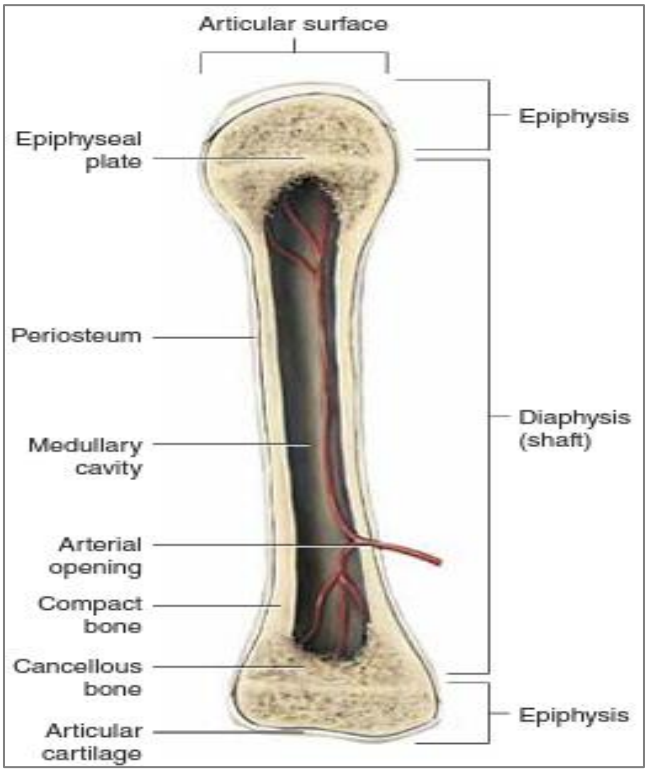
Objectives:

- Understand the function of the skeletal system.
- Understand the composition of long bones.
- Discuss whether it is safe for children to lift weights or not.
- Identify the bones of the skeletal system.

- **FUNCTION OF THE SKELETAL SYSTEM:** *(Platform Chapter title: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System – Video title: Skeletal System - An Overview - Video 1)*

What are the five functions of the skeletal system?

Long Bone Composition



Bone Landmarks: *(Platform Chapter title: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System – Video title: Bone Landmark)*

LANDMARK	DEFINITION
Tuberosity	Large bump on a bone
Process	Projection from a bone
Tubercle	Smaller bump
Spine or Spinous process	Longer and thinner projection
Condyles	Large bony knob at the end of a long bone
Epicondyles	Smaller bony knobs just above the condyles
Fossa	Smooth, hollow surface on a bone
Facet	Smaller and flatter smooth surface
Notch	Area on a bone that appears to be cut out and allows for the passage of other structures
Foramen	Similar as the notch but appear as a hole

What are the landmarks for and do we need to memorize them?

Bone Growth and Exercise *(Platform Chapter title: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System – Video title: Bone Growth and Exercise)*

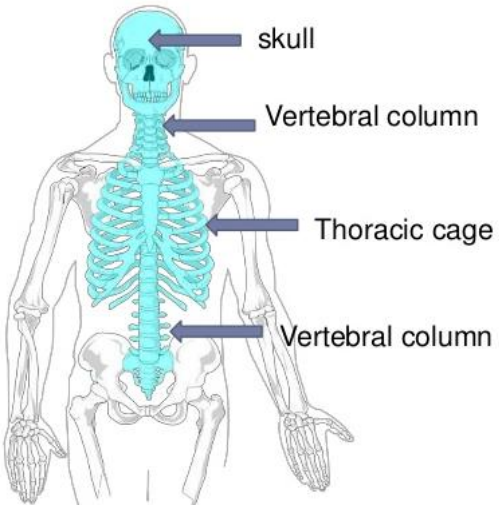
GROUP ACTIVITY

What do you know about bone remodeling, Wolf law and bone mineral density? And do you think it is safe for children to lift weights?

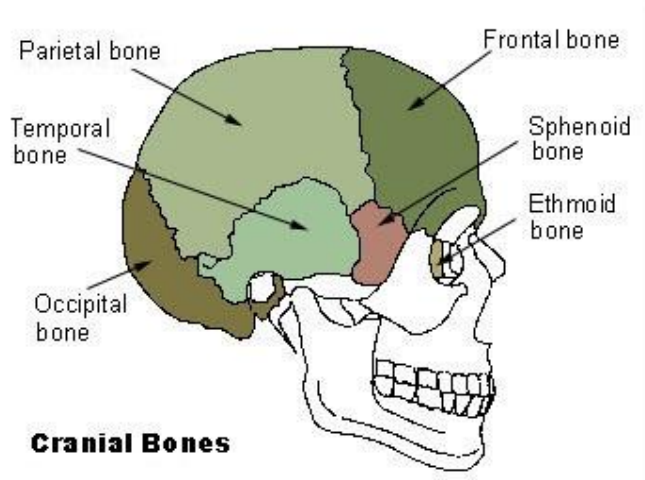


Human Skeleton (Axial and Appendicular Systems): *(Platform Chapter title: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System – Video title: Bone to Bone Approach - Axial Skeleton)*

- **Axial Skeleton:**

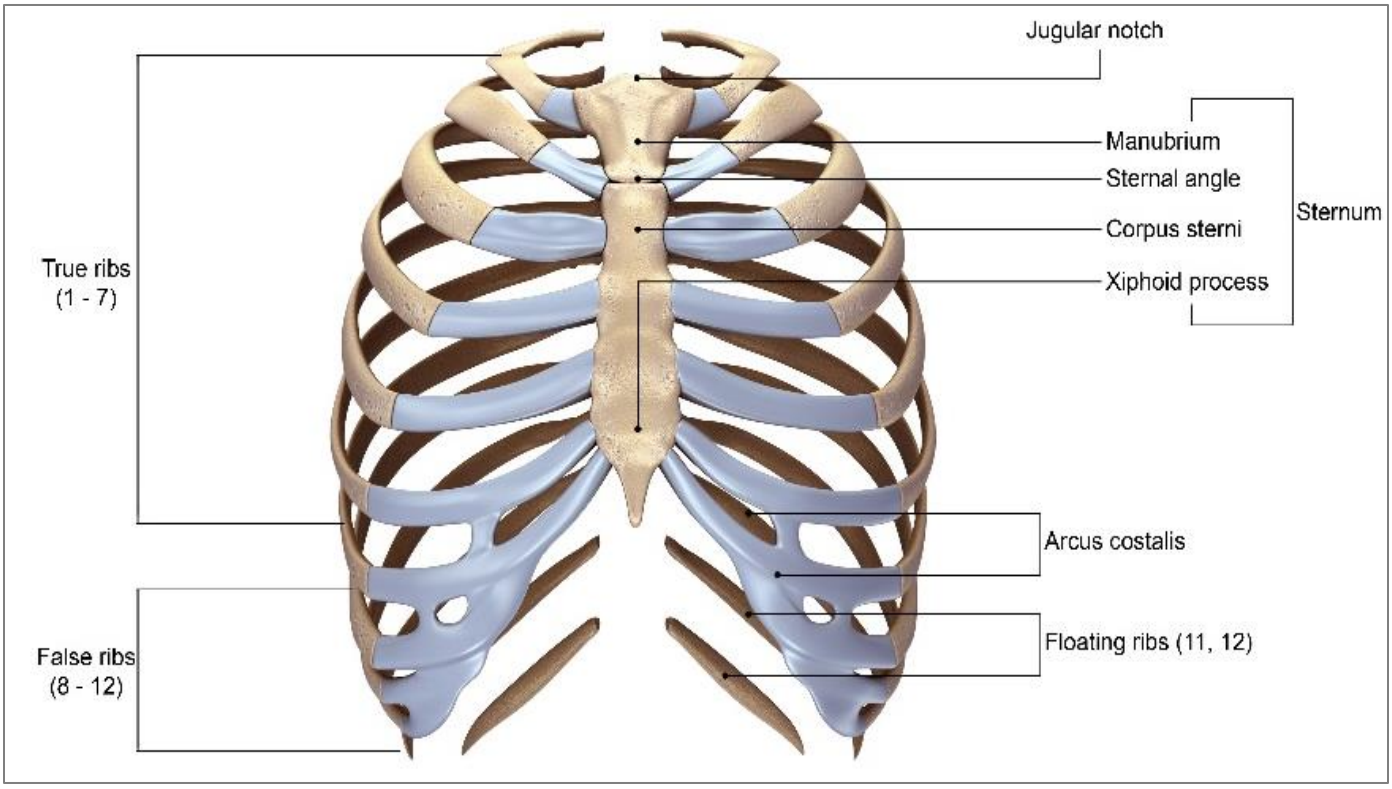


- **Skull:**



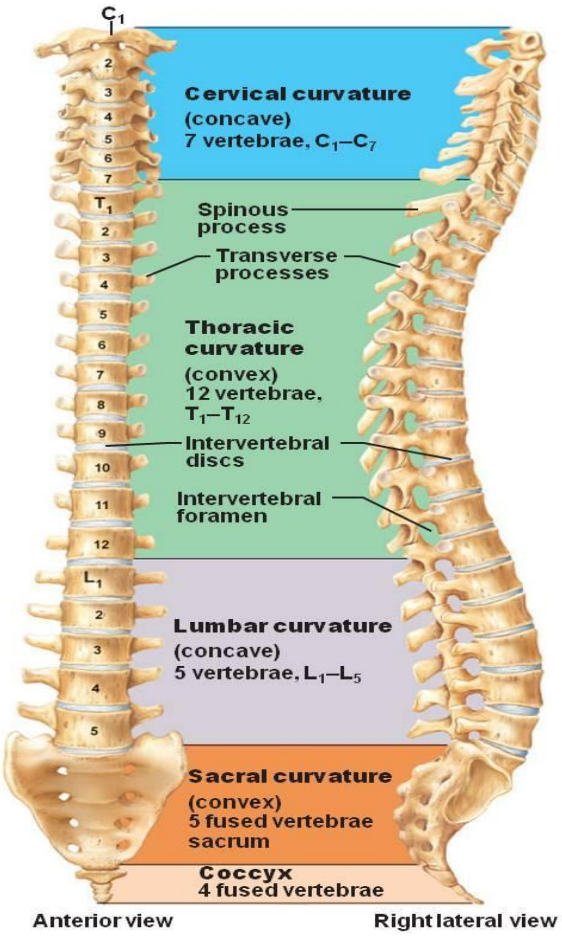
What do we need to know?

- **Thorax:**



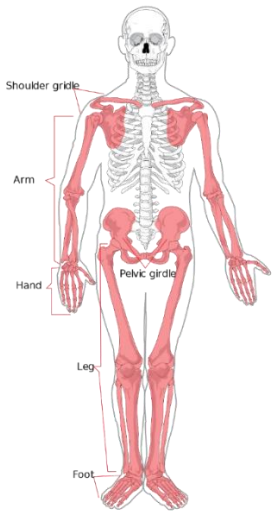
What do we need to know?

- **Spine:**

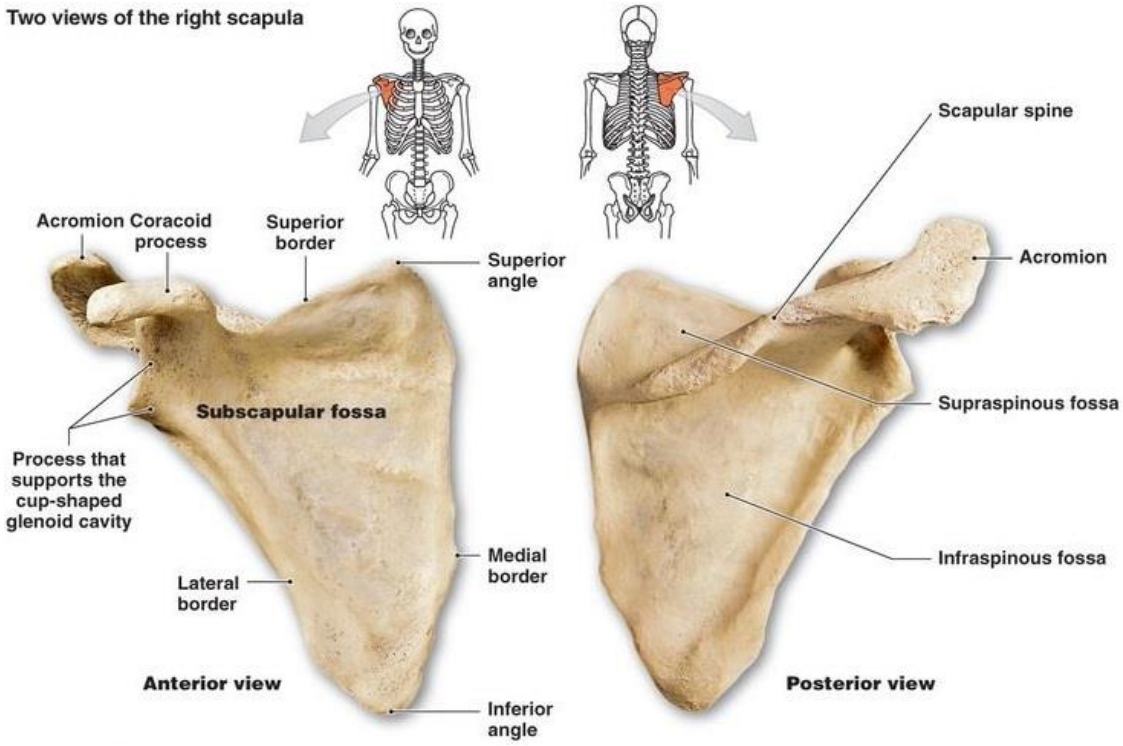


What do we need to know?

- **Appendicular Skeleton:**

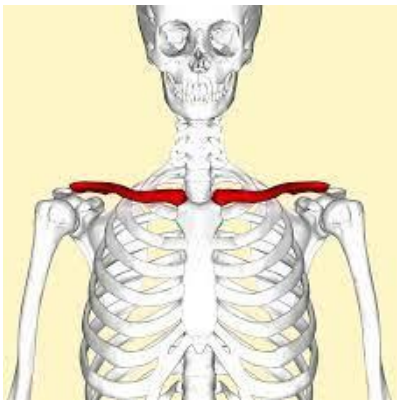


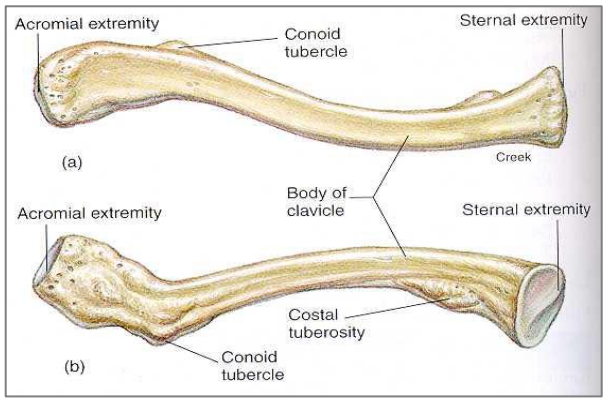
- **Scapula:**



What do we need to know?

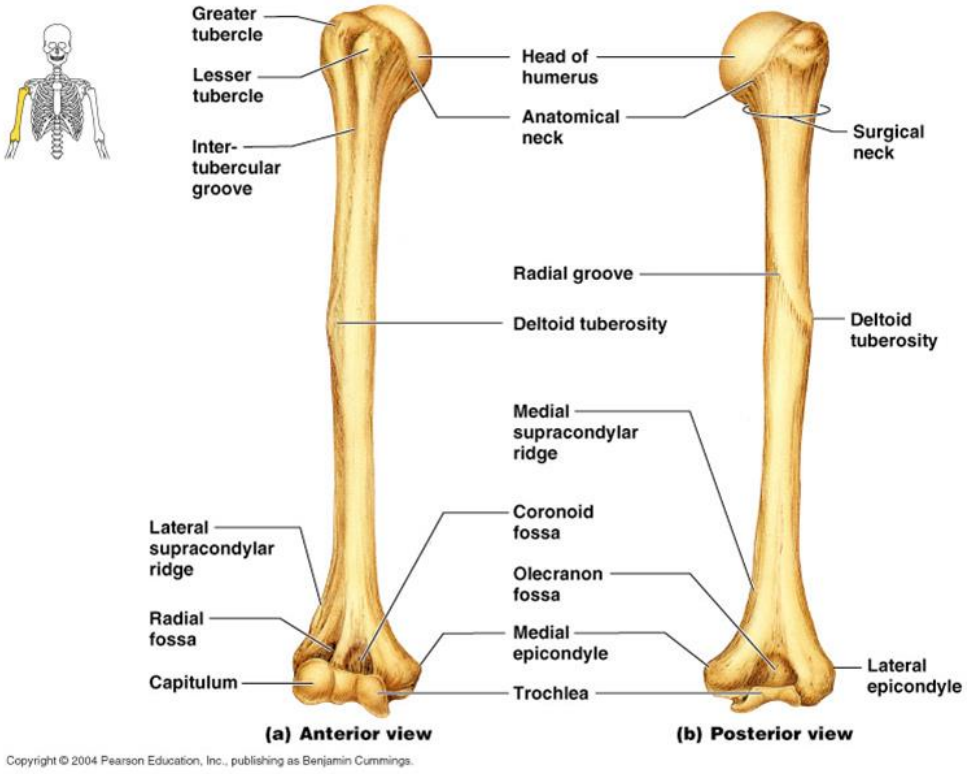
- **Clavicle:**





What do we need to know?

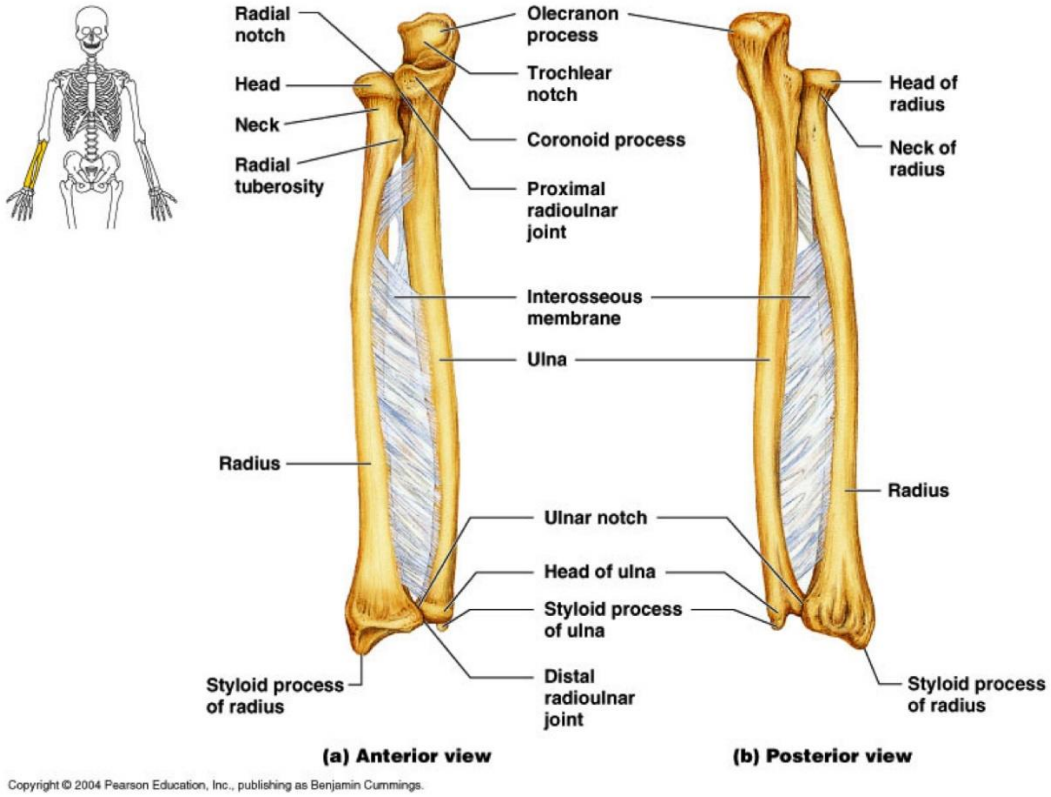
- **Humerus:**



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What do we need to know?

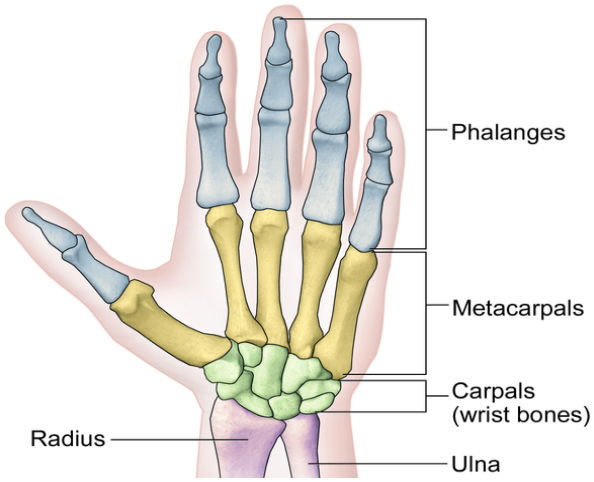
- **Radius & Ulna:**



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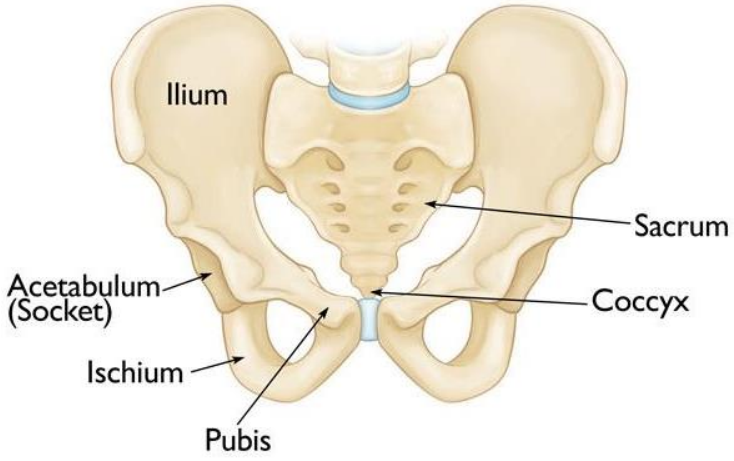
What do we need to know?

- **Hand bones:**



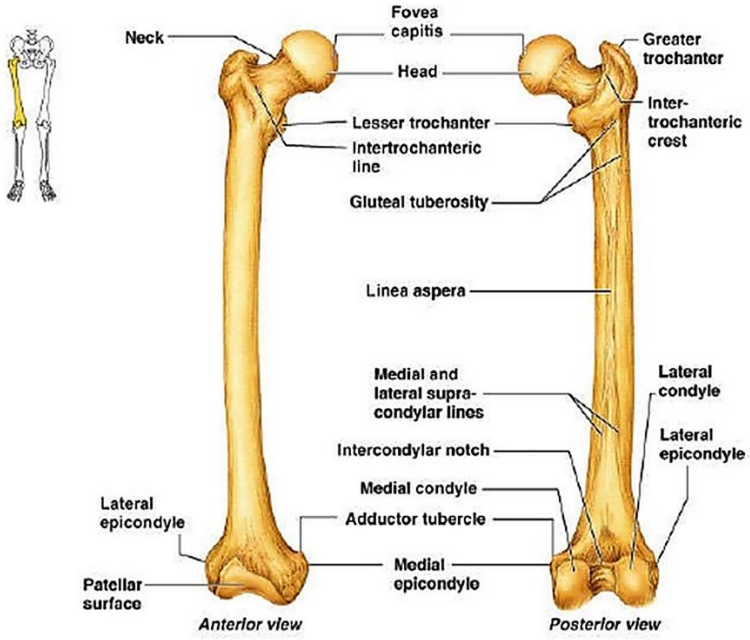
What do we need to know?

- **Pelvis:**



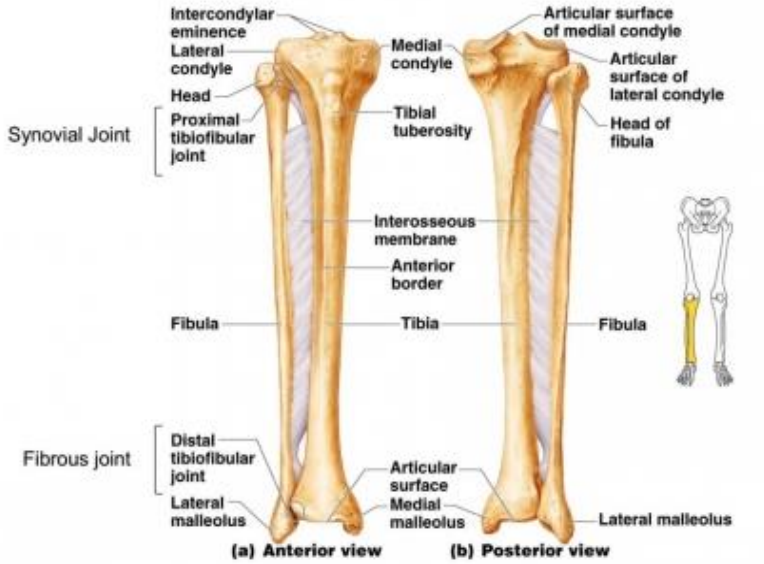
What do we need to know?

- **Femur:**



What do we need to know?

- **Tibia & Fibula:**



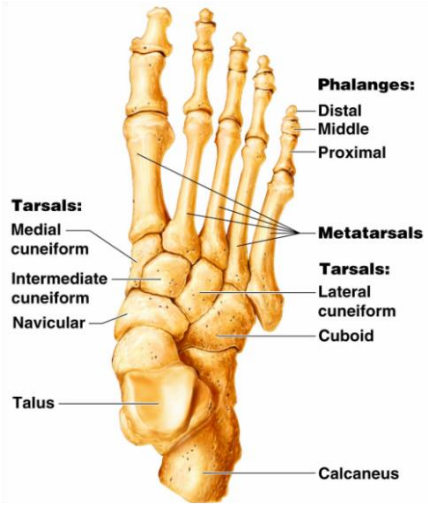
What do we need to know?

- **Patella:**



What do we need to know?

- **Feet bones:**



What do we need to know?

Is there any difference between Male vs Female Skeleton?

Ready for some challenge? Let's see who grabs the bone in question first



Consolidation:

By the end of Session 2, I am aware of:

- Functions of the skeletal system.
- Composition of long bones.
- Safety for children to lift weights.
- Bones of the skeletal system.

Post-Session 2 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 2.

Pre-Session 3 Preparation:

-Watch all videos of the Articular System from the Chapter titled: Week 2: Anatomy of the Articular System + Muscular System

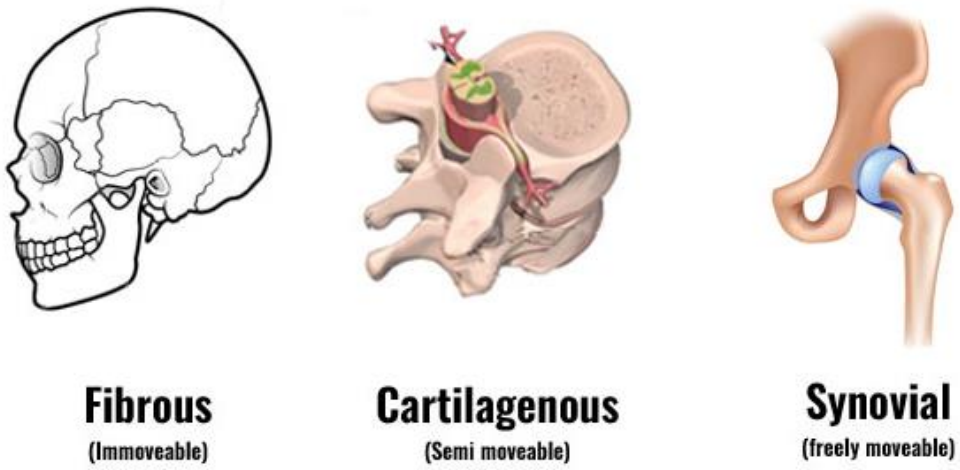
Session 3: Functional Anatomy – Articular System

LESSON 1: ANATOMY of THE ARTICULAR SYSTEM

Objectives:

- Understand the different types of joints.
- Know the main joints of the body.
- Discuss the structure of a synovial joint.
- Know the benefits of exercise on the joint's health.

- **TYPES OF JOINTS:** *(Platform: Chapter title: Week 2: Week 2: Anatomy of the Articular System + Muscular System – Video title: Intro to the Articular System)*

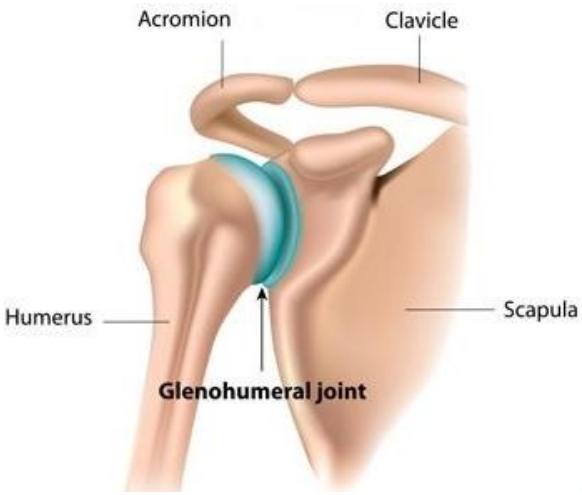


Structural type	Description	Example
Fibrous (Synarthrosis) joints	- No space between bones - Bones held together by thick connective tissue, rich in collagen	Skull Tibiofibular joint Joints between teeth and sockets
Cartilaginous (Amphiarthrosis) joints	- No space between bones - Bones held together by cartilage	Joints between ribs and sternum Pubic symphysis at the level of the pelvis

Structural type	Description	Example
Synovial (Diarthrosis) joints	- Space is present between bones - Bones held together by articular capsule - Ligaments and muscles provide support needed - Provide the greatest Range of Motion (ROM)	Hand Wrist Elbow Knee Shoulder Hip Ankle Foot

What are the main Joints of the body? Can you guess their type?

- **Glenohumeral vs Scapulothoracic Joints:**





GROUP ACTIVITY

Pair Activity: Guess which shoulder joint(s) is engaged in each of the below exercises

Front Raise?

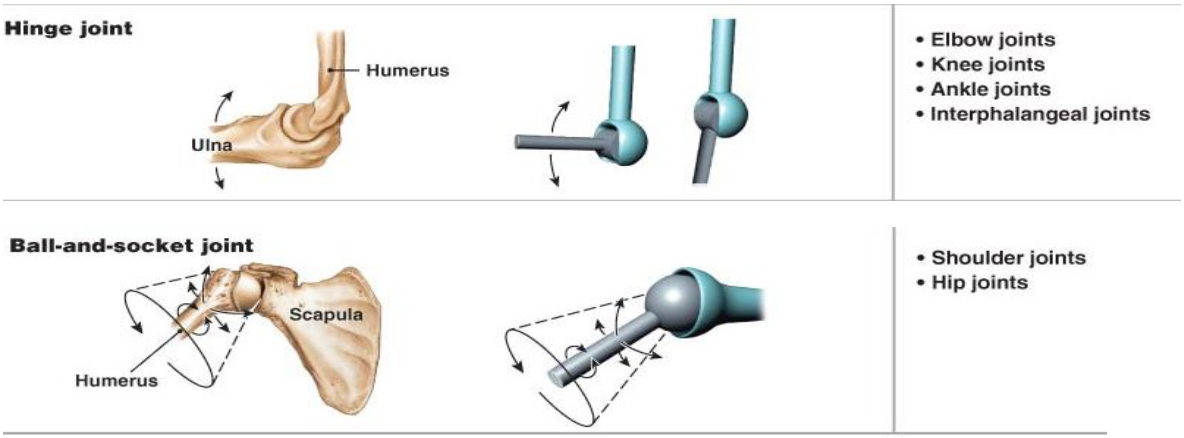
Lateral Raise?

Shrugs?

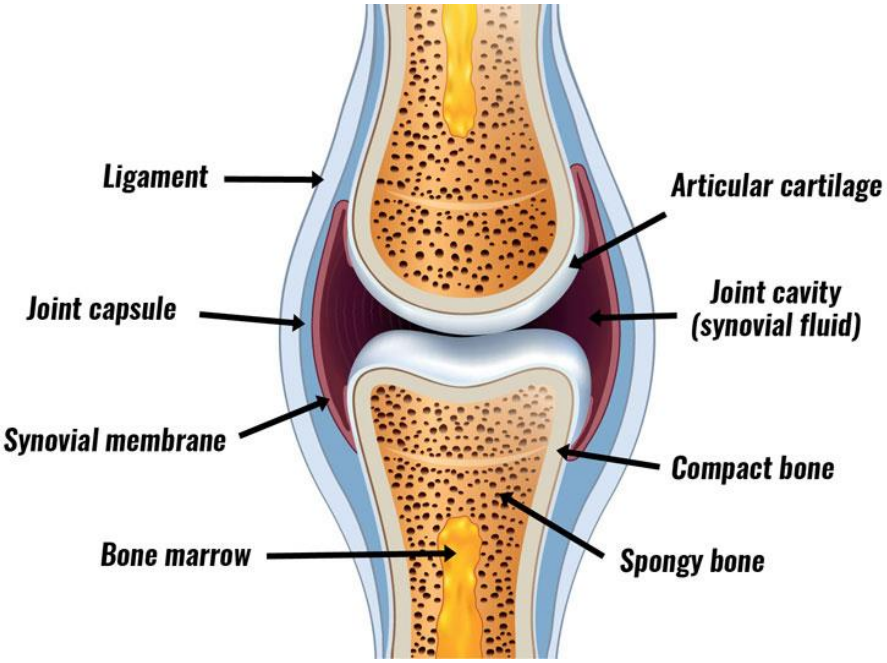
Shoulder Press?

Lat-Pulldown?

What do you need to know about the types of Synovial Joints?

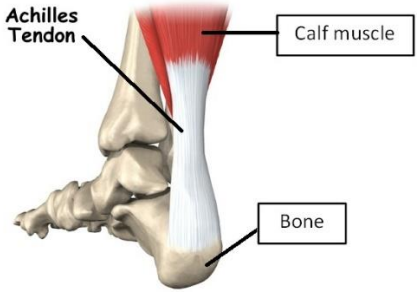


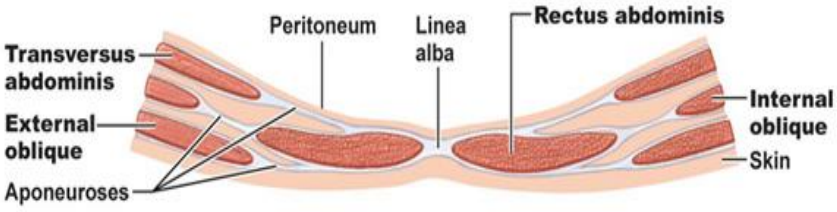
- **STRUCTURE OF SYNOVIAL JOINTS:** *(Platform: Chapter title: Week 2: Week 2: Anatomy of the Articular System + Muscular System – Video title: Connective Tissues)*



Connective Tissues: Ligaments vs Tendons vs Aponeurosis

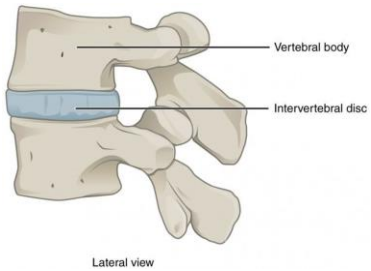






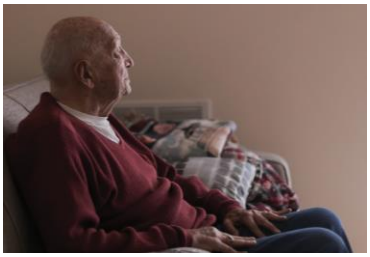
Bursa vs Synovial Membrane?

Cartilage vs Articular cartilage?





Effect of Aging Without Exercising vs Effect of Exercise on our Joints Health:





Which structures affect Joint Stability and Joint Mobility?



Based on the above, let's explain together why our shoulders are more prone to injuries compared to our hips; while our shoulders mobility is higher than our hips

Consolidation:

By the end of this lesson, I am aware of:

- Different types of joints.
- Main joints of the body.
- Structure of a synovial joint.
- Benefits of exercise on the joint's health.

LESSON 2: FUNDEMENTAL JOINT MOVEMENTS

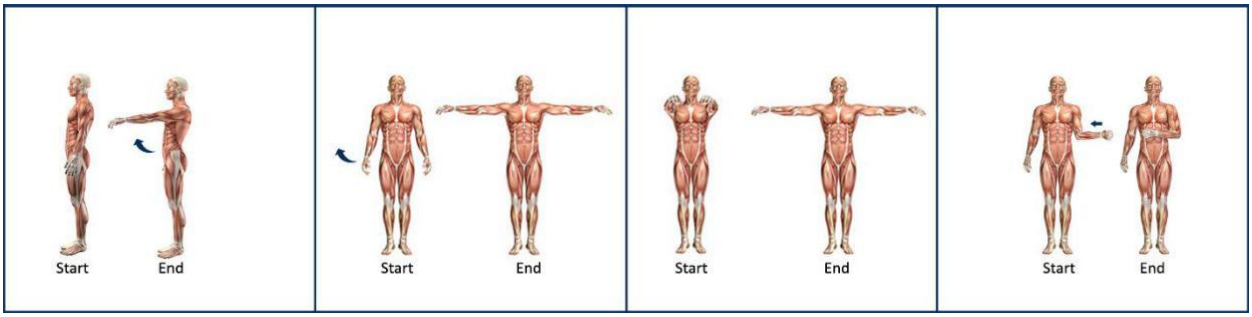
Objectives:
- Describe fundamental joints movements

- **FUNDAMENTAL JOINT MOVEMENTS:** *(Platform: Chapter title: Week 2: Week 2: Anatomy of the Articular System + Muscular System – Video title: Movements Occurring at the Main Joints of the Body)*

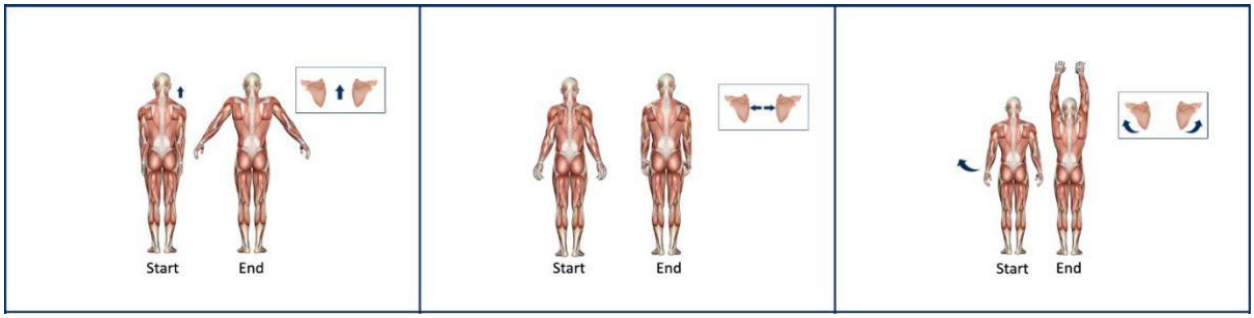
PLANE	MOTION	DESCRIPTION
Sagittal	***Flexion***	When the joint angle decreases
	Extension	When the joint angle increases
	Dorsiflexion	Decreasing joint angle at the ankle
	Plantarflexion	Increasing joint angle at the ankle
Frontal	***Abduction***	Movement away from the midline of the body
	Adduction	Movement toward the midline of the body
	Elevation	When the scapula moves upward
	Depression	When the scapula moves downward
	Protraction	When the scapula moves forward (laterally and anteriorly)
	Retraction	When the scapula moves backward (medially and posteriorly)
	Upward rotation	When the scapula rotates and moves upward (this movement happens when you are raising the arm overhead), e.g., Overhead Press
	Downward rotation	When the scapula rotates and moves downward (this movement happens when you are lowering the arm), e.g., Lat Pulldown
	Lateral flexion	When bending the spine to either side
	Inversion	When turning the sole of the foot inward or medially (movement occurs at the ankle)
	Eversion	When turning the sole of the foot outward or laterally (movement occurs at the ankle)
Transverse	***Medial/Internal rotation***	Rotation toward the midline of the body (internal rotation)
	Lateral/External rotation	Rotation toward the side of the body or away from the midline (external rotation)
	Transverse rotation	Rotation at the spine, right or left
	Pronation	Movement occurring at the elbow joint toward the prone position (i.e., Palms facing down)
	Supination	Movement occurring at the elbow joint toward the supine position
	Transverse abduction	Abduction at the transverse plane, also known as horizontal extension
	Transverse adduction	Adduction at the transverse plane, also known as horizontal flexion

GROUP ACTIVITY

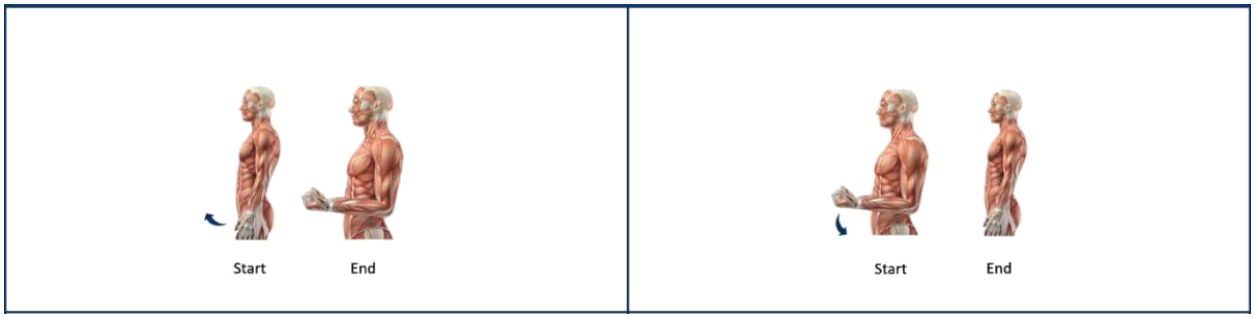
Guess the planes of motion, joints, and movements of the below pics:



Plane:	*Plane:*	*Plane:*	*Plane:*
Joint:	*Joint:*	*Joint:*	*Joint:*
Movement:	*Movement:*	*Movement:*	*Movement:*



Plane:	*Plane:*	*Plane:*
Joint:	*Joint:*	*Joint:*
Movement:	*Movement:*	*Movement:*



Plane:	*Plane:*
Joint:	*Joint:*
Movement:	*Movement:*



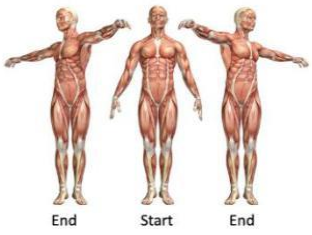


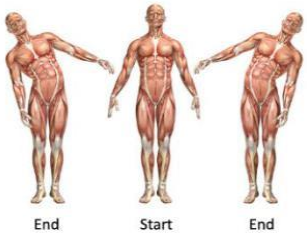
Plane:	*Plane:*
Joint:	*Joint:*
Movement:	*Movement:*





Plane:	*Plane:*
Joint:	*Joint:*
Movement:	*Movement:*





Plane:	*Plane:*
Joint:	*Joint:*
Movement:	*Movement:*

Plane:	*Plane:*
Joint:	*Joint:*
Movement:	*Movement:*

Plane:	*Plane:*
Joint:	*Joint:*
Movement:	*Movement:*

Plane:	*Plane:*
Joint:	*Joint:*
Movement:	*Movement:*

Consolidation:

By the end of Session 3, I am aware of:



- Different types of joints.
- Main joints of the body.
- Structure of a synovial joint.
- Benefits of exercise on the joint's health.
- Main movements of the body



Post-Session 3 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 2.

Pre-Session 4 Preparation:



-Watch all videos of the Muscular System from the chapter titled Week 2: Anatomy of the Articular System + Muscular System

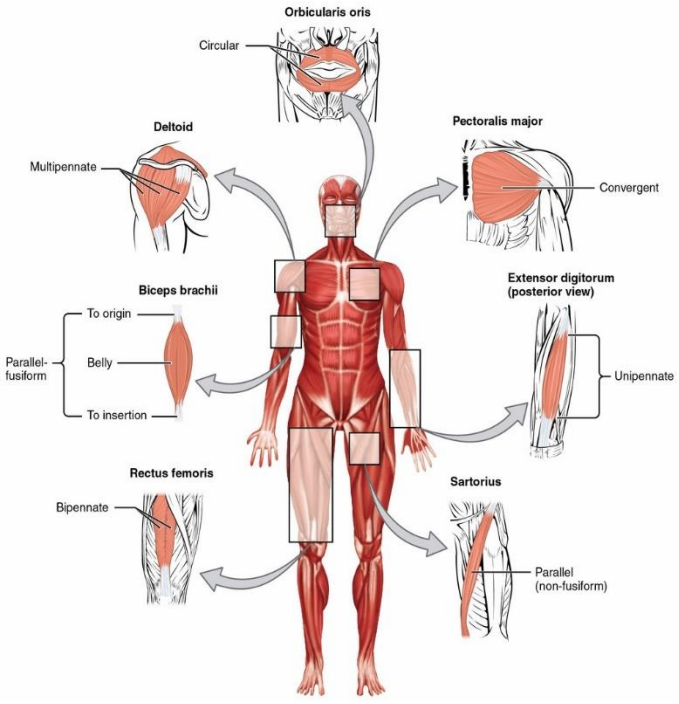
Session 4: Functional Anatomy – Muscular System

LESSON 1: FUNCTION OF THE SKELETAL MUSCLES

Objectives:
- Understand the functions of the skeletal muscles

What are the functions of the skeletal muscles?

What do we need to know about the muscle fiber arrangement?

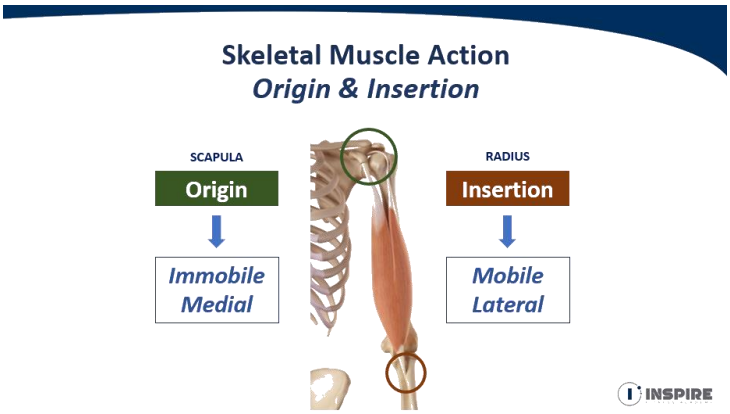


LESSON 2: ANATOMY of THE MUSCULAR SYSTEM

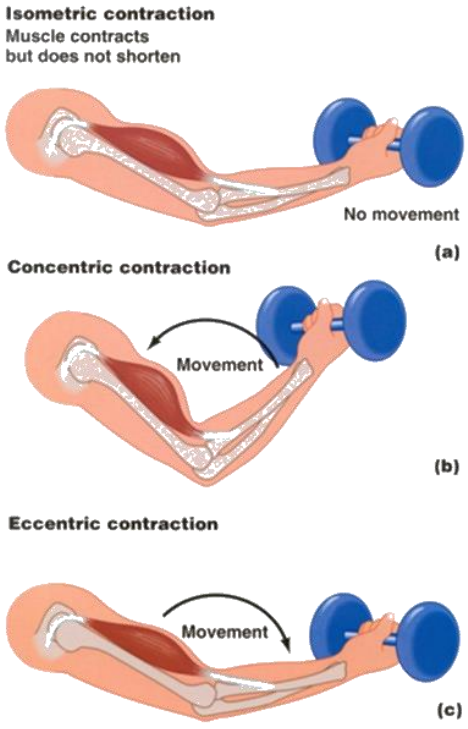
Objectives:

- Differentiate between concentric, eccentric and isometric contractions.
- Review the origin, insertion and main actions of the lower body muscles.

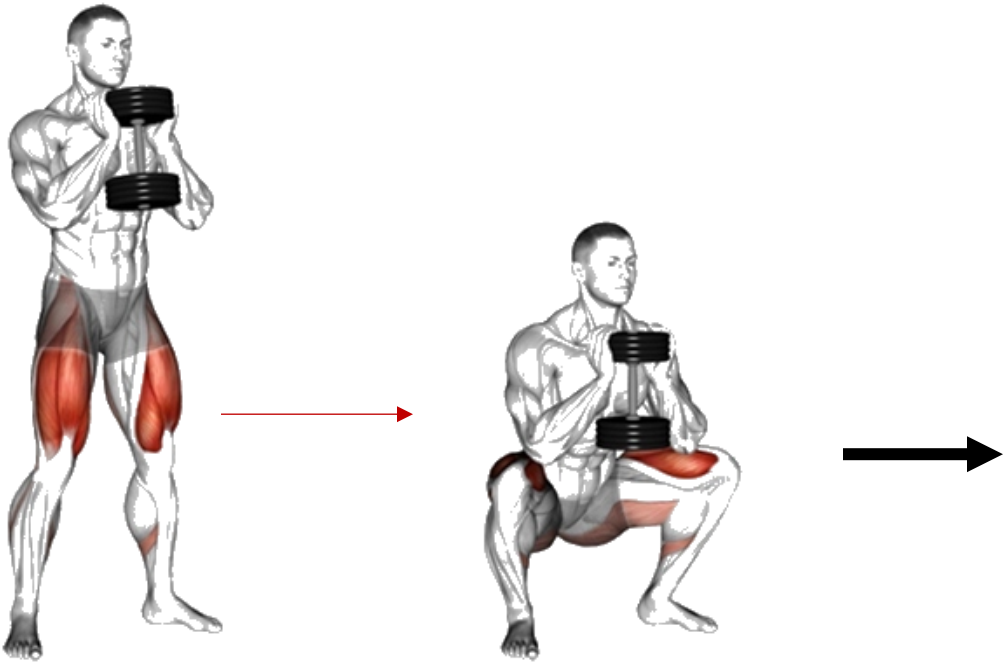
Differentiate between Muscles Origin and Insertion: *(Platform: Chapter title: Week 2: Anatomy of the Articular System + Muscular System – Video title: Introduction to the Muscular System)*

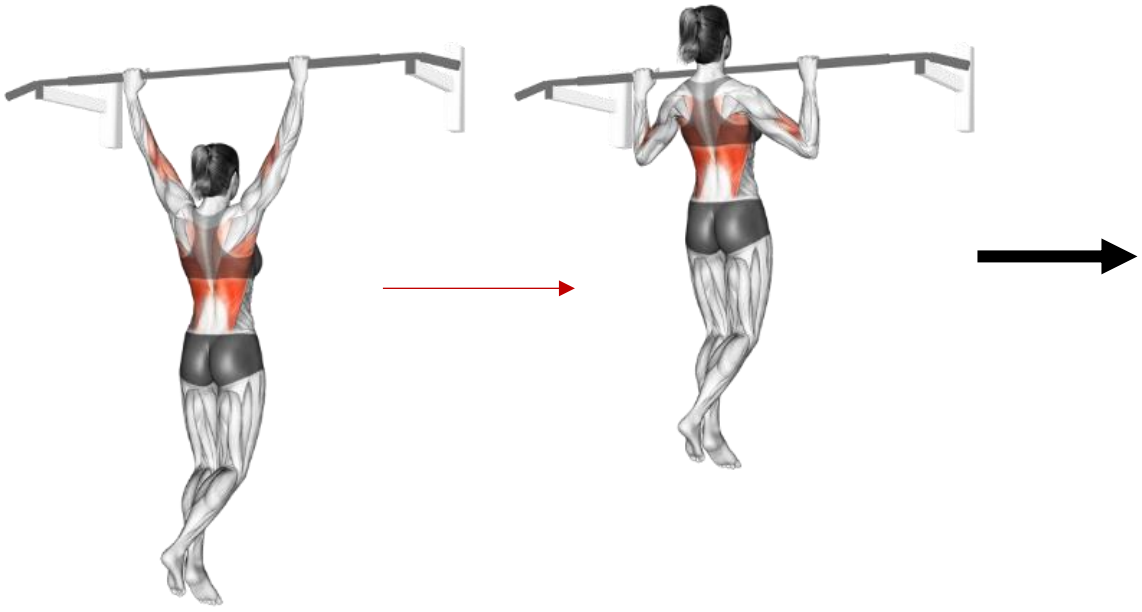


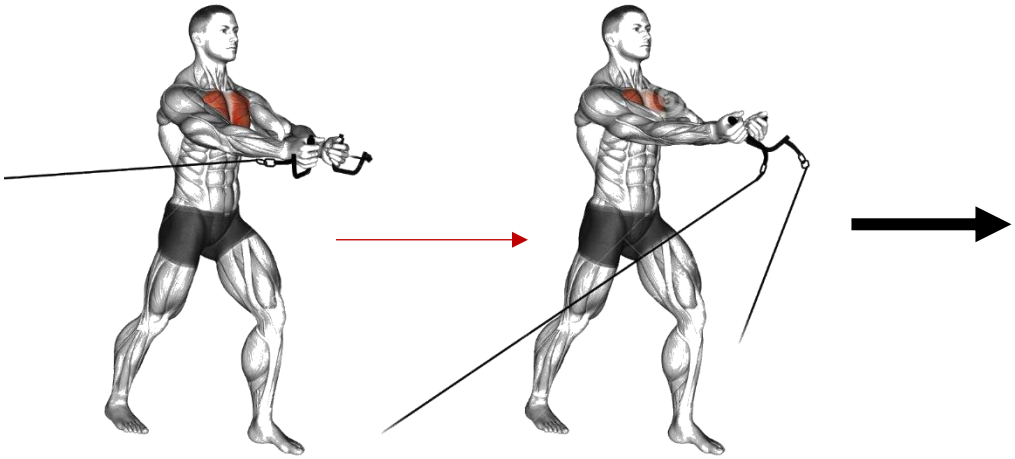
Concentric, Eccentric and Isometric contractions *(Platform: Chapter title: Week 2: Anatomy of the Articular System + Muscular System – Video title: Introduction to the Muscular System)*



Guess what type of contraction is happening in each movement:









Consolidation:

By the end of these lessons, I am aware of:

- Difference between Origin and Insertion.
- Difference between Concentric, Eccentric and Isometric contractions.

LESSON 3: UPPER BODY MUSCLES ORIGIN, INSERTION AND ACTION

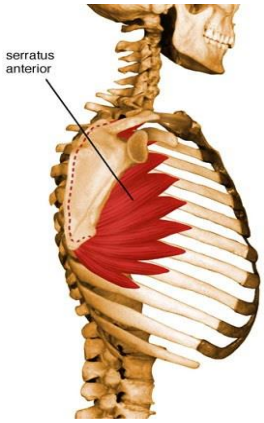
Objectives:
-Know the upper body muscles origin and insertion, to better understand their actions.

- **Pectoralis Minor:**



Origin:
Insertion:
Actions:

- **Serratus Anterior:**



Origin:
Insertion:
Actions:

- **Levator Scapulae:**



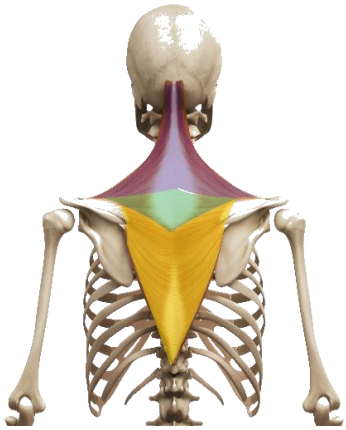
Origin:
Insertion:
Actions:

- **Rhomboids:**



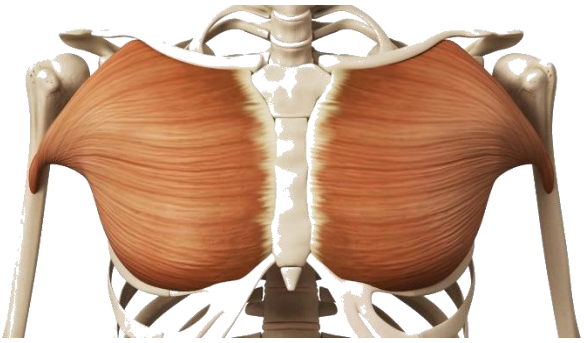
Origin:
Insertion:
Actions:

- **Trapezius: Upper – Middle – Lower**



Origin:
Insertion:
Actions:

- **Pectoralis Major:**



Origin:

Insertion:

Actions:

- **Deltoid: Anterior – Middle – Posterior**

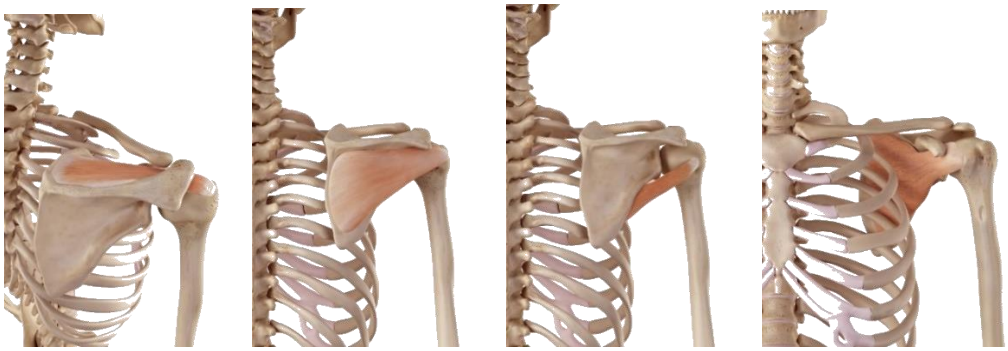


Origin:

Insertion:

Actions:

- **Rotator Cuffs: S-I-T-S**

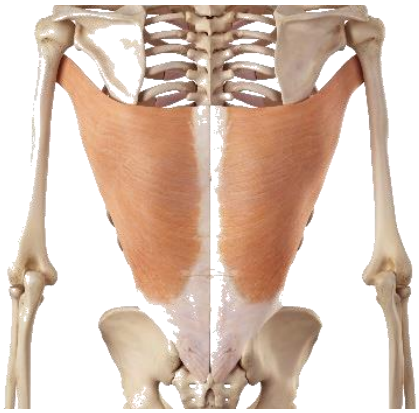


Origin:

Insertion:

Actions:

- **Latissimus Dorsi:**



Origin:

Insertion:

Actions:

- **Teres Major**:



Origin:

Insertion:

Actions:

- **Elbow Flexors:**

Biceps Brachii **Brachialis** **Brachioradialis**





Origin:
Insertion:
Actions:

- **Triceps**:



Origin:
Insertion:
Actions:

Let's have some fun to reinforce learners' understanding of joints, movements, and muscles through a structured card-based game.

Consolidation:

By the end of Session 4, I am aware of:

- Functions of the skeletal muscles
- Difference between concentric, eccentric and isometric contractions.
- Upper body muscles origin, insertion and action

Post-Session 4 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 5.

Pre-Session 5 Preparation:

-Watch all videos of the Muscular System from the Chapter titled: Week 3: Anatomy of the Muscular System + Part 1 pf the Biomechanics of Exercise and Resistance Training.

Session 5: Functional Anatomy – Muscular System

LESSON 1: MID-SECTION & LOWER BODY MUSCLES ORIGIN INSERTION & ACTION

Objectives:
-Know the mid-section and lower body muscles origin, insertion and actions.

- **Abdominal Muscles**:

Rectus Abdominis:



Origin:
Insertion:
Actions:

Internal Obliques:



Origin:
Insertion:
Actions:

External Obliques:



Origin:
Insertion:
Actions:

Transverse Abdominis:



Origin:
Insertion:
Actions:

- **Back Muscles**:

Quadratus Lumborum:

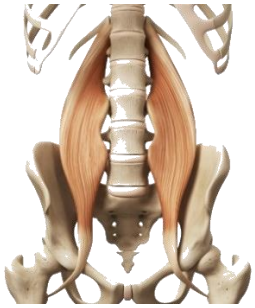


Erector Spinae:

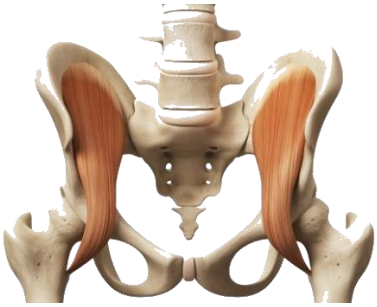


Origin:
Insertion:
Actions:

- **Iliopsoas - Iliacus & Psoas:**



Origin:
Insertion:



Origin:
Insertion:

Actions:

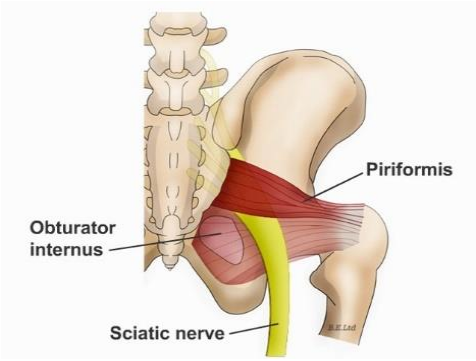
- **Gluteus Maximus, Medius and Minimus:**



Origin:
Insertion:
Actions:

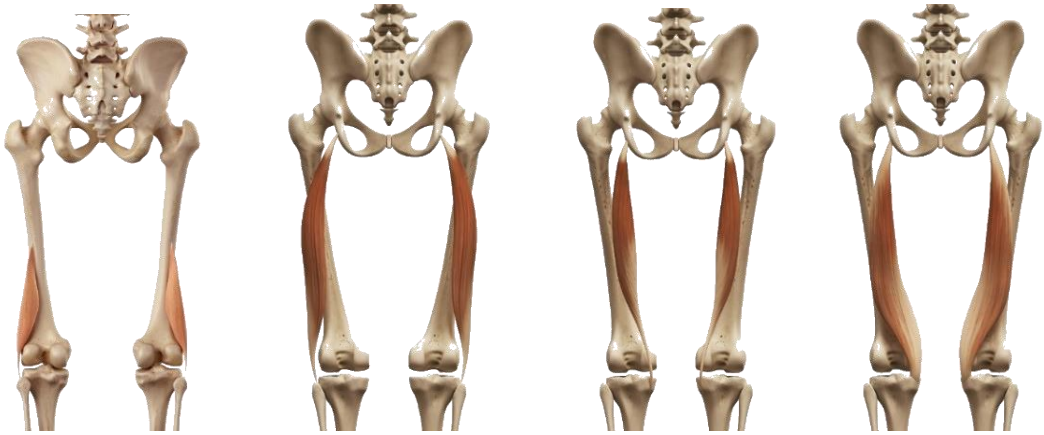
- **Piriformis**





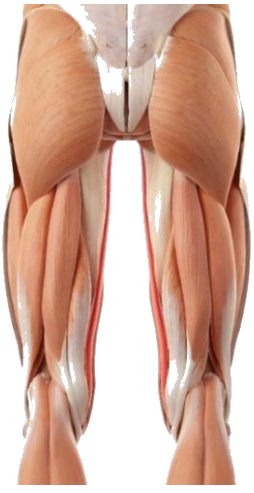
What do we need to know about the piriformis?

- **Hamstrings:**

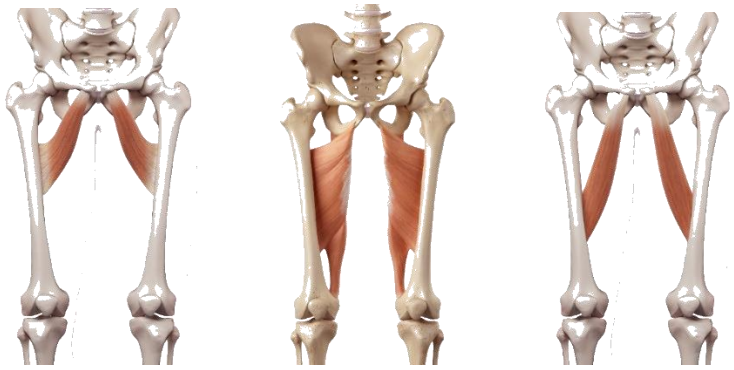


Origin:
Insertion:
Actions:

Label each of the Hamstring's muscles on this image:

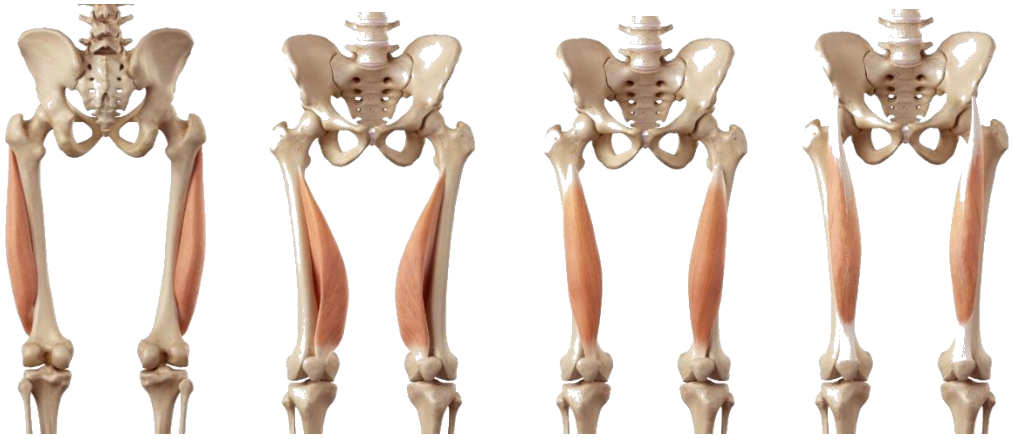


- **Hip Adductors – Adductor Brevis, Magnus and Longus:**



Origin:
Insertion:
Actions:

- **Quadriceps – Vastus Lateralis, Medialis, Intermedius and Rectus Femoris:**



Origin:
Insertion:
Actions:

Label each of the Quadricep's muscles on this image:



- **Tensor Fascia Latae:**



Origin:
Insertion:
Actions:

- **Calves – Gastrocnemius & Soleus:**

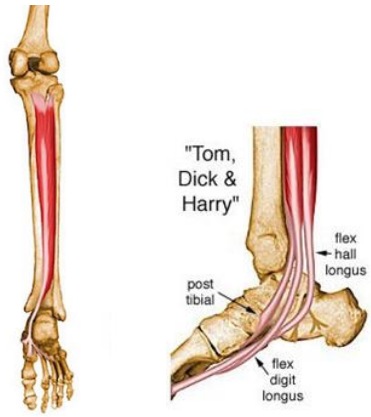


Origin:
Insertion:
Actions:

Provide an exercise focusing on gastrocnemius muscle engagement:

Provide an exercise focusing on soleus muscle engagement:

- **Tibialis Posterior**



Origin:
Insertion:
Actions:

- **Tibialis Anterior**



Origin:
Insertion:
Actions:

- **Peroneus Longus**



Origin:
Insertion:
Actions:

Let's end our session with some fun, and build an anatomical puzzle

Consolidation:
By the end of Session 5, I am aware of:
- Mid-section and Lower body muscles origin, insertion and action

Post-Session 5 Tasks:
-Review the material covered during the session.
-Prepare a brief recap of the key lessons covered to be presented in Session 6.

Pre-Session 6 Preparation:
-Watch all videos of the Biomechanics videos from the Chapter titled: Week 3: Anatomy of the Muscular System + Part 1 of the Biomechanics of Exercise and Resistance Training.

Session 6: Biomechanics

LESSON 1: IMPORTANCE OF BIOMECHANICS

Objectives:
-Understand the importance and purpose of Biomechanics for personal coaches

(Platform: Chapter title: Week 3: Anatomy of the Muscular System + Part 1 of the Biomechanics of Exercise and Resistance Training.)

What is biomechanics and how would it help you as a coach?

Can we say more / heavier is always better?

Consolidation:
By the end of this lesson, I am aware of:
-The importance of biomechanics, and why it should be studied.

LESSON 2: MECHANICAL LAWS OF MOTION

Objectives:
-Understand the three laws of motion and link them to practice in a gym / workout setting

1ST LAW: LAW OF INERTIA

An object at rest stays at rest, and an object in motion stays in motion—unless acted upon by an external force

Give a gym-related practical application of this law (Hint: Momentum)

2ND LAW: LAW OF ACCELERATION

The force acting on an object is equal to the mass of the object times its acceleration

$$Force = m \times a$$

$$Unit: Newton\ (N)\ where\ 1\ N = 1\ kg \times m/s^2$$

Give a gym-related practical application of this law (Hint: Sled)

3RD LAW: LAW OF REACTION

For every action, there is an equal opposite reaction



Give a gym-related practical application of this law (Hint: Resistance Training)



Consolidation:

By the end of this lesson, I am aware of:

- The three laws of motion and how to use them practically in a gym setting.

LESSON 3: TERMS & DEFINITIONS

Objectives:
-Gain a general overview of the main biomechanical terms and concepts.

What is the difference between Power and Strength?

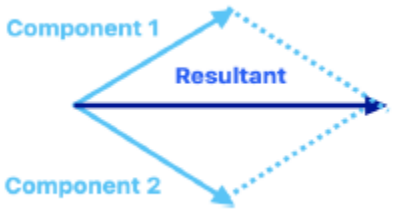
- **VECTORS HAVE MAGNITUDE AND DIRECTION**

To better understand vectors importance in biomechanics and movement analysis lets **discuss** the below cases:

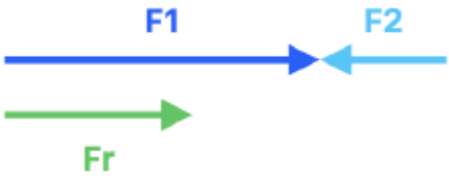
If Mounir and Samir are running at their maximum speed which is 5km per hour; who is faster?

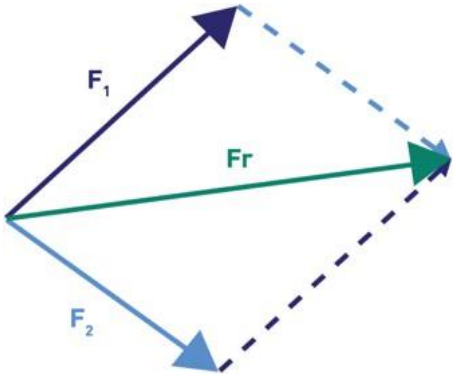
Now if you are told that Mounir is running uphill while Samir is running downhill; who do you think is faster now?

- **VECTORS IN PRACTICE: 1-Understanding Forces:**









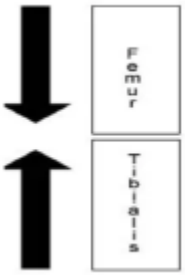
GROUP ACTIVITY

Pair Activity: Mix and Match:



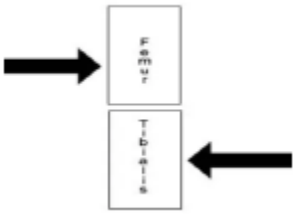
Compressive Force

Nordic Curls



Tensile Force

Injury Risk



Shear Force

Squats

- **VECTORS IN PRACTICE: 2-Different Types of Loads:**

Gravitational Load	
Elastic Load	
Cable Resistance	
Machine-Based	

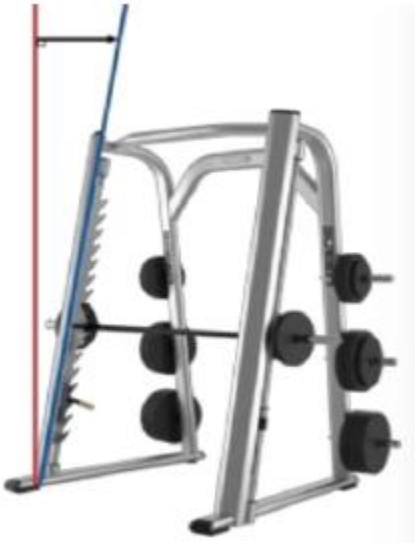
- **VECTORS IN PRACTICE: 3-Free Weights vs Smith Machine:**



Pair Activity: What is the difference between performing an exercise using free weights or using a smith machine? Answer the question while using the following words: Injury Rehab and Post-rehab, Beginners, Stability, Isolation, Muscular Effort, Vectors, and Load type.

GROUP ACTIVITY

Pair Activity: How can different foot placement engage our quadriceps or glutes more, and why?

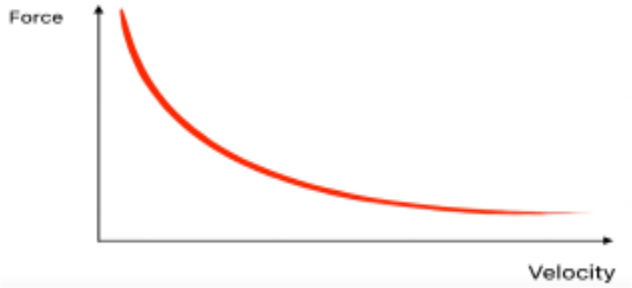


GROUP ACTIVITY

Pair Activity: What about Leg Press vs Barbell Squats? Answer the question while using the following words: Beginners, Stability, Isolation, Muscular Effort, Vectors, and Load type.



- **FORCE-VELOCITY CURVE:**



Let's See How High Can You Jump!

We need three volunteers to jump as high as they can, then repeat the same drill while holding 10kgs in each hand. Based on what you saw, and the above graph; why we cannot jump high when holding weights?

Consolidation:

By the end of Session 6, I am aware of:

-The three laws of motion along with their practical application.

-The difference between Strength and Power.

-The importance of vectors for understanding different types of forces.

-The importance of vectors for understanding different types of loads.

-The importance of vectors for comparing between free weights and machines.

-Analyze the Force-Velocity curve

Post-Session 6 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 7.

Pre-Session 2 Preparation:

-Watch all Biomechanics videos of the Chapter titled: Week 4: Part 2 of the Biomechanics of Exercise and Resistance Training + Physiology of the Muscular System

Session 7: Biomechanics – Continued

LESSON 1: MUSCULAR TORQUE

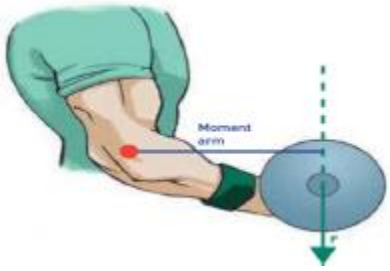


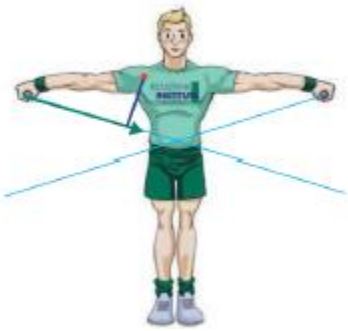
Objectives:
-Analyze exercises and exercise variations based on the concepts of Torque and Moment Arm.

- **TERMS & DEFINITIONS:**

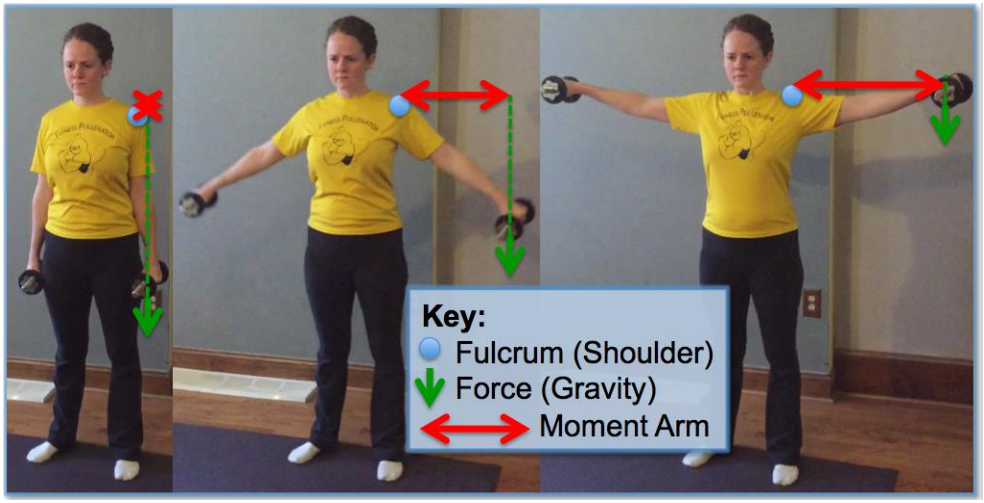
Moment or Torque	
External Moment Arm	
Axis of Rotation	
Line of Action	
Internal Moment Arm	
Lever Arm	
Load	
Resistance	

- **How to Visualize the Moment Arm in an exercise:**





- **Let's analyze the lateral raise exercise together:**



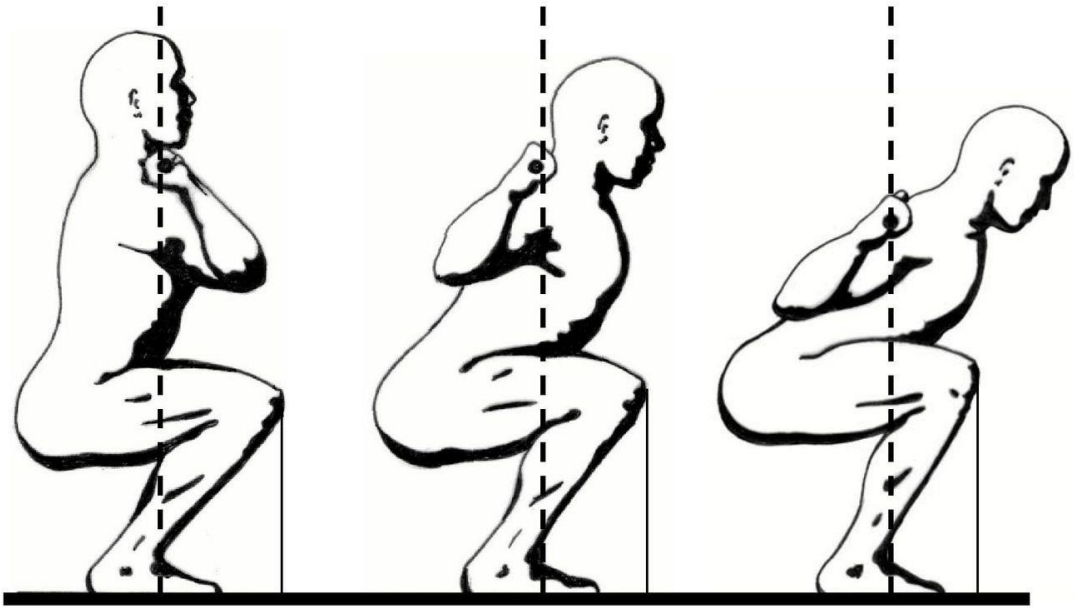
In which of the above dumbbell positions, the middle deltoids are engaged the most? And why?

Pair Activity: If a Client A is lifting a 10kg while performing lateral raises with an almost fully extended elbow vs Client B that is performing the same exercise with the same weight but with a 90 degrees flexed elbows. Knowing that both clients are training for hypertrophy, calculate each client's muscular torque to identify which client is benefiting the most form this exercise?

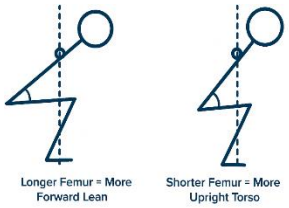
Pair Activity: If a Client A is lifting a 10kg while performing lateral raises with an almost fully extended elbow vs Client B that is performing the same exercise with a weight of 17.5kg but with a 90 degrees flexed elbows. Knowing that both clients are training for hypertrophy, calculate each client's muscular torque to identify which client is benefiting the most?

Pair Activity: If a Client A is lifting a 10kg while performing lateral raises with an almost fully extended elbow vs Client B that is performing the same exercise with a weight of 20kg but with a 90 degrees flexed elbows. Knowing that both clients are training for hypertrophy, calculate each client's muscular torque to identify which client is benefiting the most?

Pair Activity: Use what you learned about moment arm to analyze the difference between front squats, high-bar back squat and low-bar back squat?



How would having a long Femur length affect the torque of the glutes muscle when performing squats?



Consolidation:

By the end of this lesson, I am aware of:

-Concept of muscle torque and moment arm and its effect on muscular engagement.

-Practical use of moment arm and torque in analyzing different exercises such as lateral raises and squat variations

LESSON 2: RESISTANCE CURVES

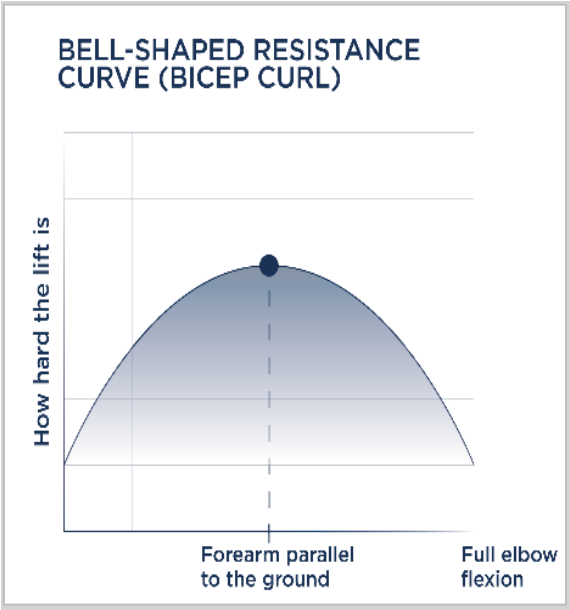
Objectives:

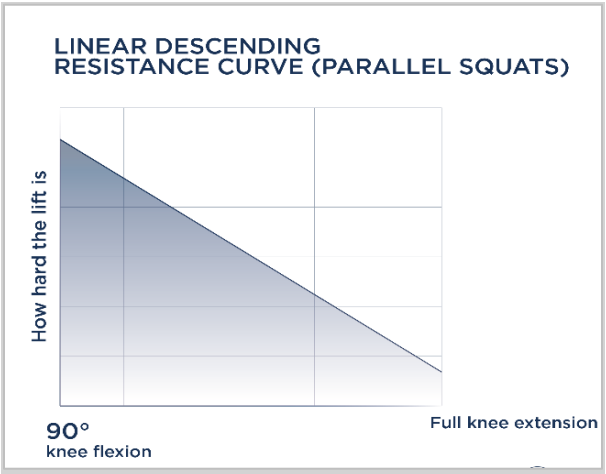
-Define strength and resistance curves.

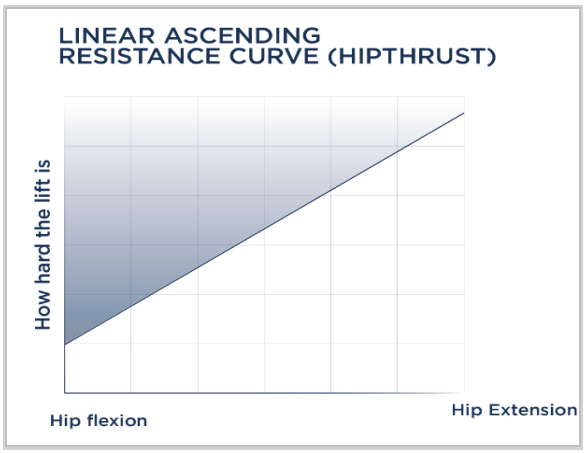
-Differentiate between all types of resistance curves.

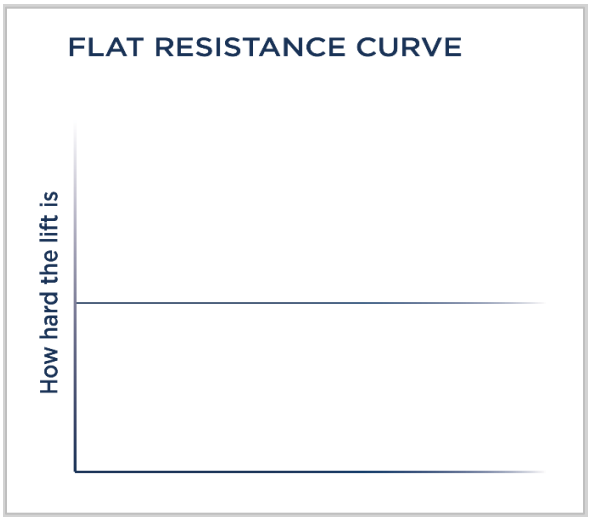
-State and analyze the sticking points of exercises.

- **RESISTANCE CURVES VS STRENGTH CURVES:**









Pair Activity: State the type of resistance curve for each of the following exercises: Chest Fly, Bench Press, Deadlift, and Skull Crusher

Pair Activity: Explain how the use of elastic bands and / or chains along with free weights can flatten the resistance curve:





Pair Activity: Does adding elastic bands or chains to free weights, always flatten the resistance curve in every performed exercise? Can you provide an example of an exercise, where adding this equipment won't flatten the curve?

- **STICKING POINT OF EXERCISES:**



What is it?



Guess the sticking point of the squat, deadlift and bench press while explaining your answer



Consolidation:
By the end of this lesson, I am aware of:
-All four types of resistance curves and how to use them practically to better understand the performed exercises
-Sticking points of the squats, deadlifts and bench press

LESSON 3: BALANCE & STABILITY



Objectives:
- Differentiate between Balance and Stability.
- Know the components of Balance and Stability.

- **BALANCE VS STABILITY:**



State the four components of balance / stability?



How should we design exercises when training elderly clients to improve their balance?



Consolidation:
By the end of this lesson, I am aware of:
-Components of balance and stability and how to use them practically when training clients to improve balance

LESSON 4: BIOMECHANICS LAB

Objectives:
-Let's practice what we learned in a practical gym setting

Each group has 3 minutes to **Perform & Answer** the task presented in each of the eight stations at the gym:

Station 1: Perform standing transverse adduction using an elastic band then perform the same movement using a dumbbell. What is the difference and why?

Station 2: Perform Lateral raises with elbows flexed vs Lateral raises using the same weight with elbows extended. What is the difference between the two?

Station 3: Perform Standing lateral raises vs Lying lateral raises using the same weight. Is there any difference?

Station 4: Perform leg raises with knees bent vs performing the same movement with knees extended. What is the difference between the two?

Station 5: Perform Chest Fly using cable machine vs dumbbells with the same amount of resistance. Is there any difference between the two pieces of equipment used?

Station 6: Perform a barbell bench press / push-up with elbows close to the torso. Does it differ from performing a barbell bench press having a 45 degrees angle at the level of the GH joint?

Station 7: Stand with both feet shoulder-width apart and move your upper body in all three planes of motion. Repeat the same movements while standing on one foot. Which one is harder and why?

Station 8: Try standing on the BOSU ball using both surfaces, which side is harder and why?

Think you're good at push-ups?! Team up with a partner and let's **PUT IT TO THE TEST!**



From a biomechanical perspective, why it felt too hard to be done?

Consolidation:

By the end of Session 7, I am aware of:

-Muscle torque and its effect on analyzing exercise variations such as lateral raises and squats

-All four types of resistance curves and how to use them practically

-Components of balance and stability and how to use them practically when training clients to improve balance

Post-Session 7 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 7.

Pre-Session 8 Preparation:

-Watch all Physiology videos of the Chapter titled: *Week 4: Part 2 of the Biomechanics of Exercise and Resistance Training + Physiology of the Muscular System*

Session 8: Physiology of the Muscular System

LESSON 1: GENERAL MUSCULAR PHYSIOLOGY

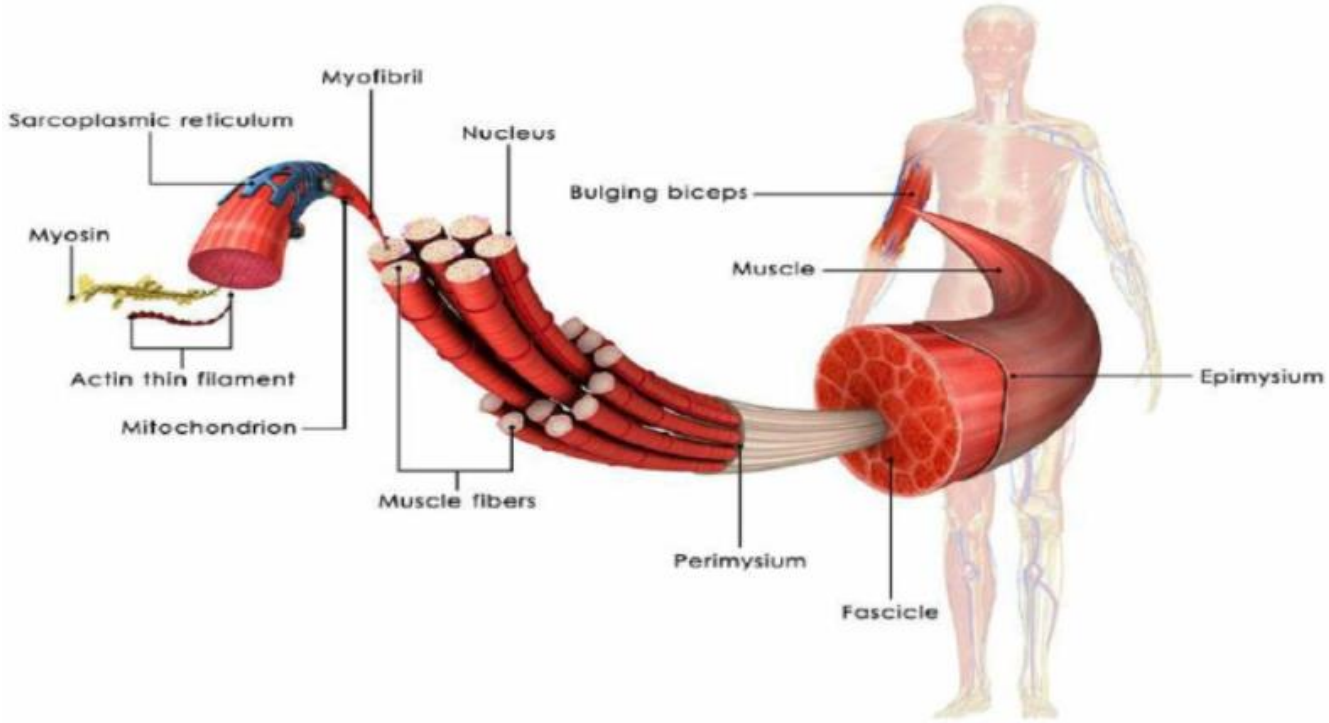
Objectives:

- Understand what the muscle is made of
- Understand the sliding filament theory behind muscle contraction
- Differentiate between different muscle fiber types
- Discuss the concept of motor units and its impact on muscle fiber types

(Platform: Chapter title: Week 4: Part 2 of the Biomechanics of Exercise and Resistance Training + ***Physiology of the Muscular System)***

- **STRUCTURE OF SKELETAL MUSCLES:**

Based on the skeletal muscle structure shown below, can you identify what muscles are made of besides proteins?

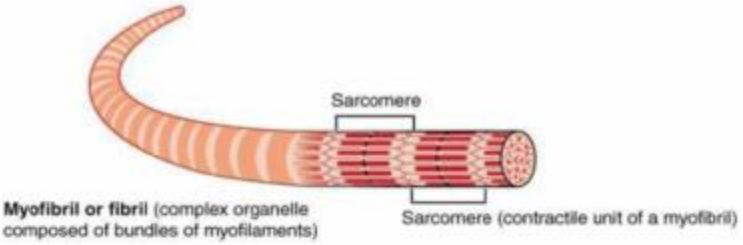


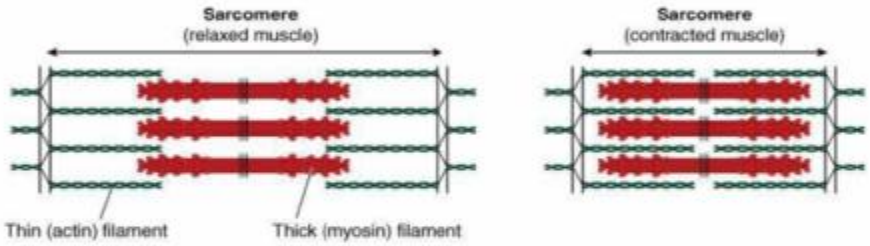
- **SLIDING FILAMENT THEORY:**

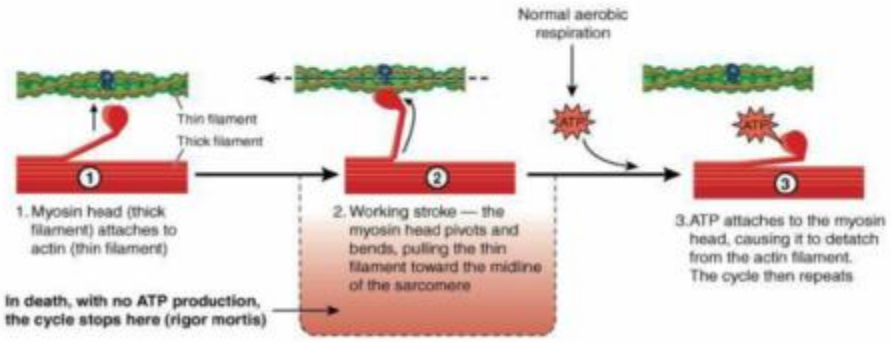
GROUP ACTIVITY

Pair activity: Mix and match the events below according to their chronological order

Calcium binds troponin	CNS stimulus through motor neuron and neuromuscular junction	Myosin pulls the Actin causing the muscle to contract
Action potential reaches sarcolemma	Myosin binds Actin	Sarcoplasmic reticulum releases calcium within the muscle
	Release of acetylcholine	







- **MUSCLE FIBER TYPES:**

GROUP ACTIVITY

Pair Activity:

1-Would running for 30 minutes five times a week lead to an increase in the size of the lower body muscles? Explain your answer

2-In the picture below, we have two athletes that train 6 days per week in order to compete in their sports; explain the difference in their muscle size:



Let's discuss the different muscle fiber types:

CHARACTERISTIC	Slow Twitch (ST)	Fast Oxidative-Glycolytic (IIA)	FAST GLYCOLYTIC (IIB)
Color	Red	White	White
Fiber diameter	Small	Large	Large
Mitochondrial availability (oxidative capacity)	High	High	Low
Glycolytic capacity	Low	High	Highest
Glycogen stores	Low	High	High
Contractile speed	Slow	Fast	Fast
Fatigue resistance	High	Moderate	Low
Myoglobin content	High	Moderate	Low
Single fiber force	Similar	Similar or slightly higher	Similar or slightly higher
Type of activity	Aerobic endurance	Strength and power	Strength and power

Which factors determine the distribution of muscle fiber types within a muscle?

Are all muscles composed of the same proportion of fiber types?

- **MOTOR UNITS:**

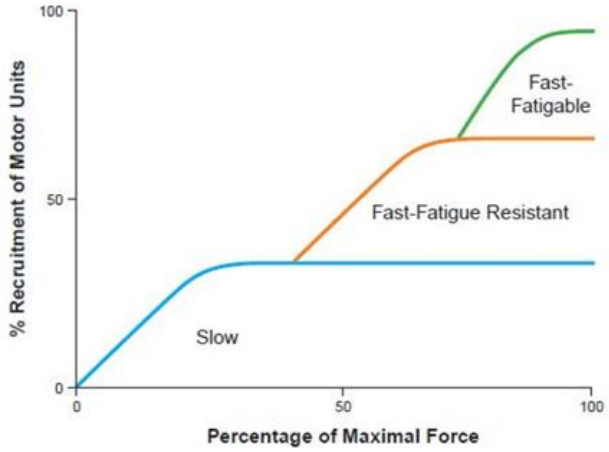
Motor Unit = Motor Neuron + Muscle Fibers it innervates or supplies

Do you think we have different motor units for each muscle or just one motor unit for each muscle? (Hint: Would you engage the same number of muscle fibers when lifting a 1Kg dumbbell vs when lifting a 20Kg dumbbell?)

Each Motor Units contains same type of muscle fibers

Do Motor Units obey the All or None Law?!

Motor Units follow the Henneman Size Principle:



What is the practical application to this Henneman Size Principle? (Hint: warm-up!)

- **Characteristics of Motor Units Types:**

CHARACTERISTIC	TYPE I Fibers	TYPE IIA Fibers	TYPE IIX Fibers
Size of motor neuron	Small	Large	Large
Fibers per motor neuron	Less or equal to 300	Greater or equal to 300	Greater or equal to 300
Motor neuron conduction speed	Slow	Fast	Fast
Contraction speed (milliseconds ms)	Slower (110ms)	Faster (50ms)	Faster (50ms)

Table 2- Main Characteristics of Different Motor Unit Types

Let's see how fast you can be in hitting that buzzer, and answering which type of muscle is mainly engaged in different exercise trainings!

Consolidation:

By the end of this lesson, I am aware of:

- The muscle micro and macrostructure.
- The steps of muscular contraction.
- Different types of muscle fibers and their practical application
- Characteristics of Motor Units

LESSON 2: ENERGY SYSTEMS

Objectives:
- Understand the three different energy systems and their practical application
- Know when do we use carbs, fat or protein to produce energy

*(Platform: Chapter title: Week 5: Physiology of the **Energy** + Cardiovascular + Respiratory Systems)*

- **ENERGY = ATP?!**

When do we, humans, need energy? Can we store energy or do we have to produce it when needed?

What is the fuel used by the muscles to produce energy?

- **ENERGY SYSTEMS:**

We have 3 different energy systems, let's fill in the details of each one according to the below table:

	ATP-Creatine Phosphate	**Anaerobic Glycolysis**	**Aerobic system**
Needs Oxygen?			
Fuel used?			
How fast is it?			
Number of produced ATPs?			
Extra Comments			

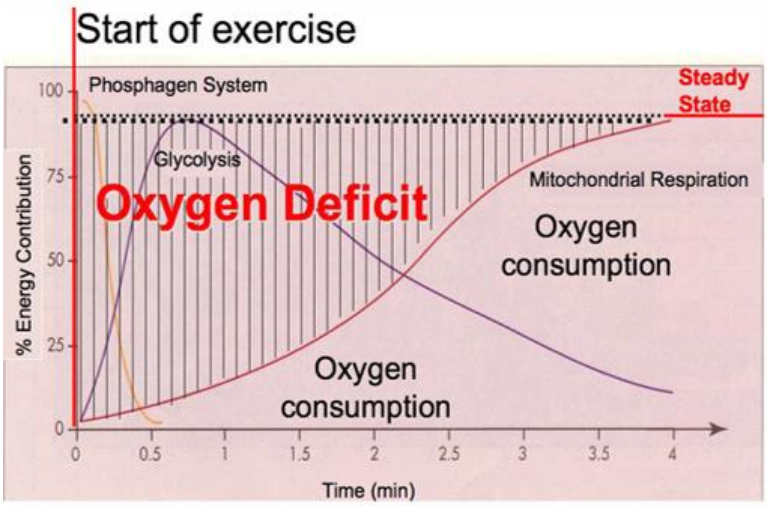
What about protein? Is it used as fuel for energy?

Training Intensity & Duration dictates which energy system to be used the most during physical activity.

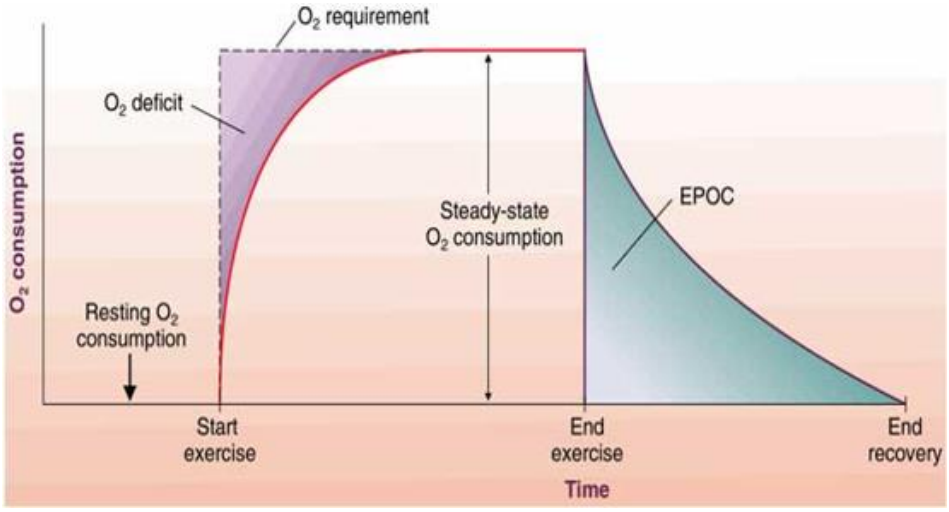
GROUP ACTIVITY

- **Case Study:**
  Client A is performing 30 minutes of low intensity cardio while client B is performing a 30 minutes high intensity interval training cardio; which client is burning more fat and why?

- **OXYGEN DEFICIT:**



- **EXCESS POST-EXERCISE OXYGEN CONSUMPTION:**



- **LACTATE THRESHOLD:**

How can we train the lactate threshold?

Ready to play some "XO" with a twist – let's move to the gym for this one!

Well Done! Let's **summarize** the game info by filling the below table all together:

Energy System	ATP Stored	ATP-CP	Anaerobic Glycolysis	Aerobic System
Duration				
Training Intensity				
Muscle Fiber Type				
Exercise				

Consolidation:

By the end of Session 8, I am aware of:

-The muscle micro and macrostructure.

-The steps of muscular contraction.

-Different types of muscle fibers and their practical application

-Different energy systems and their practical application in terms of macronutrients used

Post-Session 8 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 9.

Pre-Session 9 Preparation:

-Watch all videos of the Chapter titled: Week 5: Physiology of the Energy + Cardiovascular + Respiratory Systems

-Watch all Nervous system related videos of the Chapter titled: Week 6: Physiology of the Nervous and Endocrine Systems + Nutrition Part 1

Session 9: Physiology of the Cardiovascular, Respiratory & Nervous Systems

LESSON 1: CARDIOVASCULAR SYSTEM

Objectives:

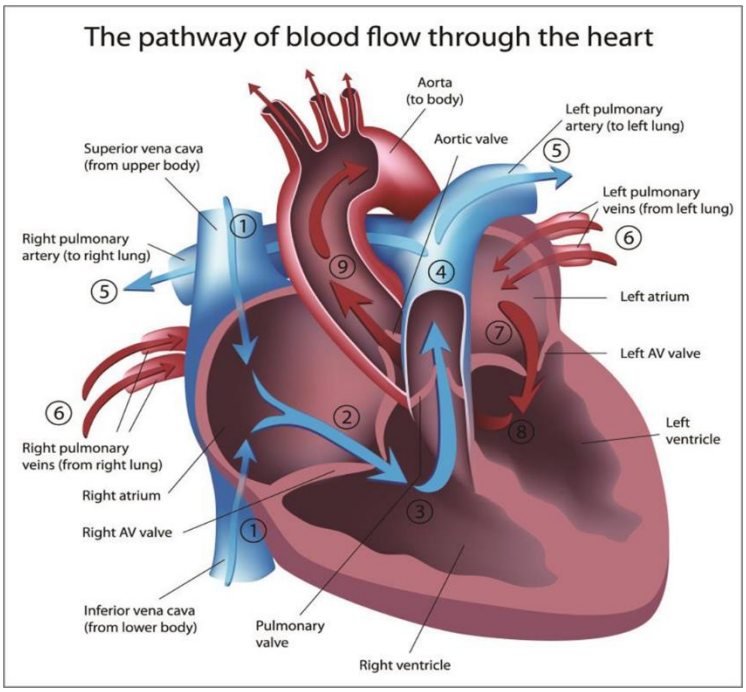
- Understand the role of the cardiovascular system.
- Understand the heart anatomy and blood circulation.
- Discuss the blood composition.
- Understand the blood pressure, heart rate, strove volume and cardiac output

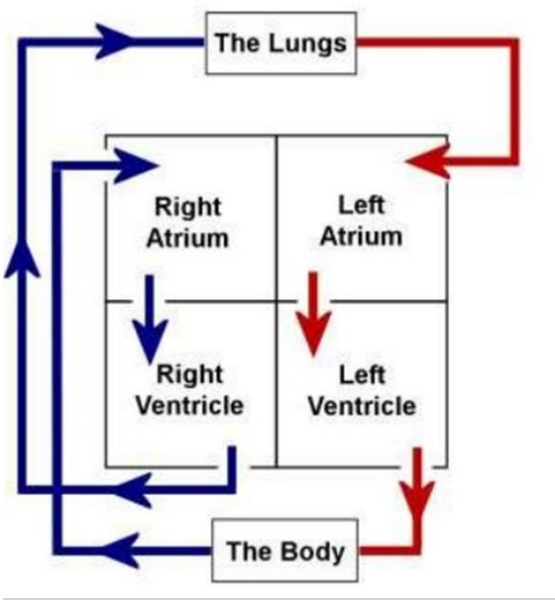
*(Platform: Chapter title: Week 5: Physiology of the **Energy** + Cardiovascular + Respiratory Systems)*

- **CARDIOVASCULAR SYSTEM:**

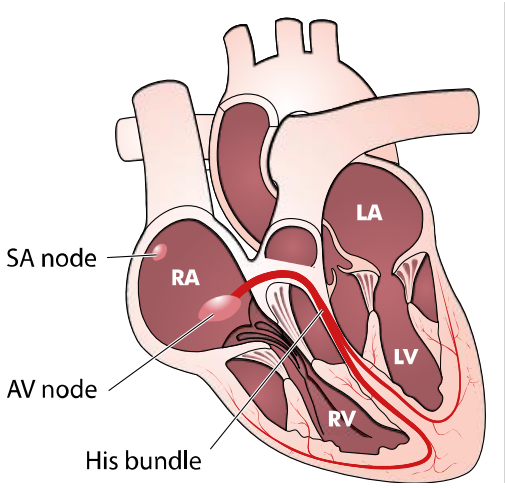
Guess the main function of the heart and blood vessels

- **Heart Anatomy & Pathway of blood through the Heart:**

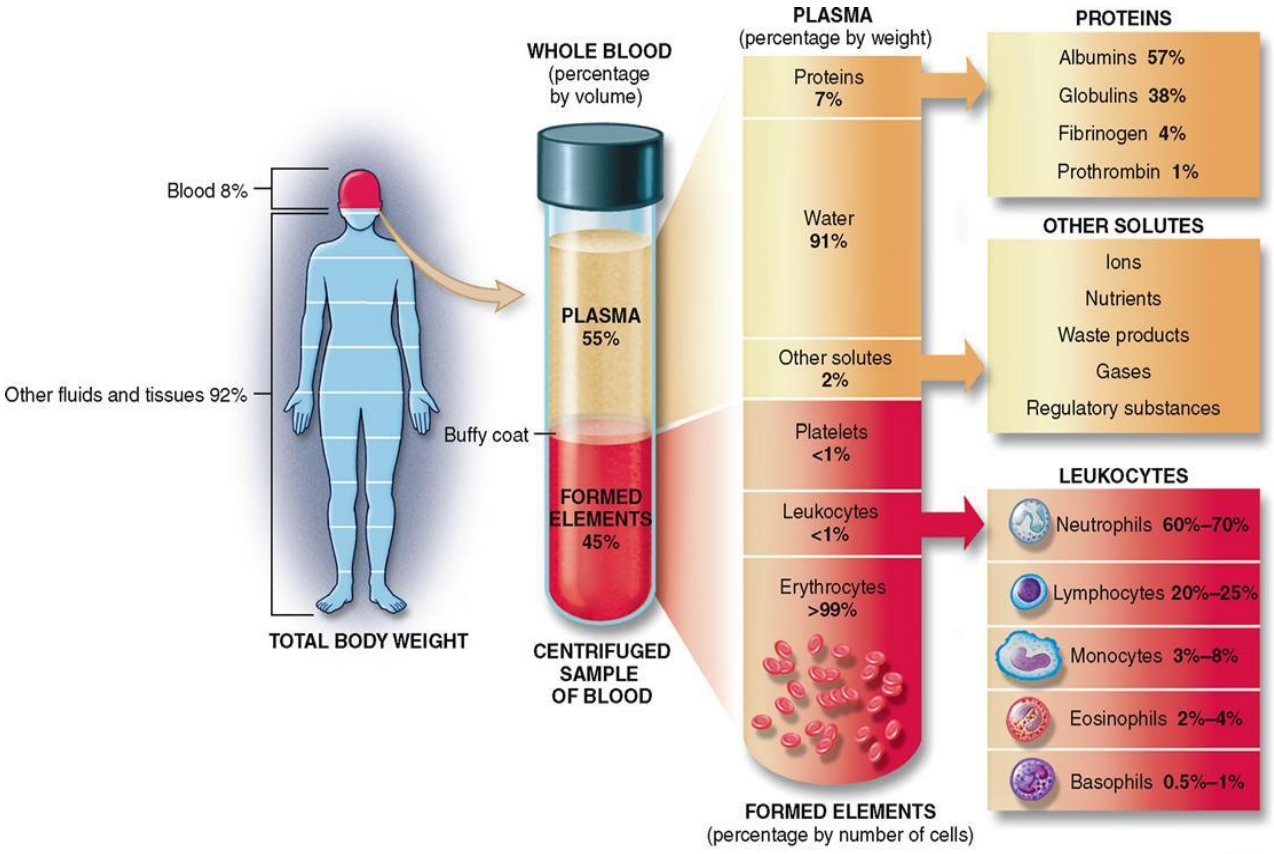




Who sends electric signals to the heart muscle to contract?

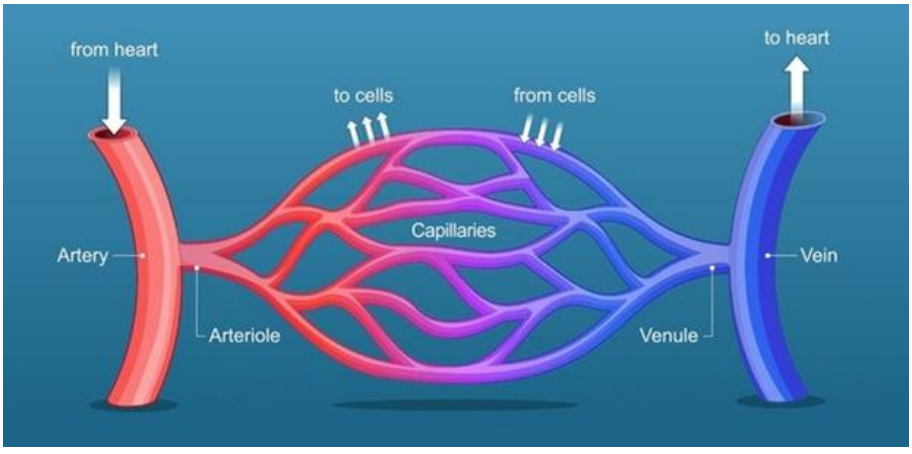


- **Blood Composition:**



Based on the above pic, why fluids are very important for exercise performance?

- **Blood Vessels:**





Pair Activity: Mix and Match

Heart Rate	Volume of blood secreted in 1 beat
Cardiac Output	Blood Pressure when Heart Ventricles Contract
Stroke Volume	Volume of blood secreted in 1 minute
Diastolic Blood Pressure	Blood Pressure when Heart Ventricles Relax
Systolic Blood Pressure	Number of heart beats per minute



Consolidation:

By the end of this lesson, I am aware of:

- Role of the cardiovascular system.
- Heart anatomy and blood circulation.
- Blood composition.
- Blood pressure, heart rate, strove volume and cardiac output

LESSON 2: RESPIRATORY SYSTEM

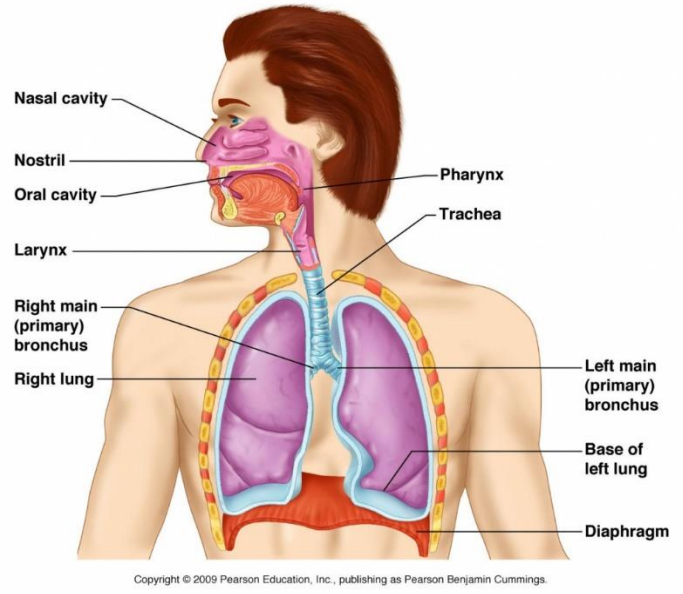
Objectives:

- Understand the role of the respiratory system.
- Understand the lungs anatomy.
- Explore the gas exchange process.
- State the accessory breathing muscles.
- Understand the Valsalva maneuver and its practical application.

*(Platform: Chapter title: Week 5: Physiology of the **Energy** + Cardiovascular + Respiratory Systems)*

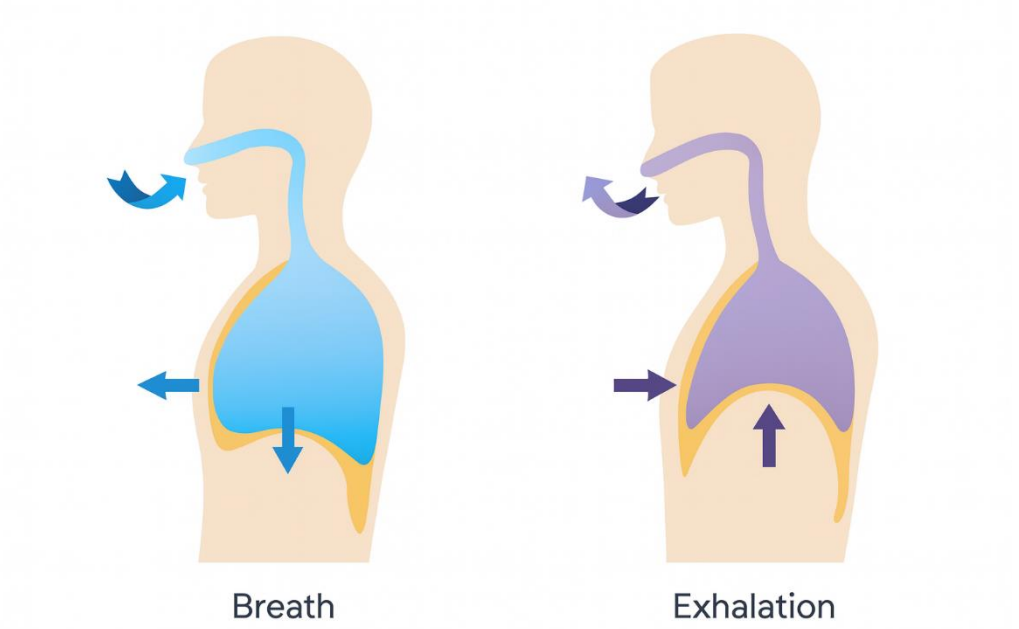
- **RESPIRATORY SYSTEM:**

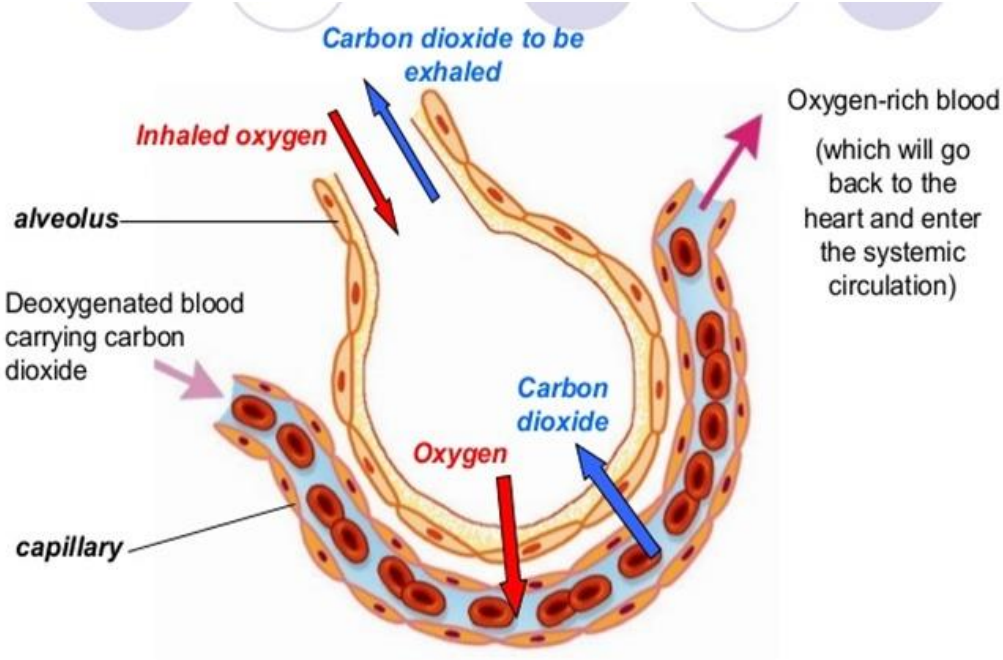
Which organs form this system?



Can you guess the number of muscles within the lungs?

Where does the gas exchange take place inside the lungs?





- **Accessory Breathing Muscles:**

	Inhalation	Resting (Passive) Exhalation	Forced Exhalation
Diaphram	✓	None	None
External Internal Coastals	✓	None	None
Accessory Muscles:	Scalene, Sternocleidomastoid, Pectoralis Minor, Serratus Anterior	None	Abdominals, Internal Intercostals

Table 9- Muscle Recruitment During Inhalation & Exhalation

- **VALSALVA MANEUVER**:

Valsalva Maneuver is holding your breath while lifting heavy weights during the concentric contraction phase. **Try guessing its advantages and disadvantages:**

- **Proper Breathing**:

How should we breathe when performing resistance training?

Consolidation:

By the end of this lesson, I am aware of:

- Role of the respiratory system.
- Lung's anatomy
- Gas Exchange process.
- Accessory breathing muscles.
- Valsalva maneuver and its practical application

LESSON 3: NERVOUS SYSTEM

Objectives:
- State the three main functions of the Nervous System.
- Explore the components of both Central and Peripheral Nervous system.
- Explore the Golgi Tendon Organ and the Muscle Spindles.

(Platform: Chapter title: Week 6: Physiology of the Nervous and Endocrine Systems + Nutrition Part 1)

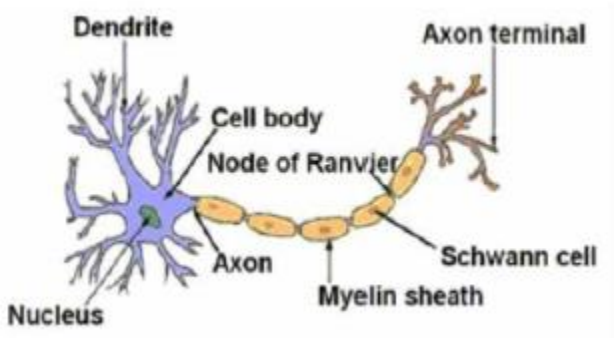
- **NERVOUS SYSTEM - Main Functions:**

- Sensory
- Integrative
- Motor

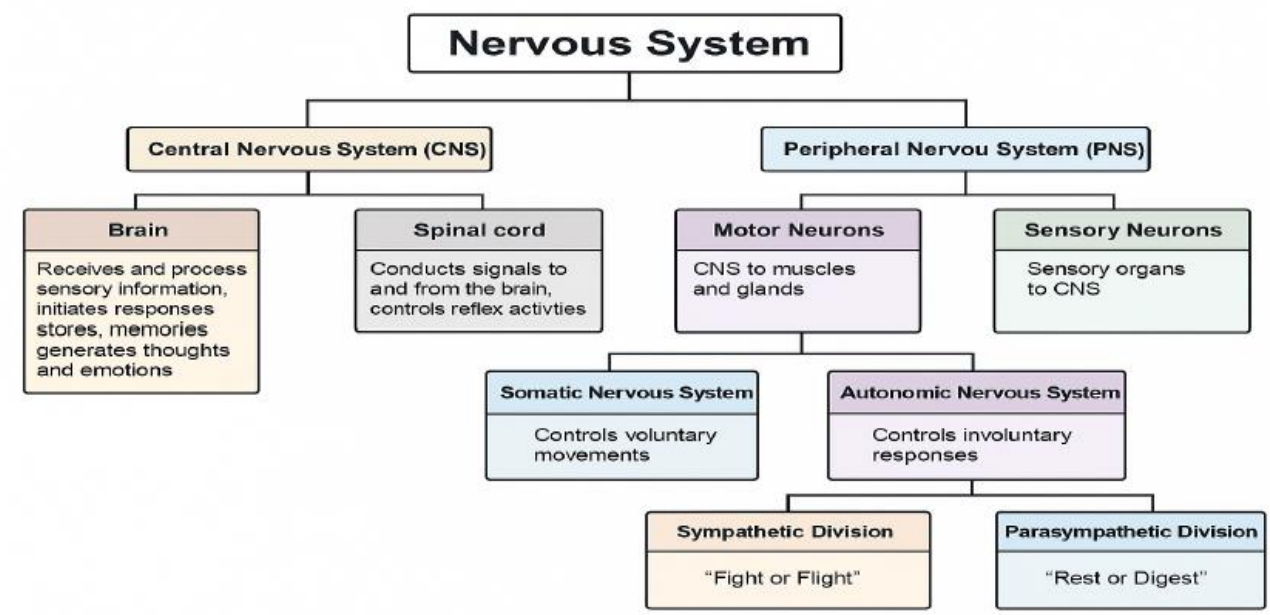
If someone is running outdoor and sees a car coming his way:
A- Eyes will inform the brain about the incoming car
B- Brain will analyze how far and fast the car is
C- Brain will send a signal to the lower body muscles either to contract faster or to relax

Assign one nervous system function for A, B & C

- **Neurons:**



- **Central vs Peripheral Nervous System:**



- **CNS – Brain:**

- **CNS – Spinal Cord:**

- **PNS – Motor vs Sensory Neurons:**

- **PNS – Somatic vs Autonomic:**

- **PNS – Sympathetic vs Parasympathetic:**

- **PNS – Sensory Neurons – Golgi Tendon Organ vs Muscle Spindles:**

Pair Activity: Match the below terms:

Golgi Tendon Organ

Muscle Spindles

Inside Tendons

Inside Muscles

Detects Tension

Detects Fiber Length

Brain Responds by Contracting the Muscle

Brain Responds by Relaxing the Muscle

Autogenic Inhibition

Reciprocal Inhibition

Consolidation:

By the end of this session, I am aware of:



- Role of the respiratory system.
- Lung's anatomy
- Gas Exchange process.
- Accessory breathing muscles.
- Valsalva maneuver and its practical application
- Main functions of the Nervous System
- Components of the CNS & PNS



Post-Session 9 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 10.



Pre-Session 10 Preparation:

-Watch all Endocrine system related videos of the Chapter titled: *Week 6: Physiology of the Nervous and* ***Endocrine*** *Systems + Nutrition Part 1*

Session 10: Physiology of the Endocrine System & Exercise Physiology of the Muscular System

LESSON 1: CARDIOVASCULAR SYSTEM

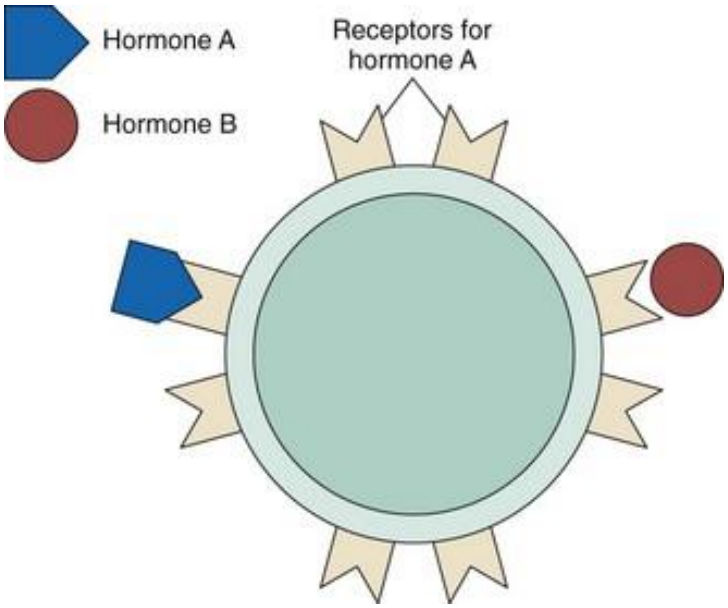
Objectives:

- Understand the difference between Hormones and Glands.
- Explore how hormones work.
- State the function of the main hormones and their impact on Exercise.
- State the Advantages and Disadvantages of Steroids.
- Be able to answer clients about taking Steroids.

(Platform: Chapter Title: Week 6: Physiology of the Nervous and Endocrine Systems + Nutrition Part 1)

- **GLANDS VS HORMONES:**

- **Key-Lock Configuration & Negative vs Positive Feedback:**



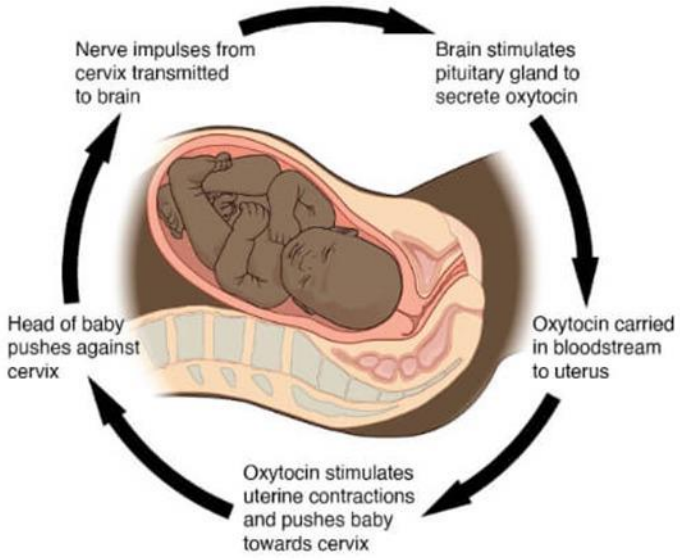


Fig: **Positive Feedback** (Childbirth)

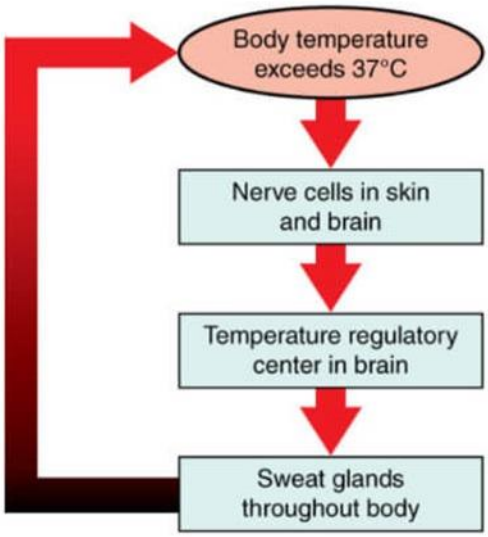


Fig: **Negative Feedback** (Temperature Regulation)

- **HORMONES FUNCTIONS AND IMPACT OF EXERCISE:**

GROUP ACTIVITY

Let's all become Instructors for this one and **explain** the function of the hormones in question along with their response to exercise:



Pair Activity: Read the provided riddle and answer it after hitting the buzzer
(Hint: riddles are about the hormones we just explained)

- Clients training for hypertrophy and strength love me, the only thing I can breakdown is fat

- I help with performance during the session by maintaining a good blood circulation despite sweat. Oh, and I like sodium

- I am the one that allows muscles to use the energy systems and produce energy during training workouts

- I help with performance during the session by increasing the cardiac output and focus levels

- I try to balance the fluids lost during exercise; kidneys are my best friend

- In small amounts I help with hypertrophy through muscle damage; in large amounts, I can cause overtraining

- I help your muscles grow by filling them with fluid

- I don’t like females!! And I increase protein synthesis

- **AGING AND HORMONES:**

Growth Hormone?

Thyroid Hormone?

Cortisol Hormone?

Aldosterone Hormone?

Epinephrine & Norepinephrine Hormones?

Insulin Hormone?

- **ADVANTAGES AND DISADVANTAGES OF STEROIDS:**

General	Male Specific	Female Specific
• Acne and greasy skin/hair • Liver dysfunction • Kidney problems • Infertility • Aggressiveness and mood swings • Atherosclerosis, heart attacks and brain strokes • Cancer/tumor (especially in the liver) and internal bleeding • Stunted growth (teenagers only)	• Gynecomastia • Testicular atrophy • Baldness • Decreased sperm production and sperm count • Prostate cancer	• Male secondary sex characteristics (thicker voice, facial hair…) • Hormonal imbalance leading to menstrual disturbances • Shrunken breasts

HGH	EPO
• Diabetes	• Blood clot formation
• Acromegaly	• Myocardial infarction
• Abnormal organ growth	• Pulmonary embolism
• Joint problems	• Hypertension
• Edema	• Tachycardia
• Hypercholesterolemia	• Edema
• Tingling and numbness	

- **<u>Our Standpoint on Steroids:</u>**

How to answer clients thinking about taking Steroids?

Consolidation:

By the end of this lesson, I am aware of:

- Hormone's function
- Main hormones function and their impact on Exercise
- Aging impact on hormones
- Advantages and Disadvantages of Steroids
- How to answer clients willing to take steroids

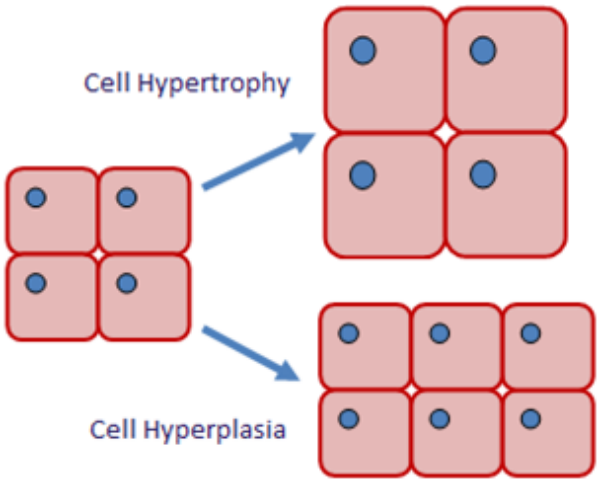
LESSON 2: EXERCISE PHYSIOLOGY – MUSCULAR SYSTEM: TYPES OF HYPERTROPHY

Objectives:

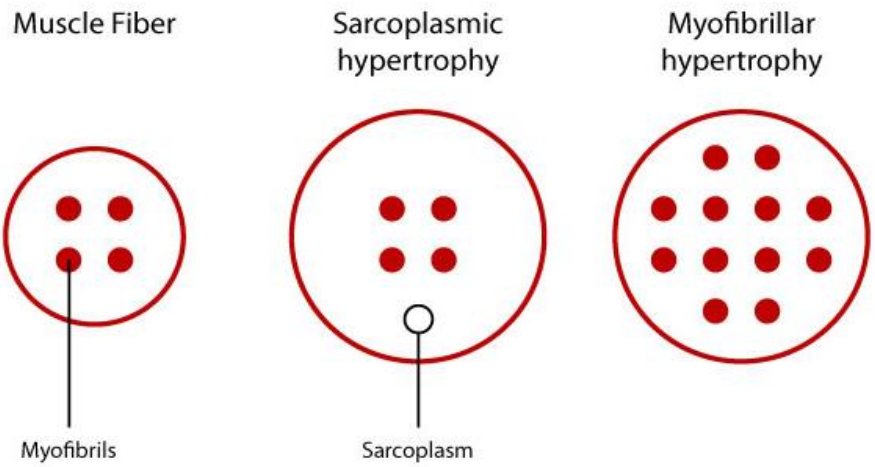
- Differentiate between Hypertrophy and Hyperplasia.
- Differentiate between Sarcoplasmic and Myofibrillar Hypertrophy.
- Differentiate between Longitudinal (stretch-mediated) and Transverse Hypertrophy
- Function of Satellite Cells

Platform: Chapter titled - Week 4: Part 2 of the Biomechanics of Exercise and Resistance Training + Physiology of the Muscular System

- **HYPERTROPHY VS HYPERPLASIA:**



- **SARCOPLASMIC HYPERTROPHY VS MYOFIBRILAR HYPERTROPHY:**



Is there a specific mode of training that favors one type of hypertrophy over the other?

With consistent training over time muscles shifts towards one of the two hypertrophy types, can you guess which one is it?

- **LONGITUDINAL (STRETCH-MEDIATED) AND TRENASVERSE HYPERTROPHY:**



Pair Activity: Mix and match the below terms:

Transverse Hypertrophy

Longitudinal Hypertrophy

Active Mechanical Tension

Passive Mechanical Tension

Sarcomeres added in Series → Longer fibers

Sarcomeres added in Parallel → Bigger fiber diameter

Eccentric Training OR Full ROM

High Effort with Low Bar

Faster Diminishing Returns (months)

Slower Diminishing Returns (years)

- **SATELLITE CELLS:**

Responsible for Muscle damage repair & Muscle Memory

What is Muscle Memory?

Satellite cells cannot be used as criteria / marker for hypertrophy since they are activated by any type of training (anything that triggers Calcium release) including cardio…

Consolidation:

By the end of this session, I am aware of:

- Hormone's function
- Main hormones function and their impact on Exercise
- Aging impact on hormones
- Advantages and Disadvantages of Steroids
- How to answer clients willing to take steroids
- Difference between Sarcoplasmic and Myofibrillar Hypertrophy.
- Difference between Longitudinal (stretch-mediated) and Transverse Hypertrophy
- Role of Satellite Cells

Post-Session 10 Tasks:

- Review the material covered during the session.
- Prepare a brief recap of the key lessons covered to be presented in Session 11.

Pre-Session 11 preparation:

- Watch all Muscular Physiology related videos of the Chapter titled: *Week 4: Part 2 of the Biomechanics of Exercise and Resistance Training + Physiology of the Muscular System*

SESSION 11: Exercise Physiology – Understand Fatigue and Muscular Hypertrophy

LESSON 1: FATIGUE TYPES



Objectives:
- Differentiate between all types of Fatigue
- Explore the mechanisms of Hypertrophy



Training under Fatigue Inhibits Hypertrophy – Do you agree with this statement and why?

- **CENTRAL NERVOUS SYSTEM FATIGUE:**

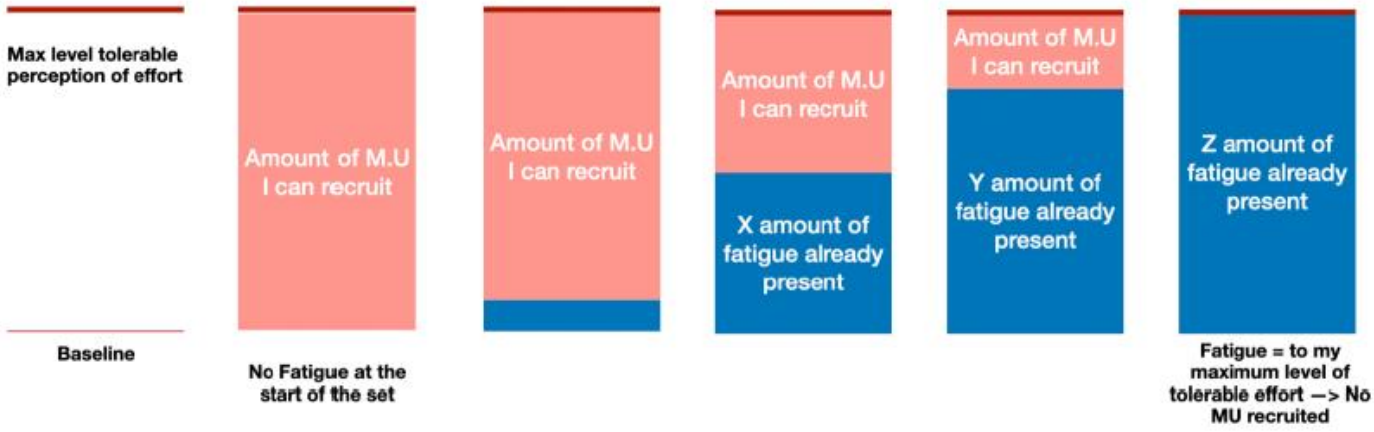
The strength of the signal secreted by the CNS to the muscles is what determine the amount of motor unit recruitment which directly affects force production. This signal strength is affected by CNS fatigue.



How would CNS Fatigue affect exercise performance and results? What if we exert maximum effort?

If this reduction in signal strength occurs at the level of the brain, it is called **Supraspinal Fatigue;** It is influenced by mental fatigue and perception of fatigue which means when we are have supraspinal fatigue we will perceive it as this the maximum we can do although our muscles are not yet fully fatigued, not yet close to failure.

If the signal strength reduction occurs at the level of the spinal cord, it is called **Spinal Fatigue.**



GROUP ACTIVITY

Can you guess the practical reasons that can increase the perception of fatigue, increasing supraspinal fatigue and or spinal fatigue?

Which type of fatigue recovers faster, the Supraspinal or the Spinal one?

Which type of fatigue affects all muscles and which one affects only the fatigued muscle itself?

Can you guess what is cross-limb neural interactions? Which type of fatigue is responsible for it?

- **PERIPHERAL NERVOUS SYSTEM FATIGUE:**

Affects the muscle itself and takes place at the level of the neuromuscular junction (even if the CNS signal reaching the muscle is strong, the force generated by the muscle will be low)



PNS fatigue has two types: Metabolite Related Fatigue and Calcium Ion Overload, can you guess which type develops and resolves quickly and which one develops slowly and gets longer to recover?

- **Metabolite Related Fatigue:**

Use of ATP leads to phosphate accumulation which does not directly affect mechanical tension but indirectly since it leads to CNS fatigue decreasing Motor Unit Recruitment.



It directly affects the velocity of muscular contraction so it directly impacts ---------- & ----------?

- **Calcium Ion Related Fatigue:**
  Caused by accumulation of Calcium within the muscle which directly affects hypertrophic stimulus by decreasing mechanical tension and increasing cellular damage which extends the recovery period



Guess the practical reasons that can increase calcium influx leading to Calcium Ion Related Fatigue?

Which training was involved in almost all types of fatigue?

Should we stay away from all trainings that lead to Fatigue?

Consolidation:

By the end of this lesson, I am aware of:

- Different types of Fatigue
- How to manage fatigue practically

LESSON 2: KEY PRINCIPLES OF HYPERTROPHY

Objectives:
- Explore the key principles of Hypertrophy
- Explore the mechanisms of Hypertrophy

- **KEY PRINCIPLES OF HYPERTROPHY:**

1- Mechanical Tension & Full Motor Unit Recruitment

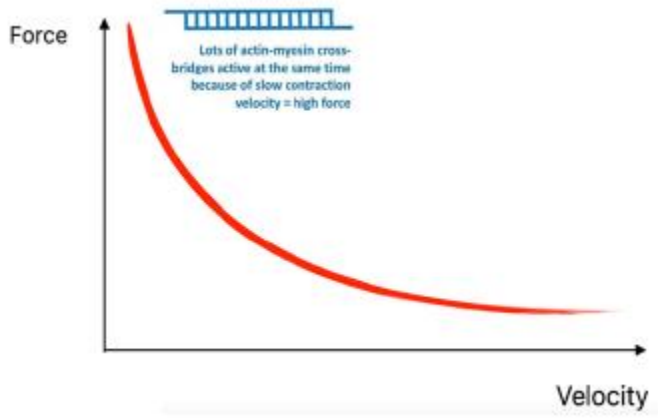
THE primary driver for hypertrophy

2- Size Principle

Motor units are recruited according to their size → Low threshold motor units experience mechanical loading at low forces while both low and high threshold motor units experience mechanical loading at high forces

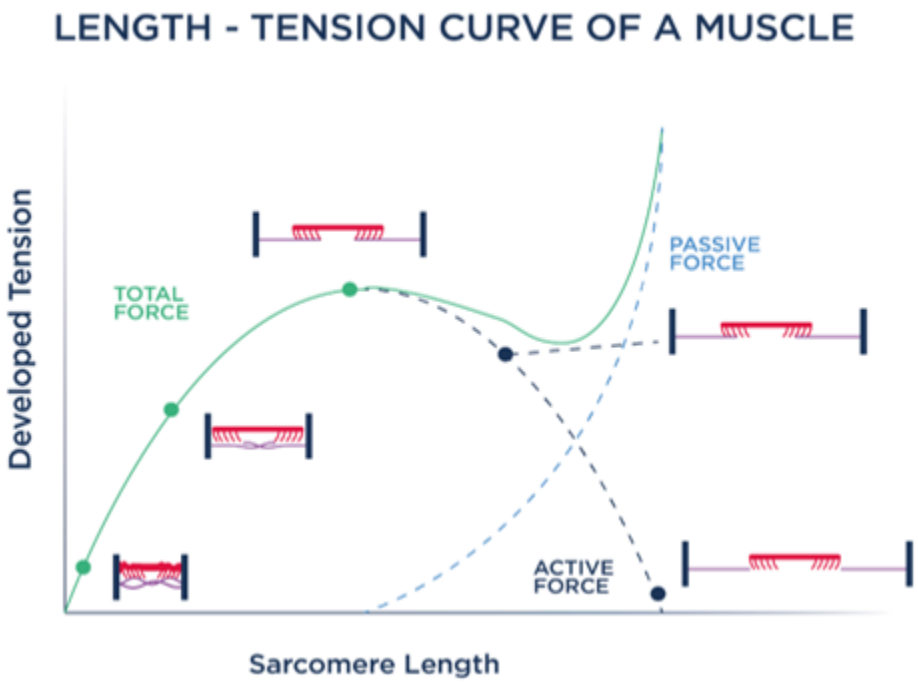
How can we translate the above practically while training for hypertrophy?

3- Force-Velocity Relationship



While training for hypertrophy, is it better to train at higher or lower velocity and why? (use the words actin, myosin and mechanical tension in your answer)

4- Length-Tension Relationship



Training at full ROM enables you to benefit from both active and passive length-tension relationship causing both cross-sectional hypertrophy and increase in fascicle length, unlike training at partial ROM

GROUP ACTIVITY

Does Muscle Damage lead to hypertrophy?

Do you think eccentric training is superior to concentric training when it comes to hypertrophy if the same training volume was used?

Too much damage can impair recovery and decrease training quality

Does Metabolic Stress (metabolite accumulation including lactate) lead to hypertrophy?

Do you think cardio or endurance training leads to metabolite accumulation?

Does Elevated GH after resistance training, lead to an increase in muscular protein synthesis and /or hypertrophy?

Does Blood Flow Restriction increase the chances of hypertrophy?

Reactive Oxygen Species & Nitric Oxide direct role in hypertrophy is weak according to studies

Anabolic Processes are linked to Mechanical Tension rather than Metabolite Accumulation

<u>Stimulating Reps Model:</u>

Hypertrophy is reached by high mechanical tension, which is governed by high motor unit recruitment achieved through training with high effort so high forces / intensity with low velocity, leading to maximum actin-myosin cross bridging formation

Consolidation:

By the end of this lesson, I am aware of:

- Key principles of hypertrophy and their practical application
- Stimulating Reps Model

LESSON 3: KEY PRINCIPLES OF STRENGTH

Objectives:
- Explore the physiological factors affecting Strength
- Explore factors affecting muscle function

Platform: Chapter titled - Week 4: Part 2 of the Biomechanics of Exercise and Resistance Training + Physiology of the Muscular System

- **KEY PRINCIPLES OF STRENGTH:**

1- Muscle Cross-sectional area

Focusing on increasing muscle size before strength-focused training is beneficial

Higher CSA leads to more actin-myosin cross-bridges → More Strength

2- Musculotendinous Stiffness

Increases both Strength and Power

3- Motor Unit Recruitment

4- Rate Coding – Firing Frequency

Increases both Strength and Power

5- Motor Unit Synchronization

More into explosive strength performance

6- Neuromuscular Inhibition

- **FACTORS AFFECTING MUSCLE FUNCTION:**

1- Agonist-Antagonist Co-contraction

Beside muscle force production, can you think of another training component that is affected by this factor?

2- Stabilizer / Synergist Activation

How would stabilizers help the agonist or main muscle?

3- Muscle Fiber Types

Can we cause a switch in our fibers from type I to type II or vice versa?

Can we cause a switch in our fibers from type IIa to type IIb or vice versa?

4- Tendon Stiffness

5- Pennation Angle

Consolidation:

By the end of this lesson, I am aware of:

- Physiological factors affecting Strength
- Factors affecting muscle function

LESSON 4: DELAYED ONSET MUSCLE SORENESS



Objectives:
- Explore the physiological causes of soreness
- Discuss its effect and how to deal with it practically

Platform: Chapter titled - Week 4: Part 2 of the Biomechanics of Exercise and Resistance Training + Physiology of the Muscular System

- **WHAT DO WE NEED TO KNOW ABOUT DOMS:**

1- DOMS causes:

Tissue damage followed by inflammation affecting both muscle fibers and connective tissue



Practically, which two factors lead to DOMS?



2- How to manage DOMS:

NSAIDs (Non-Steroidal Anti-Inflammatory Drugs)?

Cold water immersion / Ice-bath?

Massage?

Stretching?

Low Intensity physical activity / Active Rest?

3- Can we prevent DOMS?

Isometric contractions prior to eccentric training?

BCAAs?

Nutrition & Hydration?

Cool down?

Consolidation:

By the end of Session 11, I am aware of:

- Different types of Fatigue & how to manage them practically
- Key principles of hypertrophy and their practical application
- The Stimulating Reps Model
- Physiological factors affecting Strength
- Factors affecting muscle function
- Physiological causes of soreness
- Effect of soreness and how to deal with it practically

Post-Session 11 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 12.

Pre-Session 12 Preparation:

-Watch all Cardiovascular and Respiratory videos of the Chapter titled: *Week 5: Physiology of the Energy + Cardiovascular + Respiratory Systems*

SESSION 12: Exercise Physiology – Cardiorespiratory System & General Nutrition

LESSON 1: CARDIORESPIRATORY SYSTEM ACUTE RESPONSE TO EXERCISE

Objectives:
- Identify the acute effects of exercise on Heart Rate, Stroke Volume, Cardiac Output, Blood Flow, Arteriovenous Oxygen Difference, Blood Pressure and Pulmonary Ventilation

Platform: Chapter Titled: Week 5: Physiology of the Energy + Cardiovascular + Respiratory Systems

- **ACUTE VS CHRONIC RESPONSE TO EXERCISE:**

Cardiorespiratory Systems provide oxygen to muscles so they can produce energy aerobically: What role plays the Lungs, Heart, Blood and Muscles to make this possible?

Let's define each of the below terms:

- **Heart Rate:**
- **Stroke Volume:**
- **Cardiac Output:**
- **Blood Flow:**

- **Arteriovenous Oxygen Difference:**

- **Blood Pressure:**

- **Blood Lactate**

- **Pulmonary Ventilation:**



Guess the Acute changes that occurs to each of the above cardiorespiratory terms during exercise?



Consolidation:
By the end of this lesson, I am aware of:
- Acute effects of exercise on the cardiorespiratory system

LESSON 2: CARDIORESPIRATORY SYSTEM CHRONIC RESPONSE TO EXERCISE

Objectives:

- Identify the chronic effects of exercise on Heart Rate, Stroke Volume, Cardiac Output, Blood Flow, Arteriovenous Oxygen Difference, Blood Pressure, Blood Lactate and Pulmonary Ventilation

Platform: Chapter Titled: Week 5: Physiology of the Energy + Cardiovascular + Respiratory Systems

GROUP ACTIVITY

Guess the Chronic changes that occurs to each of the above cardiorespiratory terms in response to exercise?

Consolidation:

By the end of this lesson, I am aware of:

- Chronic effects of exercise on the cardiorespiratory system

LESSON 3: VENTILATORY THRESHOLD

Objectives:
- Identify the training zones based on the ventilatory threshold and their training application

What is the Ventilatory Threshold?

- **1ST VENTILATORY THRESHOLD VT1:**

- **2ND VENTILATORY THRESHOLD VT2:**

GROUP ACTIVITY

Mix and Match:

Training Below VT1 (Zone 1&2)	Focus on Endurance & Fat Oxidation
Training Between VT1 & VT2 (Zone 3)	Improve Lactate Clearance
Training Above VT2 (Zone 4&5)	Anaerobic Work
	Focus on Power & VO2max
	Aerobic Work
	Improve Sustained Performance

Consolidation:

By the end of this lesson, I am aware of:

- Ventilatory Thresholds and their practical applications

Consolidation:

By the end of Session 12, I am aware of:

- Acute effects of exercise on the cardiorespiratory system
- Chronic effects of exercise on the cardiorespiratory system
- Ventilatory Thresholds and their practical applications

Post-Session 12 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 13.

Pre-Session 13 Preparation:

-Watch all Nutrition videos of the Chapter titled: *Week 6: Physiology of the Nervous and Endocrine Systems + Nutrition Part 1*

Session 13: Nutrition

LESSON 1: BASIC NUTRITION



Objectives:
-Understand the role of nutrition in promoting health, preventing disease, and supporting the body's essential needs through balanced food choices.

*(Platform: Chapter title: Week 6: Physiology of the Nervous and Endocrine Systems + **Nutrition Part 1**)*



Why is nutrition considered an important part of maintaining good health throughout life?



How can the foods we eat every day, affect our risk of developing diseases like diabetes or heart problems?



What are the six main nutrients the body needs, and why are they all essential for our health?



What makes a diet "healthy," and how do moderation, variety, and balance help us make better food choices?



Consolidation:
By the end of this lesson, I am aware of:
-The importance of good nutrition, the main nutrients the body needs, and how healthy food choices support overall well-being

LESSON 2: MACRONUTRIENTS OVERVIEW



Objectives:
-Understand the main energy-yielding nutrients — carbohydrates, proteins, and fats — and their roles in fueling the body and maintaining health.



What is the difference between macronutrients and micronutrients?



Which nutrients give the body energy, and how much energy does each provide?



Why is water called a macronutrient even though it doesn't give energy?



How do we lose water during the day, and why do we need to replace it often?



How can you tell if you're well-hydrated, especially during or after exercise?

Nutrient	Energy (Kcal/gram)
Carbohydrates	**4 kcal**
Proteins	**4 kcal**
Fats	**9 kcal**
Alcohol (not a nutrient)	**7 kcal**
Water	**0 kcal**

List three examples of food sources for each macronutrient.

Carbohydrates	**1. _____ 2. _____ 3. _____**
Proteins	**1. _____ 2. _____ 3. _____**
Fats	**1. _____ 2. _____ 3. _____**



Alcohol provides energy but is not a nutrient, while water is a nutrient but provides no energy.

Consolidation:
By the end of this lesson, I am aware of:
-The three energy-yielding macronutrients and their caloric values.
-The role of water as a non-energy macronutrient.
-The distinction between nutrients that provide energy and those that do not.

LESSON 3: CARBOHYDRATES

Objectives:
-Understand the structure, classification, and function of carbohydrates as the body's main energy source.

- **DEFINITION:**

Carbohydrates are organic molecules composed of carbon (C), hydrogen (H), and oxygen (O) in a 1:2:1 ratio. They are the body's preferred fuel.

- **STRUCTURE & CLASSIFICATION:**

Type	Description	Examples
Monosaccharides	Single sugars — basic unit of CHO	**Glucose** (blood sugar), **Galactose** (from lactose), **Fructose** (fruits, honey)
Disaccharides	Two sugar units chemically joined	**Sucrose** (Glucose + Fructose), **Maltose** (Glucose + Glucose), **Lactose** (Galactose + Glucose)
Polysaccharides	Long glucose chains	**Starch** (plants), **Glycogen** (humans/animals), **Fiber** (non-digestible CHO)



Glycogen Storage: 1/3 in the liver (used by the body) and 2/3 in the muscles (used only by muscles).

- **FIBER TYPES:**

Type	Function	Food Sources
Soluble Fiber	Slows digestion, and decreases absorption	Oats, nuts, seeds, fruits
Insoluble Fiber	Adds bulk, helps regulate stool	Whole grains, vegetables
Certain fibers act as prebiotics, feeding beneficial gut bacteria and supporting a healthy microbiome.		

- **CARBOHYDRATE RECOMMENDATIONS:**

Recommendation Type	Recommended Value / Range	Examples of Food Sources
AMDR (Acceptable Macronutrient Distribution Range)	45–65% of total daily calories	Bread, rice, pasta, fruits, starchy vegetables
RDA (Recommended Dietary Allowance)	130 g/day (minimum to support brain function)	Whole grains, oats, fruits, legumes
Fiber (Adequate Intake – AI)	Men ≤50: 38 g/day | Men ≥51: 30 g/day Women ≤50: 25 g/day | Women ≥51: 21 g/day	Oats, nuts, seeds, vegetables, fruits

Some carbohydrate-rich food examples — Portion providing 15 g of CHO:

Bread	1 regular slice (30 g)
Rolled oats	1/3 cup (30 g)
Plain sweet biscuits	2–3 pieces (15 g total)
Rice cake	1–2 pieces (12 g)
Rice, boiled	1/3 cup (55 g)
Fresh fruit salad	3/4 cup (155 g)
Banana	1 medium
Potato	1 medium (100 g)

- **CARBOHYDRATE ATHLETIC RECOMMENDATIONS:**

Why should athletes prioritize carbohydrate intake more than sedentary individuals?

What could happen if glycogen stores are not replenished properly?

How can a trainee meet their total CHO needs if they train twice a day?

Intensity	Program	Carbohydrate Targets Per Kg of the Trainee's BW
Minimal	Light training program (low intensity or skill-based exercise)	3–5 g per kg each day
Moderate	Moderate exercise program (≈1 hr/day)	5–7 g per kg each day
High	Endurance program (1–3 hr/day of moderate- to high-intensity exercise)	6–10 g per kg each day
Very High	Extreme exercise (>4–5 hr/day of moderate- to high-intensity exercise, such as Tour de France)	10–12 g per kg each day

Role of Carbohydrates in Training:

- Preferred energy source during moderate to high-intensity exercise (≥65% VO_2 max).
- Stored as glycogen in muscles and liver (limited capacity ≈ 2000-2500 Kcal ≈ 500–625g).
- A high-carbohydrate diet helps optimize glycogen stores and delay fatigue.
- Unlike fat, glycogen stores are limited — once depleted, performance drops significantly.

Carbohydrate Timing for Athletes:

The timing of carbohydrate intake before, during, and after exercise helps maximize energy and recovery.

Timing	Recommendation	Purpose / Notes
Pre-event (1–4 hr before)	1–4 g CHO/kg	Improves performance and replenishes glycogen. Choose easily digestible carbs.
During exercise (>60–90 min)	30–60 g CHO/hr	Maintains blood glucose & spares muscle glycogen. Use gels, drinks, or fruit.
Extended endurance (>5–6 hr)	90–100 g CHO/hr	Use mix of glucose + fructose for faster oxidation and energy.
Post-exercise (1–2 hr after)	1 g CHO/kg	Maximizes glycogen resynthesis and speeds recovery.

Which exercise snack would you personally choose before your next training session or competition?

Why is maintaining sufficient carbohydrate intake important for protecting muscle protein?

What would happen to performance, if no carbohydrates were consumed during long-duration training?

If you skip carbohydrates after training, what effect could it have on your next performance?

Pre-Event Fueling	During Event Fueling
Choose familiar, high-carbohydrate foods that are easy to digest (rice, oats, bananas, toast, pasta).	Choose easily digestible carbohydrate sources, either in liquid or solid form.
Avoid high-fat or high-fiber meals that may cause discomfort.	Combining glucose and fructose enhances carbohydrate oxidation and energy replenishment.
Focus on total carbohydrate intake rather than glycemic index (GI), as GI has minimal effect on performance.	



Fill in the table below with food or drink examples you would choose before, during, and after training to meet each goal.

Training Phase	Goal	Suggested Foods / Drinks
Pre-Event Fueling	**Provide energy and fill glycogen stores before exercise.**	
During Training	**Maintain energy and prevent fatigue during exercise.**	
Post-Exercise Recovery	**Replenish glycogen and support muscle recovery.**	

GROUP ACTIVITY

- **CASE STUDY: Toufic's Morning Training**

Client Profile:

- Name: Toufic
- Age: 25 years old
- Sport: Basketball
- Weight: 75 kg
- Height: 180 cm
- Training Time: 8:00 AM – 9:00 AM
- Training Duration & Intensity: Around 2 hours, moderate to high intensity
- Daily Routine: Wakes up at 6:30 AM – Sleeps at 11:00 PM
- Goal: Performance (not concerned about slight weight gain)

Task:

Help Toufic plan his carbohydrate intake throughout the day based on his current training schedule.



GROUP ACTIVITY

CASE STUDY – Continued

Now your client Toufic is training in the afternoon instead of the morning.
His new training schedule is from 5:00 PM to 6:00 PM, while his goals and daily routine remain the same.

Task

Help Toufic adjust his carbohydrate intake and timing throughout the day based on his new afternoon training schedule.



6:30 AM *11:00 PM*

- **CARBOHYDRATE ADEQUACY & METABOLISM:**

Term / Process	Meaning
Gluconeogenesis	**The body synthesizes glucose from protein when carbohydrates are too low.**
Protein-sparing action	**Eating enough carbohydrates prevents protein from being used for energy.**
Ketosis	**Happens when fat is broken down for energy because carbs are very low.**
Ketoacidosis	**A dangerous condition caused by the presence of too many ketone bodies in blood.**



Make sure daily carbohydrate intake is sufficient to spare protein from being used for energy. Avoid restrictive low-carb or keto diets to prevent excessive ketone production and potential ketoacidosis.

Consolidation:

By the end of this lesson, I am aware of:

-The main functions and energy role of carbohydrates in the body.

-The different types and classifications of carbohydrates.

-The daily and athletic carbohydrate requirements.

-The importance of carbohydrate timing before, during, and after exercise.

-The effects of inadequate carbohydrate intake on performance and metabolism.

LESSON 4: PROTEIN

Objectives:
-Understand the role, needs, and timing of protein intake — from its structure and sources to its impact on muscle maintenance and growth, through optimal distribution and total energy balance.

- **STRUCTURE & CLASSIFICATION:**

What are proteins made of, and what makes them different from carbohydrates and fats?

What are amino acids, and how do they combine to form proteins?

What are the three types of amino acids, and how do they differ?

What are the main roles of protein in the human body?

- **TYPE OF PROTEIN: ANIMAL OR PLANT?**



DEBATE TIME!

Alright team, let's split into two groups!

- **Group 1** will defend animal proteins.
- **Group 2** will defend plant proteins.

Each group will have 5 minutes to prepare three main arguments supporting your type of protein and one counterargument to challenge the other team.
When it's time to debate, each group will have two minutes to present their arguments.

Remember — Keep it Fun, Respectful, and Evidence-Based!



Is one source truly better than the other, or do both have a place in a balanced diet?

- **Comparison between Animal and Plant proteins:**

Feature	Animal Protein	Plant Protein
Amino Acid Profile	Complete – contains all essential amino acids (EAA).	Incomplete – lacks one or more essential amino acids (EAA).
Leucine Content	Higher leucine content (≈8–11%).	Lower leucine content (≈6–8%).
Muscle Protein Synthesis (MPS)	More effective in stimulating MPS; easily reaches the leucine threshold (≈2–3 g or 0.05 g/kg bodyweight).	Requires higher portions or combinations to reach the leucine threshold and maximize MPS.
Amino Acid Availability	Amino acids are absorbed efficiently and remain elevated longer in the blood, supporting recovery and growth.	Amino acid absorption is slower and may be less complete; combining sources improves overall availability.
Protein Quality	High-quality protein with a complete amino acid balance.	Lower individually but can become complete when complemented with other plant proteins.
Examples	Eggs, poultry, dairy, fish, beef.	Beans, lentils, tofu, quinoa, nuts, seeds.



Reaching about 2–3 g of leucine per meal helps trigger muscle growth, but remember: All amino acids and how long they stay in your blood, also play a big role in recovery!

Table 2. Composition of the essential AAs in 14 common staple plant foods, g/250 kcal of food

Food	Lysine	Methionine	Tryptophan	Phenylalanine	Valine	Leucine	Isoleucine	Threonine	Histidine
Cereal/Grain									
Rice	0.16	0.1	0.05	0.24	0.27	0.37	0.19	0.16	0.1
Wheat	0.26	0.15	0.12	0.45	0.43	0.65	0.35	0.28	0.22
Maize	0.15	0.12	0.04	0.28	0.26	0.71	0.19	0.17	0.18
Sorghum	0.12	0.1	0.07	0.31	0.27	0.76	0.22	0.22	0.12
Legume									
Peanut	0.41	0.14	0.11	0.59	0.48	0.74	0.4	0.39	0.29
Chickpea	0.91	0.18	0.13	0.73	0.57	0.97	0.58	0.51	0.37
Common bean	1.21	0.27	0.21	0.96	0.93	1.41	0.78	0.74	0.49
Cowpea	1.18	0.25	0.22	1.02	0.83	1.34	0.71	0.67	0.54
Soy	1.31	0.27	0.29	1.03	0.98	1.61	0.96	0.86	0.53
Tuber/Root									
Cassava	0.29	0.07	0.08	0.22	0.25	0.3	0.21	0.2	0.12
Potatoes	0.32	0.08	0.08	0.23	0.3	0.32	0.22	0.19	0.12
Sweet potatoes	0.19	0.08	0.09	0.26	0.25	0.27	0.16	0.24	0.09
Yams	0.13	0.04	0.03	0.15	0.13	0.2	0.11	0.11	0.07
Skimmed milk	1.70	0.54	0.31	1.06	1.50	2.14	1.40	1.01	0.57

What is the difference between vegans and vegetarians? Are they able to recover properly and increase their muscle mass?

- **<u>Complementary Proteins:</u>**

Complementary proteins combine plant foods that balance each other's amino acid gaps, creating a complete protein profile:

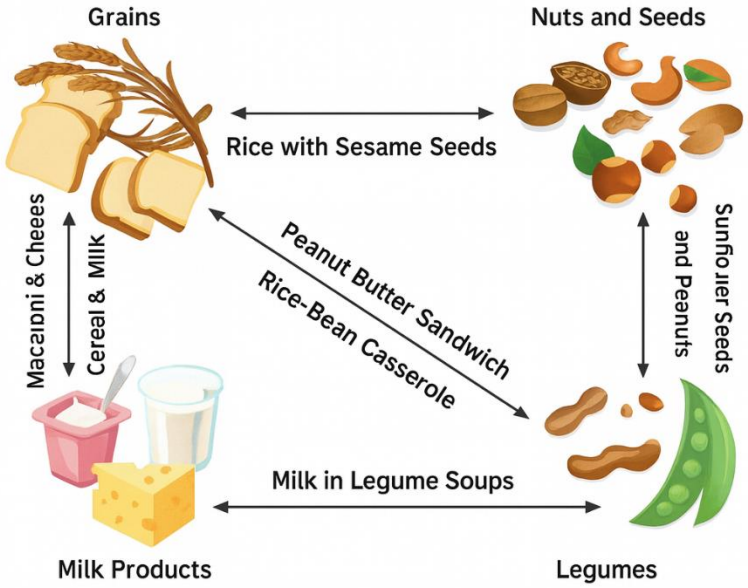
- Beans and rice

- Cereal with almond milk

- Hummus with tahini on toast

- Baked beans on toast

- Granola with grains and seeds



Work in pairs and suggest Lebanese or Mediterranean dishes that combine plant proteins (such as legumes, grains, or seeds) to form Complementary Proteins.

- **PROTEIN RECOMMENDATIONS:**

Category	Recommendation / Notes
DRI	0.8 g/kg body weight per day
AMDR	10–35% of total energy intake
Higher Needs	For infants, children, and trainees to support growth, repair, and recovery
Exercise Contribution	Proteins provide 5–15% of total energy during exercise
Food Choice	Prefer high-quality sources with healthy fat profiles

- **Estimated Protein Needs for Trainees:**

Example of Trainee or Situation	Effect	Protein needs per kg of body weight
Daily needs	Overall health and protection against lean mass loss (general and elderly population)	1.2–1.6 g/kg/day
Suggested target	Maximize resistance training adaptations in non-dieting individuals.	1.62–2.2 g/kg/day
Suggested targets	Maximizing muscle retention while dieting in relatively lean resistance training	2.3–3.1 g/kg

- **Protein Rich Foods:**

Type of Protein Source	Amount of Product Needed to Provide 20g Protein
Animal Foods	4 small eggs 500ml reduced-fat or low-fat milk 80g (2 slices) of reduced-fat cheese ~70g lean beef, lamb or pork, lean chicken
Plant Based Food	4 cups (240g) whole grain cereal 4 cups (600g) cooked rice 2 cups (440g) lentils or kidney beans

- **PLANT BASED PROTEIN RECOMMENDATIONS:**

Category	Recommendation (g/kg/day)	Notes
General population	0.8	Baseline requirement for healthy adults
Non-athletic vegetarians (eggs & dairy)	1.0	Slightly higher due to lower digestibility vs. mixed diets
Non-athletic vegans	1.4	Compensates for lower protein quality and digestibility
Athletic vegans (bulking)	2.0	Supports muscle gain during surplus
Athletic vegans (cutting)	2.7	Helps preserve muscle mass during deficit

Notes: Values are general recommendations and may need individual adjustment based on goals, body composition, and training phase.

Why increasing protein intake alone is insufficient to preserve muscle mass?



Increasing protein alone doesn't preserve muscle mass — total calories must also increase to maintain an anabolic state and spare muscle tissue.

True or False: The anabolic window only lasts 1 hour after exercise.

What does the term “anabolic door” represent in relation to protein timing?

How long does the anabolic period around a workout typically last?

How much protein per meal, is generally needed to optimize muscle protein synthesis?

Is there a limit to how much protein the body can utilize from a single meal?



Consolidation:

By the end of this lesson, I am aware of:

-How protein structure and amino acid composition determine its function and quality.

-The different dietary sources of protein and the importance of combining plant-based foods for complete amino acid profiles.

-How protein timing and distribution around training optimize muscle protein synthesis through the anabolic door concept.

LESSON 5: FAT

Objectives:
-Understand the structure, types, and functions of fats, their role in energy balance and health, and how exercise intensity and duration influence fat oxidation.

- **STRUCTURE, CLASSIFICATION AND FUNCTION:**

What makes fat the most concentrated source of energy in the body?

If fats are stored in such large amounts, why don't we rely on them as the main energy source during intense exercise?

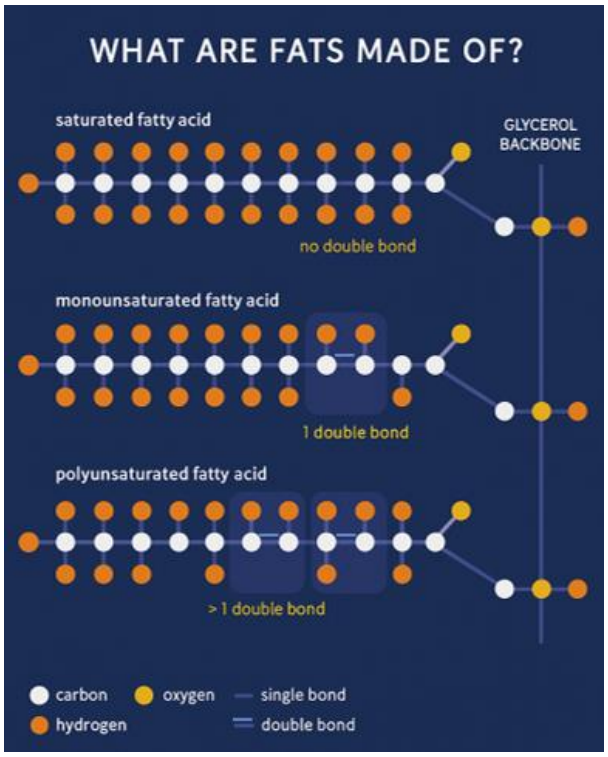
How does body fat help protect and regulate our body beyond providing energy?

Which vitamins depend on fat for their absorption, and why is this important?

How do fats contribute to the way food tastes and keeps us full?

- **TYPES OF FAT & THEIR EFFECTS:**

Type of Fat	Structure & Examples	Physical Form	Effect on Blood Lipids	Main Food Sources
Saturated Fat	No double bonds	Solid at room temp	↑ LDL and total cholesterol	Butter, meat, full-fat dairy, coconut oil
Monounsaturated Fat (MUFA)	1 double bond	Liquid	Neutral or ↓ LDL, ↑ HDL	Olive oil, avocado, nuts, peanuts
Polyunsaturated Fat (PUFA)	2+ double bonds	Liquid	↓ LDL and total cholesterol	Sunflower oil, soybean oil, fatty fish, walnuts
Trans Fat	Hydrogenated unsaturated fat	Solid	↑ LDL, ↓ HDL, ↑ CVD risk	Processed snacks, fried foods, pastries



- **LDL & HDL:**

Lipoprotein	Function	Health Impact
LDL (Low-Density Lipoprotein)	Carries cholesterol from liver to tissues	High levels → Artery buildup → ↑ heart disease risk
HDL (High-Density Lipoprotein)	Carries excess cholesterol from tissues to liver	High levels → Clears arteries → ↓ heart disease risk

- **FAT/LIPID RECOMMENDATIONS:**

Category	Recommendation / Information
AMDR (General Population)	20–35% of total calories from fat
Trainees (ACSM Recommendation)	20–35% of total energy intake
Saturated Fat	< 7% of total calories
Polyunsaturated Fat (PUFA)	Up to 10% of total calories
Monounsaturated Fat (MUFA)	Up to 20% of total calories
Minimum Fat Intake	Should not fall below 20% of total energy
Low-Fat Diet Warning	May impair absorption of fat-soluble vitamins (A, D, E, K) and reduce omega-3 & omega-6 intake
High-Fat Diet Warning	May negatively affect exercise performance and overall health

- **Fat content of some common foods:**

1 cup whole milk	7–10 g
½ medium avocado	15 g
1 tsp butter	4–6 g
1 tbsp mayonnaise	7–10 g
¼ cup sunflower seeds (dry roasted)	15 g
1 large cheeseburger	25 g

- **FAT AS EXERCISE FUEL:**

Exercise Intensity*	RER**	Fuel Type	Biochemistry***
Low (<25% VO_2 max)	0.7	Fat	$C_{16}H_{32}O_2 + 23\ O_2 \rightarrow 16\ CO_2 + 16\ H_2O$
Moderate (50% VO_2 max)	0.85	Fat + carbohydrate (increasingly carbohydrate)	Mix of palmitate and glucose usage
High (100% VO_2 max)	1.00	Carbohydrate	$C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$

** Low-intensity exercise can be very prolonged (several hours) and moderate-intensity exercise somewhat less (1–4 h), while high-intensity exercise is measured in minutes.*
*** Respiratory exchange ratio (RER) = CO_2 produced / O_2 consumed. Often used interchangeably with respiratory quotient (RQ).*
**** RER values validate biochemistry; palmitate burning ratio 16/23 = 0.70, glucose ratio 6/6 = 1.00.*

Fat Versus Carbohydrate Oxidation During Fasted Exercise of Different Intensities (Adapted from NSCA's Guide to Sport and Exercise Nutrition)

Exercise Duration	Serum Glycerol†
Prolonged, >60 min	Higher
Moderate, 30 to 60 min	Moderate
Brief, <30 min	Lower
Serum glycerol concentrations (fat breakdown and mobilization) are high during prolonged fasting and during low- to moderate-intensity prolonged exercise; greater fat mobilization relates to higher fat oxidation.	

Fat Mobilization During Fasted Exercise of Different Durations (Adapted from NSCA's Guide to Sport and Exercise Nutrition)



Client A jogging for 30min vs Client B performing interval training cardio. Which client is burning more fat and why?



What about training on a fasted stomach, does it help us lose more fat?



Consolidation:
By the end of this lesson, I am aware of:
-The different types and functions of fats in the body and diet.
-The recommended fat intake range and the impact of very low or high fat diets on health and performance.
-How exercise intensity and duration influence fat oxidation as an energy source.

LESSON 5: FLUIDS

Objectives:
-Understand the importance of hydration for performance and health, and how to apply proper strategies to maintain fluid balance before, during, and after exercise.

What do you think are the main benefits of staying properly hydrated — and what disadvantages or risks come with dehydration?

Why do you think hydration is considered one of the most important factors for both health and performance — and what risks might occur when it's neglected?

Many people say "drink as much water as possible" — do you agree or disagree? Why?

What signs do you think indicate proper hydration versus overhydration?

When you experience fatigue or muscle cramps during exercise, to what extent do you believe hydration influences these symptoms?

- **HYDRATION STRATEGIES:**

Prior To Exercise	During Exercise	Post-Exercise
• Drink around 500 ml of fluid 2 to 3 hours pre-exercise	• Drink 150 to 350 ml every 15 to 20 mins • Equivalent to 600 to 1200 ml per hour	• Measure fluid/sweat losses • For every 0.5 kg lost, drink 450 to 675 ml • Add 25% to 50% more than existing weight loss to ensure hydration • 1 L of fluid = 1 kg of BW

- **MEASURING FLUID/ SWEAT LOSSES:**

Step	Instruction
1.	Measure body weight before and 1 hour after exercise (minimal clothing, barefoot).
2.	Record fluid consumed during training.
3.	Sweat Loss Formula: Sweat loss (L) = BW before (kg) – BW after (kg) + fluid consumed (L)
4.	Sweat Rate Formula: Sweat rate (L/hr) = [Sweat loss (L) ÷ exercise time (min)] × 60

Can you think of any other practical way to measure your fluid levels?

- **COMMERCIAL SPORTS DRINKS:**

Why do you think some athletes prefer sports drinks over plain water?

In what types of training sessions might a sports drink be more beneficial than water?

Aspect	Key Points
Purpose:	• To keep the trainee hydrated and fueled during exercise.
Main Components:	• Contain carbohydrates (CHO) and electrolytes.
Best Used For:	• Prolonged exercise > 60 mins • Moderate to intermittent intensity sessions
Benefits:	• Improve performance • Maintain blood glucose and prevent hypoglycemia • Support carbohydrate oxidation • Reduce glycogen depletion and delay fatigue
Choice Depends On:	• Type and duration of exercise • Pre-event nutrition • Climatic conditions



Case Study:

Lea weighs 62 kg before her training session and 61.2 kg after.

During her 60-minute workout, she drank 400 ml of water.

1. **How much total sweat did Lea lose during her workout?**
2. **What was her sweat rate per hour?**
3. **Based on her results, how could she adjust her hydration strategy for next time?**
4. **If Lea's workout lasted longer than 60 minutes, do you think a sports drink would be more beneficial than plain water? Explain your answer?**

Consolidation:

By the end of Session 13, I am aware of:

-The role of carbohydrates as the body's main source of energy during exercise and recovery.

-The role of protein in muscle repair, recovery, and growth, along with its increased needs in trainees.

-The role of fats in providing energy, supporting hormones, and aiding vitamin absorption.

-The importance of proper hydration for maintaining performance, preventing heat illness, and replacing fluid losses effectively.

Post-Session 13 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 14.

Pre-Session 14 Preparation:

Watch the Micronutrients and Energy balance videos from the Platform Chapters Titled:

-Week 6: Physiology of the Nervous and Endocrine Systems + Nutrition Part 1

-Week 7: Fitness Nutrition + Evidence Based Coaching Part 1

Session 14: Nutrition

LESSON 1: MICRONUTRIENTS

Objectives:
-Understand the role of vitamins and minerals in maintaining health, supporting metabolism, and enhancing performance, as well as their classifications, sources, and importance in daily nutrition.

1- VITAMINS:

Since micronutrients don't provide energy, why do you think they're still vital for athletes and trainees?

What's the difference between fat-soluble and water-soluble vitamins — and why might that matter for our diet?

Type	Description	Storage	Excretion	Toxicity Risk
Fat-Soluble Vitamins	Vitamins A, D, E, and K	Stored in the liver and fatty tissues	Not easily excreted	High risk when accumulated in large amounts
Water-Soluble Vitamins	Vitamins C and B-complex	Not stored in tissues	Excreted in urine	Low risk due to regular excretion

Vitamin	RDA / AI (Men / Women)	Best Sources	Functions
A	900 µg / 700 µg	Yellow or orange fruits & vegetables, fortified oatmeal, liver, dairy products	Formation and maintenance of skin, hair and mucous membranes; helps people see in dim light; bone and tooth growth
B1 (Thiamin)	1.2 mg / 1.1 mg	Fortified cereals and oatmeal, meat, rice and pasta, whole grains, liver	Helps the body release energy from carbohydrates during metabolism; muscle growth
B2 (Riboflavin)	1.3 mg / 1.1 mg	Whole grains, green leafy vegetables, organ meat, milk, eggs	Helps the body release energy from protein, fat, and carbohydrates during metabolism
B6 (Pyridoxine)	1.3 mg / 1.3 mg	Fish, poultry, lean meat, bananas, prunes, dried beans, whole grains, avocados	Helps build body tissue and aids in metabolism of protein
B12 (Cobalamin)	2.4 µg / 2.4 µg	Meats, milk products, seafood	Aids cell development, functioning of the nervous system, and the metabolism of protein and fat
Biotin	30 µg / 30 µg	Cereal/grain products, yeast, legumes, liver	Involved in metabolism of protein, fats and carbohydrates
Choline	550 mg / 425 mg	Milk, liver, eggs, peanuts	A precursor of acetylcholine; essential for liver function
Folate (Folic Acid)	400 µg / 400 µg	Green leafy vegetables, organ meat, dried peas, beans, lentils	Aids in genetic material development; involved in red blood cell production
Niacin	16 mg / 14 mg	Meat, poultry, fish, enriched cereals, peanuts, potatoes, dairy products, eggs	Involved in carbohydrate, protein, and fat metabolism
Pantothenic Acid	5 mg / 5 mg	Lean meat, whole grains, legumes, vegetables, fruits	Helps release energy from fats and vegetables
C (Ascorbic Acid)	90 mg / 75 mg	Citrus fruits, berries, kiwis, and vegetables—especially peppers	Essential for structure of bones, cartilage, muscle, and blood vessels; helps maintain capillaries and gums and aids in absorption of iron
D	15 µg / 15 µg	Fortified milk, sunlight, fish, eggs, butter, fortified margarine	Aids in bone and tooth formation; helps maintain heart function and nervous system function
E	15 mg / 15 mg	Fortified and multigrain cereals, nuts, wheat germ, vegetable oils	Protects blood cells, body tissue, and essential fatty acids from destruction in the body
K	120 µg / 90 µg	Green leafy vegetables, fruit, dairy, grain products	Essential for blood-clotting functions

2- MINERALS:

Minerals are classified into two categories:

- **Major minerals** – **>100 mg/day** (e.g., calcium, magnesium, potassium, sodium).
- **Trace minerals** – **<100 mg/day** (e.g., iron, zinc, iodine, selenium).

Mineral	RDA / AI (Men / Women)	Best Sources	Functions
Calcium	1,000 mg / 1,000 mg	Milk and milk products	Strong bones, teeth, muscle tissue; regulates heartbeat, muscle action, and nerve function; blood clotting
Chromium	35 µg / 25 µg	Corn oil, clams, whole-grain cereals, brewer's yeast	Glucose metabolism (energy); increases effectiveness of insulin
Copper	900 µg / 900 µg	Oysters, nuts, organ meat, legumes	Formation of red blood cells; bone growth and health; works with vitamin C to form elastin
Fluoride	4 mg / 3 mg	Fluorinated water, teas, marine fish	Stimulates bone formation; inhibits or even reverses dental caries
Iodine	150 µg / 150 µg	Seafood, iodized salt	Component of hormone thyroxine, which controls metabolism
Iron	8 mg / 18 mg	Meats (especially organ meats), legumes	Hemoglobin formation; improves blood quality; increases resistance to stress and disease
Magnesium	420 mg / 320 mg	Nuts, green vegetables, whole grains	Acid/base balance; important in metabolism of carbohydrates, minerals, and sugar (glucose)
Manganese	2.3 mg / 1.8 mg	Nuts, whole grains, vegetables, fruits	Enzyme activation; carbohydrate and fat production; sex hormone production; skeletal development
Molybdenum	45 µg / 45 µg	Legumes, grain products, nuts	Functions as a cofactor for a number of enzymes in humans
Phosphorus	700 mg / 700 mg	Fish, meat, poultry, eggs, grains	Bone development; important in protein, fat, and carbohydrate utilization
Potassium	4,700 mg / 4,700 mg	Lean meat, vegetables, fruits	Fluid balance; controls activities of heart muscle, nervous system, and kidneys
Selenium	55 µg / 55 µg	Seafood, organ meats, lean meats, grains	Protects body tissues against oxidative damage from radiation, pollution, and normal metabolic processing
Zinc	11 mg / 8 mg	Lean meats, liver, eggs, seafood, whole grains	Involved in digestion and metabolism; important in development of reproductive system; aids in healing



You don’t have to completely avoid processed foods! Enjoying them in moderation alongside whole, nutrient-dense meals can help you stay consistent and maintain a healthy relationship with food.

Consolidation:

By the end of this Lesson, I am aware of:

-The role of vitamins in supporting metabolism, energy production, and tissue repair.

-The role of minerals in maintaining body structure, nerve and muscle function, and metabolic balance.

-The importance of balanced micronutrient intake to support overall health and performance.

LESSON 2: NUTRITIONAL LEARNING TOOLS AND GUIDELINES

Objectives:
-Understand the main dietary recommendations, nutrient needs, and food guides to make informed and balanced nutrition choices.

1. Dietary Guidelines:

Recommendations of the Dietary Guidelines for Americans 2015-2020:


Follow a healthy eating pattern across the lifespan.


Focus on variety, nutrient density, and amount.


Limit calories from added sugars and saturated fats and reduce sodium intake.


Shift to healthier food and beverage choices.


Support healthy eating patterns for all.

- **DRI Recommendations:**

Nutrient	Recommendations
Carbohydrates	**AMDR**: 45–65% of total calories **RDA**: 130 g/day (adults & children) — based on the minimum glucose needed by the brain
Fat	**AMDR**: 20–35% of total calories
Protein	**AMDR**: 10–35% of total calories **RDA**: 0.8 g/kg body weight (both men and women)
Fiber	**AI**: 38 g/day (men ≤50 years) **AI**: 30 g/day (men ≥51 years) **AI**: 25 g/day (women ≤50 years) **AI**: 21 g/day (women ≥51 years)
Essential Fatty Acids	Alpha-linolenic acid (**AI**): 1.6 g/day (men), 1.1 g/day (women) Linoleic acid (**AI**): 17 g/day (men), 12 g/day (women)

2. Important Vitamins and Minerals for Athletic Performance:

Sodium

- Important for fluid level and blood circulation → Performance
- Lost through perspiration.
- Inadequate replacement → hyponatremia, low performance, heat cramps, and heat-related illness.
- Individual variation in sodium loss.
- Minimal losses replaced by normal diet; high losses → increase sodium-rich foods.
- Salting foods to taste and having a variety of vegetables allows sufficient sodium levels.

Calcium

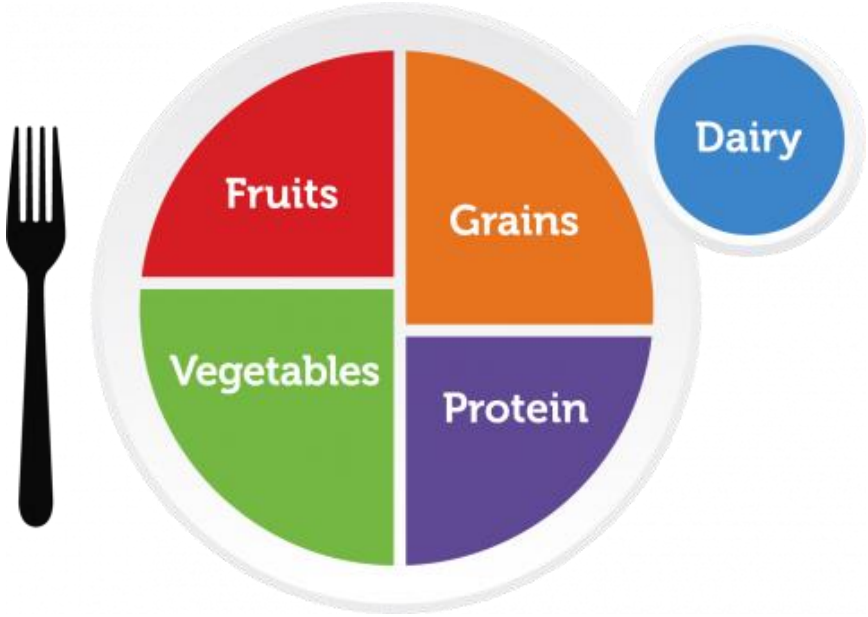
- Crucial for bone health and muscle contraction.
- Mixed evidence between calcium intake and fracture rates.
- Increasing intake: ↑ bone mineral density which ↓ fractures.
- Trainees lose calcium through perspiration.
- Recommended intake: 1000–1300 mg.
- Sources: dairy, broccoli, kale, sardines, canned salmon, fortified foods.
- Best way: 3 servings dairy/day (1 cup milk, 30 g cheese, 1 cup yoghurt).
- Fortified soy dairy products suitable.

Vitamin D

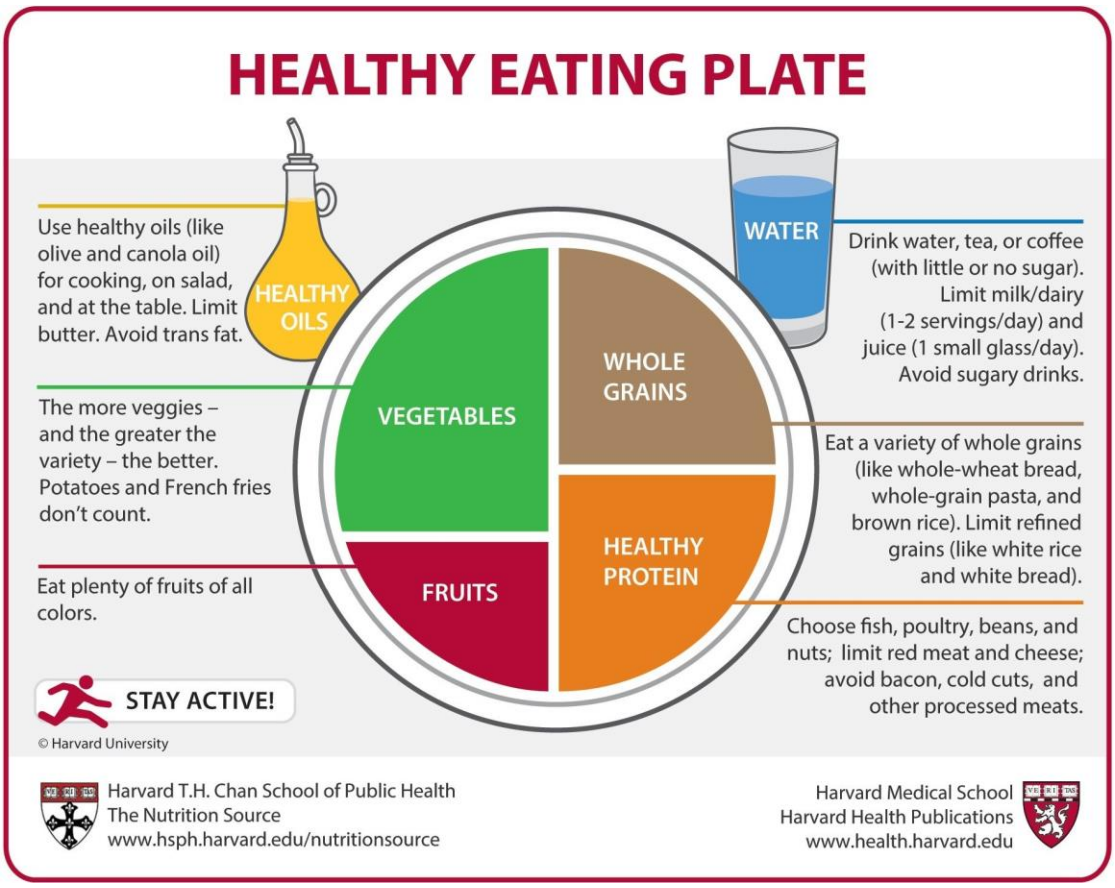
- Important for protein synthesis, calcium absorption, electrolyte metabolism, gene expression, and immune function.
- Indirectly affects performance: deficiency → muscle weakness, injury risk, delayed recovery.
- Supplement only if deficient.
- Middle-eastern women at high deficiency risk.
- RDA: 15 mcg (600 IU) for 19–50 yrs.
- Sources: sun exposure, fatty fish (salmon, tuna), beef liver, egg yolks, fortified products.
- Indoor trainees: ≥15 min sunlight daily or supplement.

Iron
▪ Component of hemoglobin & myoglobin → oxygen delivery. ▪ Low levels: ↓ O_2 to tissues → ↓ VO_2 max. → anemia if severe ▪ Daily iron needs vary by age and sex. ▪ Two forms: Heme (animal, 5–35% absorbed) & Nonheme (plant, 2–20% absorbed). ▪ Vitamin C, organic acids, meat, fish, poultry enhance nonheme absorption. ▪ Trainees should match foods for optimal iron status.

3. My Plate/Learning Tool:



Take a close look at the MyPlate (USDA) model. Based on what you see, identify three positive aspects that make it useful or effective, and three negative aspects or limitations that could be improved.



Look at the Harvard Healthy Eating Plate. What are the main differences you notice compared to MyPlate, and what do you think about it?



Work in small groups and come up with **general diet advice you would give to clients** aiming for a healthier lifestyle. Keep it **practical, simple, and realistic**; the type of advice that encourages long-term, consistent habits!

Topic	Example	Your Advice
Portion Control	Eat fries, but in moderation.	
Food Substitution	Replace potato chips with popcorn.	
Mindful Eating		
Avoiding Extreme Diets		
Physical Activity		
Drinks and Beverages		
Weight vs. Muscle Goals		

Consolidation:

By the end of this Lesson, I am aware of:

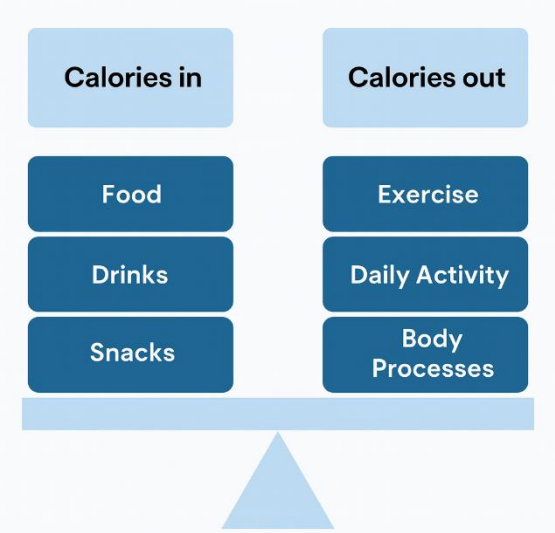
-How dietary guidelines, DRIs, and food guides support balanced eating.

-The key vitamins and minerals important for trainees.

-How to interpret nutrition labels to make healthier and more informed food choices.

LESSON 3: THE ENERGY BALANCE EQUATION

Objectives:
-Understand how the balance between calories in and calories out determines changes in body weight.
-Recognize the key components that contribute to total energy expenditure and how they affect weight loss or gain.



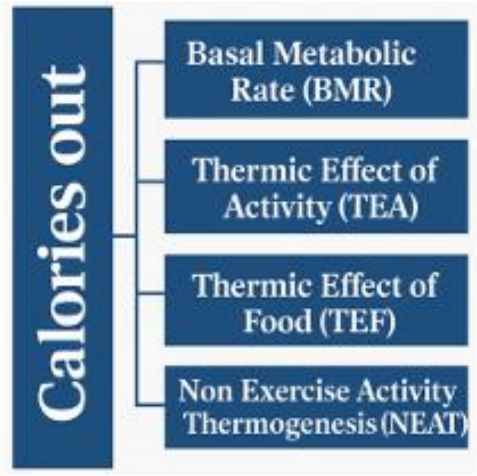
- **CALORIES IN:**

Let's Recall! How many calories does each macronutrient provide per gram?
Protein:
Carbs:
Fat:
Alcohol:

How do we calculate the total calories in a food based on its macronutrient content?

Calculate the total calories in 100 grams of chicken breast containing (~30 g protein), (~0 g carbohydrates), and (~2.5 g fat):

- **CALORIES OUT:**



Basal Metabolic Rate (BMR)	• Minimum energy our body needs to survive. • Reflects energy expended while physically and mentally at rest. • 60–70% of total energy expenditure. • Directly related to body weight; more body mass → higher BMR.
Exercise Activity Thermogenesis (EAT) / Thermic Effect of Exercise (TEE)	• Energy we burn during planned exercise such as training at the gym, walking, or biking. • Accounts for 10–15% of total energy expenditure. • Increases resting metabolic rate.
Non-Exercise Activity Thermogenesis (NEAT)	• Includes energy expended walking to work, typing, performing agricultural tasks, going up the stairs (daily life activities)... • Increases with overfeeding and decreases with underfeeding (adaptive thermogenesis).
Thermic Effect of Food (TEF)	• Energy expended to digest and absorb food. • Protein: 15–30%, Carbohydrates: 5–10%, Fat: 0–3%. • Usually accounts for 10% of total calories out.

Fill in the following regarding Energy Balance:

Calories in > Calories out	=
Calories in < Calories out	=
Calories in = Calories out	=

- **ENERGY BALANCE: WEIGHT LOSS:**

Calories In:

What makes your body weight change during the week? List every variable you can think of.

What is the only condition needed to lose body fat?

Give one example of a healthy food that can still be easy to overeat, and explain why this happens?

	
150 grams of steak (204 kcal) 200 grams carrots (69 kcal) 100 grams potato (87 kcal) 200 grams Lettuce (30 kcal)	1 slice of a large super supreme pizza
Has 374 kcal	Has 410 kcal
Protein 39 grams Carbs 35 grams Fat 5.3 grams	Protein 17 grams Carbs 36 grams Fat 23 grams

Excluding calories in and calories out, based on the pictures above, which choice is more favorable for weight loss?

How can eating more protein help your calorie balance even if total calories stay the same?

True or false: Burgers, pizza, and donuts are okay regularly if total calories are controlled. Explain your answer.

Mention two ways you can reduce your calorie intake without removing entire food groups.

Calories OUT: Let's Recall!

What determines most of your BMR, and how can resistance training indirectly influence it?

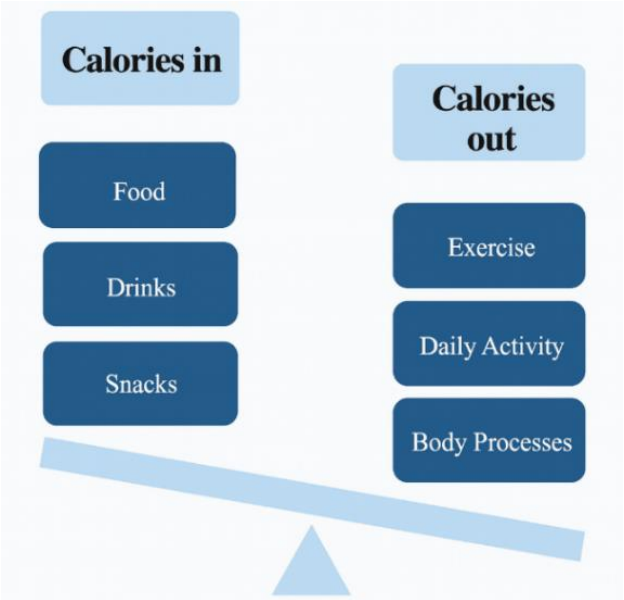
How does exercise contribute to the "calories out" part beyond just burning energy during the workout?

How can small daily movements or habits lead to big differences in total calorie expenditure?

Which macronutrient has the highest thermic effect, and how does this impact total calories burned?

Expand your Knowledge: What else can affect weight loss?

- **Sleep quality:** Poor quality slee is a contributing factor towards weight gain. It has been directly linked to altered food behaviour[103]. In fact, one study compared either 8.5 or 5.5 houts of sleep per night in combination with a reduced calorie diet found that although the total amount of weight loss was the same between groups, the shorter sleep duration group lost less fat mass and more lean body mass[104]. This means that less sleep will not stop you from loosing weight but it can skew it away from fat loss[103].
- **Water Intake:** The association between inadequate hydration and obesity, after controlling for confounding variables, has not previously been shown on a population level and suggests, that water, may deserve greater focus in weight management and clinical strategies[105].



- **ENERGY BALANCE: WEIGHT GAIN:**

Calorie Surplus → Weight Gain

Too Large Surplus → More Fat Gain

Moderate Surplus + Protein + Resistance Training → More Muscle

- **Quality of Weight Depends On:**

Condition	What Happens
Excessive Caloric surplus (over 500 kcal) & mostly carbs + fats	Most weight gained from fat
Moderate surplus (≤ 500 kcal) + adequate protein + resistance training	Most weight gained from muscle with some fat
Large surplus without resistance training	More fat gain, not necessarily more muscle gain

- **Clean Bulk vs. Dirty Bulk:**

Clean Bulk (~500 kcal increase):

• Gradual, controlled increase
• High protein focus
• Best for lean mass gain

Dirty Bulk (too high & uncontrolled):

• Excessive food intake
• "I'm on a bulk" mindset
• Leads to unnecessary fat gain

-"With the low protein diet, more than 90% of the extra energy was stored as fat. Because there was no change in lean body mass" - Bray et al., 2012

Based on what we discussed, why doesn't a very large surplus always lead to more muscle gain?



Consolidation:

By the end of this lesson, I am aware of:

-How energy intake and expenditure determine body weight changes.

-That a deficit leads to fat loss, while a moderate surplus with resistance training promotes muscle gain.

-The influence of protein intake, sleep, and hydration on overall body composition.

LESSON 4: BODY COMPOSITION

Objectives:
-Understand how to apply energy balance and macronutrient calculations to design nutrition plans that support specific body composition goals

- **Body Recomposition:**

What does body recomposition mean, and which factors influence it?

In which type of individuals is body recomposition most likely to occur according to studies?

How does the size of a calorie deficit affect muscle mass during recomposition?

What are the two key elements required to optimally gain muscle during recomposition?

1- Protein Recommendations:

Purpose	Protein Intake
For health and protection against lean mass loss (general and elderly population)	1.2 - 1.6 g/kg/day
To maximize resistance training adaptations in non-dieting individuals	1.62 - 2.2 g/kg/day of body weight
To maximize muscle retention while dieting in relatively lean resistance training subjects	2.3 - 3.1 g/kg of fat free mass

2- Energy Balance Equation: How to calculate the Calories IN



- **Reading Food Labels:**

What information can you learn from this nutrition label, and which part do you find most useful?

What are some challenges people face when trying to understand food labels?

Do you think nutrition labels help people make healthier choices? Explain your answer

- **Understanding Food Labels:**

Category	Main Point
Serving Size	Always check the serving size — nutrition values are based on it. Eating more means higher calories and nutrients.
Calories & Calories from Fat	Shows how much energy and how many calories come from fat.
Nutrients to Limit	Watch for sodium, cholesterol, and trans-fat.
Fiber	Choose foods with about 5 g or more per serving.
Vitamins & Minerals	Pick foods rich in vitamins and minerals.
% Daily Value	Based on a 2000 kcal diet: <5% = low, >20% = high. Helps compare products.

Let's do some calculations using the above food label if a client eats 42g of this product?

How can Artificial Intelligence (AI) tools and apps assist clients in tracking their food intake, and how accurate are they in estimating calories and nutrients?

(MyFitnessPal, Cronometer, Lose It!, Yazio, ChatGPT, Fitbit, Apple Health, Samsung Health)

- **Kitchen Scale:**

Practical Example

Meal:

- 120 g cooked rice

- 150 g grilled chicken breast

- 1 cup salad (with lemon dressing)

Food	Amount	Carbs (g)	Protein (g)	Fat (g)	Calories
Rice	120 g	35 g	3 g	0 g	160 kcal
Chicken	150 g	0 g	45 g	3 g	210 kcal
Salad	1 cup	5 g	1 g	3 g	60 kcal
Total		40 g	49 g	6 g	≈ 430 kcal

Energy Balance Equation: How to calculate the Calories OUT (TDEE)

TDEE = BMR x AF

a- Calculate BMR:

Harris–Benedict Equation

For Men: BMR = 88.36 + (13.4 × weight in kg) + (4.8 × height in cm) − (5.7 × age in years)

For Women: BMR = 447.6 + (9.2 × weight in kg) + (3.1 × height in cm) − (4.3 × age in years)



Katch–McArdle Formula

BMR = 370 + (21.6 × Lean Body Mass in kg)

Lean Body Mass (LBM) = Body Weight × (1 − Body Fat %)



Mifflin St Jeor

Males: 10 x Wt (kg) + 6.25 x Ht (cm) - 5 x Age + 5

Females: 10 x Wt (kg) + 6.25 x Ht (cm) - 5 x Age – 161

Activity Factor Table for TDEE Calculation



Activity Level	Description	Activity Factor
Sedentary	Little or no exercise	1.2
Light Activity	Light exercise/sports 1 to 3 days per week	1.375
Moderate Activity	Moderate exercise/sports 3 to 5 days per week	1.55
Very Active	Hard exercise/sports 6 to 7 days per week	1.725
Extra Active	Very hard exercise/sports 6 to 7 days per week and a physically demanding job	1.9



Practice by using your weight, height, age, gender, and activity level to **calculate** your TDEE:

- **Energy Adjustments for Fat Loss and Muscle Gain:**

Goal	TDEE Adjustment	Effect
Fat Loss (Slow Cut)	TDEE - 10%	Slower fat loss, more muscle mass preserved
Fat Loss (Aggressive Cut)	TDEE - 25%	Faster fat loss, higher risk of losing muscle mass
Muscle Gain	TDEE + 10%	Allows muscle gain while limiting fat mass (going for higher than 10% increases the chance of fat gain… its individualized depending on the client's goals

You can lose fat (not weight) and gain muscle when eating at maintenance calories if you are consuming a high protein diet coupled with a resistance training program!

3- How to Calculate Macronutrient Intake:

Macronutrient	Calories per gram	Calculation Method
Protein	4 kcal/g	Calculated according to goals by factor of body weight: 0.8–2.5 g/kg
Fat	9 kcal/g	Around 20–35% of total calorie intake
Carbohydrates	4 kcal/g	Calculated from remaining calories after subtracting protein and fat from TDEE.
Formula: Carbohydrate intake (g) = TDEE – { (calories from protein) + (calories from fat) }		



- **Case Study:**

Client Information:

- Name: Ahmad
- Age: 27 years
- Weight: 80 kg
- Height: 178 cm
- Occupation: Office job (mostly sedentary during the day)
- Training: 4 times per week (mix of resistance and cardiovascular training)
- Goal: Lose body fat while maintaining muscle mass
- Current concern: Low energy levels and difficulty maintaining strength

1- **Calculate Ahmad's BMR.**
2- **Estimate his TDEE based on his activity level.**
3- **Determine how many calories he should consume to support his goal.**
4- **Calculate his daily protein, fat, and carbohydrate needs.**
5- **Suggest nutrition advice that could help Ahmad achieve his goal effectively.**
6- **Discuss how sleep, hydration, and daily movement might influence his progress.**
7- **If Ahmad's progress slows down, what adjustments would you recommend?**

Consolidation:

By the end of Session 14, I am aware of:

-The essential roles of micronutrients and their importance for overall health.

-The key dietary guidelines that help maintain balanced nutrition.

-How the energy balance equation and the relationship between calories in and calories out affect body weight.

-How to calculate BMR, TDEE, and macronutrient distribution to support body composition goals.

Post-Session 14 Tasks:

-Review the material covered during the session.

-Apply what you learned in this session by using your own goals to create a personalized plan

Pre-Session 15 Preparation:

Watch the nutritional supplements video from the Platform Chapter Titled: Week 7: Fitness Nutrition + Evidence Based Coaching Part 1

Session 15: Nutrition

LESSON 1: SUPPLEMENTS & DOPING

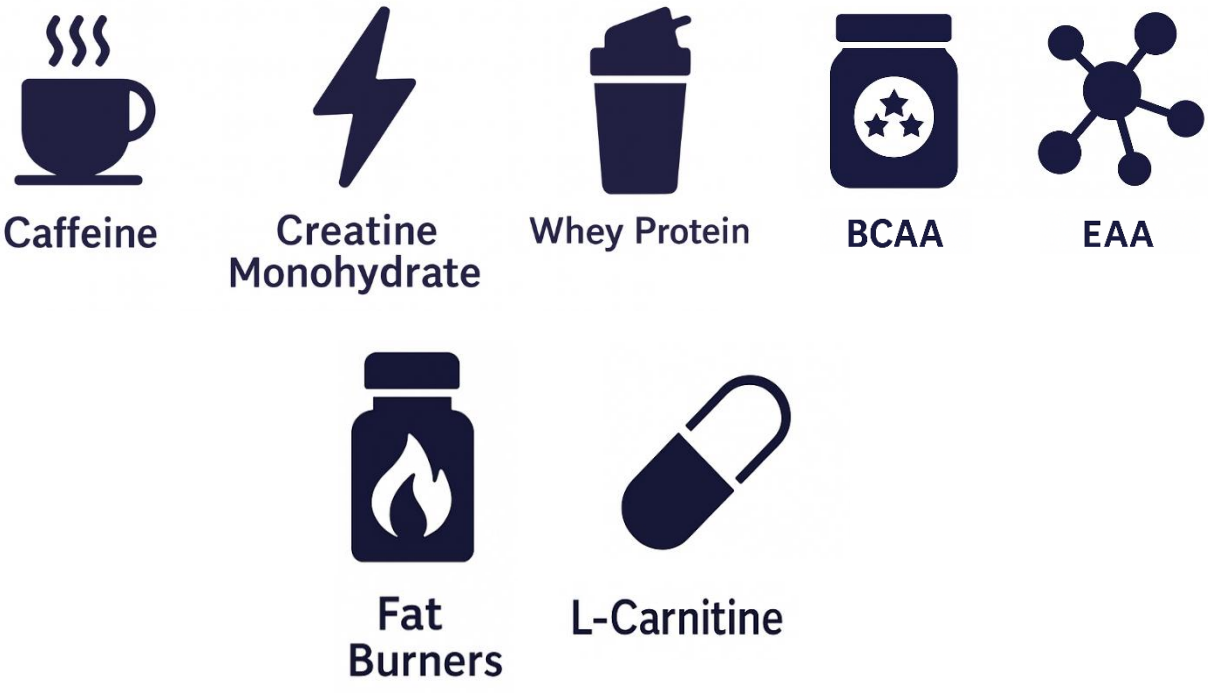
Objectives:
-Understand the mechanisms and effects of commonly used ergogenic supplements.
-Recognize evidence-based recommendations for their safe and effective use.
-Identify the potential health risks associated with anabolic androgenic steroids.

- **CATEGORIES AND EXAMPLES of ERGOGENIC AIDS:**

CATEGORY	EXAMPLES
Mechanical aids	Nasal strips to improve airflow to the lungs etc.
Pharmacological aids	Androgenic steroid hormones (and their derivatives); substances that exert a pharmacological effect.
Nutritional aids	Carbohydrate loading, sports drinks, caffeine intake and the consumption of other substances commonly available in the food supply
Psychological aids	Hypnosis, relaxation techniques, motivational techniques etc.
Physiological aids	Blood doping, sauna, massage etc.

GROUP ACTIVITY

- **SUPPLEMENT BREAKDOWN:**



1. **Discuss and list the benefits, drawbacks, and key facts about your assigned supplement.**
2. **Prepare a short summary (2–3 minutes) to present to the class.**
3. **Highlight how it supports performance, recovery, or overall health.**
4. **Be ready to share one common myth or misconception about it.**

1- CAFFEINE:



- **Caffeine: Mechanisms, Effects & Guidelines**

Category	Main Points / Key Findings
Mechanism of Action	• Increases catecholamines (adrenaline, noradrenaline) → ↑ heart rate, cardiac output, and fat mobilization. • Inhibits phosphodiesterase → ↑ lipolysis and glycogen sparing. • Enhances Na^+/K^+ pump activity → maintains muscle contraction efficiency. • Increases calcium release in muscles → improved force production.
CNS Effects	• Acts as adenosine antagonist → ↑ alertness, wakefulness, HR, BP, and dopamine. • ↓ perceived pain and exertion. • ↑ motor unit firing → better power output and fatigue resistance.
PNS Effects	• Boosts calcium release in muscle → stronger contractions and delayed fatigue.
Performance Impact	• ↑ strength, power, and sprint performance (especially in caffeine-naïve individuals). • Enhances submaximal work volume (6–12 reps).
Fat Oxidation	• ↑ fat oxidation short term but minimal long-term fat loss effect. • Fat loss requires a caloric deficit.
Dosage	• 3–6 mg/kg bodyweight ideal for performance. • ~100 mg (new users), 200 mg (fat oxidation), 300–400 mg (strength/power). • No added benefit above 5–6 mg/kg.
Timing	• Peak blood levels: 30–60 min after intake. • Consume 30–90 min before exercise.
Tolerance & Withdrawal	• Tolerance develops with daily use → reduced ergogenic effects. • Break for ~1 month to reset tolerance. • Possible withdrawal: headaches, irritability, fatigue. • IOC limit: urinary caffeine >12 mg/mL (~13.5 mg/kg dose).



Caffeine's performance effects fade with daily use… BUT one month off can bring them back!

- **Caffeine Content of Common Beverages and Products:**

Item	Serving Size (ml)	Caffeine Dose (Mg)
Brewed coffee	148 ml	~100
Decaffeinated coffee	148 ml	<15
Black tea	237 ml	47
Green tea	237 ml	29
Most energy drinks	473 ml	160
Coffee ice cream	118 ml	10–30
Hot Chocolate	473 ml	25
Chocolate bar	47 g	9
Cocoa powder	1 tbsp.	8
Chocolate milk	355 ml	2
Most caffeine tablets	1 serving	200
Nespresso Kazaar capsule	1 capsule, makes 1 oz.	120
Caffeinated gum	1 piece	~50
Most caffeinated OTC medications	1 serving	65–130
7-Up, Fanta, ginger ale, or Sprite	355 ml	0
Cola/Pepsi Regular	355 ml	34–36
Cola/Pepsi Diet	355 ml	46–69
Most energy drinks	473 ml	160
Monster Energy	473 ml	160
Red Bull	237 ml	80

2- CREATINE:



- **Creatine: Mechanisms, Benefits & Guidelines:**

Category	Main Points / Key Findings
What is Creatine?	• Chemical naturally found in muscles, derived from glycine, arginine, and methionine. • Well-researched supplement (>700 human studies). • Allowed by IOC, NCAA, and professional sports.
Sources	• Found in meat and fish (e.g., 1.1 kg beef = 5 g creatine; 1 kg chicken = 3.4 g). • Around 30% degrades when meat is cooked medium-well. • Average intake: 1.08 g/day (men), 0.64 g/day (women). • Body also synthesizes ~1 g/day (mainly liver, kidneys, pancreas).
How It Works	• Stored as phosphocreatine in muscles, which helps regenerate ATP for short, high-intensity efforts. • Increased phosphocreatine = more rapid ATP resynthesis during explosive exercise.
Performance Effects	• Improves performance by: – Increasing phosphocreatine stores – Faster phosphocreatine resynthesis – Decreasing lactate accumulation – Enhancing glycogen storage. • Indirectly supports muscle growth by allowing more total work during training.
Muscle Building	• Linked with increases in lean body mass, power output, muscle thickness, and resting metabolic rate. • In older adults, improves lean mass and bone density when combined with resistance training.
Endurance Sports	• Little to no benefit on aerobic or sub-maximal endurance activities. • Ineffective for exercise lasting >150 seconds, as energy relies mainly on oxidative phosphorylation.
Safety	• Safe long-term; no negative effects on kidney function or blood parameters. • Possible minor effects: water retention, GI discomfort, or acute weight gain. • Water retained mainly inside the muscle (not under skin).

Category	Main Points / Key Findings
Forms of Creatine	• Creatine monohydrate = gold standard (most studied, effective, affordable). • Other forms (pyruvate, citrate, nitrate, magnesium chelate) show no superior performance benefits. • Buffered, HCl, or ethyl ester forms are less effective or degraded.
Timing	• Consistency is more important than timing — must accumulate in muscle before benefits appear. • May be best post-workout with carbs + protein (~47 g carbs + 50 g protein → increased retention).
Loading Methods	• Option 1: 20 g/day (0.3 g/kg) for 5–7 days → then 3–5 g/day (0.03 g/kg). • Option 2: 3–5 g/day consistently → reach same saturation in ~4 weeks.
Creatine & Caffeine Interaction	• Evidence inconclusive, some studies show no negative interference; others show neutral results. • Caffeine + creatine may enhance HIIT performance. • No strong reason to avoid taking them together.

3- WHEY PROTEIN:



- **Whey Protein: Mechanisms, Effects & Guidelines:**

Category	Main Points / Key Findings
What is it?	• Fast-digesting, high-quality milk protein rich in essential amino acids (EAAs) and leucine. • 25 g whey typically provides ~2.7 g leucine, sufficient to stimulate Muscle Protein Synthesis (MPS).
Efficacy	• Protein supplementation enhances gains in muscle size and strength with resistance training; effects plateau around ~1.6 g/kg/day total protein. • Whey's rapid digestion and higher leucine content stimulate MPS more than slower proteins like casein in acute studies.
Practical Dosing	• Per meal: 20–40 g high-quality protein or ~0.25 g/kg, targeting 700–3000 mg leucine and a full EAA profile. • Daily: ~1.6–2.2 g/kg/d for active individuals; beyond ~1.6 g/kg/d shows diminishing returns for hypertrophy.
Timing	• Distribute protein doses every 3–4 hours across the day; pre- or post-workout both effective because the exercise-induced anabolic window is broad.
Safety/Notes	• Generally safe in healthy adults. Choose isolate if lactose sensitive.

Type	Protein %	Lactose	Digestion Speed	Cost	Ideal For
Whey Concentrate	70–80%	Moderate	Moderate	$	General fitness use
Whey Isolate	90%+	Low	Fast	$$	Less fat & carbs, Lactose-sensitive individuals
Hydrolyzed Whey	90%+	Very Low	Very Fast	$$$	Less fats & carbs, Recovery, sensitive digestion



4- BCAAs:

- **Branched-Chain Amino Acids (BCAAs): Mechanisms, Effects & Guidelines:**

Category	Main Points / Key Findings
What are they?	• The essential amino acids leucine, isoleucine, valine provided without the other EAAs.
Efficacy	• Review evidence indicates BCAA alone cannot support or maximally stimulate MPS because all EAAs are required. • Some acute trials show modest increases in post-exercise MPS vs placebo, but responses are smaller than with complete proteins/EAAs and do not demonstrate superior training outcomes.
Use Case / Practical Take	• If total protein/EAA intake is already sufficient, adding isolated BCAA is unlikely to add benefit over protein/EAAs.
Safety/Notes	• Generally well tolerated at common supplemental doses; prioritize meeting total daily protein/EAA targets.



5- EAAs:

- **Essential Amino Acids (EAAs): Mechanisms, Effects & Guidelines:**

Category	Main Points / Key Findings
What are they?	• The nine amino acids the body cannot synthesize; provided in supplements or complete proteins.
Efficacy	• Oral EAA ingestion after resistance exercise switches muscle from net protein breakdown to net synthesis in humans. • Nonessential amino acids are not required to stimulate the anabolic response; adequate EAAs are sufficient.
Dosing Used in Studies (examples)	• Acute trials have used a wide range; e.g., 40 g EAA solution in classic post-exercise study produced net positive muscle protein balance. • Practical programming often achieved via 20–40 g of high-quality protein supplying sufficient EAAs and ~2–3 g leucine.
Timing	• Effective when taken near training; overall daily distribution every 3–4 h supports repeated MPS pulses.
Safety/Notes	• Generally safe in healthy adults; product quality varies.

6- L-CARNITINE:



- **L-Carnitine: Mechanisms, Benefits & Guidelines:**

Category	Main Points / Key Findings
What is L-Carnitine?	• Nonessential nutrient synthesized in the liver and kidneys from amino acids lysine and methionine. • The body produces enough to meet daily needs, so supplementation is not required. • Found naturally in meat, dairy products, fish, and poultry. • Exists in two forms (D and L); the L-form is the active one found in food.
Role in Energy Production	• Plays a critical role in energy production. • Component of enzymes that transport fatty acids into mitochondria for β-oxidation (fat burning).
Performance Effects	• Research over the past 20 years does not support improvements in athletic performance. • Does not increase oxygen use or muscle carnitine content. • May be beneficial in very prolonged endurance events, as it can spare glycogen and reduce lactic acid production.
Side Effects	• At doses of approximately 3 g/day, may cause nausea, vomiting, abdominal cramps, diarrhea, and a fishy odor in urine and breath.
Weight-Loss Findings	• Meta-analysis showed initial weight-loss effect with L-carnitine, but results fade over time. • Indicates it may help at first but does not sustain long-term weight-loss.

- **Selected Food Sources of Carnitine:**

Food	Milligrams (mg)
Beef steak, cooked, 4 ounces	56–162
Ground beef, cooked, 4 ounces	87–99
Milk, whole, 1 cup	8
Codfish, cooked, 4 ounces	4–7
Chicken breast, cooked, 4 ounces	3–5
Ice cream, ½ cup	3
Cheese, cheddar, 2 ounces	2
Whole-wheat bread, 2 slices	0.2
Asparagus, cooked, ½ cup	0.1



Cons:

-Diuretics

-Increase heart rate (some contain high caffein & stimulant dosages)

-Might lead to digestive track issue

Pros:

-Adherence (the fact that they paid for something might help someone stick to diet)

-Might lead to MINOR weight changes



7- ANABOLIC ANDROGENIC STEROIDS:

- **Anabolic Androgenic Steroids (AAS): Mechanisms, Effects & Risks:**

Category	Main Points / Key Findings
What Are AAS?	• Synthetic variations of the male sex hormone testosterone. • Around 20 different AAS products exist in the illegal drug market. • Used since the 1970s by trainees to enhance performance. • Taken orally or through intramuscular injection.
Performance Effects	• Short-term use increases strength gains by 5–20% of initial strength. • No effect on endurance performance, even though it increases erythropoiesis (red blood cell production).
Adverse Effects	• Disturbs endogenous testosterone production. • Elevates blood pressure (hypertension). • Decreases HDL cholesterol levels. • Causes dangerous effects on heart structure and function. • Increases coagulation leading to thrombosis. • Induces aggression and hostility. • Causes mood disturbances such as depression. • Leads to reverse anorexia syndrome (body dissatisfaction and low self-esteem).
Medical Use	• In medicine, AAS are prescribed only for specific clinical cases.



Imagine your client tells you they are considering taking steroids to gain muscle faster.
As a fitness professional, how should you respond?

How to Respond if a Client Mentions Steroid Use

- ✓ Explain the potential benefits objectively (e.g., faster muscle gain, improved recovery).
- ✓ Clearly outline the serious risks (e.g., hormonal imbalance, heart and liver issues, mood changes).
- ✓ Keep your explanation neutral and evidence-based, without judging or promoting.
- ✓ Stay within your professional scope; never recommend, sell, or prescribe steroids.
- ✓ **If the client insists, advise them to consult a sports medicine doctor before making any decision!**



Consolidation

By the end of Session 15, I am aware of:

-The main ergogenic supplements and their impact on performance.

-The difference between scientifically supported and non-effective supplementation.

-The role of proper dosage and consistency in maximizing benefits.

-The adverse effects and dangers linked to anabolic androgenic steroids.



Post-Session 15 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 16.



Pre-Session 16 Preparation:

Watch the Evidence based chapter videos from the Platform Chapter Titled: Week 7: Fitness Nutrition + Evidence Based Coaching Part 1

Have access to the book (hard copy or online through the platform)

Session 16: Evidence Based Practice

LESSON 1: HOW TO BECOME AN EVIDENCE BASED COACH



Objectives:

- Define evidence-based coaching.
- Describe the three pillars of evidence-based coaching.
- Understand the Hierarchy of Evidence.
- Identify the strengths and weaknesses of each study type.

- **THREE PILLARS OF EVIDENCE-BASED COACHING:** *(Platform: Chapter title: Week 7: Fitness Nutrition + Evidence Based Coaching Part 1)*



Do you think reading studies is enough for a coach to become an evidence-based one?

- **HIERARCHY OF EVIDENCE:**





Pair Activity: Each group has to read one study type, understand it well and inform his classmates about its strengths and limitations (both learners of each group must present the study to their classmates)

Study Type & Definition	Study Strength Points	Study Limitations
In-Vitro Studies		
Animal Studies		
Expert Opinions		
Case Reports & Series		

Study Type & Definition	Study Strength Points	Study Limitations
Cross-Sectional Studies		
Case Control Studies		
Cohort Studies		

Study Type & Definition	Study Strength Points	Study Limitations
Randomized Control Trial		
Systematic Review		
Meta-Analysis		

To sum-up, what is the main difference between Observational, Experimental and Review studies?

Pair Activity: Which expert opinion has stronger evidence and why?

Expert Opinion A:

One of the most stubborn myths in rehab and training is you must keep your knees behind your toes when you squat! When this myth started is hard to establish but one of the first papers that mentions it was back in 1978 (ref: McLaughlin et al (1978) Research Quarterly). They stated keeping a more vertical tibia reduces the shear forces within the knee joint, and this message has since been repeated in many other studies — one of the most well-known being Fry et al (2003), Journal of Strength and Conditioning Research. In this study they found squatting with the knees behind the toes reduced forces in the knee joint by 22–28% compared to squats with the knees over the toes! BUT... and this is often forgotten about...

They also found it increased the forces in the hip joint by over 100%! It is physiologically impossible to do a full squat without letting your knees go past your toes. So many physios and trainers recommend avoiding full squats, instead suggesting partial range or squatting only to parallel to reduce the risk of knee pathology and injury. However, as common and popular as this advice is by many well-known insta-trainers and therapists, it is ignorant and foolish. There is simply no evidence that deep knee-over-toe squats are any more dangerous or risky than partial or parallel squats. In fact, research has shown that deep squats are effective for the protection against injuries, and that squats do not contribute to an increased risk of knee injury (ref: Hartmann et al (2013) Sports Med). Now that's not to say everyone has to do deep squats, or that partial squats can never be used — saying this is just as ignorant and foolish as those that say the opposite! When it comes to squatting, we must recognize that there are many ways to do it, and deciding on the type, depth, load, and positioning of the squat will depend on variables such as the individual's anatomy and ability. There is no one best way to squat — just many ways to squat!

Expert Opinion B

Sometimes the body is very acidic, which causes the nerves to become more sensitive to pain. If you eat the standard American diet, chances are you are too acidic. If you are too acidic, it can adversely affect several bodily functions, including your nervous system, which controls everything. Other types of vinegar stay acidic when ingested; however, raw organic apple cider vinegar becomes alkaline when you ingest it. That's one aha moment for you. Apple cider vinegar is alkaline-forming because of its "ash" content, which means when it is digested, what is left over becomes ash. When you check the pH of that ash and dissolve it in water, the content is alkaline. Whenever our body digests anything, it undergoes oxidation, and the end result determines whether the product was alkaline or acid. Some other foods, such as celery, spinach, and figs, are also very alkalizing. Water is one of the most alkalizing things you can put into your body. Once your digestive system starts to calm down, then you're on the road to extreme health. One foolproof way to determine whether your body is in an acidic state is to measure your body's pH levels. It's a fairly simple process to test your urine and saliva for pH levels: Take a strip of pH test paper (available in any drug or health food store). Pass the paper through your urine stream first thing in the morning. Different colors on the chart represent different pH levels. A urine pH level of 6.5 to 7.0 would be a great morning measurement. To test your saliva, moisten the pH strip with your saliva as soon as you wake up — before brushing your teeth or eating. Ideally, your morning saliva pH should be 7.1 to 7.5. If your body is too acidic, the urine pH levels will be below 6.5. You need to understand what's causing those levels: diet, environmental toxins, stress, insufficient water, constipation, pinched nerves, and many other factors can contribute to acidity.



Ready to play the Who Am I Game:

1- In order to have better results, I can control the variables present by asking study participants to perform specific exercises, eat specific types of food. Who am I?

2- I'm retrospective, quick, and great for rare injuries, but I rely on memory and old data — which can be biased. Who am I?

3- I come from social media blogs, coaches, and magazines. I can be helpful and practical, but I'm subjective, and bias is my middle name. You shouldn't trust me without stronger evidence. Who am I?

4- I help isolate tiny biological processes, you can't rely totally on me since I am not a human study, I only take place in a Lab. Who am I?

5- I offer detailed stories about unique patient recoveries, like a pro athlete's ligament rehab. Useful? Yes. Generalizable? Not really — I'm too small and lack controls. Who am I?

6- I can be prospective or retrospective. I reveal incidence and long-term effects, but I am not randomized and its very hard for me to get blinded. Who am I?

7- I answer broad questions without doing new tests. I save you time and judge evidence quality, but if I select poor studies, my conclusions fall apart. Who am I?

8- I reveal prevalence and associations not cause-and-effect relationships. If someone trains more and has fewer injuries, I can't tell you if training prevented them or not — I just saw it at one point in time. Who am I?

9- I provide one powerful effect size from many experiments and I am in love with numbers. If I include biased studies, the whole result gets biased. Who am I?

10- I offer controlled environments to test interventions safely, but species differences mean that my results don't always apply to people. Who am I?

11- I can describe one athlete's rehab success — or a few similar cases. I'm great for spotting new ideas, but I can't prove cause and effect and my sample is tiny. Who am I?

12- I follow groups over time — exposed vs. unexposed — like tracking athletes to see if training style predicts injury. I show risk, but I'm expensive and some people can drop out of my study or change their behavior. Who am I?

13- I statistically combine data from multiple studies to produce the strongest overall conclusion — but I am only as strong as the studies are. Who am I?

14- I take a snapshot in time — measuring both behavior and outcome all at once, like checking how many athletes currently have jumper's knee. I show relationships, not causation. Who am I?

15- I randomly assign participants to different training programs. I'm the gold standard for finding cause-and-effect — but I'm costly and can be unethical. Who am I?

16- A famous coach writes an article recommending 8–12 reps for muscle growth, but doesn't cite research — only experience. You might like me, but I'm the lowest level of evidence among observational studies. Who am I?

17- I compare people with a condition to people without it — then look back in time to see what might have caused it, like early specialization and injuries. But I can't prove causation. Who am I?

18- I help researchers study biological mechanisms on living organisms when humans can't be tested yet. My downside is that sometimes what works on me doesn't work in humans. Who am I?

19- I allow scientists to examine cells and tissues under controlled conditions, but I can never involve actual organisms. Who am I?

20- Instead of running an experiment, I evaluate and summarize many high-quality studies using strict criteria without going into numbers. If the included research is weak, I become weak too. Who am I?

Consolidation:

By the end of Lesson 1, I am aware of:

-The three pillars of evidence-based coaching.

-Different types of studies, their strengths and limitations.

LESSON 2: IDENTIFY THE ANATOMY OF A RESEARCH PAPER OR STUDY

Objectives:
-Understand each part of the research paper.

- **ANATOMY OF A PAPER:** *(Platform: Chapter title: Week 8: Evidence-Based Coaching Part 2 + Coaching Behavioral Change Part 1)*

Parts of the Paper	Key Points to Look For
Title	
Abstract	

Parts of the Paper	Key Points to Look For
Introduction - General Topic - Hypothesis - Literature Review - Research Question	
Methods Used - Study Design - Participants - Interventions - Data Collection - Statistical Analysis	

Parts of the Paper	Key Points to Look For
Results	
Discussion	
Conclusion	



Consolidation:
By the end of Lesson 2, I am aware of:
-The parts and subparts of a research paper and how to benefit from them.

LESSON 3: UNDERSTANDING STUDY BIAS

Objectives:
- Describe the study biases

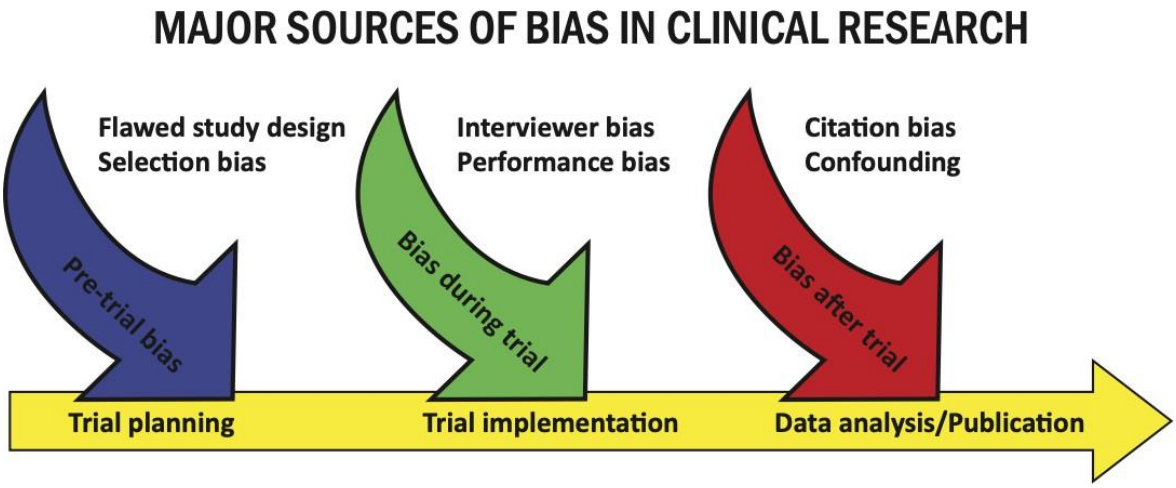
- **STUDY BIAS:** *(Platform: Chapter title: Week 8: Evidence-Based Coaching Part 2 + Coaching Behavioral Change Part 1)*

GROUP ACTIVITY

Group Activity: Mix & Match

Reporting Bias	Chosen participants do not accurately reflect the population of interest
Confounding Bias	When researchers unintentionally influence the outcomes of a study based
Publication Bias	Common in studies that require skill-based measurements. Variations in how the experimenter conducts the trial—for instance, different levels of experience or attention—may impact the results
Selection Bias	In experimental trials, participants may change their behavior due to their awareness of being studied
Performance Bias	When researchers manipulate the data or outcomes of a study or even the interpretation of the results to align with their expectations or desired results
Experimental Bias	When an external factor, which is not accounted for in the study, influences both the exposure and the outcome, leading to a false association
Participant's Bias	When studies with negative or non-significant results are less likely to be published than studies that report positive or significant outcomes

To Sum-up the Biases:



Group Activity: Identify the bias in each of the below examples

1- In a study investigating the relationship between smoking and lung cancer, researchers chose a sample of people that smokes on special occasions such as holidays or weddings

2- Researches did not include some articles they read in their systematic review

3- In a study detecting the impact of various diets on body composition; with different individuals performing body fat measurements, with some having more experience or using more accurate methods than others.

4- In a study examining different training programs, one group was to asked to train using the most recent trending type of workout while the other group was told to train using an old school one

5- In a study where participants know that they are being tested to see which group reaches best hypertrophy results.

6- In a study showing a link between lung cancer and drinking coffee, researchers did not take into consideration the fact that most coffee drinkers smoke while having coffee.

7- In a study where there were some results were discarded since they do not support the hypothesis.

Where can we practically find credible studies with the least possible number of biases and red flags?



PUBLICATION DATE

- 1 year
- 5 years
- 10 years
- Custom Range

TEXT AVAILABILITY

- Abstract
- Free full text
- Full text

ARTICLE ATTRIBUTE

- Associated data

ARTICLE TYPE

- Books and Documents
- Clinical Trial
- Meta-Analysis
- Randomized Controlled Trial
- Review
- Systematic Review

See all article type filters

Additional filters +

Clear applied filters

It is closely related to gout and other metabolic diseases. To develop a **safe** and effective dietary supplementation for alleviating HUA, we investigated the effects of **whey protein** hydrolysate (WPH) on HUA and associated renal dysfunction and explored their u ...

2 Cite

Improved Safety, Bioavailability and Pharmacokinetics of Zidovudine through Lactoferrin Nanoparticles during Oral Administration in Rats.

Kumar P, Lakshmi YS, C B, Golla K, Kondapi AK.

PLoS One. 2015 Oct 13;10(10):e0140399. doi: 10.1371/journal.pone.0140399. eCollection 2015.

PMID: 26461917 Free PMC article.

The bioavailability and tissue distribution of AZT is higher in blood followed by liver and **kidney**. AZT-lactonano causes the improvement of pharmacokinetic profile as compared to soluble AZT; a more than 4 fold increase in AUC and AUMC in male and female rats. ...AZT-lacto ...

3 Cite

Postprandial leucine and insulin responses and toxicological effects of a novel **whey protein** hydrolysate-based supplement in rats.

Toedebusch RG, Childs TE, Hamilton SR, Crowley JR, Booth FW, Roberts MD.

J Int Soc Sports Nutr. 2012 Jun 6;9(1):24. doi: 10.1186/1550-2783-9-24.

PMID: 22672725 Free PMC article.

The purpose of this study was: aim 1) compare insulin and leucine serum responses after feeding a novel hydrolyzed **whey protein** (WPH)-based supplement versus a **whey protein** isolate (WPI) in rats during the post-absorptive state, and aim 2) to perform a ...

4 Cite

Bovine lactoferrin and piroxicam as an adjunct treatment for lymphocytic-plasmacytic gingivitis stomatitis in cats.

Hung YP, Yang YP, Wang HC, Liao JW, Hsu WL, Chang CC, Chang SC.

Vet J. 2014 Oct;202(1):76-82. doi: 10.1016/j.tvjl.2014.06.006. Epub 2014 Jun 10.

PMID: 24973002 Free article. Clinical Trial.

There was no detectable influence on liver or **kidney** function during a 12-week assessment. It was concluded that combining oral bLf spray and piroxicam was **safe** and might be used to decrease the clinical signs of caudal stomatitis in cats....

5 Cite

Lactoferrin acts as an adjuvant during influenza vaccination of neonatal mice.

Sherman MP, Pritzl CJ, Xia C, Miller MM, Zaghouani H, Hahm B.

Biochem Biophys Res Commun. 2015 Nov 27;467(4):766-70. doi: 10.1016/j.bbrc.2015.10.067. Epub 2015 Oct 22.

PMID: 26478433 Free PMC article.

Neonates given lactoferrin had no anaphylaxis when challenged four weeks later. Lactoferrin is a **safe**



Group Activity: Read the study shared by your course instructor titled: “Whey protein adverse effects systemic review” and give your opinion about its credibility and quality.



Consolidation:
By the end of Lesson 2, I am aware of:
-The study biases and how to use them practically when reading a study.

LESSON 4: STATISTICAL ANALYSIS

Objectives:
- Understand the key terms and statistics commonly used in applied exercise science research

- **KEY TERMS AND STATISTICS:** *(Platform: Chapter title: Week 8: Evidence-Based Coaching Part 2 + Coaching Behavioral Change Part 1)*

Mean	Sum of all values divided by the total number of values (Average)
Standard Deviation	Measures the variation or spread of the data around the mean
P-value	Probability that any observed difference between groups is due to chance. In other words, it determines whether the findings of a study are statistically significant. $p \leq 0.05$ means there is less than 5% chance that results are due to random error meaning that results are statistically significant. p-value between 0.051 and 0.099, it is often described as "approaching significance" suggesting that the authors believe there is likely a meaningful difference worthy of further investigation.

R-value	Measures the strength and direction of the relationship between two variables: 0.35 or less: Weak 0.36 to 0.67: Moderate 0.68 to 0.89: Strong 0.90 or more: Very strong Ranges from -1 to +1: +1 indicates a perfect positive correlation (when one variable increases, the other also increases). -1 indicates a perfect negative correlation (when one variable increases, the other decreases). 0 indicates no correlation.
R-Squared	Describes the proportion of variance in the dependent variable that can be explained by the independent variable(s)
T-test	Compares the means of two groups to determine if there is a statistically significant difference between them.
Student's t-test:	Used when comparing two unrelated groups (e.g., comparing two different training programs).
Paired t-test:	Used when comparing the same group under different conditions (e.g., comparing performance before and after an intervention in a crossover design).

Effect Size	Quantifies the magnitude of a difference between groups or conditions. It's particularly useful for determining whether a difference is meaningful, beyond just being statistically significant. ES = (Mean 2 – Mean 1) / Pooled Standard Deviation = (Mean 2 – Mean 1) / (SD 2 + SD 1) / 2 < 0.20 = Trivial effect 0.20 - 0.59 = Small 0.60 - 1.19 = Moderate 1.20 - 1.99 = Large ≥ 2.00 = Very large
Percentage Change	Describes the relative change in a group's performance or characteristic from pre- to post-study. While it is useful, it should always be presented alongside other statistical measures, as percentage change alone can be misleading.



Consolidation:

By the end of this Session, I am aware of:

- The three pillars of evidence-based coaching.
- Hierarchy of Evidence
- Strengths and weaknesses of each study type
- Different types of study biases
- Anatomy of a research paper
- Key terms and statistics commonly used in applied exercise science research



Post-Session Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented next session.



Pre-Session Preparation:

Watch all videos of the Coaching Behavioral Change chapter found on the platform under chapter titled:

-Week 8: Evidence-Based Coaching Part 2 + Coaching Behavioral Change Part

Session 17: Coaching for Behavior Change

LESSON 1: FOUNDATIONS OF COACHING

Objectives:
-Understand the shift from training to coaching.
-Recognize the main principles of health and wellness coaching.
-Identify the traits and communication skills of an effective coach.
-Apply strategies that encourage autonomy and reduce the righting reflex.

(Platform: Chapter title: Week 8: Evidence-Based Coaching Part 2 + ***Coaching Behavioral Change Part 1****)*

- **ROLE OF A COACH:**

What is the difference between a "Trainer" and a "Coach"?

In fitness, success rarely depends only on program design or knowledge of anatomy… it depends on a trainer's ability to influence behavior. This chapter shifts your focus from being a "trainer who instructs" to a "coach who facilitates change".

A coach doesn't just give answers, they guide clients to find their own*.*

- **Trainer vs. Coach:**

Trainer	Coach
Operates in a directive mode	**Works collaboratively**
Tells clients what, how, and when to do things	**Facilitates self-discovery and problem-solving**
Acts as the expert; client as recipient	**Builds a partnership; client as active participant**
Focuses on teaching skills	**Focuses on behavioral adherence and motivation**
Relies on instruction and control	**Emphasizes empowerment and intrinsic motivation**
Often achieves short-term compliance	**Promotes sustainable lifestyle change**

Why do most fitness results fade when the focus is only on "what to do" instead of "why to do it"?

- **DEFINING COACHING:**

Training builds skills. Coaching builds meaning. When clients connect goals to personal values, change becomes sustainable.

Training	Coaching
Skill-focused	Meaning-focused
Instructor-driven	Client-centered
Solves problems for clients	Helps clients find their own solutions
Short-term compliance	Long-term autonomy

- **HEALTH & WELLNESS COACHING PRINCIPLES:**

Principle	Application in Practice
Values-driven dialogue	Helping clients connect behaviors to deeply held values (e.g., "I want to exercise because I want to be a strong role model for my children").
Unconditional positive regard	Accepting clients without judgment, even when they relapse or struggle.
Collaborative agenda-setting	Working with, rather than on clients; enhancing autonomy and ownership.

- **QUALITIES OF A COACH: CHARACTER VS PERSONALITY:**

Character (Builds Trust)	Personality (Builds Rapport)
Integrity	**Humor**
Authenticity	**Enthusiasm**
Humility	**Energy**
Consistency	**Extroversion**



Coach In Action!

Rewrite the following trainer statements using a coaching approach that reflects empathy, values, and collaboration.

Trainer Statement	Rewrite as a Coach
You need to stop skipping your workouts.	
You're being lazy.	
If you really wanted results, you'd be more disciplined.	
I'll tell you what works best for you.	
You just have to make more time.	

- **COMMUNICATION SKILLS:**

Effective coaching starts with effective communication. A great coach listens more than they talk, aiming to understand, not to reply.

Core communication skills:

- Active listening (listening to understand, not to reply).
- Open-ended questioning.
- Reflective summarizing.
- Non-verbal presence (eye contact, posture, attentiveness).

- **THE RIGHTING REFLEX:**

The righting reflex is the automatic urge to correct, advise, or "fix" a client's behavior. Although well-intentioned, it often triggers resistance and reduces autonomy.

Righting Reflex Response	Coaching Response
Client: "I don't have time to exercise." **Trainer:** "You just need to manage your time better."	**Client:** "I don't have time to exercise." **Coach:** "Tell me more about what a typical day looks like for you."

- **QUALITIES OF AN EFFECTIVE COACH VS. INEFFECTIVE COACH**

Effective Coach	Ineffective Coach
Listens actively	Talks more than listens
Evokes client's values	Imposes own agenda
Builds trust through integrity	Relies on personality alone
Controls righting reflex	Gives unsolicited advice
Focuses on sustainable change	Focuses on quick results

- **Case Example: Rami**

Rami, a 34-year-old office worker, joins a gym seeking weight loss. His trainer tells him to train 5 days a week, cut out carbs, and run 30 minutes daily. After 3 weeks, Rami quits, citing "too much pressure."

How would you react in this situation?

A coaching approach would explore Rami's values ("Why is weight loss important to you?"), identify barriers ("Tell me more about your work schedule"), and co-create a manageable plan (2–3 sessions/week, gradual dietary shifts). Over time, Rami internalizes his own reasons for change, leading to sustained progress.



Consolidation:

By the end of this lesson, I am aware of:

-How coaching builds meaning, motivation, and self-discovery.

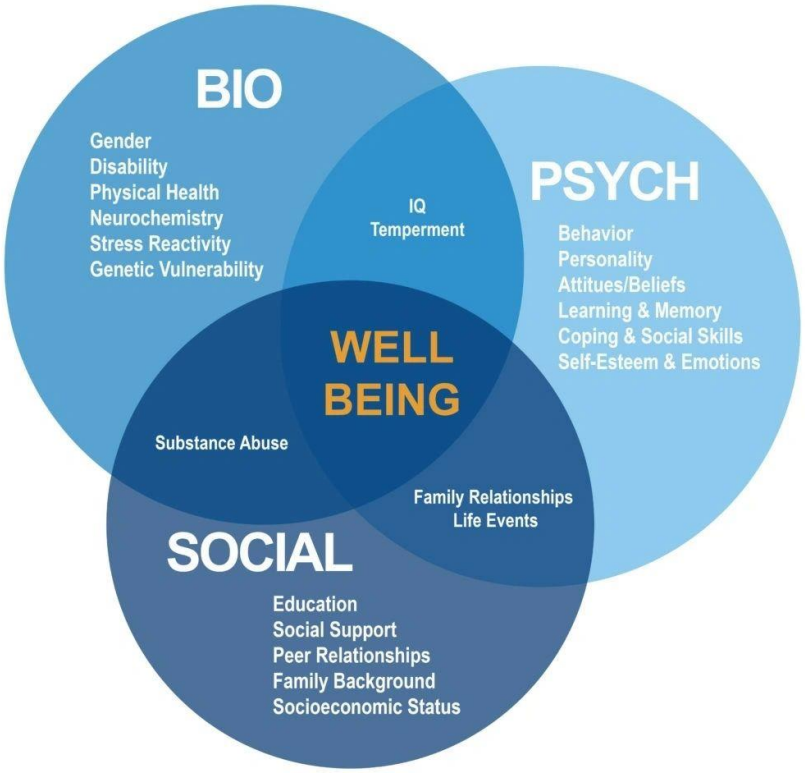
-The core principles that make coaching client-centered.

-The importance of empathy, character, and effective communication in lasting change.

LESSON 2: COACHING TOOLS

Objectives:
-Understand how to apply the Biopsychosocial (BPS) Model in coaching.
-Recognize the role of Emotional Intelligence (EI) in building trust and communication.
-Identify how DISC personality styles help adapt coaching to different clients.

1- THE BIOPSYCHOSOCIAL (BPS) MODEL:



APPLICATION IN COACHING:

Domain	Examples
Biological	**Cardiovascular fitness, injury history, genetic predispositions.**
Psychological	**Motivation, beliefs about exercise, past experiences, stress levels.**
Social	**Family or peer support, work environment, socioeconomic factors.**

- **Applying the BPS Lens:**

▪ Ask: “What challenges do you face outside the gym that make it harder to exercise?”
▪ Assess: Include not only VO_2 max, but also sleep, stress, and social support.
▪ Plan: Add strategies like stress management, time efficiency, or family engagement.



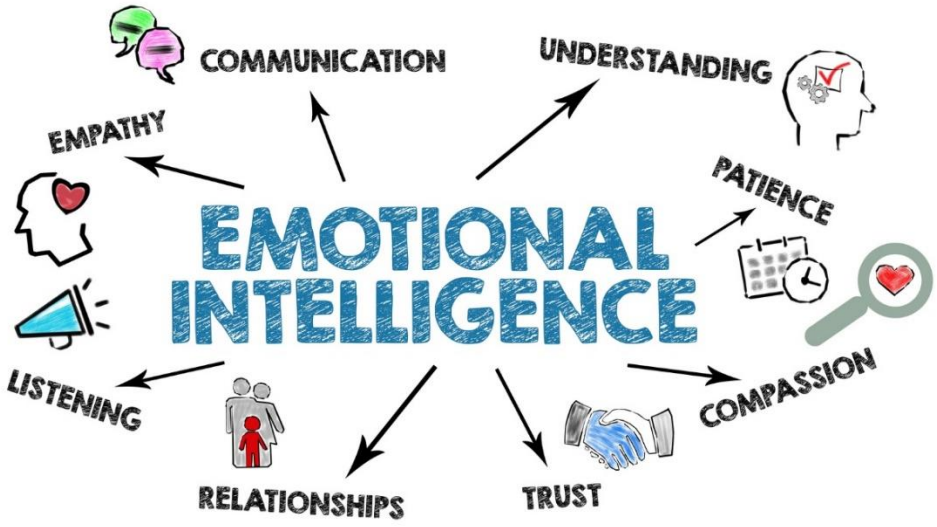
CASE STUDY:

Clara is a 28-year-old new mother who gave birth seven months ago. She is eager to return to exercise, tone her body, and increase her energy levels, but currently identifies as a beginner with medical clearance to exercise. Physically, she presents with a weakened core, occasional wrist discomfort, and mild lower-back discomfort. Her lifestyle is demanding, as she is caring for her infant full-time and experiencing disrupted sleep.

Although she is motivated to feel strong again, she often feels overwhelmed and struggles with self-prioritization, experiencing guilt when taking time for herself. She is worried that she has “lost her fitness”. She has limited childcare, her partner works long hours, and her schedule is inconsistent. While her family is emotionally supportive, they are not able to provide consistent hands-on help.

Guide Clara using the BPS model by:
1- Identifying the biological, psychological and social factors in her case
2- Trying to find a solution for each of the above factors accordingly

2- EMOTIONAL INTELLIGENCE (EI):



Emotional Intelligence (EI) is the ability to recognize, understand, and manage emotions in yourself and others. It includes four main domains that support effective coaching:

EI Domain	Description
Self-awareness	**Recognizing your own emotions, and their impact.**
Self-management	**Regulating emotions and adapting to challenges.**
Social awareness	**Understanding others' emotions (empathy).**
Relationship management	**Building rapport and resolving conflict.**

EMPATHY TYPES:

Type of Empathy	Description
Cognitive	**Understanding another's perspective.** *("I get what you're saying").*
Emotional	**Feeling alongside another person**. *("I feel your frustration").*
Compassionate	**Combining understanding and feeling with supportive action.** *("I see your struggle, let's find a way forward together").*

Compassionate empathy is the most effective in coaching because it balances connection with constructive support.

EMOTIONAL INTELLIGENCE IN COACHING PRACTICE:

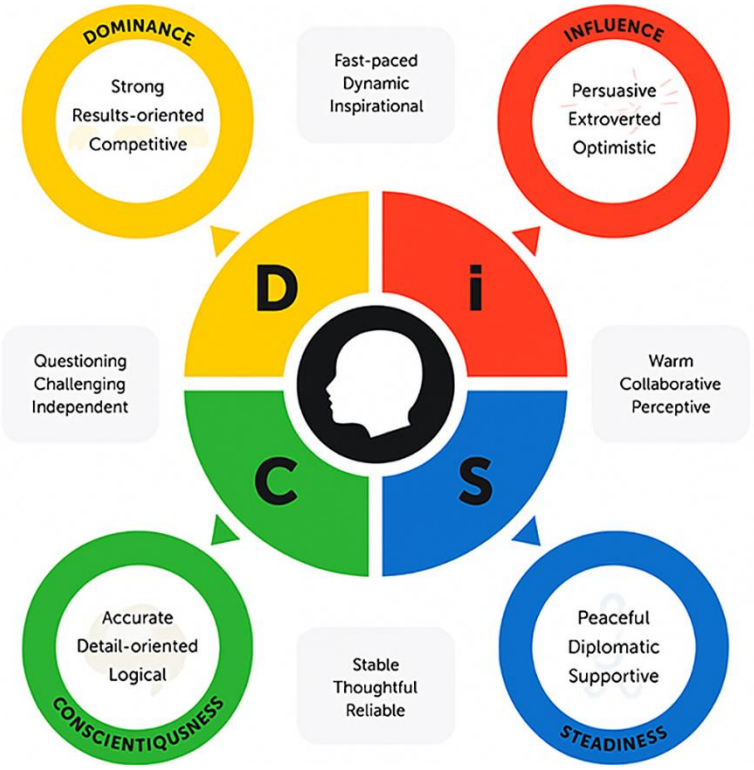
EI Domain	Example in Coaching
Self-awareness	Coach notices impatience and slows down.
Self-management	Coach avoids giving advice, asks questions instead.
Social awareness	Coach senses discouragement behind a client's words.
Relationship management	Coach builds rapport by affirming small successes.



GROUP ACTIVITY:

Practice emotional intelligence while answering the given scenarios of the instructor.

3- DISC PERSONALITY STYLES:



STYLES AND COACHING TIPS:

Style	Main Traits	How to Coach
D – Dominance	Assertive, direct, results-driven	Be clear, set goals, focus on progress.
I – Influence	Energetic, social, enthusiastic	Use motivation, celebrate effort, keep sessions engaging.
S – Steadiness	Patient, loyal, calm	Give reassurance, keep structure, avoid sudden changes.
C – Conscientiousness	Careful, logical, detail-oriented	Explain the 'why,' show data, allow reflection time.

Tip: Match your coaching style to the client's personality. A **D** may lose interest with too much detail, a **C** may feel uneasy without enough structure, an **S** would prefer structured routine and needs reassurance, while an **I** would always look for engaging sessions and encouragement.

WORKING WITH DISC MODEL PERSONALITIES:

- **D:** Focus on performance results.
- **I:** Give positive energy and encouragement.
- **S:** Keep things consistent and predictable.
- **C:** Provide clear explanations and logic.

- **Case Example: Sarah**

Sarah, 28, an accountant with a high C profile, felt anxious when told, "Let's just jump into some exercises and see how would you perform." A better approach was, "We'll start with a quick mobility check to guide your personalized plan." This gave Sarah clarity and confidence, helping her stay consistent.



Guess the personality type, and suggest how to deal with the scenario in question:

❖ Scenario 1:
Client enters the training room smiling and say: "Hey coach! I brought my friend today — she wants to try this out too! By the way, did you see my post about yesterday's workout?" Client talks a lot, makes jokes, and love when everyone joins in.

What personality type are we dealing with?

How would you react as a coach?

❖ Scenario 2:

Client walks into the gym and say confidently: “Let’s skip the small talk — what’s today’s target? I want to beat last week’s record.”

When answering, your client interrupts you occasionally, asks for the previous results, and get impatient when the coach starts explaining too many details.

What personality type are we dealing with?

How would you react as a coach?

❖ Scenario 3:

Your client arrives quietly, greets the coach, and ask: “Can we keep the same routine as last week? It helps me stay consistent.”

Client prefers a calm pace, follows instructions carefully, and dislike sudden changes or chaos.

What personality type are we dealing with?

How would you react as a coach?

❖ Scenario 4:

Your client starts directly by asking the following questions: “What’s the goal of this exercise exactly? How does it activate the posterior chain?”

Your client wants to see data, asks for clarification, and takes time to process before acting.

What personality type are we dealing with?

How would you react as a coach?

Consolidation:

By the end of this lesson, I am aware of:

-How the BPS, EI, and DISC models guide client-centered coaching.

-The importance of empathy, communication, and adaptability in improving results.

-How using structured coaching tools supports long-term motivation and change.

LESSON 3: MODELS OF BEHAVIORAL CHANGE

Objectives:
-Understand how behavior change occurs through different stages and levels.
-Recognize the role of self-efficacy in motivation and adherence.
-Apply behavior change tools like TTM and self-efficacy strategies to real coaching situations.

1- TRANSTHEORETICAL MODEL (TTM):



STAGES OF CHANGE:

Stage	Description
Precontemplation	No intention to change; unaware or in denial.
Contemplation	Aware but unsure; weighing pros and cons.
Preparation	Ready to act soon; taking small steps.
Action	Actively changing behavior (e.g., starting exercise).
Maintenance	Sustaining change for 6+ months, preventing relapse.
Termination	Behavior is fully integrated; no risk of relapse.

PROCESSES OF CHANGE:

The TTM identifies how people move from one stage to another using both cognitive and behavioral strategies:

- **Consciousness raising:** Learning more about benefits.
- **Self-reevaluation:** Connecting behavior to personal values.
- **Counterconditioning:** Replacing unhealthy behaviors.
- **Reinforcement management:** Rewarding positive progress.

STAGES OF CHANGE AND COACHING STRATEGIES:

Stage	Coaching Focus	Example
Precontemplation	Raise awareness	Share benefits of small activity changes.
Contemplation	Explore ambivalence	Discuss pros and cons of starting.
Preparation	Build confidence	Co-create a simple starter plan (e.g., 2x/week).
Action	Support adherence	Track progress, solve barriers.
Maintenance	Prevent relapse	Celebrate milestones, create coping plans.

- **Case Example: Ahmed:**

Ahmed, 42, says: "I know I should exercise, but I'm too busy with work." He is in Contemplation: aware but unsure. Instead of giving him a program immediately, the coach explores: "What might you gain from regular exercise? What could you give up?" By shifting his decisional balance, Ahmed moves toward Preparation.



For each scenario, guess from what stage to what stage the coach is trying to help the client move according to the TTM

1. The coach shows the client a simple, non-judgmental fitness assessment to raise awareness about inactivity risks.

2. The coach helps the client set a clear goal and schedule one workout next week.

3. The coach sends a motivational message after each session to celebrate effort and reinforce progress.

4. The coach invites the client to attend a free “Movement for Beginners” class just to observe.

5. The coach reviews the client’s training log and adjusts the program to suit the client’s busy work schedule

6. The coach provides educational material about how inactivity affects mood and health.

7. The coach suggests scheduling workouts with a friend for accountability and social support.

8. The coach helps the client who missed sessions reflect on barriers and plan realistic adjustments.

9. The coach praises the client for maintaining the habit for 4 months and introduces a new challenge.

10. The coach uses motivational interviewing to explore the client's conflicting thoughts about joining a gym.

2- SELF-EFFICACY:

In your own words, what does self-efficacy mean?



SOURCES AND COACHING APPLICATIONS:

Source of Self-Efficacy	Description	How to Apply in Coaching
Mastery experiences	Past success builds confidence.	Start with achievable tasks (e.g., bodyweight before weighted exercises).
Vicarious experiences	Seeing others like you succeed.	Share stories of clients with similar backgrounds.
Verbal persuasion	Encouragement from trusted people.	Give specific, genuine feedback ("Your form improved today").
Physiological states	Viewing physical effort positively.	Reframe sensations ("This breathlessness means your heart is getting stronger").
Emotional states	Managing stress and anxiety effectively.	Include mindfulness or stress-reduction techniques.
Imaginal experiences	Visualizing yourself performing successfully.	Use visualization before new challenges.

- **Coaching for Self-Efficacy:**

- **Ask:** "When was a time you overcame a challenge like this before?"
- **Normalize setbacks:** "Everyone struggles when they start, but that doesn't mean you can't succeed."
- **Visualize success:** "Imagine yourself finishing this set strong...what does that look like?"

Give three examples where self-efficacy is present, and another three examples where it is low.

Consolidation:

By the end of Session 17, I am aware of:

-How the Transtheoretical Model (TTM) guides clients through the stages of change.

-How building self-efficacy increases confidence and long-term consistency.

-How to adapt coaching strategies to each client's mindset, stage, and environment.

-How combining all coaching tools, BPS, Emotional Intelligence, DISC, TTM, and Self-Efficacy, creates a complete, client-centered approach to behavior change.

Post-Session 17 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 18.

Pre-Session 18 Preparation:

Watch the remainder videos of the behavioral change models on the Platform Chapter Titled: Week 9: Coaching Behavioral Change Tools + Process of Personal Fitness Coaching

Session 18: Coaching for Behavior Change – Continued

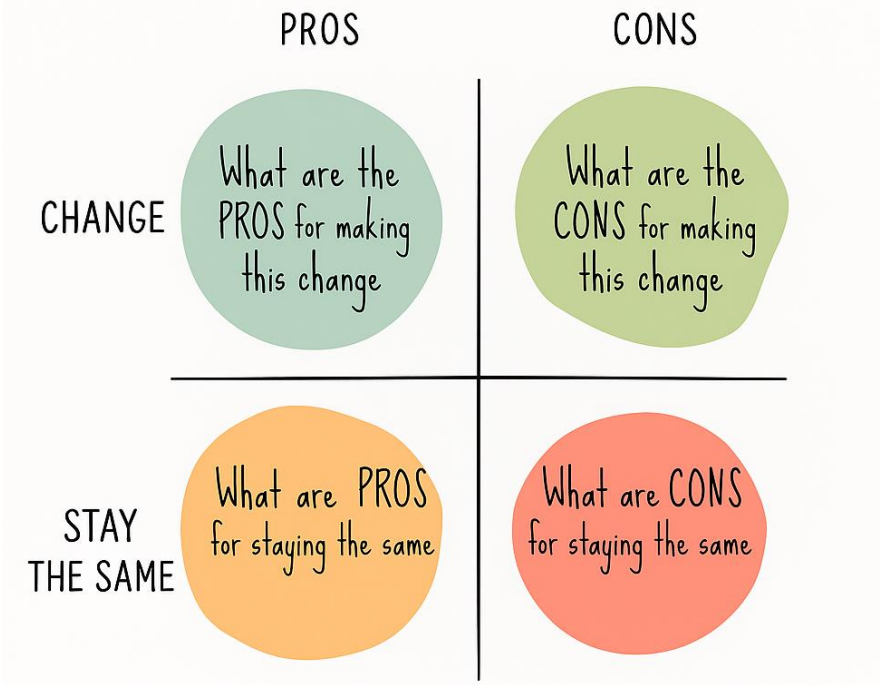
LESSON 1: MODELS OF BEHAVIORAL CHANGE

Objectives:
-Understand the key behavior change models used in coaching.
-Recognize how each model supports motivation, reflection, and long-term adherence.
-Learn how to apply each model (Decisional Balance, Readiness Ruler and HBM) in real coaching situations.

(Platform: Chapter title: Week 9: Coaching Behavioral Change Tools + Process of Personal Fitness Coaching)

3- DECISIONAL BALANCE & READINESS RULER:

- **Decisional Balance:**



Decisional balance means weighing the pros and cons of change. At first, barriers (cons) often feel stronger than benefits (pros). Coaching helps shift this balance by highlighting meaningful advantages and reframing barriers.

Your Own Balance Sheet!

List your top 3 pros and 3 cons of making a change in your lifestyle or training routine.

Pros	Cons

- **Readiness Ruler:**

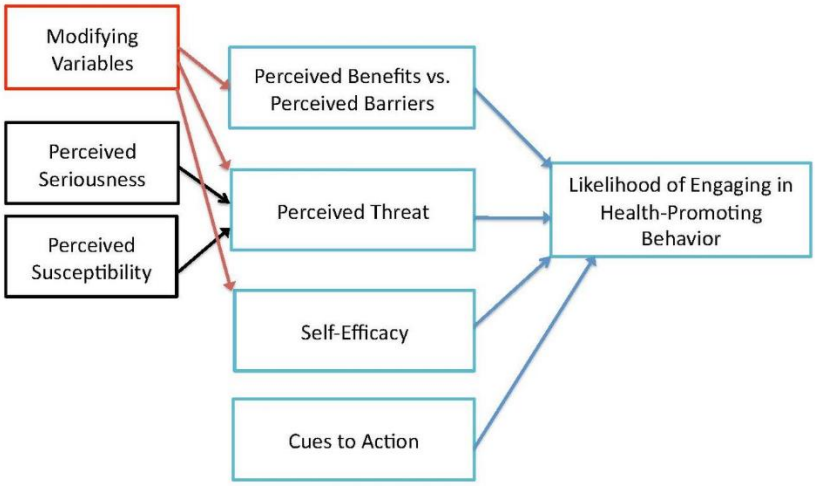


A client tells you, "I'd say I'm about a 5 out of 10 when it comes to starting exercise." What could you ask next to help the client move closer to action, and why?

- **Sample Coaching Questions Using the Readiness Ruler:**

Ruler Type	Example Question	Purpose
Readiness	Why a 6 and not a 3?	Elicit motivation.
Confidence	What would help move you from a 5 to a 7?	Build strategies.
Importance	How important is this change for you right now?	Clarify values.

4- HEALTH BELIEF MODEL (HBM):



What factors make someone more likely to take action for their health according to the Health Belief Model?

How can a coach use questions about risk, benefits, and barriers to help a client make informed choices?

What types of "cues to action" can motivate clients to start or maintain healthy behaviors?

Why might the Health Belief Model be more effective when combined with other models like TTM or Self-Efficacy?

- **Case Example: Maya**

Maya, 50, felt “healthy enough” and avoided exercise. After a doctor mentioned prediabetes, her coach asked:
- What worries you about diabetes?
- How could exercise help?
- What gets in the way?
Seeing exercise as a way to protect her health motivated Maya to begin and stay consistent.

Watch the following video:
https://www.youtube.com/watch?v=HwJ9LxopBag

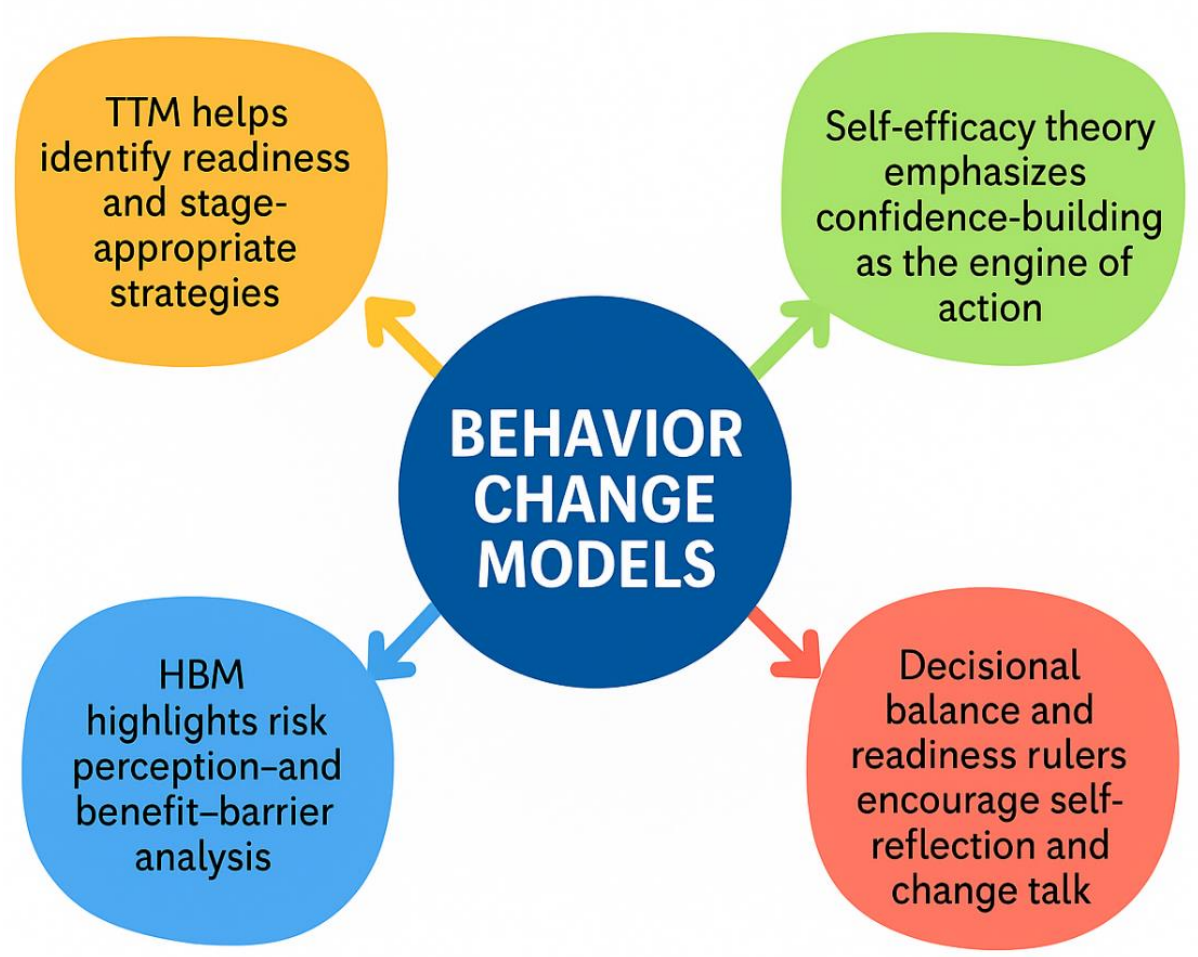


Why did they use helium?



Why did they show them the person who is suffering from cancer?

Let's wrap up all Behavioral Change Models!



Consolidation:

By the end of this lesson, I am aware of:

-How different models (TTM, Self-Efficacy, Decisional Balance, and HBM) explain why and how people change behaviors.

-How to use these models together to improve motivation, confidence, and decision-making in clients.

-The value of combining behavioral science with coaching skills to guide lasting lifestyle change.

LESSON 2: MOTIVATION & CHANGE

Objectives:
-Understand how rewards and feedback affect behavior.
-Identify different types of motivation.
-Recognize how to build long-term motivation using SDT.
-Use mindfulness, values, and purpose in coaching.

GROUP ACTIVITY

List 5 reasons why some clients stay motivated to exercise over time and 5 to why others lose motivation after a few weeks. Then, **label** each reason as either: External or Internal.

1- OPERANT CONDITIONING & MOTIVATION SCIENCE:

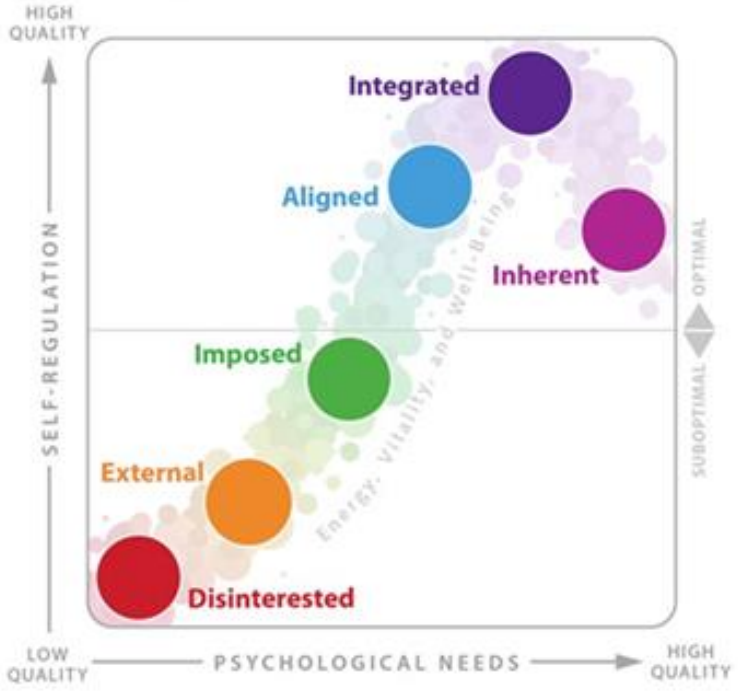
	PUNISHMENT (DECREASING BEHAVIOR)	REINFORCEMENT (INCREASING BEHAVIOR)
Positive (adding)	Ask the client to do 20 pushups or 1 minute wall sits if they showed up late to the session.	Give them a free session if they're always showing up early.
Negative (removing)	Remove 15 minutes from the session duration if they showed up late.	Train the client in a private room if they feel intimidated or shy to train in the presence of other clients.

Concept	Coaching Application
Role of Reinforcement	-Rewarded behaviors repeat, punished ones fade. -Start with extrinsic rewards (praise, progress). -Build intrinsic motivation (enjoyment, success). *Example: "You added 5kg, great job!"*
Limitations of Rewards	-Rewards help start, not sustain behavior. -Too much reward = pressure, lower performance. -Clients chasing approval often quit when it ends.
Practical Coaching Tips	-Begin with short-term rewards. -Shift focus to enjoyment and health. -Avoid guilt or punishment.

- **Case Example: Omar**

Omar joins a gym for a company wellness challenge with cash prizes. He trains hard for 8 weeks but stops once it ends. A coach emphasizing intrinsic rewards like better sleep, stress relief, and family role-modeling can help him stay consistent.

2- SPECTRUM OF MOTIVATION: SELF-DETERMINATION THEORY (SDT)

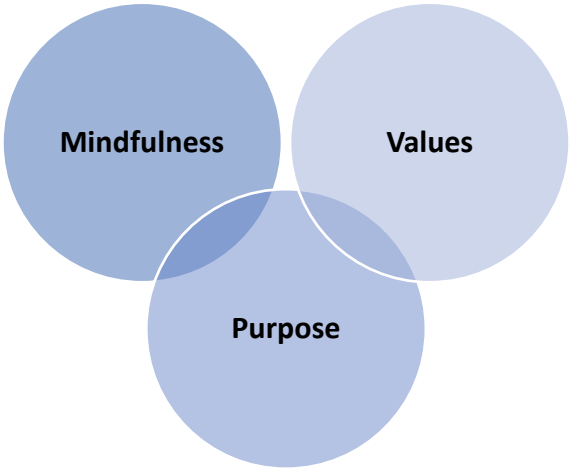


Concept	Coaching Application
Overview	-Motivation ranges from no motivation to full self-determination. -Quality of motivation matters more than quantity.
Motivation Continuum	**1. Amotivation:** "I don't care about exercising." → No intention. **2. External:** "My doctor told me to." → Driven by reward/punishment. **3. Introjected:** "I'd feel bad if I skipped." → Guilt or internal pressure. **4. Identified:** "It improves my energy." → Personal value recognized. **5. Integrated:** "Exercise is part of who I am." → Aligned with identity. **6. Intrinsic:** "I love how it makes me feel." → Done for enjoyment.
Key Insight	-Clients with identified, integrated, or intrinsic motivation are more likely to sustain physical activity.
Basic Needs in SDT	- Autonomy: choice and control. - Competence: feeling capable. - Relatedness: feeling supported and connected. → Supporting these needs promotes self-determined motivation.

- **Examples of Coaching Language Across the Motivation Spectrum:**

Motivation Type	Client Statement	Coaching Response
External	"My doctor told me to."	"It's great you want to take care of your health. What matters most to you about this?"
Introjected	"I'll feel guilty if I don't."	"What would exercising give you beyond avoiding guilt?"
Identified	"I want more energy."	"How will more energy impact your daily life?"
Integrated	"Fitness is who I am."	"That identity can anchor you even during tough times."
Intrinsic	"I just love it."	"That joy will fuel your consistency. How can we keep it fun?"

3- MVPS OF SELF-DETERMINED MOTIVATION:



Concept	Coaching Application
Overview	-MVPs help clients move toward long-term motivation: Mindfulness, Values, Purpose.
Mindfulness	-Focus on the present moment without judgment. -Builds awareness, calm, and better focus. -Use mindful breathing or body check-ins.
Mindful Coaching Tips	-Start with awareness if the client is distracted. -End with reflection: "What part of today's workout did you enjoy most?"
Values	-Link exercise to what truly matters to the client. -Example: "Let's train to stay active for your kids."
Purpose	-Go beyond goals to the bigger why. -Example: "To live with energy and inspire others."

- **Case Example: Lina**

Lina, 37, struggles with consistency. She connects her value of independence and purpose of staying strong for future adventures. Exercise becomes an investment in her future self, not a chore.

GROUP ACTIVITY

COACHING FOR LASTING MOTIVATION!

Case 1 – Rana, 32 years old

Rana started training 6 weeks ago to lose weight for her wedding. She was consistent at first but is now skipping sessions, saying she feels tired and isn't seeing results. Her coach notices she only feels motivated when praised or compared to others.

Case 2 – Omar, 40 years old

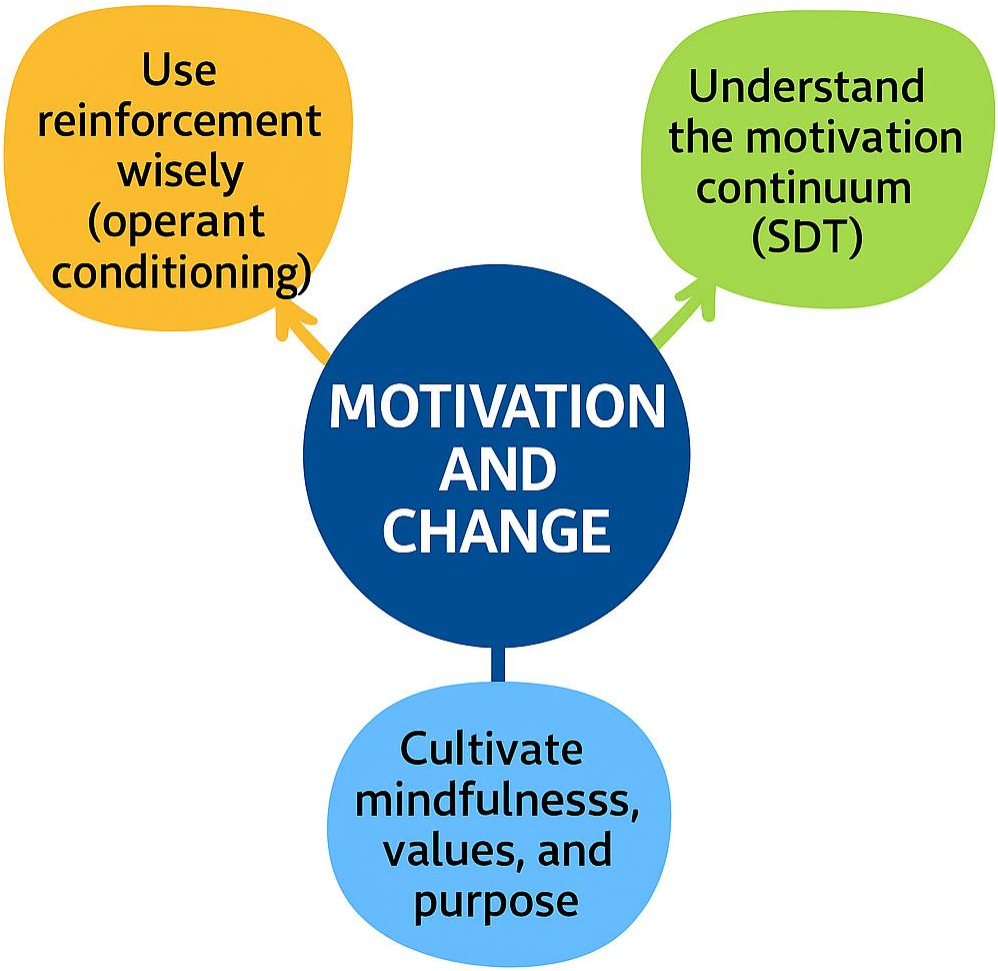
Omar joined the gym after his company launched a wellness challenge with cash prizes. He trained regularly for 8 weeks but stopped once the challenge ended. His coach wants to help him find motivation beyond external rewards.

Case 3 – Lina, 37 years old

Lina often starts programs but never finishes them. She says exercise feels like a "chore." Through discussion, she mentions wanting to stay strong and travel freely in her 60s.

1. How can the coach apply reinforcement and punishment to shape this client's behavior?
2. What type of motivation does this client currently show according to SDT?
3. How can the coach guide the client toward more self-determined motivation (identified, integrated, intrinsic)?
4. What Mindfulness, Values, and Purpose strategies could the coach use in the next session?

Let's wrap up Motivation and Change!



Consolidation:

By the end of this lesson, I am aware of:

-How reinforcement shapes behavior.

-How motivation can shift from external to internal.

-How supporting autonomy, competence, and relatedness keeps clients consistent.

-How MVPs help clients find meaning and sustain motivation.

LESSON 3: MOTIVATIONAL INTERVIEWING (MI)

Objectives:
-Understand how to apply motivational interviewing techniques to build rapport, evoke motivation, and guide clients toward self-directed behavior change.

When was the last time you tried to convince someone to change, and how did it go?

Based on that experience, what do you think "Motivational Interviewing" means in a coaching context?

- **THE SPIRIT OF "MI":**

Element	Meaning	Example in Action
Collaboration	Working with the client, not on them. The relationship is a partnership rather than an expert–recipient dynamic.	"Let's figure this out together."
Acceptance	Respecting client autonomy, practicing unconditional positive regard, and affirming their worth.	"I respect that you're not ready yet."
Compassion	Acting in the client's best interests, prioritizing their well-being over personal agenda.	"I want to understand what matters most to you."
Evocation	Drawing out the client's own reasons for change rather than imposing external ones.	"What would make this change meaningful for you?"

- **CORE SKILLS (OARS):**

Motivational Interviewing uses four core communication skills known as OARS:

O – Open-ended questions

A – Affirmations

R – Reflective listening

S – Summaries

Each skill helps create meaningful, two-way conversations that build motivation and trust.



- **<u>Your Turn to Apply the OARS Skills:</u>**

Client Case:
Lea, 32, works full-time and says she's been feeling tired and unmotivated to exercise. She mentions that she starts a workout plan every few months but quickly stops once her schedule gets busy. Lea adds, "I know I should work out, but honestly, I just don't have the energy anymore."

Now, apply each of the OARS skills below:

1. Open-ended Question

_____

_____

2. Affirmation

_____

_____

3. Reflective Listening

_____

_____

4. Summary

_____

_____

- **OVERCOMING THE RIGHTING REFLEX:**

The righting reflex is the natural urge coaches have to correct behavior by giving advice, solutions, or arguments. Although it comes from good intentions, it often creates resistance instead of motivation.

- **Example:**

Righting Reflex Response:

Client: "I just don't have time to work out."
Coach: "You need to make time — health should be your priority."

MI Response:

Client: "I just don't have time to work out."
Coach: "Time feels scarce right now. What would need to change for exercise to fit in?"



- **Your Turn to Rewrite It!**

Below are common statements clients might say. Rewrite each one using an MI-based response instead of the righting reflex.

1. Client: "I'm just not motivated lately."

_____

_____

2. Client: "I always end up eating junk when I'm stressed."

_____

_____

3. Client: "There's no point in starting again; I'll probably quit."

_____

_____

- **THE "MI" PROCESS:**

From Building Connection to Creating Action.

1. Engaging
Building trust and rapport.

2. Focusing
Narrowing the conversation toward a specific change.

3. Evoking
Drawing out the client's own reasons for change (change talk).

4. Planning
Developing a clear, collaborative action plan.



- **<u>Your Turn to Apply the MI Process</u>**

Read the client case below and write one possible question or action you would use in each stage.

Client Case:
Rami, 28, says he feels inconsistent with his training and can't stick to a routine. He says he wants to "get back on track" but doesn't know where to start.

Engaging:

_____

Focusing:

_____

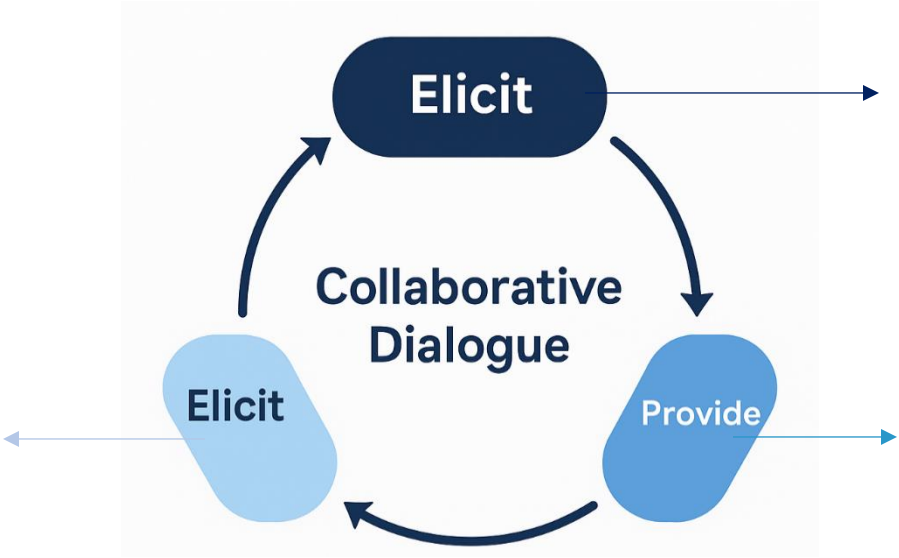
Evoking:

_____

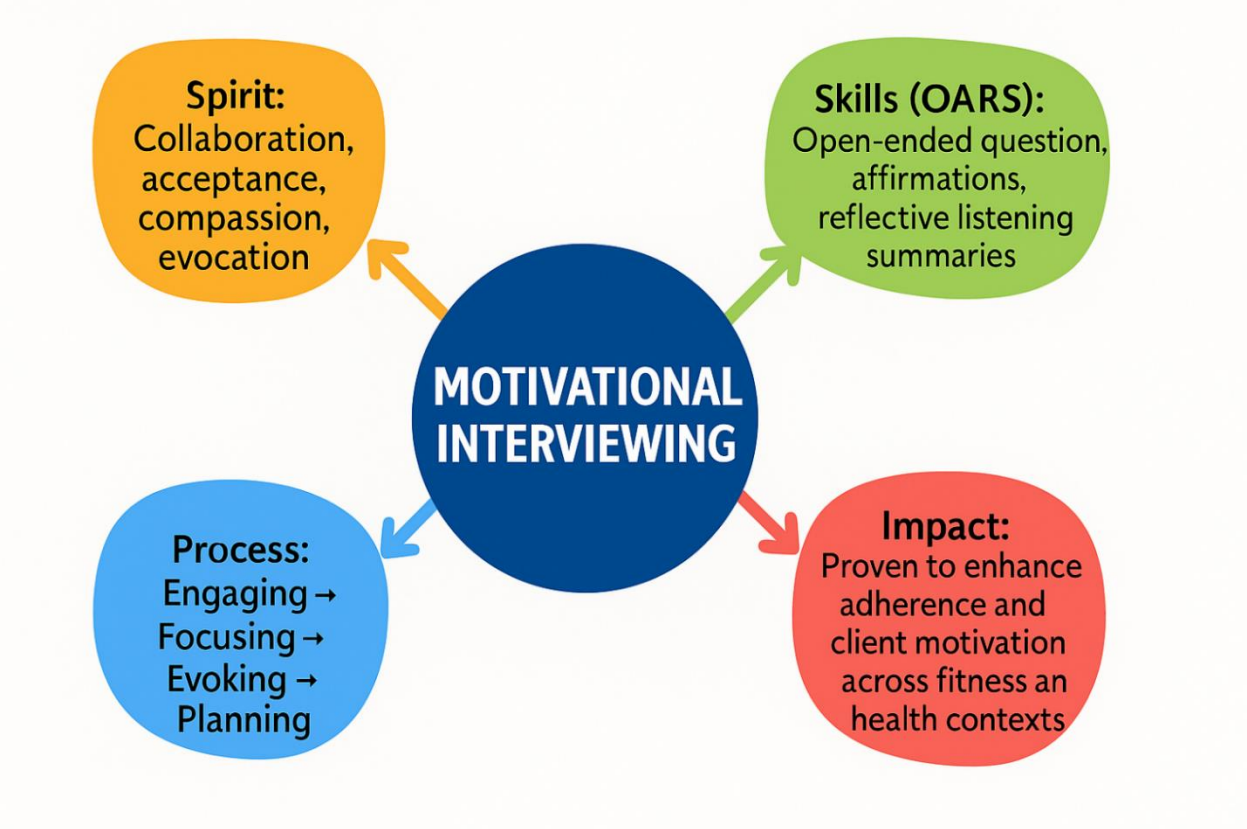
Planning:

_____

- **PRACTICAL APPLICATION: ELICIT–PROVIDE–ELICIT:**



Let's wrap up the Motivational Interviewing Process!



Consolidation:

By the end of Session 18, I am aware of:

-What Motivational Interviewing is and why it is used in coaching.

-The core components of MI, including the spirit (collaboration, acceptance, compassion, evocation) and the OARS skills.

-How to avoid the righting reflex and guide conversations using the MI processes: engaging, focusing, evoking, and planning.

-How to apply the Elicit–Provide–Elicit technique to support motivation and improve client adherence.

Post-Session 18 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 19.

Pre-Session 19 Preparation:

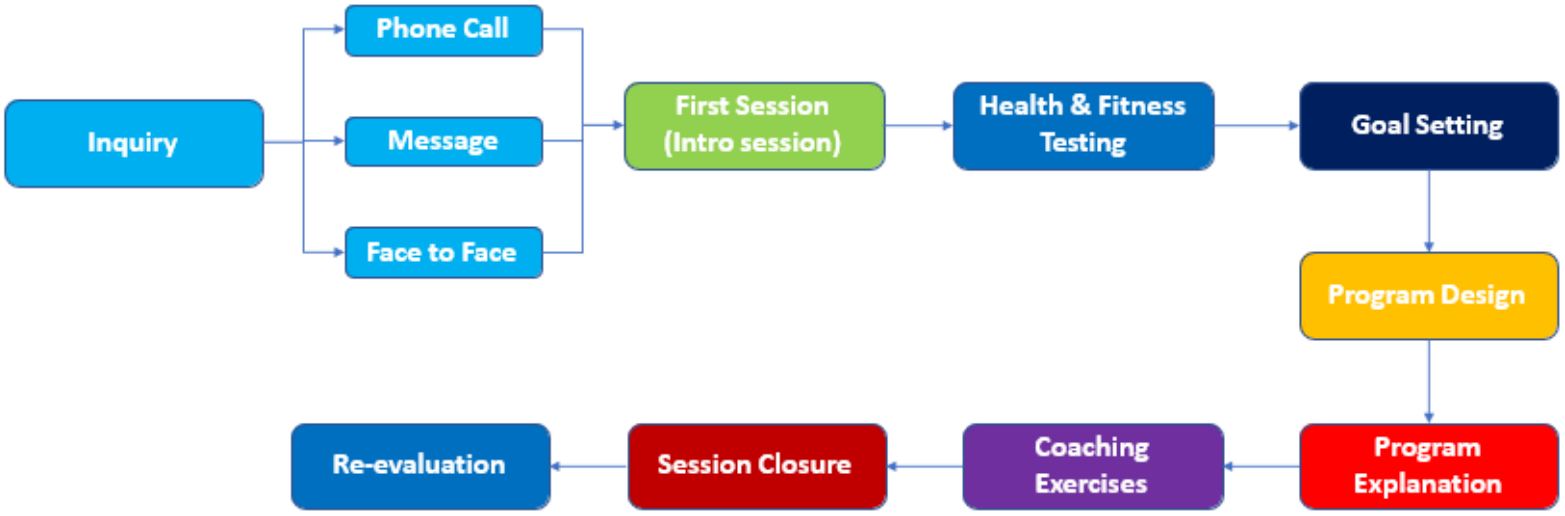
Watch the following videos – Platform Chapter Titled:

Week 9: Coaching Behavioral Change Tools + **Process of Personal Fitness Coaching**

Session 19: Coaching for Behavior Change – Continued

LESSON 1: THE COACHING PROCESS IN PERSONAL TRAINING

Objectives:
-Apply the full coaching process: from building trust and ensuring safety to setting meaningful goals, explaining programs clearly, providing effective feedback, and reinforcing progress through closure and re-evaluation.



1- INQUIRY: THE FIRST CONTACT

The first interaction sets the tone for trust and connection.

Trainer approach: *"We have great packages, which one do you want?"*
Coach approach: *"Tell me what you're hoping to achieve by working with me."*

Tips: respond promptly and politely, show genuine interest, ask "What made you reach out today?", and avoid overselling.



How can you give a good 1st impression?

2- THE FIRST SESSION

Building rapport enhances adherence and trust.

- Warm greeting, positive body language.
- Share your philosophy: "My role is to support you, not just to train you."
- Ask open-ended questions; let the client talk more.

-Confidentiality and Safety: Explain that health information is confidential and handled professionally.
-Health Screening: Use evidence-based screening tools (PAR-Q+, medical clearance if needed) to ensure safety and build trust.

3- HEALTH & FITNESS TESTING

Emphasize that the process matters more than the numbers. Tests provide benchmarks to track progress and program effectiveness.

Best practices:
- Explain the purpose of each test.
- Use clear, simple language.
- Frame results as a baseline, not a judgment.

Case Example (Nadine, 35): Instead of saying, "Let's see how much fat you have," say, "This gives us a snapshot of where you're starting so we can track progress together."

4- GOAL SETTING

Specific, challenging goals improve adherence.

SMART Goals:

• Specific • Measurable • Attainable • Relevant • Time-bound.

Example: *"I will walk 30 minutes, 5 days a week, for the next month."*

Beyond SMART: Add personal meaning for stronger motivation (*e.g., "Be active to play with my children", "Improve my fitness level so I can go on hikes with my husband" …).*



A client shows up stating that her main long-term goal is to lose weight. Design 2 SMART short-term goals for your client?

5- PROGRAM EXPLANATION: KISS PRINCIPLE

Explain the program clearly before applying it. This builds trust, reduces stress, and sets realistic expectations.

Trainer approach: *"We'll do mesocycles with progressive overload targeting hypertrophy."*
Coach approach: *"We'll start with simple workouts to build strength and gradually increase challenge."*

Always apply KISS (Keep It Simple, Silly). Avoid overloading clients with technical details. Customize the plan based on goals, readiness, and personality.

6- COACHING EXERCISES & FEEDBACK

Stages of Motor Learning:

- Cognitive – learning what to do; clear demos, simple cues, frequent feedback.
- Associative – refining technique; more specific feedback and practice variation.
- Autonomous – skill becomes automatic; less feedback, more challenge.

Feedback Principles:

- Specificity – pinpoint what to adjust.
- Timeliness – give feedback soon after performance.
- Balanced – start with positives, then corrections.
- Quantity – 1–2 key points at a time.
- Fading – reduce feedback as clients improve.
- Mode – use verbal, visual, or kinesthetic cues.



Your client performs a squat with her knees caving inward, how would you give her a good feedback regarding her performance?

7- SESSION CLOSURE & RE-EVALUATION

Closing the Session:

- Summarize progress.
- Affirm effort: “I saw you push through even when tired — that shows resilience.”
- Give clear next steps.

Re-Evaluation:

- Reassess every 4–6 weeks.
- Celebrate small wins and highlight improvements beyond the scale.



Pair Activity – Role Play: Applying the Coaching Process

Scenario:
Your client Rami, 40, recently rejoined the gym after several failed attempts to stay consistent. He says, "I always start strong but stop after a few weeks. Maybe I'm just not motivated enough."

Instructions:
1. Work in pairs — one as the coach, one as Rami (the client).
2. Go through each stage of the coaching process as if it were a real session.
3. Switch roles after completing all stages.
4. Use the space in the table to take short notes after the role play.

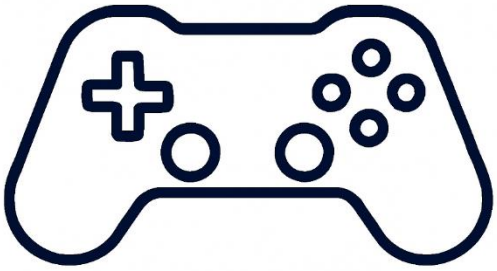
Make it fun! You must choose different questions, answers, or examples than your partner. Creativity counts!

Stage	Coach's Task	Notes / Response
Inquiry	Start the conversation and explore why Rami reached out.	
First Session	Build rapport, ask open questions, and gather information.	

Stage	Coach's Task	Notes / Response
Goal Setting	Help Rami create realistic and meaningful goals.	
Program Explanation	Present your plan simply and clearly using the KISS principle.	
Feedback & Coaching	Give constructive and motivating feedback during the session.	
Session Closure	Reinforce effort, summarize progress, and set next steps.	

Let’s wrap up the Coaching Process in Personal Training!





Plank the Process!

- **Work in 2 teams.**
- **One player of each team holds a plank while the instructor asks a question.**
- **If the answer is correct, run and tap your friend to switch; if not, hold the plank longer!**
- **Switch players each turn until your team completes all the stages.**
- **The first team to finish correctly wins!**

Consolidation:
By the end of Session 19, I am aware of:
-How the coaching process flows from first contact to follow-up, ensuring a structured and client-centered approach.
-The importance of trust, communication, and clarity at each stage.
-How using SMART and value-based goals, the KISS principle, and feedback tailored to learning stages supports client motivation and long-term adherence.

Post-Session 19 Tasks:
-Review the material covered during the session.
-Prepare a brief recap of the key lessons covered to be presented in Session 20.

Pre-Session 20 Preparation:
Watch the following videos – Platform Chapter Titled:
Week 10: Exercise Techniques and Methods - Lower Body Muscles Part 1

Session 20: Intro to Functional Anatomy

LESSON 1: TERMS & DEFINITIONS

Objectives:
- State the common anatomical terms and definitions.
- Go over the anatomy of the skeletal system.

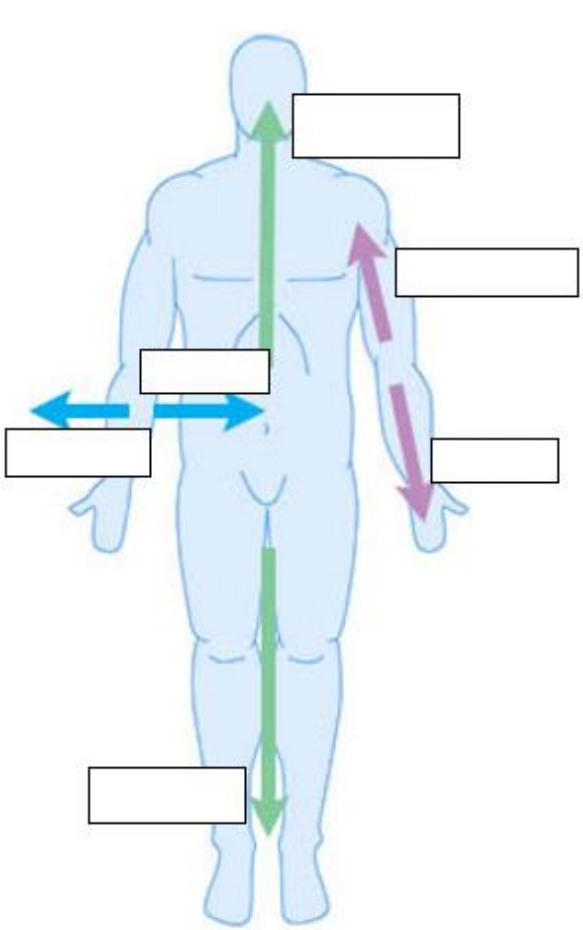
TERMS & DEFINITIONS: *(Platform: Chapter title: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System – Video title Anatomical Position & Directional Terms) (Refer to the manual, page 5)*

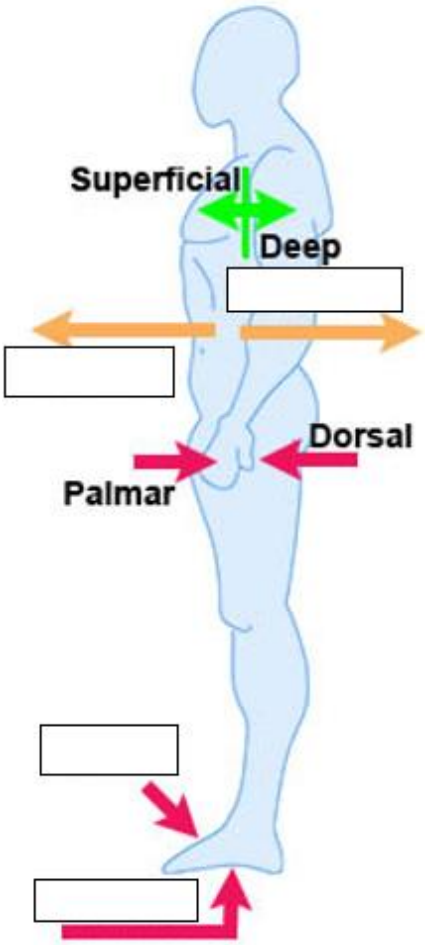
Term	Definition	Example
Anterior	Front of the body	Anterior Delt
Posterior	Back of the body	Posterior Delt
Superior	Toward the head	Knee is superior to the ankle
Inferior	Away from the head	Ankle is inferior to the knee
Medial	Toward the midline of the body	Medial/Internal Rotation
Lateral	Away from the midline of the body	Lateral/External Rotation
Dorsal	Top of the foot	Dorsiflexion
Plantar	Bottom of the foot	Plantar Flexion
Supine	Lying face up	Lying on the spine
Prone	Lying face down	Prone Leg Curl
Proximal	Extremity close to the midline of the body	Top of the humerus
Distal	Extremity far from the midline of the body	Bottom of the humerus
Superficial	Close to the surface	Rectus Abdominis
Deep	Below the surface	Transverse Abdominis
Arm	Region from the shoulder to the elbow	
Forearm	Region from the elbow to the wrist	
Thigh	Region from the hip to the knee	
Leg (Lower Leg)	Region from the knee to the ankle	
Ipsilateral	On the same side	Side Plank
Contralateral	On the opposite side	Resisted Side Bending
Unilateral	On one side	Single Leg Squat
Bilateral	On both sides	Squat
Palmar	Anterior ventral surface of the hand	
Hyper	Above normal	Hyper lordosis - Hyperglycemia
Hypo	Below normal	Hypoglycemia

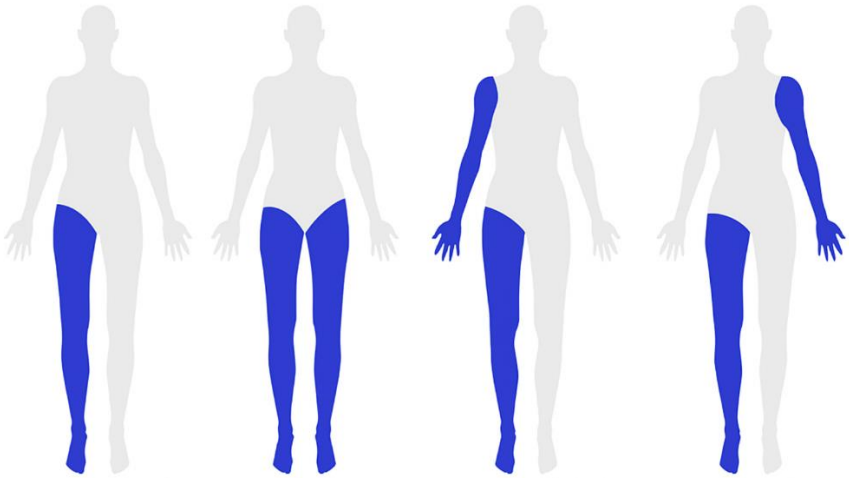
GROUP ACTIVITY

Fill in the blanks using the below terms:

Anterior – Bilateral – Dorsal – Superior – Unilateral – Inferior – Posterior – Proximal – Ipsilateral – Lateral – Medial – Contralateral – Plantar – Distal







LESSON 2: ANATOMY of THE SKELETAL SYSTEM

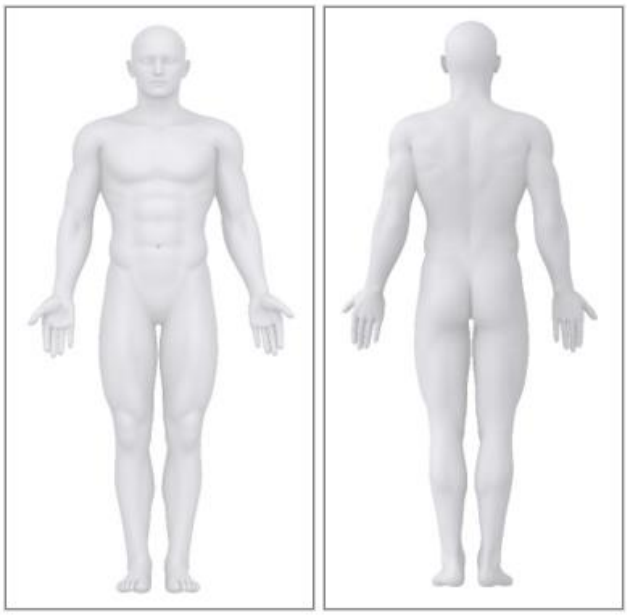
Objectives:

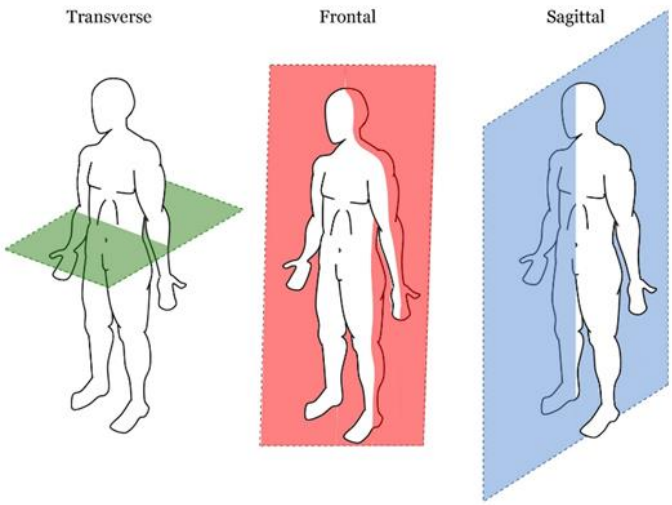
-Understand the planes of motion and the actions that occur in each.

-Understand the main anatomical terms and definitions.

-Identify the bones of the skeletal system.

ANATOMICAL POSITION & PLANES OF MOTION: *(Platform: Chapter title: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System – Video title: Anatomical Position & Directional Terms / Planes of Motion) (Refer to the manual, page 3)*





HUMAN SKELETON (AXIAL AND APPENDICULAR SYSTEMS): *(Platform Chapter title: Week 1: Introduction to Personal Fitness Coaching + Anatomy of the Skeletal System – Video title: Bone to Bone Approach - Axial Skeleton) (Refer to the manual, page 13-21)*



Let us see how fast you are in grabbing the bone in question :D

Consolidation:

By the end of Lessons 1 and 2, I am aware of:

-The different planes of motion and the types of movement that occur in each.

-The key anatomical terms and definitions used in movement and training.

-The major bones of the skeletal system.

LESSON 3: ANATOMY of THE ARTICULAR & MUSCULAR SYSTEMS

Objectives:

- Describe the fundamental joint movements.
- Differentiate between concentric, eccentric and isometric contractions.
- Review the origin, insertion and main actions of the lower body muscles.

FUNDEMENTAL JOINTS MOVEMENTS: *(Platform: Chapter title: Week 2: Week 2: Anatomy of the Articular System + Muscular System – Video title: Movements Occurring at the Main Joints of the Body) (Refer to the manual, page 28)*

PLANE	MOTION	DESCRIPTION
Sagittal	***Flexion***	When the joint angle decreases
	Extension	When the joint angle increases
	Dorsiflexion	Decreasing joint angle at the ankle
	Plantarflexion	Increasing joint angle at the ankle
Frontal	***Abduction***	Movement away from the midline of the body
	Adduction	Movement toward the midline of the body
	Elevation	When the scapula moves upward
	Depression	When the scapula moves downward
	Protraction	When the scapula moves forward (laterally and anteriorly)
	Retraction	When the scapula moves backward (medially and posteriorly)
	Upward rotation	When the scapula rotates and moves upward (this movement happens when you are raising the arm overhead), e.g., Overhead Press
	Downward rotation	When the scapula rotates and moves downward (this movement happens when you are lowering the arm), e.g., Lat Pulldown
	Lateral flexion	When bending the spine to either side
	Inversion	When turning the sole of the foot inward or medially (movement occurs at the ankle)
	Eversion	When turning the sole of the foot outward or laterally (movement occurs at the ankle)
Transverse	***Medial/Internal rotation***	Rotation toward the midline of the body (internal rotation)
	Lateral/External rotation	Rotation toward the side of the body or away from the midline (external rotation)
	Transverse rotation	Rotation at the spine, right or left
	Pronation	Movement occurring at the elbow joint toward the prone position (i.e., Palms facing down)
	Supination	Movement occurring at the elbow joint toward the supine position
	Transverse abduction	Abduction at the transverse plane, also known as horizontal extension
	Transverse adduction	Adduction at the transverse plane, also known as horizontal flexion

QUIZ BATTLE TIME!

4 teams. 4 questions each.

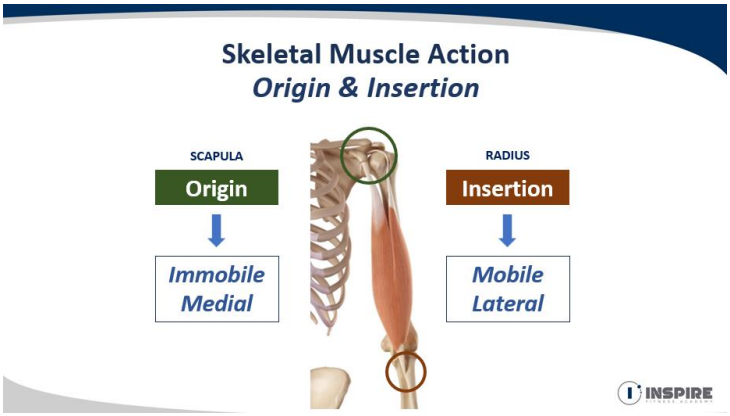
Pick your opponent and put their knowledge to the test!

- Answer solo = **1 point**
- Need backup? Team helps = **0.5 point**

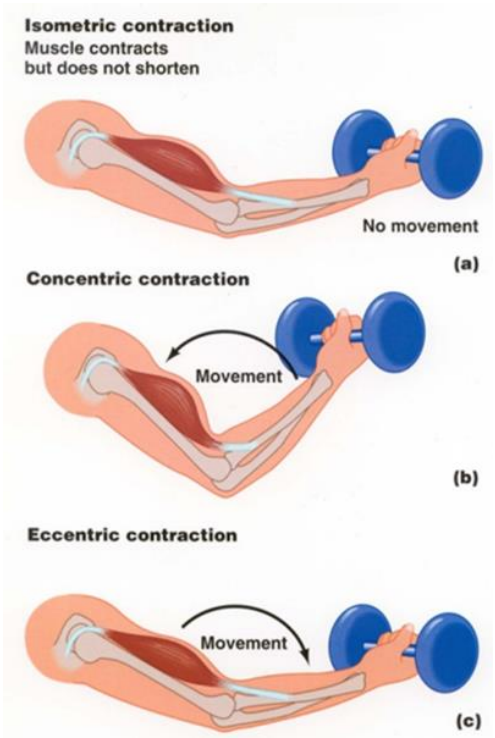
Let the challenge begin!

- **ANATOMY of THE MUSCULAR SYSTEM**

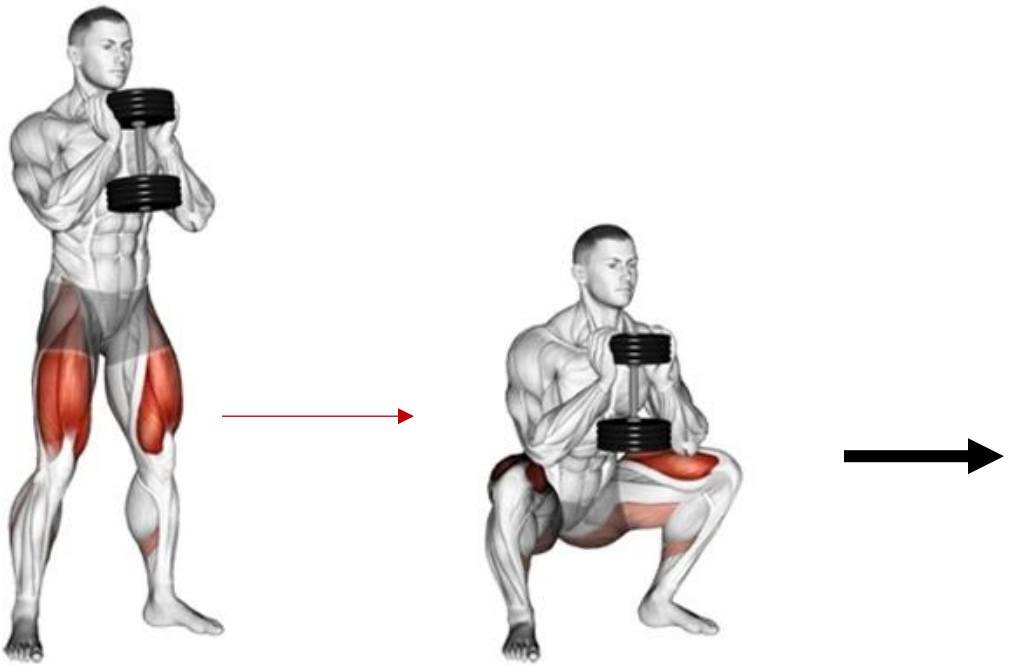
Differentiate between Muscles Origin and Insertion: *(Platform: Chapter title: Week 2: Anatomy of the Articular System + Muscular System – Video title: Introduction to the Muscular System)*

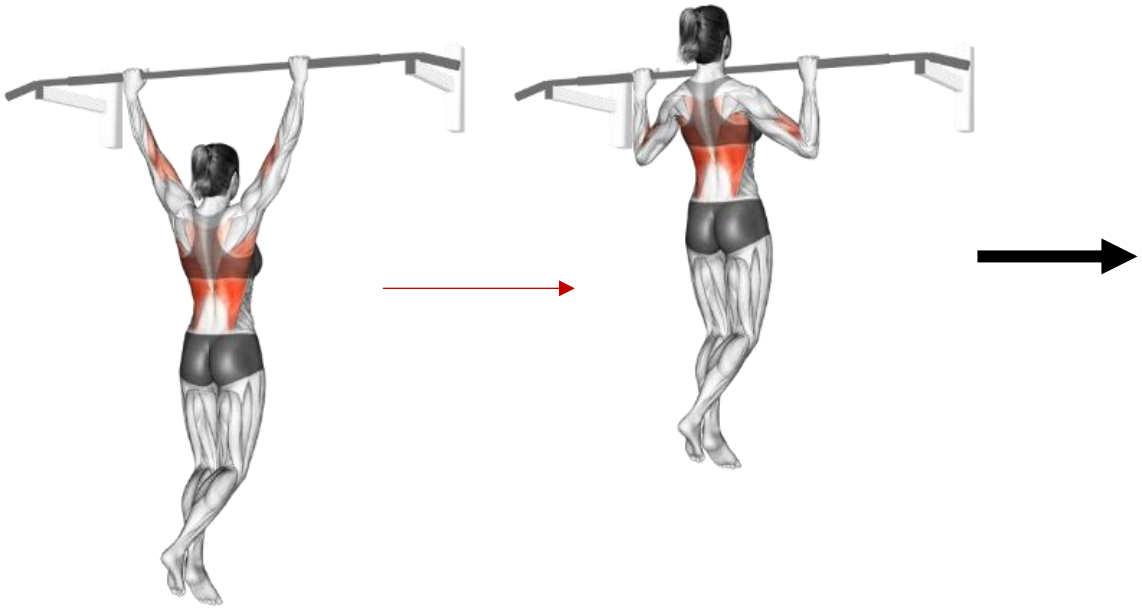


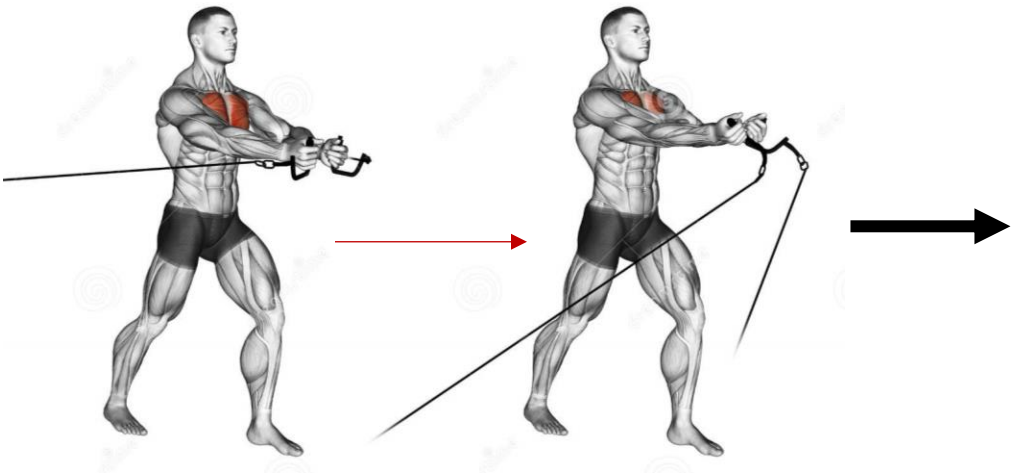
Concentric, Eccentric and Isometric contractions *(Platform: Chapter title: Week 2: Anatomy of the Articular System + Muscular System – Video title: Introduction to the Muscular System)*



Guess what type of contraction is happening in each movement:









REVIEW OF THE LOWER BODY MUSCLES ORIGIN, INSERTION AND ACTIONS: *(Platform: Chapter Titled: Week 3: Anatomy of the Muscular System + Part 1 of the Biomechanics of Exercise and Resistance Training – One video for each muscle) (Refer to the manual, page 48-55)*

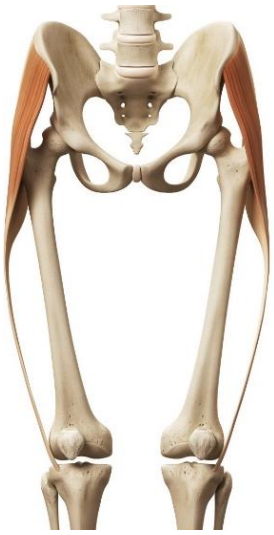
Based on the what we know so far and the below muscle pictures, let's try to **guess** each muscle's origin, insertion and action:

- **Gluteus Maximus, Medius and Minimus:**



Origin:
Insertion:
Actions:

- **Tensor Fascia Latae:**



Origin:
Insertion:
Actions:

- **Iliopsoas - Iliacus & Psoas:**



Origin:
Insertion:

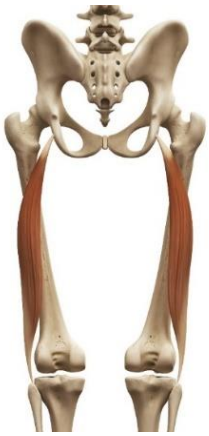


Origin:
Insertion:

Actions:

- **Hamstrings:**



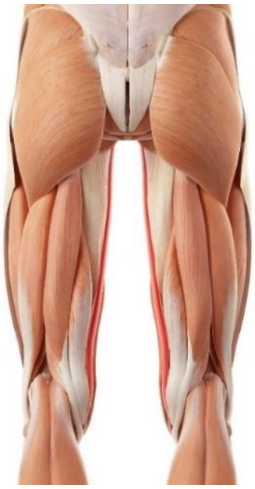






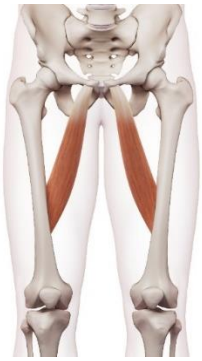
Origin:
Insertion:
Actions:

Label each of the Hamstring's muscles on this image:



- **Hip Adductors – Adductor Brevis, Magnus and Longus:**





Origin:
Insertion:
Actions:

- **Quadriceps – Vastus Lateralis, Medialis, Intermedius and Rectus Femoris:**






Origin:
Insertion:
Actions:

Label each of the Quadricep's muscles on this image:



- **Calves – Gastrocnemius & Soleus:**



Origin:
Insertion:
Actions:

Provide an exercise focusing on the gastrocnemius muscle engagement:

Provide an exercise focusing on the soleus muscle engagement:

<u>"Muscle Charades" Game:</u>

Pick a muscle card and read it carefully: it contains the origin, insertion, and action of a muscle. Your goal is to help your teammates guess the correct muscle, but here's the challenge:

- You're only allowed to say the words "origin," "insertion," and "action."
- After that, you can demonstrate the movement or point to the area on the body, but no extra words allowed!
- Oh! And you have one minute to do all that!

Let's see who can guess it right!

Consolidation:

By the end of Session 20, I am aware of:

-The fundamental joint movements.

-The difference between concentric, eccentric and isometric contraction

-The main lower body muscles, including their origin, insertion, and action.

Post-Session 20 Tasks:

-Review the material covered during the session.

-Prepare a brief recap of the key lessons covered to be presented in Session 2.

Pre-Session 21 Preparation:

-Review all materials covered in Session 20

-Watch all videos of the Chapter titled: Week 10: Exercise Techniques and Methods - Lower Body Muscles Part 1

Session 21: Exercise Techniques and Methods – Lower Body – Squats

Session 20 Recap + Q&A:

LESSON 1: BODYWEIGHT SQUAT MOTION ANALYSIS

Objectives:
- Analyze the movement conducted to identify the main muscles engaged
- Perform the bodyweight squat exercise in a proper technique
- Discuss the Squat common mistakes and propose short-term and long-term solutions

- **BODYWEIGHT SQUAT:** *(Platform: Chapter Titled: Week 10: Exercise Techniques and Methods - Lower Body Muscles Part 1) (Refer to the manual, page 185)*

GROUP ACTIVITY

Perform the motion analysis for the Squat Exercise:

	Static/Dynamic	Movement	Muscles Engaged
Spine			
Hip			
Knee			
Ankle			

Practice the Squat exercise following the below checklist in a coach / client scenario:

Client	Checklist
✓	Feet shoulder-width apart
	Brace your core
	Externally rotate your hips
	Keep the spine neutral NOT hyperextended
	Lead with the hip and knees directly follows
	Knees should follow the toes
	Keep your feet in full contact of the floor
	Aim is to reach parallel and later on to go below parallel. If this position cannot be obtained yet, stop when the trunk begins to round or flex forward, or when the heels rise off the floor, or knees cave inward

Instructor's feedback:

Squats Common Mistakes:



Identify common squat mistakes, **understand** their possible causes, and **learn** how to correct them acutely and chronically by filling the below table:

Common Mistake	Short-Term Modification(s)	Long-Term Modification(s)
Toes off the floor		
Heels off the floor		
Knees caving inward		
Leading the movement with the knees		
Arching the lower-back on the way down		

Common Mistake	Short-Term Modification(s)	Long-Term Modification(s)
Bending the back too much		
Neck does not remain neutral		
What about Knees passing toe-line?		
What about butt-wink or hip posterior pelvic tilt taking place at the end of the eccentric phase?		

Consolidation:

By the end of this lesson, I am aware of:

-How to perform the squat with proper technique.

-The most common squat mistakes and how to correct them as a coach.

-How to analyze the squat using motion analysis principles.

LESSON 2: COACHING THE BODYWEIGHT SQUAT & HIGH-BAR BACK SQUAT

Objectives:
- Coach the bodyweight squat exercise to a beginner client.
- Coach the high-bar back squat exercise to a beginner client.

LET'S PRACTICE AGAIN:

Practice the Squat exercise in a client / coach / assessor role play scenario using the Tell-Show-Do coaching method. Assessors, **grab** your pens and **rate** the work based on the above checklist:

Instructor's feedback

PRACTICE AGAIN & AGAIN while changing groups and introducing the High Bar Back Squat

Consolidation:

By the end of Session 21, I am aware of:

-How to coach the squat exercise to a beginner client.

Post-Session 21 Tasks:

-Practice performing and coaching the Bodyweight & High Bar Back Squat exercises.

-Prepare a brief recap of the lessons covered to be presented in Session 22.

Pre-Session 22 Preparation:

-Review all materials covered in Session 21.

Session 22: Exercise Techniques and Methods – Lower Body – Squats Variations

Session 21 Recap + Q&A:

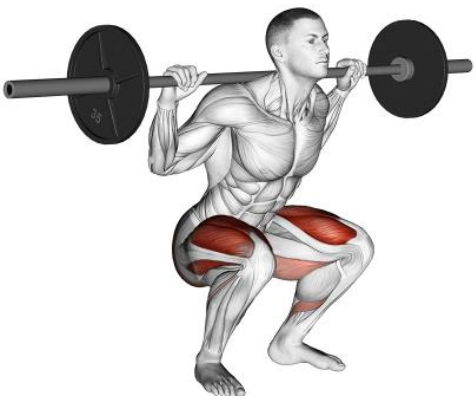
LESSON 1: COACHING THE SQUAT VARIATIONS

Objectives:
-Understand the differences between common squat exercise variations.
-Know how to perform and coach each squat variation effectively.

- **SQUAT VARIATIONS:**

Practice the Squat Variations in a client / coach / assessor role play scenario

Squat Variation:	Bar/Weight Position:	Starting Rack Height:
High-Bar Back Squat	Bar on trapezius	Upper chest level
Low-Bar Back Squat	Bar on posterior deltoid	Upper chest level
Front Squat	Bar on anterior deltoid	Upper chest level
Goblet Squat	Dumbbell/Kettlebell at chest level (elbows flexed)	Not required
Dumbbell Squat	Dumbbells/Kettlebell at hip level (elbows extended)	Not required
Sumo Squat	Wide stance, toes out (weight position varies)	Depends on equipment



High Bar Squat



Low Bar Squat



Goblet Squat



Front Squat



Sumo Squat



Dumbbells Squat

Instructor's feedback

Switch teams and **practice** the movements again.

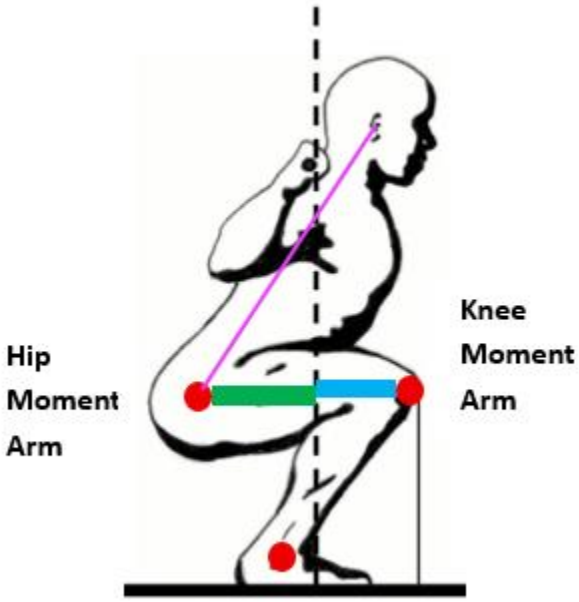
GROUP ACTIVITY

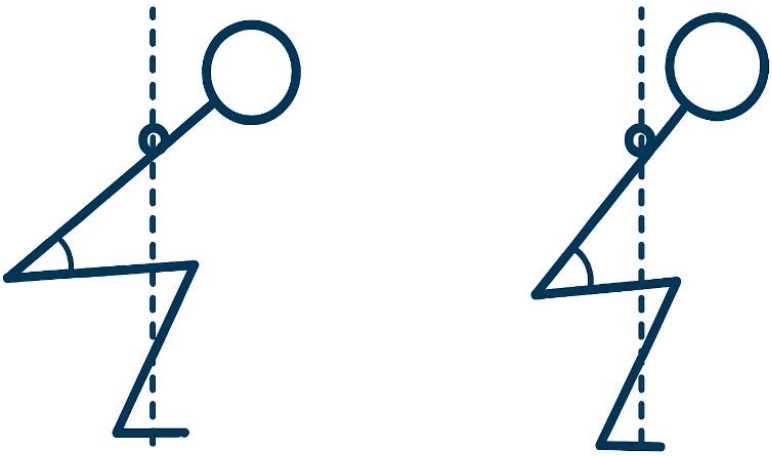
Analyze the differences between each and every variation:

Biomechanics Review: *(Chapter Titled: Week 4: Part 2 of the Biomechanics of Exercise and Resistance Training + Physiology of the Muscular System – Video Titled: Terms and Definitions Part 2) (Refer to the manual page 63)*

Define the Following Terms:

Moment or Torque:	
Moment Arm:	



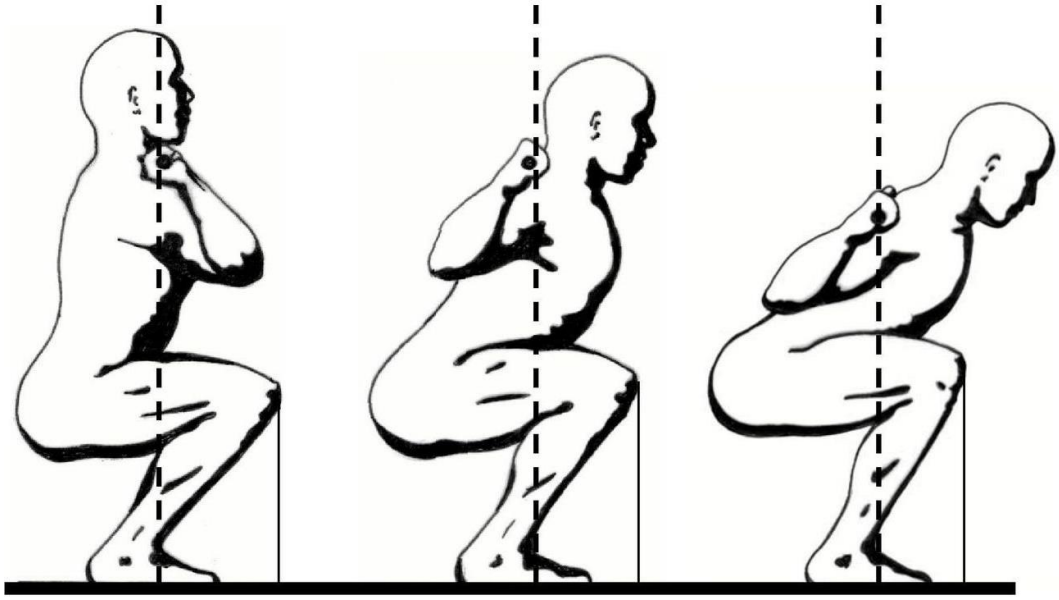


Longer Femur = More Forward Lean

Shorter Femur = More Upright Torso

GROUP ACTIVITY

Explain how each squat variation changes the movement and muscle engagement from a biomechanical point of view.



(Platform: Chapter Titled: Week 4: Part 2 of the Biomechanics of Exercise and Resistance Training + Physiology of the Muscular System – Video Titled: Front Squat vs High-Bar Squat vs Low-Bar Squat)

Consolidation:

By the end of Session 22, I am aware of:

-The different Squat variations.

-How to differentiate between these variations.

-How to perform and coach these variations.

Post-Session 22 Tasks:

-Practice performing and coaching all squat exercise variations.

-Prepare a brief recap of the lessons covered to be presented in Session 23.

Pre-Session 23 Preparation:

Watch the following videos on the platform:

-All videos of the Chapter titled: *Week 11: Exercise Techniques and Methods for Lower Body Muscles – Part 2*

Session 23: Exercise Techniques and Methods – Lower Body – Lunges

Session 22 Recap + Q&A:

LESSON 1: COACHING THE LUNGES VARIATIONS



Objectives:
-Perform and coach the static lunge exercise.
-Perform and coach various lunge exercise variations.
-Analyze and identify the key differences between each variation.

- **LUNGES:** *(Platform: Chapter Titled: Week 11: Exercise Techniques and Methods - Lower Body Muscles Part 2) (Refer to the manual page 191)*



Practice the static lunge in a coach / client / assessor role play scenario referring to the below checklist:

Client	Checklist
	Stand with feet hip width and toes pointing straightforward.
	Maintain the core bracing technique.
	Take a large step forward with the non-dominant leg first.
	As the working (front) leg steps forward, the supporting (rear) leg's heel should elevate off the floor and shift pressure towards the working front leg.
	Most of the pressure should remain on the working leg while the supporting leg should handle only a miniature portion of the pressure (balance).
	Move the hips slightly backwards and then drop downwards while simultaneously bending the knees and keeping the chest upright.
	Go down until the backward knee touch the floor then stand back up into the starting position.
	In the split squat/static lunge, none of the legs is lifted off of the floor.
	Switch legs and repeat all the above steps

Instructor's feedback

Practice the Lunges Variations in a coach / client / assessor role play scenario:

Lunge Variation	Key Characteristics
Reverse Lunge	Step backward with dominant leg, use front leg to pull body back to start position. Alternate legs.
Forward Lunge	Step forward with non-dominant leg, push floor with front leg to return. Alternate legs.
Side Lunge	Step to the side with non-dominant leg, push floor to return to start. Alternate legs.
Walking Lunge	Same as forward lunge but continue stepping forward instead of returning to starting position.
Bulgarian Lunge	Start seated, step forward with front leg, then elevate rear leg on bench for the entire set. Switch legs.

Instructor's feedback

Switch teams and **practice** the movements again.

GROUP ACTIVITY

Analyze the differences between the lunge exercise variations after performing a motion analysis using the table below:

	Spine	Hip	Knee
Reverse Lunges			
Forward Lunges			

	Spine	Hip	Knee
Side Lunges			
Walking Lunges			
Bulgarian Lunges / Split Squats			

Consolidation:
By the end of this lesson, I am aware of:
-The different Lunges exercise variations.
-How to differentiate between these variations.
-How to perform and coach each variation effectively.

LESSON 2: FORMATIVE ASSESSMENT



Objectives:
-Review the main lower body exercises performed so far.
-Get a sneak peek of what is required for the practical exam.



Formative Assessment – Lower Body Muscle Exercises:

In pairs, prepare and demonstrate the following exercises—Barbell Back Squat, and Static Lunges; in a coach/client role-play scenario. The demonstration will be presented to the entire group, followed by feedback on technique, coaching, and execution.

Instructor's Feedback



Consolidation:
By the end of Session 23, I am aware of:
-The different Lunges variations.
-How to differentiate between these variations.
-How to perform and coach these variations.
-The main points I need to focus on while filming my practical exam.



Post-Session 23 Tasks:
-Practice performing and coaching all lunges exercise variations.
-Prepare a brief recap of the lessons covered to be presented in Session 23.



Pre-Session 24 Preparation:
Watch the following videos on the platform:
-All videos of the Chapter titled: *Week 11: Exercise Techniques and Methods for Lower Body Muscles – Part 2*

Session 24: Exercise Techniques and Methods – Lower Body – Deadlifts

Session 23 Recap + Q&A:

LESSON 1: COACHING DEADLIFT VARIATIONS



Objectives:
-Perform and coach the conventional deadlift exercise.
-Perform and coach the deadlift variations.
-Analyze and differentiate between these variations.

- **CONVENTIONAL DEADLIFT:** *(Platform: Chapter Titled: Week 11: Exercise Techniques and Methods - Lower Body Muscles Part 2) (Refer to the manual page 191)*



Practice the Conventional Deadlift exercise in a client / coach / assessor role play scenario:

Client	Checklist
	Bar should be above floor level (use either steppers or bumper plates)
	Place the bar over your mid-foot with feet hip-width apart
	Go down by hinging your hips backward to the max, then bend your knees until you reach and grab the bar
	Keep going until your tibia touches the bar
	Choose the grip you prefer (pronated, supinated or mixed)
	Chest up, brace your core, and engage your lats (imagine as if you are breaking the bar into two pieces or bringing your elbows towards your pockets)
	Lift the bar up by pushing the floor away with the feet and driving the hips forward
	Keep the bar very close to the body
	Finish the move by fully extending your hips without hyperextending your back

Instructor's feedback



Switch teams and **practice** the movement again. Assessors, **grab** your pens and **rate** the work based on the above checklist:

Instructor's feedback

Practice AGAIN &AGAIN while switching teams, this time by **adding weights**.

Instructor's feedback

Practice the Deadlift Variations in a client / coach / assessor role play scenario

Deadlift Variation	Key Characteristics
Romanian Deadlift	Start with bar at hip level on rack. Knees slightly bent (~15°). Lower bar slowly by hinging at the hips until you feel a strong stretch in the hamstrings. Keep back straight throughout.
Stiff Leg Deadlift	Same movement as Romanian Deadlift but keep the knees fully extended during the entire exercise. Focus is more on hamstrings and lower back.
Sumo Deadlift	Start with feet wider than shoulders and toes pointing out. Grip the bar with hands shoulder-width apart. Arms vertical, elbows extended. Keep chest up and hips low before lifting.
B Stance Deadlift	Unilateral version of Romanian or Stiff Leg Deadlift. One foot slightly behind for balance (heel raised). Load is mainly on the front leg. Great for improving balance and unilateral strength.



Romanian Deadlift



Sumo Deadlift



Stiff-Leg Deadlift



Conventional Deadlift

Instructor's feedback

Switch teams and **practice** the movements again. Assessors, don't forget, have your pens ready and **rate** the work based on the above checklist.

GROUP ACTIVITY

Analyze the differences between each deadlift variation by performing a movement analysis based on the table below:

	Spine	Hip	Knee
Squat			
Conventional deadlift			

	Spine	Hip	Knee
Romanian deadlift			
Stiff leg Deadlift			
Sumo Deadlift			

Consolidation:

By the end of Session 24, I am aware of:

-The Deadlifts exercise and its common variations.

-How to differentiate between these variations.

-How to perform and coach these variations.

Post-Session 24 Tasks:

-Practice performing and coaching all deadlift exercise variations.

-Prepare a brief recap of the lessons covered to be presented in Session 24.

Pre-Session 25 Preparation:

Review the following videos on the platform:

-All videos of the Chapter titled: Week 11: Exercise Techniques and Methods for Lower Body Muscles – Part 2

Session 25: Exercise Techniques and Methods – Lower Body – Hip Thrusts, Hip Adductors vs Abductors exercises, and Calf Raises

Session 24 Recap + Q&A:



Practice the Deadlift Variations in a client / coach / assessor role play scenario.

Instructor's Feedback

LESSON 1: HIP THRUSTS vs GLUTE BRIDGES



Objectives:
-Practice the Hip Thrust and Glute Bridge exercises.
-Analyze the movement in each of the performed exercises.

- **HIP THRUST & GLUTE BRIDGE:** *(Platform: Chapter Titled: Week 11: Exercise Techniques and Methods - Lower Body Muscles Part 2) (Refer to the manual page 194)*

Client	Checklist
	Use a bench preferably around knee height, make sure that the bench is stable.
	If using weights, make sure to use bumper plates, if not use steppers: roll the bar over your legs until the bar is at the level of your hips.
	Place the entire shoulder blades on the bench.
	Place the feet forward to create a 90-degree angle at the knee. Your body will be parallel to the floor once the hips are fully extended.
	Look at the bar at all times to form a posterior pelvic tilt which will engage the glutes more. Looking up to the ceiling may lead to hyperextension of the lumbar spine shifting the tension towards the back muscles.
	Lower the weight, then push through the feet to lift your hips fully up.

Practice the Hip Thrust and Glute Bridges in a client / coach / assessor role play scenario:

Aspect	Hip Thrust	Glute Bridge
Starting Position	Upper back supported on a bench, feet flat on the floor	Back flat on the floor, knees bent, feet flat
Surface Used	Bench + floor	Floor only
Range of Motion	Greater - hips drop below knee level	Limited - hips move from floor to neutral
Back Support	Yes - upper back on bench	No - back stays on floor
Equipment Used	Barbell or dumbbell often used	Usually, bodyweight or light resistance

Instructor's Feedback

Switch teams and **practice** the movements again.

GROUP ACTIVITY

Analyze the differences between the Hip Thrust and Glute Bridge exercises after performing a motion analysis using the table below:

	Spine	Hip	Knee	Ankle
Hip Thrust				
Glute Bridge				

Perform the hip thrust while moving your knees outward when lifting your hips up. What is the purpose of such variation?

Perform the hip thrust while moving your feet forward in the starting position before lifting your hips up. What is the purpose of such variation?

Consolidation:

By the end of Lesson 1, I am aware of:

-The difference between Hip Thrust and Glute Bridge exercises.

-How to perform and coach both exercises.

-The common variations of the Hip Thrust.

LESSON 2: HIP ADDUCTORS vs HIP ABDUCTORS



Objectives:
-Perform and coach hip adductor and hip abductor exercises.
-Analyze the movement patterns in each of the performed exercises.

- **HIP ADDUCTORS:** *(Platform Chapter Titled: Week 11: Exercise Techniques and Methods - Lower Body Muscles Part 2)*



<u>"Spin the Bottle" game:</u>

In small groups, come up with several exercises to target the hip adductors muscles using the following equipment: Bodyweight / Cable machine / Swiss ball / Elastic bands / Sliders / Bench

- ✓ *Part 1: Bottle Flip Challenge*
  Flip the bottle; whoever it lands on becomes the **Coach** and must demonstrate one of the exercises.
- ✓ *Part 2: Partner Practice*
  Pair up and perform all selected adductor and abductor exercises together.

- **HIP ABDUCTORS:** *(Platform: Chapter Titled: Week 11: Exercise Techniques and Methods - Lower Body Muscles Part 2)*



<u>"Spin the Bottle" game:</u>

In small groups, come up with several exercises to target the hip abductors muscles using the following equipment: Bodyweight / Cable machine / Ankle weights / Elastic bands / Bench / Box

- ✓ *Part 1: Bottle Flip Challenge*
  Flip the bottle; whoever it lands on becomes the **Coach** and must demonstrate one of the exercises.
- ✓ *Part 2: Partner Practice*
  Pair up and perform all selected adductor and abductor exercises together.

Consolidation:

By the end of Lesson 2, I am aware of:

-How to perform and coach various Hip Adductor and Hip Abductor exercises using different types of equipment.

LESSON 3: NORDIC CURLS, REVERSE NORDIC CURLS & CALF RAISES

Objectives:
-Perform and analyze the differences between the Nordic Curl and Reverse Nordic Curl exercises.
-Perform and analyze the differences between Standing and Seated Calf Raise exercises.

- **NORDIC CURL:** *(Refer to the manual page 195)*

Client	Checklist
	Kneel on a padded surface while anchoring your ankles under a heavy object, bar, or with a partner holding them down
	Keep your body upright from knees to head (neutral spine, glutes engaged, core braced)
	Slowly lean forward from the knees as far as you can, keeping hips extended (don't bend the hips)
	Arms can be used to catch yourself when you can't resist anymore. Push off lightly with your hands and use your hamstrings to pull yourself back up
	Advanced athletes may attempt to return without using the hands (concentric phase)

- **REVERSE NORDIC CURL:**

Client	Checklist
	Kneel on a mat
	Keep your body upright from knees to head (neutral spine, glutes engaged, core braced)
	Slowly lean backward from the knees as far as you can, keeping hips extended (don't bend the hips)
	Pull yourself back up into the starting position

Practice both exercises, and **discuss** the differences between them by analyzing the movement of each using the table below:

	Spine	Hip	Knee
Nordic Curl			
Reverse Nordic Curl			

- **STANDING CALF RAISES:** *(Platform Chapter Titled: Week 11: Exercise Techniques and Methods - Lower Body Muscles Part 2) (Refer to the manual page 196)*

Client	Checklist
	Stand feet hip-width and knees extended either on the floor or on an elevated platform
	If the elevated platform (stepper) is chosen, place the ball of the foot on the edge of the stepper
	Position the stepper close to a stable object or support that you can hold onto for balance
	Lift yourself up
	Hold for 1-2 seconds at the top
	Lower yourself controllably back down slowly and then lift yourself back up again

- **SEATED CALF RAISES:**

Perform the same calf raise movement, but while seated. Keep your knees bent at 90 degrees, feet flat on the floor, and lift your heels by pressing through the balls of your feet.



Practice both exercises, and **discuss** the differences between them:

Standing Calf Raises	Seated Calf Raises



Consolidation:
By the end of Lesson 24, I am aware of:
-How to perform and coach the Nordic Curl and Reverse Nordic Curl.
-The key differences between both exercises.
-How to perform and coach the Standing and Seated Calf Raises.
-The key differences between both variations.



Post-Session 24 Tasks:
Practice performing and coaching all deadlift exercise variations.
Prepare a brief recap of the lessons covered to be presented in Session 25.



Pre-Session 25 Preparation:
Watch all videos of the Chapter titled: *Week 12: Exercise Techniques and Methods for Upper Body Muscles*.

🎉 YOU CRUSHED LEG WORKOUTS! 🦵🔥

Congratulations you have squatted, lunged, deadlifted… and conquered!
Your legs are stronger, your technique sharper, and your energy unmatched.

🏅 Certified Lower Body Beast 🏋️‍♀️



CONGRATULATIONS!

Session 26: Exercise Techniques and Methods – Upper Body – Chest Muscles

Session 25 Recap + Q&A:

LESSON 1: FORMATIVE ASSESSMENT



Objectives:
-Review the main lower body exercises.
-Get a sneak peek of what is required for the practical exam.



Formative Assessment – Lower Body Muscle Exercises:

In pairs, prepare and demonstrate the following exercises—Barbell Back Squat, Conventional Deadlift, Hip Thrust—in a coach/client role-play scenario. The demonstration will be presented to the entire group, followed by feedback on technique, coaching, and execution.

Instructor's Feedback



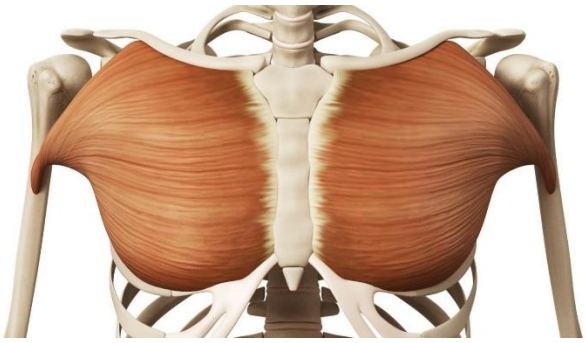
Consolidation:
By the end of Lesson 1, I am aware of:
-The main points I need to focus on while filming my practical exam.

LESSON 2: REVIEW OF THE CHEST, SHOULDERS AND TRICEPS MUSCLES ORIGIN, INSERTION AND ACTION

Objectives:
-Review the muscle anatomy of the chest, shoulders, and triceps to better understand their actions.

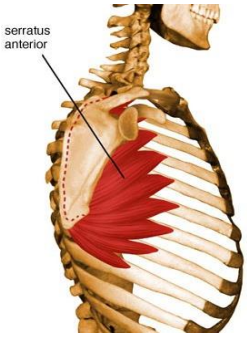
Based on the what we know so far and the below muscle pictures, let's try to **guess** each muscle's origin, insertion and action:

- **Pectoralis Major:**



Origin:
Insertion:
Actions:

- **Serratus Anterior:**



Origin:
Insertion:
Action:

- **Deltoid: Anterior – Middle – Posterior**



Origin:
Insertion:
Actions:

- **Rotator Cuffs: S-I-T-S**






Origin:
Insertion:
Actions:

- **Triceps**:



Origin:
Insertion:
Actions:



“Muscle Charades” Game:

Pick a muscle card and read it carefully: it contains the origin, insertion, and action of a muscle. Your goal is to help your teammates guess the correct muscle, but here's the challenge:

- You are only allowed to say the words “origin,” “insertion,” and “action.”
- After that, you can demonstrate the movement or point to the area on the body, but no extra words allowed!
- Oh! And you have one minute to do all that!

Let’s see who can guess it right!



Consolidation:

By the end of Lesson 2, I am aware of:

-The main actions of the chest, shoulders, and triceps muscles.

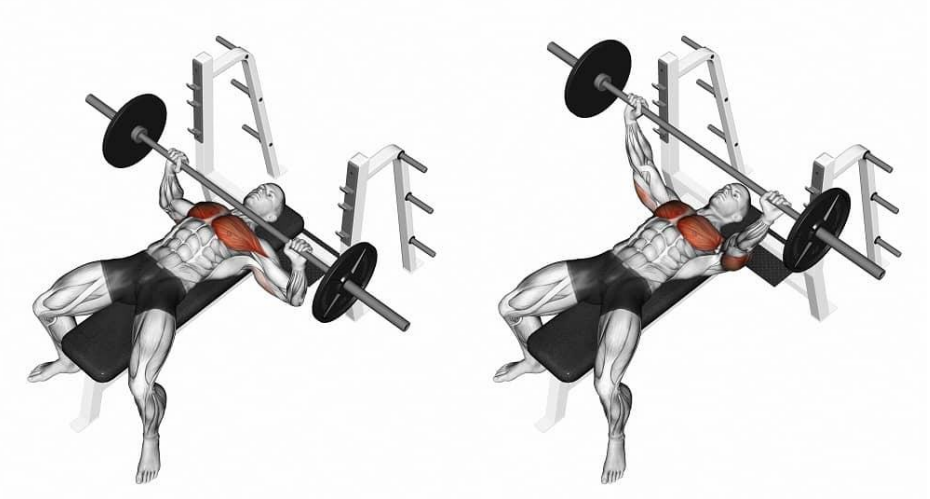
LESSON 3: COAHCING THE MAIN CHEST EXERCISES

Objectives:
-Perform and coach the main chest exercises.
-Analyze and differentiate between these exercises.

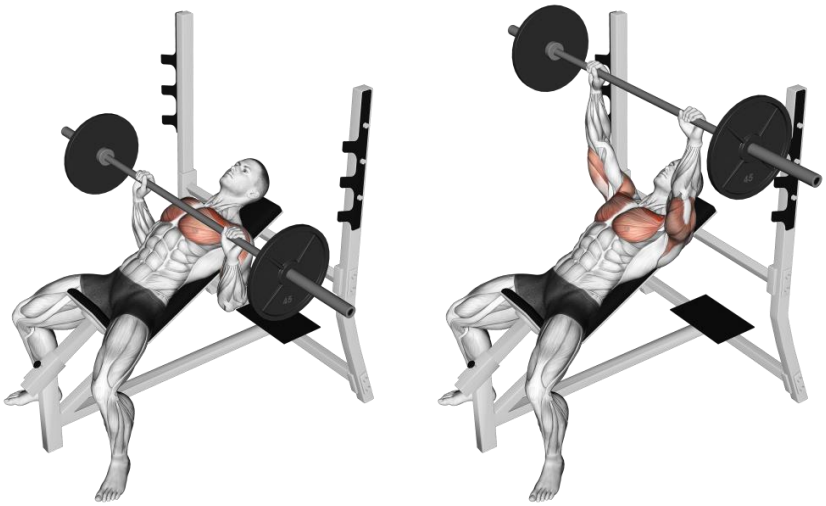
- **CHEST PRESS:** *(Platform: Week 12: Exercise Techniques and Methods for Upper Body Muscles):*

Client	Checklist
	The bar is evenly balanced on both sides
	Eyes below the bar
	Head – upper back – glutes in contact with the bench, while the feet in contact with the floor
	Grab the bar slightly wider than shoulder width
	The bar is lowered to the nipple line
	Push the bar up and slightly backward until elbows are slightly unlocked
	Re-rack the bar safely

Incline Chest Press	Same movement as the flat chest press, but performed on an inclined bench. Lower the weights toward your upper chest, then press them back up to the starting position.
Cable Press	Grasp the handles and position your hands at mid-chest level (aligned with your chest or shoulders). Keep a neutral wrist position and engage your core. Press forward, fully extending your arms at chest height.
Incline Cable Press	Grasp the handles and position your hands at upper-chest or shoulder level (slightly above shoulder height). Maintain a neutral wrist position and engage your core. Press forward, extending your arms diagonally upward.
Decline Cable Press	Grasp the handles and position your hands just below chest level (handles slightly above shoulder level). Keep a neutral wrist position and engage your core. Press forward, extending your arms diagonally downward.



Chest Press



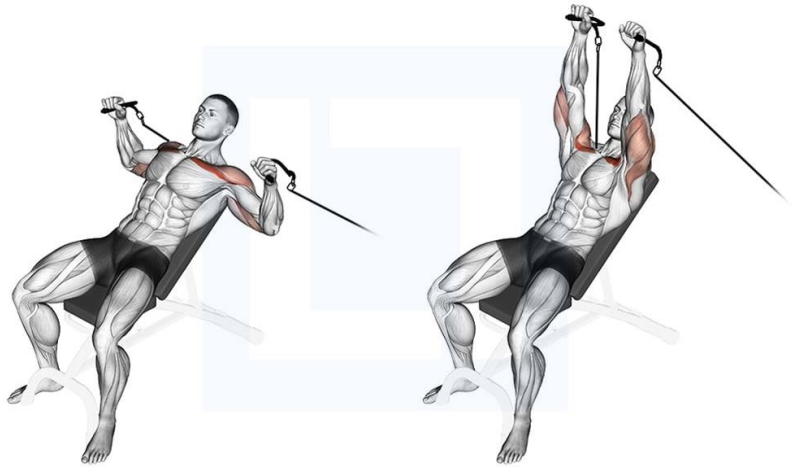
Incline Chest Press



Cable Chest Press



Decline Cable Press

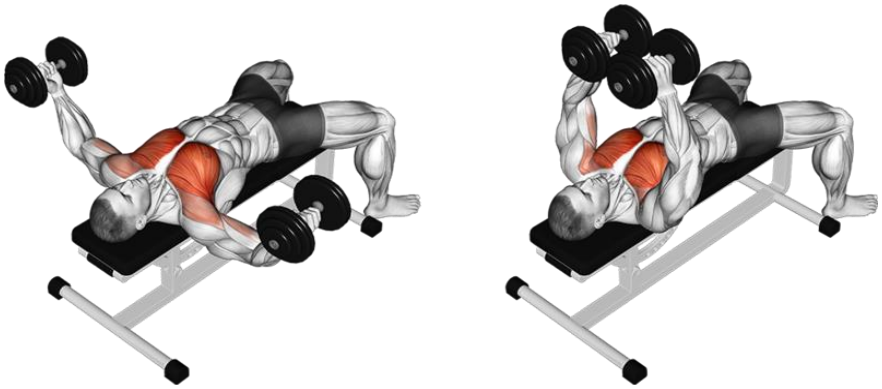


Incline Cable Press

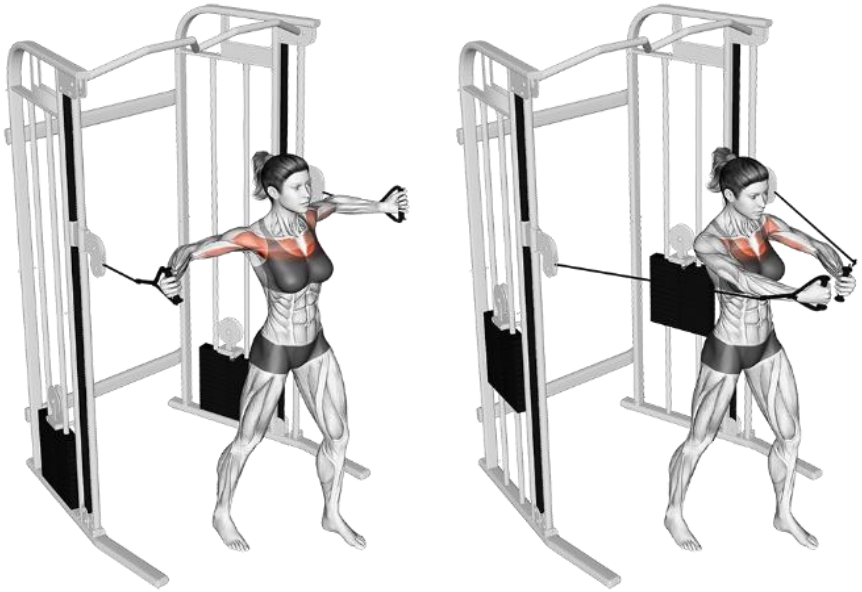
- **CHEST FLY:** *(Refer to the manual page 199)*

Client	Checklist
	Grasp the dumbbells in a neutral position and place them just above your knees
	Sit on the edge of the bench with the dumbbells positioned over the knees and lift the weight up while going into a supine position with the help of your lower body
	Extend elbows above the chest by pushing the dumbbells up
	Slightly flex the elbows pointing them out to the side.
	Lower the dumbbells creating the widest arc possible until the dumbbells are leveled with the shoulders or chest
	Raise the dumbbells up toward each other in a wide arc back going back to the starting position

Standing Cable Fly	Set the cable handles slightly above shoulder level. With a slight bend in your elbows, slowly bring your hands together in front of your body as if hugging someone. Keep one foot in front of the other for balance. Pause for one second when your hands meet, then slowly return to the starting position.
Standing Incline Cable Fly	Same movement as the Standing Chest Fly, but set the cable handles between knee and hip level. Start with your hands down and out to the sides, then slowly bring them diagonally upward until they meet at or just above head level.
Standing Decline Cable Fly	Same movement as the Standing Chest Fly, but position the cable handles around the level of the top of your head. Start with your hands up and out to the sides, then slowly bring them diagonally downward until they meet below chest level, around the abdominal area.



Chest Flies



Standing Cable flies



Standing Decline Cable flies



Standing Incline Cable flies

Practice all-Chest Press and Cable Fly exercise variations in a client / coach / assessor role play scenario (assessors, do not forget your pens).

Instructor's Feedback

Switch teams and **practice** the movements again.

Instructor's Feedback



Analyze the differences between the performed chest exercises after performing a motion analysis using the table below:

Exercise	GH	ST	Elbow	Spine
Chest Press				
Incline Chest Press				
Decline Chest Press				
Chest Fly				
Incline Chest Fly				
Decline Chest Fly				

- **PUSH UP:** *(Refer to the manual page 199)*

Client	Checklist
	Lie down on the floor with core engaged and hands shoulders width apart
	Tuck the elbows slightly in to create a 45-degree angle between the humerus and the torso and move your body downward
	Keep going until your chest is close to the floor
	While maintaining the core bracing technique, push up to extend the elbows as if trying to bounceoff the floor back into the starting position

Perform the push-up exercise and **answer** the questions below:

What is the difference between push-ups and chest press exercises?	
What happens when you do push-ups with your elbows very close to your torso?	
State some regressions for the push-up exercise and practice them.	
State some progressions for the push-up exercise and practice them.	

Consolidation:
By the end of Session 26, I am aware of:
-The main actions of the chest, shoulders, and triceps muscles.
-How to perform, coach, and analyze the most common chest muscle exercises.

Post-Session 26 Tasks:
-Practice performing and coaching all chest muscle exercises.
-Prepare a brief recap of the lessons covered to be presented in Session 26.

Pre-Session 27 Preparation:
Watch all videos of the Chapter titled: *Week 12: Exercise Techniques and Methods for Upper Body Muscles*.

Session 27: Exercise Techniques and Methods – Upper Body – Shoulders & Triceps Muscles

Session 26 Recap + Q&A:

LESSON 1: COAHCING THE MAIN SHOULDER MUSCLES EXERCISES

Objectives:
-Perform and coach the main shoulder muscle exercises.
-Analyze and differentiate between these exercises.

- **SHOULDER PRESS:** *(Platform Week 12: Exercise Techniques and Methods for Upper Body Muscles):*

Client	Checklist
	Dumbbells to be slightly wider than shoulder width
	Do not arch your back while pushing the dumbbells up
	Keep pressing up until you reach full elbow extension (keep elbows slightly unlocked)
	Allow the elbows to slowly flex to lower the weight back down to upper chest level
	Perform the movement in the scapular plane
	Keep your wrists directly over your elbows so that both joints are in a straight vertical line throughout the movement

- **FRONT RAISE:**

Client	Checklist
	Engage your core and keep it engaged throughout the exercise.
	Lift the dumbbells up with your elbows slightly bent.
	Adopt a neutral grip to better target the anterior deltoids.
	Raise the dumbbells until your arms are parallel to the floor (shoulder level).
	Lower the dumbbells slowly back to the starting position.

- **LATERAL RAISE:**

Client	Checklist
	Engage your core and keep it engaged throughout the exercise.
	Lift the dumbbells laterally (to the sides) in the scapular plane.
	Slightly flex the elbows and keep them fixed in that position.
	Adopt a pronated grip to better engage the middle deltoids.
	Raise the dumbbells until your arms are parallel to the floor (shoulder level).
	Lower the dumbbells slowly back to the starting position.

- **REVERSE FLY:**

Client	Checklist
	Sit on a bench facing the backrest angled at 45 degrees or less
	Allow your arms to hang towards the floor with your elbows slightly flexed
	Slowly raise the dumbbells up and out to the sides until the arms / dumbbells reach shoulders level
	Adopt a reverse grip for a more efficient engagement of the Posterior Deltoids
	Bring the dumbbells slowly back to the starting position

- **FACE PULL:**

Client	Checklist
	Adjust the cable handles till approximately shoulders level or slightly higher
	Engage the core and keep it engaged till the end of the exercise
	Take a step backward to create some tension on the cable
	Pull the rope toward your forehead while keeping your elbows high and out to the sides
	Keep pulling until the rope almost touches your forehead without shrugging
	Slowly allowing your arms to extend forward to return to the starting position



Practice all Shoulder muscles exercises in a client / coach / assessor role play scenario (assessors, do not forget to take notes):

Instructor's Feedback



Switch teams and **practice** the movements again.

Instructor's Feedback

GROUP ACTIVITY

Analyze the differences between the performed shoulder exercises after performing a motion analysis using the table below:

Exercise	GH	ST	Elbow
Shoulder Press			
Front Raise			
Lateral Raise			
Reverse Fly			
Face Pull			

Consolidation:
By the end of Lesson 1, I am aware of:
-How to perform, coach, and analyze the most common shoulder muscle exercises.

LESSON 2: COAHCING THE MAIN TRICEPS EXERCISES

Objectives:
-Perform and coach the main shoulder muscle exercises.
-Analyze and differentiate between these exercises.

- **TRICEPS CABLE PUSHDOWN:** *(Platform Week 12: Exercise Techniques and Methods Upper Body Muscles):*

Client	Checklist
	Stand up facing the cable machine and grab the ropes
	Engage the core and keep it engaged till the end of the exercise
	Pull the rope downward while keeping your shoulders fixed at neutral position
	Keep pulling until you reach full elbow extension (slightly unlocked)
	Slowly reverse the movement, till you reach the starting position

- **TRICEPS CABLE PUSHDOWN VARIATION:**

Client	Checklist
	Repeat same movement while taking one step backward (away from the cable machine) while keeping your shoulders in a neutral position – This is your starting position

- **LYING TRICEPS EXTENSION:**

Client	Checklist
	Lie on the bench in a supine position
	Position the dumbbells / Bar above the chest with wrists, elbows and shoulders aligned vertically above each other
	Keep the upper arms stationary, and flex the elbows slowly to lower the dumbbells (or the bar)toward your face / skull
	Push the dumbbells (or the bar) upward by extending the elbows back to the starting position

- **TRICEPS KICKBACK:**

Client	Checklist
	Lean your body forward while grabbing the dumbbell rack with one arm for stability
	Engage the core and keep it engaged till the end of the exercise
	Grab the dumbbell with the other arm and lift your elbow up till it reaches shoulder level
	Keep the upper arm stationary, and extend the elbow to lift the dumbbells up till it reaches same level with elbow and shoulder
	Slowly reverse the movement to return to the starting position

- **TRICEPS DIPS:**

Client	Checklist
	Sit on the edge of a stable bench with your hands placed next to your hips, fingers pointing forward.
	Slide your glutes off the bench and position your feet slightly in front of you, knees bent at a 90-degree angle
	Lower your body by bending your elbows, keeping them close to your torso, until your upper arms are parallel to the ground.
	Keep your back close to the bench and maintain a straight posture throughout the movement
	Push through your hands to extend your arms and raise your body back to the starting position

- **OVERHEAD TRICEPS EXTENSION:**

Client	Checklist
	Set the pulley at the highest position on the cable machine
	Stand facing away from the machine
	Grasp the rope handle with an overhand grip (palms facing each other)
	Lean forward and extend your arms forward by extending your elbows
	Keep your upper arms stationary and parallel to the floor
	Slowly allow the cable to pull your arms backward then push the cable forward by extending your elbows and repeat the same movement



Practice all Triceps muscles exercises in a client / coach / assessor role play scenario:

Instructor's Feedback



Switch teams and **practice** the movements again.

Instructor's Feedback



Analyze the differences between the performed triceps exercises after performing a motion analysis using the table below:

	Shoulder	Elbow
Triceps Cable Pushdown		
Triceps Cable Pushdown Variation		
Lying Triceps Extension		
Triceps Kickback		
Overhead Triceps Extension		
Triceps Dips		
How can you make the triceps dips exercise more challenging?		

Consolidation:

By the end of Session 27, I am aware of:

How to perform, coach, and analyze the most common triceps exercises.

Post-Session 27 Tasks:

Practice performing and coaching shoulders and triceps muscle exercises.

Prepare a brief recap of the lessons covered to be presented in Session 27.

Pre-Session 28 Preparation:

Watch all videos of the Chapter titled: *Week 12: Exercise Techniques and Methods for Upper Body Muscles*.

Session 28: Exercise Techniques and Methods – Upper Body – Back Muscles

Session 27 Recap + Q&A:

LESSON 1: FORMATIVE ASSESSMENT



Objectives:
-Review the main chest, shoulder, and triceps exercises.
-Get a sneak peek of what is required for the practical exam.



Formative Assessment – Chest, Shoulder & Triceps Muscles Exercises:

In pairs, prepare and demonstrate one of the following exercises—Bench Press, Chest Fly, Shoulder Press, and Lying Triceps Extension — in a coach/client role-play scenario. The demonstration will be presented to the entire group, followed by feedback on technique, coaching, and execution.

Instructor's Feedback



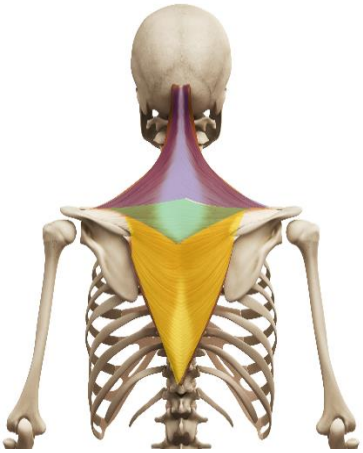
Consolidation:
By the end of Lesson 1, I am aware of:
-The main points I need to focus on while filming my practical exam.

LESSON 2: REVIEW OF THE BACK AND ELBOW FLEXORS ORIGIN, INSERTION AND ACTIONS:

Objectives:
-Review the muscle anatomy of the back and elbow flexors to better understand their actions *(Platform: Week 13: Exercise Techniques and Methods for the Back and Elbow Flexors Muscles) (Refer to manual page 40-43):*

Based on the what we know so far and the below muscle pictures, let's try to **guess** each muscle's origin, insertion and action:

- **Trapezius: Upper – Middle – Lower**



Origin:
Insertion:
Actions:

- **Rhomboids**:



Origin:
Insertion:
Action:

- **Latissimus Dorsi:**



Origin:
Insertion:
Actions:

- **Elbow Flexors:**

Biceps Brachii **Brachialis** **Brachioradialis**







Origin:
Insertion:
Actions:



“Muscle Charades” Game:

Pick a muscle card and read it carefully: it contains the origin, insertion, and action of a muscle. Your goal is to help your teammates guess the correct muscle, but here's the challenge:

- You’re only allowed to say the words “origin,” “insertion,” and “action.”
- After that, you can demonstrate the movement or point to the area on the body, but no extra words allowed!
- Oh! And you have one minute to do all that!

Let’s see who can guess it right!



Consolidation:

By the end of Lesson 2, I am aware of:

-The main actions of the back and elbow flexor muscles.

LESSON 3: BACK MUSCLES EXERCISES



Objectives:

-Perform and coach the main back muscle exercises.

-Analyze and differentiate between these exercises.

- **LAT PULLDOWN:** *(Platform: Week 12: Exercise Techniques and Methods for Upper Body Muscles):*

Client	Checklist
	Grasp the bar with a closed, pronated grip.
	The grip should be wider than shoulder width
	Sit down on the seat facing the machine while positioning the thighs under the pads with the feet flat on the floor
	Lean the torso slightly backward and fully extend the elbows
	Pull the bar down toward the upper chest without jerking your torso (controlled movement)
	Allow the elbows to slowly extend back to the starting position

- **BENT-OVER BACK ROW:**

Client	Checklist
	Lift the bar up using a conventional deadlift technique
	Lift the weight up and remain in a leaned forward position of around 60 degrees with respect to the floor
	Row the bar towards your hip (back pocket) by focusing on reducing the angle between the humerus and the torso
	Extend the shoulder in that way as to let the barbell touch the body while squeezing the back muscles
	Once the bar touches the torso, release it back down completely without the need for it to touch the floor or rack

- **SINGLE-ARM ROW:***(Refer to manual page 201)*

Client	Checklist
	Use the bench or the dumbbell rack to lean forward while having your spine neutral and parallel to the floor
	Grasp the dumbbell with a closed, neutral grip
	Hang the dumbbell with elbows fully extended
	Pull the dumbbell up towards the hip, keeping the elbow close to the body
	Lower the dumbbell back to the starting position

- **STRAIGHT ARM PULLDOWN:**

Client	Checklist
	Set the pulley at the highest position on the cable machine
	Stand facing the machine with your feet shoulder-width apart
	Grasp the bar or handles with an overhand grip (palms facing down)
	Lean forward while engaging your core and having your arms extended overhead with elbows slightly bent
	Keeping your arms straight, pull the bar or handles down towards your thighs
	Go back to the starting position in a controlled way and stop when you feel a stretch in your lats (wrists to be slightly higher than shoulder level)

- **INVERTED ROW:**

Client	Checklist
	Lie in a supine position underneath the bar
	Grab the bar in an overhand grip slightly wider than shoulder width
	Position your body so that your heels are on the ground, and your torso is in a straight line from your head to your heels by engaging your core
	Pull your chest up toward the bar by flexing your elbows and retracting your shoulders
	Go back slowly to the starting position while keeping your body straight throughout the entire duration of the exercise

- **SHRUGS:** *(Refer to manual page 202)*

Client	Checklist
	Stand upright and hold the dumbbells to your sides
	Raise the shoulders straight up toward the ears with scapula retracted



Practice all Back muscles exercises in a client / coach / assessor role play scenario:

Instructor's Feedback



Switch teams and **practice** the movements again.

Instructor's Feedback

GROUP ACTIVITY

Analyze the differences between the exercises using motion analysis, following the table provided; and **answer** the questions below:

	GH	ST	Elbow	Spine
Single Arm Row				
Bent Over Row				
Lat Pulldown				
Straight Arm Pulldown				
Inverted Row				

What is the difference between pulling the bar with elbows wide vs close to your torso? Perform both movements and analyze the difference between the two:

What is the difference between pulling the bar with a supinated vs pronated grip? Perform both movements and analyze the difference between the two:

What is the difference between pulling the dumbbell up or backward while performing the single arm back row exercise? Perform both movements and analyze the difference between the two:

What is the difference between performing the Lat-pulldown exercise with a wider pronated grip vs neutral and close grip vs supinated and close grip? Perform the other variations and analyze the difference between all three:

What are good regressions for the pull-ups exercise?



Consolidation:
By the end of Session 28, I am aware of:
-How to perform, coach, and analyze the most common back muscles exercises.



Post-Session 28 Tasks:
-Practice performing and coaching back and elbow flexor muscle exercises.
-Prepare a brief recap of the lessons covered to be presented in Session 28.



Pre-Session 29 Preparation:
Watch all videos of the Chapter titled: *Week 14: Exercise Techniques Methods for the Abdominal Muscles + Flexibility Training*.

Session 29: Exercise Techniques and Methods – Upper Body – Elbow Flexors + Mid-Section

Session 28 Recap + Q&A:

LESSON 1: ELBOW FLEXORS EXERCISES



Objectives:
-Perform and coach the main elbow flexors muscles exercises.
-Analyze and differentiate between these exercises.

- **BICEPS CURL:** *(Platform: Week 12: Exercise Techniques and Methods for Upper Body Muscles):*

Client	Checklist
	Grasp the dumbbells (or the bar) with a supinated grip
	Stand erect with knees and elbows slightly bent
	Flex the elbows until the dumbbells (or the bar) are near the anterior deltoids
	Do not jerk the body or swing the dumbbells (or the bar) upward
	Descend the dumbbell (or the bar) until you reach starting position

- **REVERSE CURL:**

Same as Biceps Curl exercise while holding the dumbbells or the bar in a pronated grip

- **HAMMER CURL:**

Same as Biceps Curl exercise while holding the dumbbells or the bar in a neutral grip

- **PREACHER CURL:**

Client	Checklist
	Adjust the preacher curl bench so that the angled pad supports your upper arms while your feet are firmly planted on the ground
	Sit on the preacher curl bench with your chest pressed against the pad and your armpits resting on the top edge
	Grasp the barbell or dumbbells with an underhand grip (palms facing upward) and hands shoulder-width apart
	Start with your arms fully extended, allowing them to hang perpendicular to the ground
	Keeping your upper arms pressed against the pad, exhale and curl the weight upward by flexing your elbows
	Continue the curl until your biceps are fully contracted and the weight is close to your shoulders
	Slowly return to the starting position

- **INCLINED-BENCH BICEPS CURL:**

Client	Checklist
	Adjust the incline bench to a 45-degree angle
	Start with your arms fully extended, allowing them to hang straight down (shoulders in an extended position)
	Curl the dumbbells or barbell upward by flexing your elbows while making sure not to go into shoulder flexion
	Continue the curl until your biceps are fully contracted and the weight is close to your shoulders
	Slowly return to the starting position



Practice all Elbow Flexors muscles exercises in a client / coach / assessor role play scenario:

Instructor's Feedback



What are the **differences** and specific goals of each exercise performed?



Consolidation:
By the end of this lesson, I am aware of:
-The main elbow flexors exercises and variations along with the differences between them.

🎉 UPPER BODY BEAST! 💪🔥

Push-ups, pull-ups, presses, and rows… you crushed them all.
Your shoulders are solid, your arms are pumped, and your chest is built like a FORTRESS

🏅 Certified Upper Body Beast 🏋️



CONGRATULATIONS!

LESSON 2: FORMATIVE ASSESSMENT



Objectives:
-Review the main back and elbow flexor exercises.
-Get a sneak peek of what is required for the practical exam.



Formative Assessment – Upper Body Muscle Exercises:

In pairs, prepare and demonstrate one of the following exercises— Back Row and Lat-Pulldown—in a coach/client role-play scenario. The demonstration will be presented to the entire group, followed by feedback on technique, coaching, and execution.



Consolidation:
By the end of this lesson, I am aware of:
-The main points I need to focus on while filming my practical exam.

LESSON 3: REVIEW THE ABDOMINALS AND CORE MUSCLES ORIGIN, INSERTION AND ACTIONS

Objectives:

-Review the muscle anatomy of the abdominals and core muscles to better understand their actions**:**

(Platform: Chapter titled: Week 14: Exercise Techniques and Methods for the Abdominal Muscles + Flexibility Training) (Refer to the manual page 46-47):

Based on the what we know so far and the below muscle pictures, let's try to **guess** each muscle's origin, insertion and action:

Rectus Abdominis:



Origin:
Insertion:
Actions:

Internal Obliques:



Origin:
Insertion:
Actions:

External Obliques:



Origin:
Insertion:
Actions:

Transverse Abdominis:



Origin:
Insertion:
Actions:

Quadratus Lumborum:



Erector Spinae:



Origin:
Insertion:
Actions:

Consolidation:

By the end of this lesson, I am aware of:

-The main core muscles origin, insertion and actions.

LESSON 4: PRACTICE THE MAIN CORE MUSCLES EXERCISES

Objectives:
-Perform and coach the main core muscle exercises.
-Analyze and differentiate between these exercises.

- **CRUNCHES:** *(Platform: Chapter titled: Week 14: Exercise Techniques and Methods for the Abdominal Muscles + Flexibility Training) (Refer to the manual page 46-47):*

Client	Checklist
	Lie down on your back while bending your knees with your feet flat on the ground
	Keep your arms relaxed at your sides or crossed over your chest
	Engage your core muscles
	Lift your upper body up until your shoulder blades are completely off the ground
	Avoid pulling on your neck or straining your back.
	Return to the starting position allowing your shoulder blades to touch the ground while maintaining tension in your abdominals

- **PLANK:**

Client	Checklist
	Lie prone on a floor mat with your elbows close to your sidesand right under your shoulders,palms down, and hands facing ahead
	Extend your legs and lift your torso
	Keep your legs and torso stiff
	Keep your hips neutral or in a posterior pelvic tilt position and hold that position

- **SIDE PLANK:**

Client	Checklist
	Lie on one side with your legs extended and stacked on top of each other
	Place your elbow directly under your shoulder, forming a straight line from shoulder to elbow
	Lift your hips off the ground while keeping your torso and legs stiff
	Keep your body in a straight line and maintain a neutral or slightly posterior pelvic tilt

- **RUSSIAN TWIST:**

Client	Checklist
	Start by sitting on the floor, with your knees bent and feet flat on the ground
	Lean back so yourupper body is at a 45-degree angle to the floor while maintaining a neutral spine
	Put your hands together and rotate all the way over to one side, then do the same in the other direction (or hold a medicine ball and rotate it)
	Tapping your hands on the floor beside you or allow the medicine ball to touch the floor on either side of your body

- **ABS ROLLOUT:**

Client	Checklist
	Place a matt on the floor or a cushion under your knees and go down to the floor on both knees
	Hold the ab rollout wheel firmly with both hands
	Create a posterior pelvic tilt and contract your abdominal muscles well as you roll the wheel forward
	As you roll down, try to reach a point where your head is between your arms facing the floor
	Roll back to the starting position and keep on repeating the movement

- **LEG RAISES:**

Client	Checklist
	Place a matt on the floor and lie supine with your hip extended.
	Go for a posterior pelvic tilt through removing the natural gap created by the arc in the back.
	Lift your legs off the floor then lower them back down while maintaining a posterior pelvic tilt.
	Right before your feet touch the floor, reverse the movement and lift your legs up again.

- **IYTWO SCAPULAR RETRACTORS:**

Client	Checklist
	Lie down prone on a mat
	lift your arms straight up in front of you, forming an "I" shape with your
	Retract your shoulder blades, squeezing them together and hold for few seconds
	Return to the starting position (rest your arms on the floor) then repeat
	Repeat the same exercise forming a “Y”, “T”, “W and an “O” shapes

- **DEAD BUG:**

Client	Checklist
	Lie down on your back on an exercise mat or floor
	Bend your knees and lift your legs, so your knees are directly above your hips
	Bend your arms at the elbows, with your hands pointing towards the ceiling
	Maintain a posterior pelvic tilt or a neutral hip (do not allow any form of high arch at the level of the lower back)
	Extend your right leg, lowering it towards the ground while simultaneously lowering your left arm overhead
	Lower your leg and arm only as far as you can while maintaining proper form and without arching your back Pause briefly in the extended position, feeling the tension in your core then return to the starting position then repeat the movement on the opposite side

- **BIRD DOG:**

Client	Checklist
	Begin on your hands and knees, with your hands directly under your shoulders and your knees directly under your hips
	Engage the core, keep the spine (including the neck) and hips neutral
	Extend your right leg straight back, keeping it parallel to the ground while extending your left arm straight forward, reaching it out in front of you
	Avoid arching or rounding your back. Keep your core engaged to stabilize your body in a proper alignment
	Hold the lifted position for a few seconds then return to the starting position and repeat the movement on the opposite side

- **SUPERMAN:**

Client	Checklist
	Lie face down on an exercise mat or floor with your arms extended overhead
	Simultaneously lift your arms, chest, and legs off the ground (you can start by lifting the arms and chest only)
	Hold the lifted position for a brief moment then return to the starting position

Perform all the abdominal and core exercises, then **discuss** the muscles engaged and the types of muscle contractions involved in each variation:

Instructor's Feedback

Consolidation:

By the end of Session 29, I am aware of:

-How to perform, coach, and analyze the most common elbow flexors, abdominal and core muscle exercises.

Post-Session 29:

-Practice performing and coaching elbow flexors, core and abdominal muscles exercises.

-Prepare a brief recap of the lessons covered to be presented in Session 28.

Pre-Session 30 Preparation:

Watch all videos of the Chapter titled: *Week 14: Exercise Techniques Methods for the Abdominal Muscles + Flexibility Training*.

🎯 CORE COMMANDER! 🧱🔥

Planks, twists, leg raises, and crunches—you powered through every rep.
Your abs are activated, your posture is solid, and your core is strong like steel!

🏅 Certified Midsection Master 💥🧍



CONGRATULATIONS!

LET'S END THIS CHAPTER WITH SOME FUN, COMPETITION & LEARNING:



Get ready for the Muscle Match-Up Challenge!

- You'll find cards with muscle names, and others with their origin, insertion, and action.
- All cards are shuffled and spread out; your mission is to find the correct matches!
- Work as a team to connect each muscle name with its correct origin, insertion, and action.
- Once your team has a full match, run to the buzzer and hit it to lock in your answer.
- You'll earn a point for every correct match. Let's see which team knows their anatomy best!

Welcome to the Exercise Challenge Room!

Get ready to team up and take on 4 tough-but-fun stations:

- Station 1: 200 bodyweight squats (with proper technique!)
- Station 2: 100 push-ups
- Station 3: 200 crunches
- Station 4: 50 pull-ups

After each station, your team will earn a clue to help you guess a mystery exercise. The team that completes all stations and guesses the exercise the fastest… WINS!
You'll be in teams of at least 4 people, and you can plan your strategy however you want:
– All work together at the same station to complete its task
– Or split up and each tackle one station
– You can take breaks—reps don't need to be done all at once
Let the challenge begin!

Session 30: Flexibility Training

Session 29 Recap + Q&A:

LESSON 1: UNDERSTANDING STRETCHING: TYPES AND BENEFITS

Objectives:
-Identify and explain the different types of stretching
-Identify the benefits of stretching

- **INTRODUCTION TO FLEXIBILITY:** *(Platform: Chapter title: Week 14: Exercise Techniques and Methods for the Abdominal Muscles + Flexibility Training)*

Based on what you know, what are the different types of stretching, and how do they differ in terms of how and when they are applied during a training session?

- **TYPES of STRETCHING:**

Match each type of stretching in the first column with its correct definition from the second column:

Inter-set Stretching	Performed during the rest period between sets of an exercise, usually targeting the same muscle group being trained.
PNF Stretching	Involves moving into a comfortable stretch and holding the position for a set time; can be active or passive.
Ballistic Stretching	An extreme form of dynamic stretching involving rapid, bouncing movements at end range of motion.
Static Stretching	An external force (e.g., partner or equipment) assists in creating the stretch; commonly used in rehab settings.
Active Stretching	Combines active and passive techniques, involving a pre-contraction of the muscle before stretching; commonly used in physical therapy.
Dynamic Stretching	Incorporates movement along with muscle tension development to enhance flexibility while maintaining neuromuscular control.
Passive Stretching	The individual generates the force for the stretch by actively moving the joint through its range of motion.

- **FOAM ROLLING:**

Answer the following questions related to Foam Rolling (SMR):

What does SMR stand for and what does it involve?	
How does foam rolling affect range of motion (ROM)?	
Does foam rolling negatively affect strength or power performance?	
When is the best time to use foam rolling during a training session?	
Is foam rolling recommended as a post-workout recovery tool?	

- **BENEFITS of STRETCHING:**

GROUP ACTIVITY

State the benefits of stretching. **Use** all of the following terms:

Injury – Hypertrophy – Training – Performance – Muscle Soreness – Tendon – Daily life activity – Stretch Tolerance – Joints – Range of Motion

Based on the information discussed, which type(s) of stretching are most appropriate during the warm-up and which are better suited for the cool-down?

Compare the Stretching v/s the Resistance training effect on the range of motion? Do we actually need to stretch?

Consolidation:

By the end of this lesson, I am aware of:

- The different types of stretching, and when to use each.
- The benefits of flexibility for fitness, performance, and daily life.

LESSON 2: PROPER STRETCHING TECHNIQUE



Objectives:
- Perform stretches with proper technique, breathing, and body positioning.
- Identify the needed stretching Frequency, Intensity, Duration and Type

- **GUIDELINE FOR PROPER STRETCHING:**

✓ **Warm-Up First**

✓ **Gradual Progressive Overload (GPO)**

✓ **Breathing**

✓ **Positioning**

✓ **FITT Principle for Flexibility: Frequency**

✓ **FITT Principle for Flexibility: Intensity**

✓ **FITT Principle for Flexibility: Time**

✓ **FITT Principle for Flexibility: Type**



Consolidation:
By the end of this lesson, I am aware of:
- The proper technique to perform the stretches.
- The Frequency, Intensity, Time and Type of stretching needed.

LESSON 3: FLEXIBILITY IN PRACTICE

Objectives:

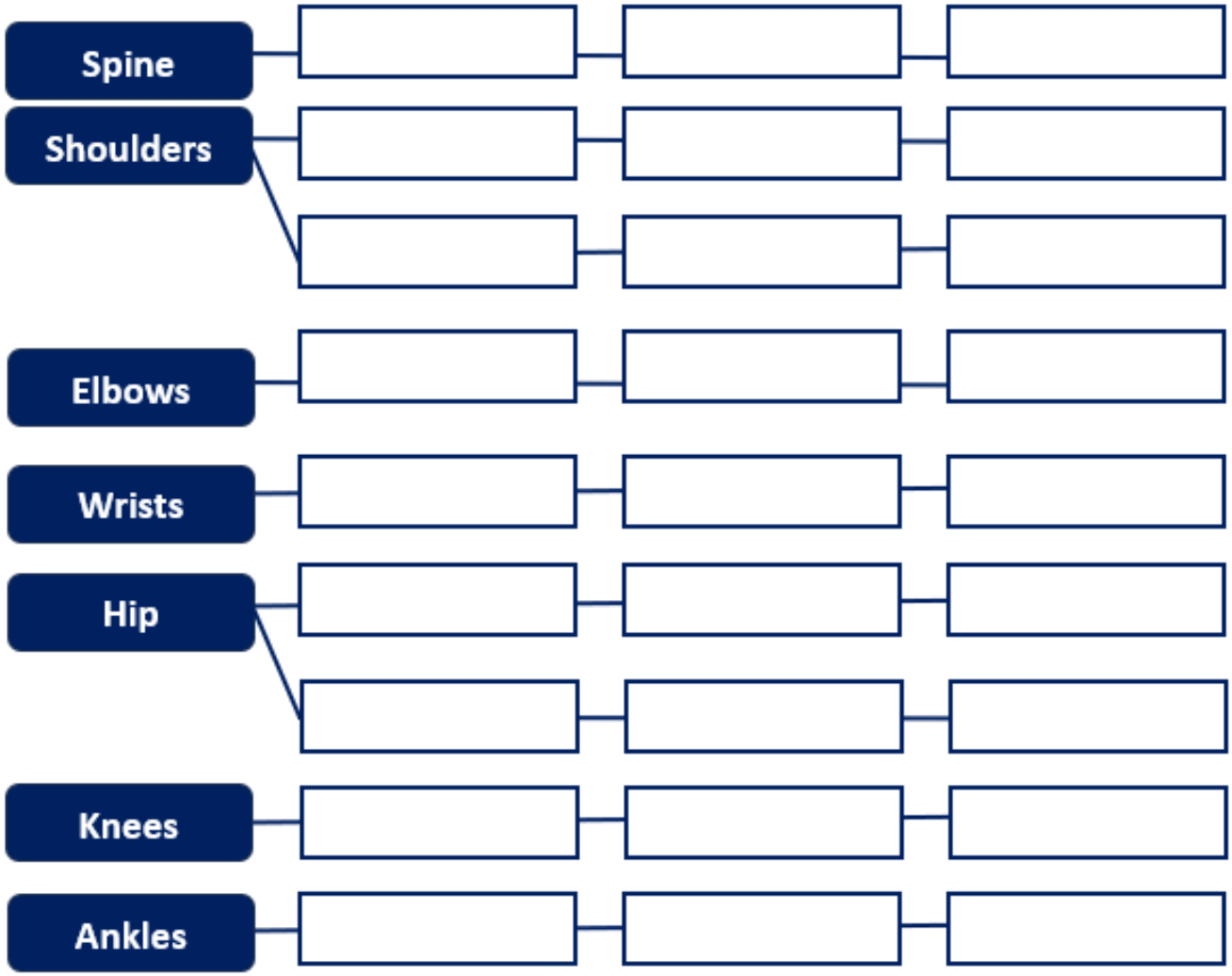
-Practice a variety of static and dynamic stretches targeting all major muscle groups.

-Coach and correct stretching technique in a practical setting.

GROUP ACTIVITY

Design a full-body warm-up and **coach** it to a partner in a client/coach role-play. Switch roles after, ensuring each learner uses a different warm-up routine to promote variety.

(**Note** down the stretches or mobility drills used for each joint area in the spaces provided):



Instructor's feedback:



Answer the following questions regarding your program above:

Did it target all body muscles and joints?	
How long was it?	
Can we come-up with drills involving several muscles and joints at the same time?	
Can we use drills combining cardio and dynamic stretching?	

GROUP ACTIVITY

State the muscle stretched in each of the below pictures:

(Platform: Chapter title: Week 14: Exercise Techniques and Methods for the Abdominal Muscles + Flexibility Training – One video for each muscle)





























GROUP ACTIVITY

Practice the above stretches and **create** several static stretches targeting each of the muscles listed below:

- Plantar Fascia
- Calf Muscles
- Hamstrings
- Quadriceps
- Adductors
- Abductors
- Glutes and Piriformis
- Hip Flexors
- Abdominals
- Quadratus Lumborum / Erector Spinae
- Latissimus Dorsi
- Scapular Retractors
- Pectoralis Major and Anterior Deltoid
- Posterior Deltoid
- Biceps Brachii
- Triceps
- Forearms

GROUP ACTIVITY

<u>"Spin the Wheel" game:</u>

Take turns spinning the wheel; each time, the selected learner will stand in the center and coach the group through a stretch while everyone else performs it.



Consolidation:
By the end of this session, I am aware of:
-How to perform a variety of static and dynamic stretches for all major muscle groups.
-How to coach and correct stretching techniques effectively.

Post-Session 30:
Practice full body static and dynamic stretches and go over the material discussed to prepare a recap to be presented in the upcoming session

Pre-Session 31:
-Watch all videos of the Chapter titled: *(Week 15: Fitness Assessments)*

🎉 YOU NAILED YOUR FLEXIBILITY TRAINING! 🤸‍♂️🧘

You've mastered both dynamic stretches to warm up your body and static stretches to improve mobility and relaxation.

Your muscles are more supple, your range of motion is enhanced, and you're moving with greater freedom than ever before.

🏅 Certified Stretching Champion! 🦋



CONGRATULATIONS!

Session 31: Fitness Assessment

Session 30 Recap + Q&A:

LESSON 1: IMPORTANCE OF FITNESS TESTING



Objectives:
-Understand the importance and purpose of fitness testing.
-Identify when and why to perform different fitness assessments.
-Identify general fitness tests used to measure different components of fitness.



Answer the following questions about the purpose and use of fitness testing in a client session:
(Platform: Chapter Titled: Week 15: Fitness Assessments)

What fitness components can we measure through fitness testing? Give couple of examples

Why do we conduct fitness assessments? What is their main purpose?

Should we perform all fitness assessments for every client?

Is it necessary to complete all assessments in a single session?

Should they be repeated every once in a while? If Yes, when would you repeat them? If No, why?



Consolidation:
By the end of this lesson, I am aware of:
-The importance of fitness testing and why it should be performed.
-When to conduct fitness assessments during a client's journey.
-Common fitness tests used to assess different fitness components.

LESSON 2: HEALTH RELATED ASSESSMENT

Objectives:
-Understand the importance of conducting health-related assessments before starting a training program.
-Identify the main tools used for health screening.

- **Health History Questionnaire HHQ**

Pair up and **complete** the Health History Questionnaire (HHQ) and the Get-Active Questionnaire together. **Take** notes of your partner's responses as if you were their coach:

Personal and Emergency Information:

Name: Date:

Age: Sex: M F Phone number:

Email: Date of birth:

Marital status: Single Married Divorced Widowed Other:

Occupation:

Person to contact in case of emergency:

Name: Phone number:

Lifestyle and Habits

1. Are you currently on a diet (e.g., keto, caloric restriction, detox…)? Y N *(if no, skip to 5)*

2. If yes, is your diet prescribed by a physician for a medical reason? Y N

3. Name of physician:

4. Briefly describe your diet:

5. Caffeine intake: None Coffee Soda Tea Other:

- Number of cups/cans per day?

6. Do you currently smoke tobacco or stopped smoking less than six months ago? Y N

- Type of tobacco: Cigarettes Hookah Pipe Nicotine chew Cigars

- Describe quantity of tobacco consumed per day or week:

7. Alcohol: Y N

- Describe kind and quantity of alcohol consumed per day or week :

Medical Information

Primary health care provider:

Name: Phone number:

1. When was the last time you were seen by a physician?

2. Have you had any kind of surgery in the past two years? Y N

- If yes, briefly explain:

3. Are you currently taking any medication? Y N

- If yes, please list:

4. Are you currently taking any supplements or vitamins? Y N

- If yes, please list:

5. Please indicate by ticking the box ✓ if you have any of the following:

- ☐ Alcohol abuse
- ☐ Anemia
- ☐ Anesthetic complication
- ☐ Anxiety disorder
- ☐ Arthritis
- ☐ Asthma
- ☐ Autoimmune problems
- ☐ Birth defects
- ☐ Bladder problems
- ☐ Bleeding disease
- ☐ Blood clots
- ☐ Blood transfusion(s)
- ☐ Bowel disease
- ☐ Breast cancer
- ☐ Cervical cancer
- ☐ Colon cancer
- ☐ Depression
- ☐ Diabetes
- ☐ Growth/development disorder
- ☐ Hearing impairment
- ☐ Heart attack
- ☐ Heart disease
- ☐ Hepatitis A
- ☐ Hepatitis B
- ☐ Hepatitis C
- ☐ High blood pressure
- ☐ High cholesterol
- ☐ HIV/AIDS
- ☐ Hives
- ☐ Kidney disease
- ☐ Liver cancer
- ☐ Lung cancer
- ☐ Lung/respiratory disease
- ☐ Mental illness
- ☐ Migraines
- ☐ Osteoporosis
- ☐ Prostate cancer
- ☐ Rectal cancer
- ☐ Reflux/GERD
- ☐ Seizures/convulsions
- ☐ Severe allergy
- ☐ Sexually transmitted disease
- ☐ Skin cancer
- ☐ Stroke/CVA of the brain
- ☐ Substance abuse
- ☐ Suicide attempt
- ☐ Thyroid problems
- ☐ Ulcer
- ☐ Visual impairment
- ☐ Other, please mention:
- ☐ NONE OF THE ABOVE

Musculoskeletal injury/pain

- ☐ Carpal tunnel syndrome
- ☐ Fracture
- ☐ Golfer's elbow
- ☐ Greater trochantric bursitis
- ☐ Groin pain
- ☐ Hernia
- ☐ Ligament sprain
- ☐ Low back pain
- ☐ Meniscus tear
- ☐ Muscle strain
- ☐ Neck pain
- ☐ Patellar tendinitis
- ☐ Patellofemoral syndrome
- ☐ Rotator cuff injury
- ☐ Shoulder dislocation
- ☐ Tennis elbow

- **Get Active Questionnaire:**



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Get Active Questionnaire

CANADIAN SOCIETY FOR EXERCISE PHYSIOLOGY –
PHYSICAL ACTIVITY TRAINING FOR HEALTH (CSEP-PATH®)

Physical activity improves your physical and mental health. Even small amounts of physical activity are good, and more is better.

For almost everyone, the benefits of physical activity far outweigh any risks. For some individuals, specific advice from a Qualified Exercise Professional (QEP – has post-secondary education in exercise sciences and an advanced certification in the area – see csep.ca/certifications) or health care provider is advisable. **This questionnaire is intended for all ages – to help move you along the path to becoming more physically active.**

☐ I am completing this questionnaire for myself.

☐ I am completing this questionnaire for my child/dependent as parent/guardian.

PREPARE TO BECOME MORE ACTIVE

The following questions will help to ensure that you have a safe physical activity experience. Please answer **YES** or **NO** to each question before you become more physically active. If you are unsure about any question, answer **YES**.

YES	NO	
		1 Have you experienced **ANY** of the following (A to F) **within the past six months**?
○	○	**A** A diagnosis of/treatment for heart disease or stroke, or pain/discomfort/pressure in your chest during activities of daily living or during physical activity?
○	○	**B** A diagnosis of/treatment for high blood pressure (BP), or a resting BP of 160/90 mmHg or higher?
○	○	**C** Dizziness or lightheadedness during physical activity?
○	○	**D** Shortness of breath at rest?
○	○	**E** Loss of consciousness/fainting for any reason?
○	○	**F** Concussion?
○	○	2 Do you currently have pain or swelling in any part of your body (such as from an injury, acute flare-up of arthritis, or back pain) that affects your ability to be physically active?
○	○	3 Has a health care provider told you that you should avoid or modify certain types of physical activity?
○	○	4 Do you have any other medical or physical condition (such as diabetes, cancer, osteoporosis, asthma, spinal cord injury) that may affect your ability to be physically active?

NO to all questions: go to Page 2 – ASSESS YOUR CURRENT PHYSICAL ACTIVITY

YES to any question: go to Reference Document – ADVICE ON WHAT TO DO IF YOU HAVE A YES RESPONSE

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PAGE 1 OF 2

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Get Active Questionnaire

ASSESS YOUR CURRENT PHYSICAL ACTIVITY

Answer the following questions to assess how active you are now.

1 During a typical week, on how many days do you do moderate- to vigorous-intensity aerobic physical activity (such as brisk walking, cycling or jogging)? ____ DAYS/WEEK

2 On days that you do at least moderate-intensity aerobic physical activity (e.g., brisk walking), for how many minutes do you do this activity? ____ MINUTES/DAY

For adults, please multiply your average number of days/week by the average number of minutes/day: ____ MINUTES/WEEK

Canadian Physical Activity Guidelines recommend that adults accumulate at least 150 minutes of moderate- to vigorous-intensity physical activity per week. For children and youth, at least 60 minutes daily is recommended. Strengthening muscles and bones at least two times per week for adults, and three times per week for children and youth, is also recommended (see csep.ca/guidelines).

GENERAL ADVICE FOR BECOMING MORE ACTIVE

Increase your physical activity gradually so that you have a positive experience. Build physical activities that you enjoy into your day (e.g., take a walk with a friend, ride your bike to school or work) and reduce your sedentary behaviour (e.g., prolonged sitting).

If you want to do **vigorous-intensity physical activity** (i.e., physical activity at an intensity that makes it hard to carry on a conversation), and you do not meet minimum physical activity recommendations noted above, consult a Qualified Exercise Professional (QEP) beforehand. This can help ensure that your physical activity is safe and suitable for your circumstances.

Physical activity is also an important part of a healthy pregnancy.

Delay becoming more active if you are not feeling well because of a temporary illness.

DECLARATION

To the best of my knowledge, all of the information I have supplied on this questionnaire is correct.
If my health changes, I will complete this questionnaire again.

I answered NO to all questions on Page 1

Sign and date the Declaration below

I answered YES to any question on Page 1

Check the box below that applies to you:

- ☐ I have consulted a health care provider or Qualified Exercise Professional (QEP) who has recommended that I become more physically active.
- ☐ I am comfortable with becoming more physically active on my own without consulting a health care provider or QEP.

Name (+ Name of Parent/Guardian if applicable) [Please print] | Signature (or Signature of Parent/Guardian if applicable) | Date of Birth

Date | Email (optional) | Telephone (optional)

With planning and support you can enjoy the benefits of becoming more physically active. A QEP can help.

☐ Check this box if you would like to consult a QEP about becoming more physically active.
(This completed questionnaire will help the QEP get to know you and understand your needs.)

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- **ACSM Risk Stratification**

Risk Stratification Scoring

Positive Risk Factors	Defining Criteria	Points
Age	Men ≥ 45 years, Women ≥ 55 years	+1
Family History	Myocardial infarction, coronary revascularization, or sudden death before 55 years of age in father of other 1st degree male relative or before 65 years of age in mother or other 1st degree female relative	+1
Cigarette Smoking	Current cigarette smoker or those who quit within the previous six months, or exposure to environmental tobacco smoke (i.e., secondhand smoke)	+1
Sedentary Lifestyle	Not participating in at least 30 minutes of moderate-intensity physical activity on at least three days/week for at least three months	+1
Obesity	Body mass index ≥30 kg/m^2 or waist girth >102 cm (40 inches) for men >88 cm (35 inches) for men	+1
Dyslipidemia	Low-density lipoprotein (LDL) cholesterol ≥ 130mg/dL (3.37 mmol/L) or high-density lipoprotein (HDL) cholesterol <40mg/dL (1.04mmol/L) or currently on lipid-lowering medication; If total serum cholesterol is all that is available, use serum cholesterol >200 mg/dL (5.18mmol/L)	+1
Prediabetes	Fasting plasma glucose ≥100 mg/dL (5.50 mmmol/L) but <126 mg/dL (6.93 mmol/L) or impaired glucoe tolerance (IGT) where a two-hour oral glucose tolerance test (OGTT) value is ≥140 mg/dL (7.70 mmol/L), but <200 mg/dL (11.00mmol/L)	+1
Negative Risk Factors	**Defining Criteria**	**Points**
High HDL Cholesterol	≥60 mg/dL (1.55 mmol/L)	-1

Total CVD Risk Score: _____

Signs & Symptoms to Screen For:

1. **Chest discomfort (angina)**
    - Pain, tightness, heaviness, or pressure in the chest, neck, jaw, arms, or other areas that may be ischemic in origin.

2. **Unusual shortness of breath (dyspnea)**
    - At rest or with mild exertion (not explained by conditioning level).

3. **Dizziness or syncope (fainting)**
    - Especially during or immediately after exercise.

4. **Orthopnea or paroxysmal nocturnal dyspnea**
    - Difficulty breathing when lying flat, or sudden shortness of breath at night.

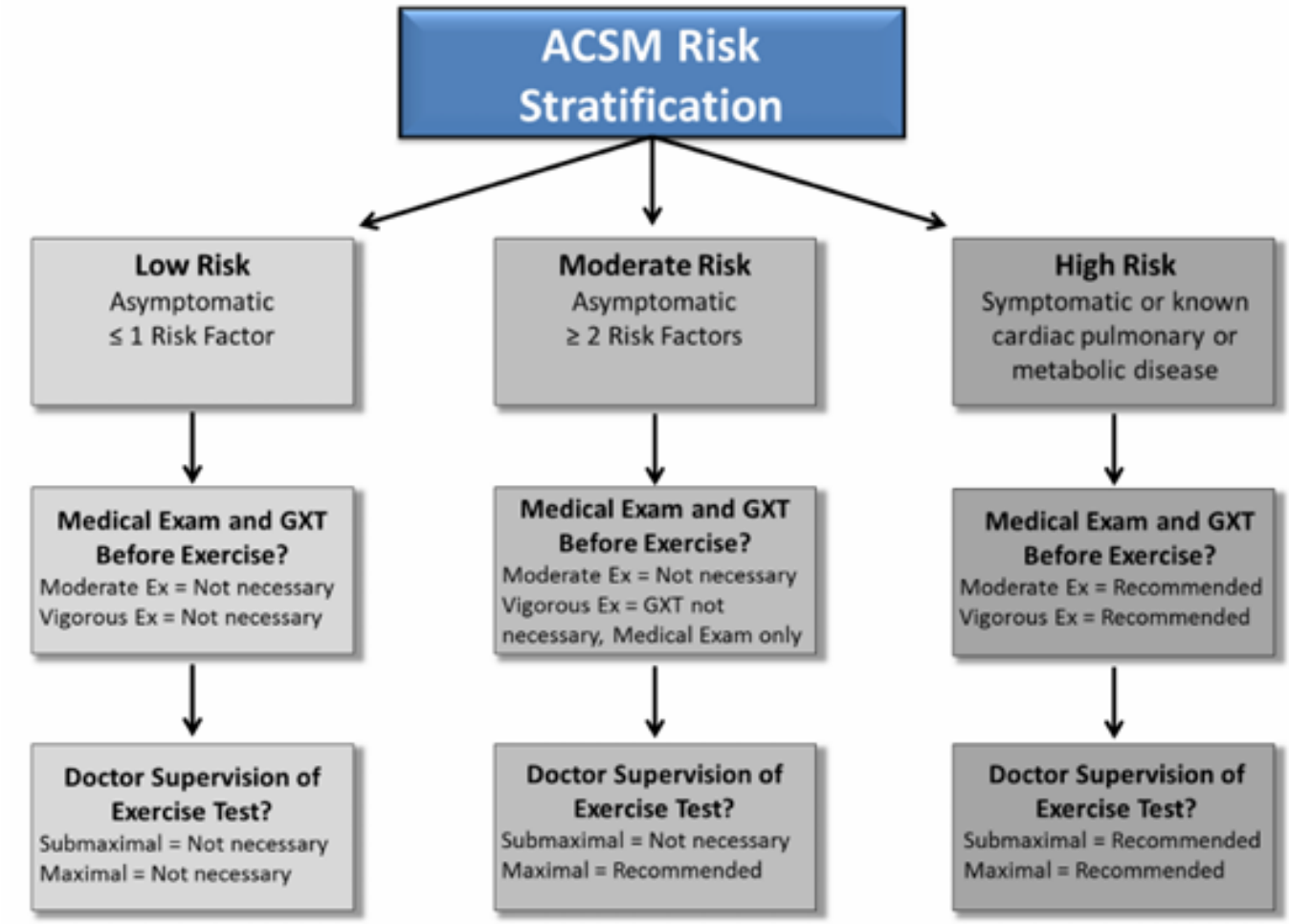
5. **Ankle edema (swelling)**
    - Especially bilateral or when not due to injury.

6. **Palpitations or tachycardia**
    - Noticeably rapid or irregular heartbeat.

7. **Intermittent claudication**
    - Pain, cramping, or discomfort in the legs with exertion (relieved by rest), indicating poor circulation.

8. **Known heart murmur**
    - Especially if previously undiagnosed or unexplained.

9. **Unusual fatigue or shortness of breath**
    - Occurring during usual activities (not just at vigorous exertion).



GXT = Graded Exercise Test

Pair up and **apply** the ACSM Risk Stratification on the below case:

Client: Hadi, 47 years old

- **Background:**
  - Works in sales, moderately active lifestyle (walks daily for ~30 minutes, 5 days/week)
  - Wants to start **jogging and weight training** at the gym
- **Health Status:**
  - No known cardiovascular, metabolic, or renal disease
  - Blood pressure: **130/82 mmHg**
  - Cholesterol slightly elevated (borderline, no medication)
  - BMI: **27 kg/m²**
  - Non-smoker

- **Exercise History:**
    - **Currently active** (meets ACSM definition: 30+ min moderate activity, 3+ days/week, for 3+ months)
- **Symptoms:**
    - No chest pain, dizziness, or shortness of breath with normal activity
- **Goal:**
    - Progress from moderate walking to **vigorous exercise (jogging, circuit training)**



What do you think about these forms?



Consolidation:
By the end of this lesson, I am aware of:
-The importance of conducting a health-related assessment before starting any training program.
-The key screening tools used to assess a client's readiness for exercise, including the HHQ, Get Active Questionnaire, and ACSM Risk Stratification.

LESSON 3: MOVEMENT SCREENS

Objectives:
-Gain a general overview of movement screens that replicate everyday functional movements.

What is the purpose of these tests?

Sit and stand

Bend and lift

Single-leg movements

Horizontal push

Vertical push

Horizontal pull

Trunk rotation

Consolidation:
By the end of this lesson, I am aware of:
-The main movement screens that reflect everyday functional movements and how they can help identify limitations or imbalances.

LESSON 4: MOBILITY & FLEXIBILITY

Objectives:
-Understand the difference between mobility and flexibility.
-Practice basic mobility and flexibility drills targeting major joints and muscle groups.

MOBILITY ASSESSMENTS OVERVIEW *(Platform: Chapter Titled: Week 15: Fitness Assessments)*

- **Shoulder Mobility:**

Test Name	Assessment Criteria
Shoulder Flexion, Abduction, and External Rotation	Aim to reach for the medial and upper border of the contralateral scapula
Shoulder Extension, Adduction, and Internal Rotation	Aim to reach for the inferior angle of the contralateral scapula
Shoulder Extension Test	Client must be able to extend each hand 50-60 degrees off the floor
Shoulder Flexion Test	Client should be able to reach 170-180 degrees of shoulder flexion
Shoulder Internal and External Rotation Test	Client should reach 90 degrees of external rotation and 70 degrees of internal rotation

- **Hip Mobility:**

Test Name	Assessment Criteria
Thomas Test	Check whether the thigh is in contact with the bench and the knee is relaxed in 80-90 degrees of flexion
Passive Straight Leg Raise Test (PSLR)	Client's hips must achieve at least 80 degrees of flexion before the pelvis tilts posteriorly with the knee straight

- **Ankle Mobility:**

Test Name	Assessment Criteria
Dorsiflexion Lunge Test (Knee to Wall Test)	Client should reach the wall with their knee while the big toe is 11 cm away from the wall; heel must stay on the floor and hip should not rotate



Pair up and **test** each other's mobility in a coach / client role play. **Record** the client's results and **specify** which tight muscle is preventing them from reaching the expected range:

Mobility Test	**Client Result (Angle / Ability)**	**Able to Perform? (Yes/No)**	**Notes on Tightness / Limitation**
Shoulder Flexion, Abduction, and External Rotation (A)			
Shoulder Extension, Adduction, and Internal Rotation (B)			
Shoulder Extension Test (C)			
Shoulder Flexion Test (D)			
Shoulder Int. & Ext. Rotation Test (E)			
Thomas Test (Hip)			
PSLR – Passive Straight Leg Raise (Hip)			
Dorsiflexion Lunge Test – Knee to Wall (Ankle)			



What is the purpose of the above tests?

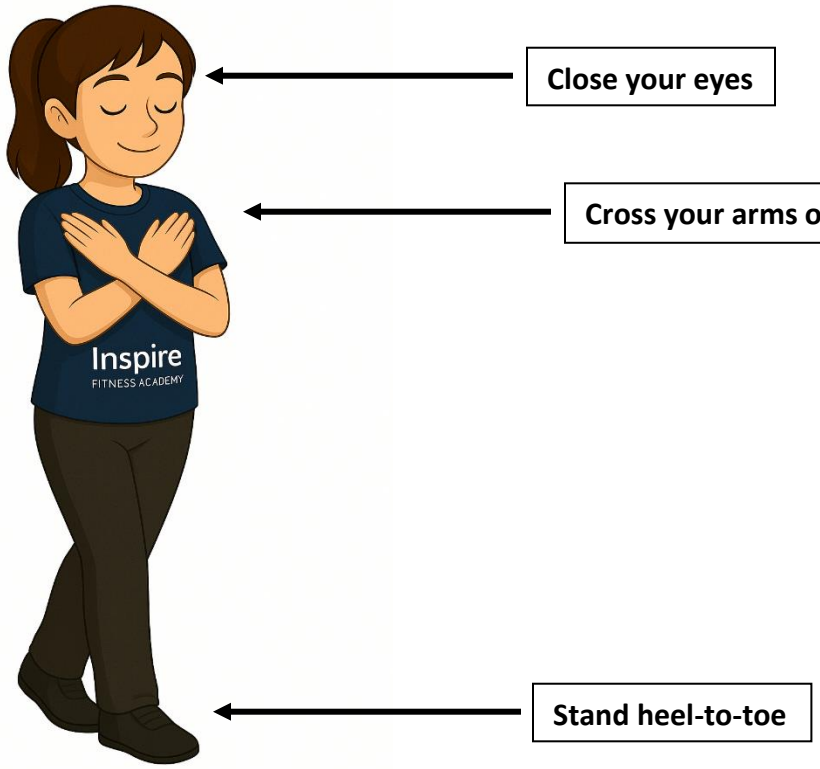


Consolidation:
By the end of this lesson, I am aware of:
-How to perform and coach basic flexibility and mobility assessments.
-Which tight muscles may be preventing the client from reaching full range of motion.

LESSON 5: BALANCE TESTING

Objectives:
-Perform and coach the Sharpened Romberg Test to a client.

- **SHARPENED ROMBERG TEST:** *(Chapter Titled: Week 15: Fitness Assessments – Video: Balance Test)*



➔ **Hold this position for up to 60 seconds!**

Expected results:

- Less than 30 seconds → Inadequate static balance ability
- 30-60 seconds → Good static balance capability
- 60 seconds → Very good static balance capability

LET'S ALL PRACTICE IT TOGETHER!

Consolidation:

By the end of this session, I am aware of:

- Importance of fitness tests.
- How to perform the health history questionnaire, movement screening, flexibility and balance testing

Post-Session 31:

Practice the fitness tests discussed with a friend or family member, and go over the material discussed to prepare a recap to be presented in the upcoming session

Pre-Session 32:

- Watch all remainder fitness testing videos *(Week 15: Fitness Assessments)*

Session 32: Fitness Assessment

Session 31 Recap + Q&A:

LESSON 1: BODY COMPOSITION

Objectives:
-Recognize the most common methods used to assess body composition.
-Practice different body composition assessment methods.

- **BODY MASS INDEX:** *(Platform: Chapter Titled: Week 15: Fitness Assessments – Video: Body Composition)*

BMI= $\frac{Weight\ (Kg)}{Height^{2}\ (m^{2})}$

BMI RANGE	WEIGHT CLASSIFICATION
<18.5	Underweight
18.5-24.9	Normal weight
25-29.9	Overweight
30-34.9	Grade I obese
35-39.9	Grade II obese
>40	Grade III obese

Calculate your BMI using the formula provided above:

BMI= $\frac{Weight\ (Kg)}{Height^{2}\ (m^{2})}$ = _____ = ... Kg/m^2

What is a major limitation of using BMI as a measure of health, and why is it important to assess body fat percentage separately?

- **CIRCUMFERENCE MEASUREMENTS:**

Anatomical Site	Client Position	Where to Measure
Arm	Stand upright, arms relaxed at sides	Midway between shoulder and elbow
Forearm	Arms slightly away, palms forward	Largest part of the forearm
Chest (upper)	Arms slightly away from the body	Around upper chest, near the armpits
Waist	Stand upright, arms relaxed	Narrowest point between ribs and hips
Hips	Feet hip-width apart	Widest point of the hips
Thigh (upper)	Weight shifted to rear foot	Thickest part of the front thigh
Calf	Weight shifted to rear foot	Widest part of the calf

Pair up and **measure** the body circumferences at each anatomical site, then take notes of the results in the table provided:

Anatomical Site	Results
Arm	
Forearm	
Chest (upper)	
Waist	
Hips	
Thigh (upper)	
Calf	

Calculate your client's measurements below, while also interpreting their results:

- **Waist Circumference:**
  The waist circumference should be less than 102cm for men and 88cm for women.
  Waist Circumference: _____

RISK CATEGORY	FEMALES	MALES
Very low	<70cm	<80cm
Low	70-89cm	80-99cm
High	90-109cm	100-120cm
Very high	>110cm	>120cm

- **Waist-to-Height Ratio (WHtR):**
  "Keep your waist circumference less than half your height" (< 0.5 for all populations).
  Waist-to-height Ratio = Waist circumference (cm) / Height (cm)
  Waist-to-height Ratio = _____

- **Waist-to-Hip Ratio (WHR):**
  The waist-to-hip ratio should be less than 0.95 for men and 0.86 for women.
  Waist-to-hip = waist circumference(cm) / hip circumference(cm)
  Waist-to-hip = _____

  Note the below:

Waist-to-height ratio → Waist circumference → Waist-to-hip ratio → Body mass index

- **SKINFOLD MEASURMENT:**

What is the purpose of this assessment?

	If your client is a man	If your client is a woman	Measurement (1st attempt)	Measurement (2nd attempt)	Average
1- Pinch skinfold **2- Place caliper** **3- Read dial** **4- Release caliper** **5- Release pinch**	Chest	Triceps			
	Abdomen	Waist			
	Thigh	Thigh			
	TOTAL:		*(Sum of all three measurements)* →		



Team up and use the caliper to **measure** the listed body sites. After calculating the TOTAL, refer to the appropriate conversion tables to **determine** your client's estimated body fat percentage:

PERCENT BODY FAT ESTIMATIONS FOR WOMEN - JACKSON AND POLLOCK FORMULA									
AGE GROUPS									
Sum of	Under 22	23-27	28-32	33-37	38-42	43-47	48-52	53-57	Over 57
23-25	9.7	9.9	10.2	10.4	10.7	10.9	11.2	11.4	11.7
26-28	11.0	11.2	11.5	11.7	12.0	12.3	12.5	12.7	13.0
29-31	12.3	12.5	12.8	13.0	13.3	13.5	13.8	14.0	14.3
32 34	13.6	13.8	14.0	14.3	14.5	14.8	15.0	15.3	15.5
35-37	14.8	15.0	15.3	15.5	15.8	16.0	16.3	16.5	16.8
38-40	16.0	16.3	16.5	16.7	17.0	17.2	17.5	17.7	18.0
41-43	17.2	17.4	17.7	17.9	18.2	18.4	18.7	18.9	19.2
44-46	18.3	18.6	18.8	19.1	19.3	19.6	19.8	20.1	20.3
47-49	19.5	19.7	20.0	20.2	20.5	20.7	21.0	21.2	21.5
50-52	20.6	20.8	21.1	21.3	21.6	21.8	22.1	22.3	22.6
53-55	21.7	21.9	22.1	22.4	22.6	22.9	23.1	23.4	23.6
56-58	22.7	23.0	23.2	23.4	23.7	23.9	24.2	24.4	24.7
59-61	23.7	24.0	24.2	24.5	24.7	25.0	25.2	25.5	25.7
62-64	24.7	25.0	25.2	25.5	25.7	26.0	26.7	26.4	26.7
65-67	25.7	25.9	26.2	26.4	26.7	26.9	27.2	27.4	27.7
68-70	26.6	26.9	27.1	27.4	27.6	27.9	28.1	28.4	28.6
71-73	27.5	27.8	28.0	28.3	28.5	28.8	29.0	29.3	29.5
74-76	28.4	28.7	28.9	29.2	29.4	29.7	29.9	30.2	30.4

77-79	29.3	29.5	29.8	30.0	30.3	30.5	30.8	31.0	31.3
80-82	30.1	30.4	30.6	30.9	31.1	31.4	31.6	31.9	32.1
83-85	30.9	31.2	31.4	31.7	31.9	32.2	32.4	32.7	32.9
86-88	31.7	32.0	32.2	32.5	32.7	32.9	33.2	33.4	33.7
89-91	32.5	32.7	33.0	33.2	33.5	33.7	33.9	34.2	34.4
92-94	33.2	33.4	33.7	33.9	34.2	34.4	34.7	34.9	35.2
95-97	33.9	34.1	34.4	34.6	34.9	35.1	35.4	35.6	35.9
98-100	34.6	34.8	35.1	35.3	35.5	35.8	36.0	36.3	36.5
101-103	35.3	35.4	35.7	35.9	36.2	36.4	36.7	36.9	37.2
104-106	35.8	36.1	36.3	36.6	36.8	37.1	37.3	37.5	37.8
107-109	36.4	36.7	36.9	37.1	37.4	37.6	37.9	38.1	38.4
110-112	37.0	37.2	37.5	37.7	38.0	38.2	38.5	38.7	38.9
113-115	37.5	37.8	38.0	38.2	38.5	38.7	39.0	39.2	39.5
116-118	38.0	38.3	38.5	38.8	39.0	39.3	39.5	39.7	40.0
119 121	38.5	38.7	39.0	39.2	39.5	39.7	40.0	40.2	40.5
122-124	39.0	39.2	39.4	39.7	39.9	40.2	40.4	40.7	40.9
125-127	39.4	39.6	39.9	40.1	40.4	40.6	40.9	41.1	41.4
128-130	39.8	40.0	40.3	40.5	40.8	41.0	41.3	41.5	41.8

PERCENT BODY FAT ESTIMATIONS FOR MEN - JACKSON AND POLLOCK FORMULA									
AGE GROUPS									
Sum of Skinfolds (mm)	Under 22	23-27	28-32	33-37	38-42	43-47	48-52	53-57	Over 57
8-10	1.3	1.8	2.3	2.9	3.4	3.9	4.5	5.0	5.5
11-13	2.2	2.8	3.3	3.9	4.4	4.9	5.5	6.0	6.5
14-16	3.2	3.8	4.3	4.8	5.4	5.9	6.4	7.0	7.5
17-19	4.2	4.7	5.3	5.8	6.3	6.9	7.4	8.0	8.5
20-22	5.1	5.7	6.2	6.8	7.3	7.9	8.4	8.9	9.5
23-25	6.1	6.6	7.2	7.7	8.3	8.8	9.4	9.9	10.5
26-28	7.0	7.6	8.1	8.7	9.2	9.8	10.3	10.9	11.4
29-31	8.0	8.5	9.1	9.6	10.2	10.7	11.3	11.8	12.4
32-34	8.9	9.4	10.0	10.5	11.1	11.6	12.2	12.8	13.3
35-37	9.8	10.4	10.9	11.5	12.0	12.6	13.1	13.7	14.3
38-40	10.7	11.3	11.8	12.4	12.9	13.5	14.1	14.6	15.2
41-43	11.6	12.2	12.7	13 3	13.8	14.4	15.0	15.5	16.1
44-46	12.5	13.1	13.6	14.2	14.7	15.3	15.9	16.4	17.0
47-49	13.4	13.9	14.5	15.1	15.6	16.2	16.8	17.3	17.9
50-52	14.3	14.8	15.4	15.9	16.5	17.1	17.6	18.2	18.8
53-55	15.1	15.7	16.2	16.8	17.4	17.9	18.5	19.1	19.7
56-58	16.0	16.5	17.1	17.7	18.2	18.8	19.4	20.0	20.5
59 61	16.9	17.4	17.9	18 5	19.1	19.7	20.2	20.8	21.4
62-64	17.6	18.2	18.8	19.4	19.9	20.5	21.1	21.7	22.2

65-67	18 5	19.0	19.6	20 2	20.8	21.3	21.9	22.5	23.1
68-70	19.3	19.9	20.4	21.0	21.6	22.2	22.7	23.3	23.9
71-73	20.1	20.7	21.2	21.8	22.4	23.0	23.6	24.1	24.7
74-76	20.9	21.5	22.0	22.6	23.2	23.8	24.4	25.0	25.5
77-79	21.7	22.2	22.8	23.4	24.0	24.6	25.2	25.8	26.3
80-82	22.4	23.0	23.6	24.2	24.8	25.4	25.9	26.5	27.1
83 85	23.2	23.8	24.4	25.0	25.5	26.1	26.7	27.3	27.9
86-88	24.0	24.5	25.1	25.7	26.3	26.9	27.5	28.1	28.7
89 91	24.7	25.3	25.9	26.5	27.1	27.6	28.2	28.8	29.4
92-94	25.4	26.0	26.6	27.2	27.8	28.4	29.0	29.6	30.2
95-97	26.1	26.7	27.3	27.9	28.5	29.1	29.7	30.3	30.9
98-100	26.9	27.4	28.0	28.6	29.2	29.8	30.4	31.0	31.6
101-103	27.5	28.1	28.7	29.3	29.9	30.5	31.1	31.7	32.3
104-106	28.2	28.8	29.4	30.0	30.6	31.2	31.8	32.4	33.0
107-109	28.9	29.5	30.1	30.7	31.3	31.9	32.5	33.1	33.7
110-112	29.6	30.2	30.8	31.4	32.0	32.6	33.2	33.8	34.4
113-115	30.2	30.8	31.4	32.0	32.6	33.2	33.8	34.5	35.1
116-118	30.9	31.5	32.1	32.7	33.3	33.9	34.5	35.1	35.7
119-121	31.5	32.1	32.7	33.3	33.9	34.5	35.1	35.7	36.4
122-124	32.1	32.7	33.3	33.9	34.5	35.1	35.8	36.4	37.0
125-127	32.7	33.3	33.9	34.5	35.1	35.8	36,4	37.0	37.6

Essential body fat:

- Men: 3-5%
- Women: 8-13%
- Suggested not to go below 4-6% for men and below 10% for women

Consolidation:

By the end of this lesson, I am aware of:

-How to perform body composition tests and interpret their results

LESSON 2: BODY COMPOSITION TEST

Objectives:
-Understand what is the bioelectrical impedance analysis (BIA) and how it works.

- **BIOELECTRICAL IMPEDANCE ANALYSIS (BIA):**

Discuss the following questions to better understand how BIA works:

What is Bioelectrical Impedance Analysis (BIA)?	
What does it measure? Why is it used in fitness or health assessments?	
What factors can affect the accuracy of a BIA reading?	
Is BIA equally reliable for everyone? Why or why not?	
What are some rules or preparation steps to follow before doing a BIA test?	
What are the advantages and limitations of using BIA to assess body composition?	
What does body fat %, lean mass, etc. tell us about the client's health status?	

Consolidation:
By the end of this lesson, I am aware of:
-How BIA works, who it's suitable for, and how to prepare the client for accurate results.

LESSON 3: MUSCULAR STRENGTH

Objectives:
-Understand the difference between 1RM and estimated 1RM testing, and learn how to perform them safely with a client.

- **1-RM TEST:** *(Platform: Chapter Titled: Week 15: Fitness Assessments – Video Titled: Muscular Endurance & Muscular Strength)*

Set	Repetitions	LOAD	REST
1	10-15	20-30% of expected 1-RM	60 sec
2	5-10	40-50% of expected 1-RM	1-2 min
3	3-5	60-70% of expected 1-RM	2-3 min
4	1-3	80-90% of expected 1-RM	3-5 min or as much as needed
5-7	1	Actual 1-RM	3-5 min or as much as needed

- **ESTIMATED 1-RM TEST:**

$$\textbf{1-RM} = \frac{\boldsymbol{Weight\ Lifted}}{(\mathbf{1.0278 - 0.0278\ x\ }\boldsymbol{Reps})}$$

Calculate the estimated 1RM for your client who is currently doing squats in the following manner:
Set 1: 60kg for 12 reps
Set 2: 80kg for 10 reps
Set 3: 90kg for 6 reps
Set 4: 85kg for 8 reps

What is the purpose of these assessments?

Consolidation:
By the end of this lesson, I am aware of:
-How to perform and estimate a 1-RM test, and why it's important for designing safe and effective strength training programs.

LESSON 4: MUSCULAR ENDURANCE

Objectives:
- Understand what muscular endurance is and practice it through simple tests.

COMPETITION TIME!

In teams, two members from each team will compete in each category.

- The team with the highest combined score earns **1 point.**
- The player with the highest individual score earns **1 point.**

Let the best team win!

- **SQUAT TEST:**

MALES						
Age	18-25	26-35	36-45	46-55	56-65	65+
Excellent	>49	>45	>41	>35	>31	>28
Good	44-49	40-45	35-41	29-35	25-31	22-28
Above average	39-43	35-39	30-34	25-38	21-24	19-21
Average	35-38	31-34	27-29	22-24	17-20	15-18
Below Average	31-34	29-30	23-26	18-21	13-16	11-14
Poor	25-30	22-28	17-22	13-17	9-12	7-10
Very Poor	<25	<22	<17	<9	<9	<7

FEMALES						
Age	**18-25**	**26-35**	**36-45**	**46-55**	**56-65**	**65+**
Excellent	>43	>39	>33	>27	>24	>23
Good	37-43	33-39	27-33	22-27	18-24	17-23
Above average	33-36	29-32	23-26	18-21	13-17	14-16
Average	29-32	25-28	19-22	14-17	10-12	11-13
Below Average	25-28	21-24	15-18	10-13	7-9	5-10
Poor	18-24	13-20	7-14	5-9	3-6	2-4
Very Poor	<18	<20	<7	<5	<3	<2

- **PUSHUP TEST:**

% RANK	20-29 YEARS		30-39 YEARS		40-49 YEARS		50-59 YEARS		60+ YEARS	
	M	F	M	F	M	F	M	F	M	F
90	57	42	46	36	36	28	30	25	26	17
80	47	36	39	31	30	24	25	21	23	15
70	41	32	34	28	26	20	21	19	21	14
60	37	30	30	24	24	18	19	17	18	12
50	33	26	27	21	21	15	15	13	15	8
40	29	23	24	19	18	13	13	12	10	5
30	26	20	20	15	15	10	10	9	8	3
20	22	17	17	11	11	6	9	6	6	2
10	18	12	13	8	9	2	6	1	4	0

- **CRUNCH TEST:**

		Women- AGE (YEARS)					
Rating	**%Rating**	**18-25**	**26-35**	**36-45**	**46-55**	**56-65**	**66+**
Excellent	100	91	70	74	73	63	54
	95	76	60	60	57	55	41
	90	68	54	54	48	44	34
Good	85	64	50	48	44	42	33
	80	61	46	44	40	38	32
	75	58	44	42	37	35	31
Above Average	70	57	41	38	36	32	29
	65	54	40	36	35	30	28
	60	51	37	35	33	27	26
Average	55	48	36	32	32	25	25
	50	44	34	31	31	24	22
	45	41	33	30	30	23	21
Below Average	40	38	32	28	28	22	20
	35	37	30	24	27	20	18
	30	34	28	23	25	18	16
Poor	25	33	26	22	23	15	13
	20	32	24	20	21	12	11
	15	28	22	19	19	11	10
Very Poor	10	25	20	16	13	8	9
	5	24	17	14	9	7	8
	0	11	7	4	2	1	0

		Men- AGE (YEARS)					
Rating	**%Rating**	**18-25**	**26-35**	**36-45**	**46-55**	**56-65**	**66+**
Excellent	100	99	80	79	78	77	66
	95	83	68	65	68	63	55
	90	77	62	60	61	56	50
Good	85	72	58	57	57	53	44
	80	66	56	52	53	49	40
	75	61	53	48	52	48	38
Above Average	70	57	52	45	51	46	35
	65	54	46	44	47	43	32
	60	52	44	43	44	41	31
Average	55	49	51	39	41	39	30
	50	46	38	36	39	36	27
	45	43	37	33	36	33	26
Below Average	40	41	36	32	33	32	24
	35	40	34	31	32	31	23
	30	37	33	29	29	28	22
Poor	25	35	32	28	25	25	21
	20	33	30	25	24	24	19
	15	29	26	24	21	21	15
Very Poor	10	27	21	21	16	20	12
	5	23	17	13	11	17	10
	0	14	7	6	6	5	5

Consolidation:

By the end of this session, I am aware of:

- Body composition, muscular strength and muscular endurance tests.
- How to perform and interpret the mentioned assessments results.

Post-Session 32:

Practice the fitness tests discussed with a friend or family member, and go over the material discussed to prepare a recap to be presented in the upcoming session

Pre-Session 33:

- Watch all remainder fitness testing videos *(Week 15: Fitness Assessments)*

Session 33: Fitness Assessment – Continued

Session 32 Recap + Q&A:

LESSON 1: CARDIOVASCULAR ENDURANCE

Objectives:
- Understand what cardiovascular endurance is and practice it through simple tests.

- **TALK TEST:** *(Chapter Titled: Week 15: Fitness Assessments – Video: Cardiovascular Endurance – Talk Test)*

Sing a Song or Counting Talk Test:

Talk Test	Exercise Intensity
Client can talk comfortably	Low
Client has difficulty talking	High

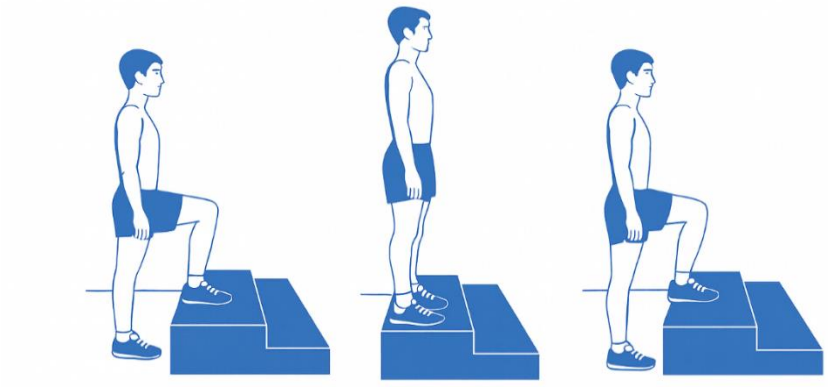
What is the purpose of this test?

GROUP ACTIVITY

- **STEP TEST:** *(Chapter Titled: Week 15: Fitness Assessments – Video: Cardiovascular Endurance – Step Test)*

Pair up with a partner and **practice** taking each other's heart rate. Then, using a stepper, assess one another's fitness level as a coach / client role play.

Set the metronome at 96 clicks per minute, **allow** the client to step up on a 30cm height stepper and keep performing the movement for 3 minutes then **take** their heart rate for one full minute. Once done, use the below tables to identify your client's fitness level:



MEN							
	% Rating	Age					
		18-25	26-35	36-45	46-55	56-65	65+
Excellent	100	50	51	49	56	60	59
	95	71	70	70	77	71	74
	90	76	76	76	82	77	81
Good	85	79	79	80	87	86	87
	80	82	83	84	89	91	91
	75	84	85	88	93	94	92
Above Average	70	88	88	92	95	97	94
	65	90	91	95	99	99	97
	60	93	94	98	101	100	102
Average	55	95	96	100	103	103	104
	50	97	100	101	107	105	106
	45	100	102	105	111	109	110
Below Average	40	102	104	108	113	111	114
	35	105	108	111	117	115	116
	30	108	110	113	119	117	118
Poor	25	111	114	116	121	119	121
	20	114	118	119	124	123	123
	15	119	121	124	126	128	126
Very Poor	10	124	126	130	131	131	130
	5	132	134	138	139	136	136
	0	157	161	163	159	154	151

WOMEN							
	% Rating	Age					
		18-25	26-35	36-45	46-55	56-65	65+
Excellent	100	52	58	51	63	60	70
	95	75	74	77	85	83	85
	90	81	80	84	91	92	92
Good	85	85	85	89	95	97	96
	80	89	89	92	98	100	98
	75	93	92	96	101	103	101
Above Average	70	96	95	100	104	106	104
	65	98	98	102	107	109	108
	60	102	101	104	110	111	111
Average	55	104	104	107	113	113	116
	50	108	107	109	115	116	120
	45	110	110	112	118	118	121
Below Average	40	113	113	115	120	119	123
	35	116	116	118	121	123	125
	30	120	119	120	124	127	126
Poor	25	122	122	124	126	129	128
	20	126	126	128	128	131	129
	15	131	129	132	132	135	133
Very Poor	10	135	134	137	137	141	135
	5	143	141	142	143	147	145
	0	169	171	169	171	174	155

- **RPE (Rate of Perceived Exertion):**

RPE is a simple way to measure exercise intensity based on how hard you feel you are working. Lower RPE values mean lighter exercise, and higher values mean more intense exercise:

GOALS	RPE	Estimated Straight Line %HR
Aerobic Endurance Training	3-4	60-70
Anaerobic Endurance Training	4-6	70-75
Muscular Endurance/Strength	6-7	75-80
Fast Race/Peak Performance	9-10	90-100



Consolidation:

By the end of this lesson, I am aware of:

-What cardiovascular endurance is and can practice it through the Step Test and Talk Test.

LESSON 4: PRACTICAL APPLICATIONS – Case Studies



Objectives:

- Identify suitable fitness tests and assessments for a given client profile.
- Determine an appropriate schedule for conducting assessments.



Answer the following cases:

Case Study I: Diana:

Diana is a 25 years old accountant who can't wait to start working out. Diana has been going to the gym for the past 7 years, but with great inconsistency. The last time Diana worked out was around one year ago, and she has been contemplating about going to the gym ever since but her work schedule didn't allow her to take action. Along with the busy work schedule, Diana couldn't maintain a healthy diet and took many poor decisions with fast food which made her gain a significant amount of weight and fat. Now that Diana has settled in her job, she can train a minimum of three days per week and is willing to return to her fit and healthy lifestyle she used to have. Diana has assigned you with the task of being her personal trainer, and she is excited to see her progress.



- **What fitness tests and assessments will you apply with Diana?**



- **How would you schedule your assessments?**

Case Study II: Julia

Julia is a 19 years old college athlete. She plays in the football team as an attacker. Julia is very passionate about playing football and is seriously considering pursuing a career as a professional football player. However, Julia complains about her height preventing her from scoring headers, and her jump doesn't help her a lot. She compensates her poor heading skills with being very agile and accurate with her shooting. Julia trains regularly at your gym, and has requested that you start training her. She is obsessed with improving her jump, as well as increasing her endurance so she can sustain her performance for entire 90 minutes matches.



• What fitness tests and assessments will you apply with Julia?



• How would you schedule your assessments?

Case Study III: Adam

Adam is a 35 years old architect whose success is recognized internationally. He is a very committed worker, to the extent that he has neglected his health for the sake of his career. Adam believes in the "live fast, die young" lifestyle, but his recent visit to the doctor made him consider otherwise. Adam's doctor warned him that if he doesn't start pursuing a healthy lifestyle, he may develop several metabolic and cardiovascular conditions. Adam has consulted your services as a personal trainer, and mentions that he wants to work out to improve his health and reduce his weight as the doctor warned him that he is stage I obese, according to the Body Mass Index (BMI) calculation. Adam is not very excited about working out and he clearly states that if it wasn't for the doctor's strict recommendation, he wouldn't have considered working out.



• What fitness tests and assessments will you apply with Adam?



• How would you schedule your assessments?

Spot the Mistake!

Each group chooses 3 movement screens and 3 mobility assessments. For each one, they decide if the client will commit 0, 1, or 2 mistakes and write these down on paper.

Learners are then paired with someone from another group in a client–coach role-play. The coach's task is to spot the mistake made by the client, then they switch roles.

Let the Best Team Win!

Consolidation:
By the end of this lesson, I am aware of:
-How to identify suitable fitness tests and assessments for a given client profile and how to determine an appropriate schedule for conducting them.

🎉 YOU MASTERED FITNESS TESTING! 🦾💪

You've stretched, balanced, and pushed your limits—acing mobility, flexibility, endurance, and strength tests with confidence!
Your foundation in assessing performance is solid, your eye for detail is sharper, and you're ready to measure, track, and elevate every component of fitness like a true pro.

🏅 Certified Fitness Testing Pro 📊



CONGRATULATIONS!

Session 34: Introduction to Strength & Conditioning

Session 33 Recap + Q&A:

LESSON 1: INTRODUCTION to STRENGTH & CONDITIONING – Terms & Definitions

Objectives:

-Understand the difference between Fitness and Athleticism.
-Define the Health-Related Components of Fitness (HRCF).
-Define the Performance-Related Components of Fitness (PRCF).
-Identify the difference between HRCF and PRCF.

- **FITNESS & ATHLETICISM:** *(Platform Week 16: Introduction to Strength and Conditioning + Program Design: Training Principles) (Refer to the manual page 245-248)*

GROUP ACTIVITY

Label each statement either as "Fitness" or "Athleticism":

Statement	Answer
Involves the ability to perform daily physical tasks	
Refers to sport-specific physical qualities	
Focuses on general physical capacity and well-being	
Includes speed, agility, and coordination	
Refers to overall health	
Involves competition and performance	

- **TERMS & DEFINITIONS – Health -Related vs Performance-Related Components of Fitness:**

Read each definition and **match** it with the correct term in the blank space provided while specifying while the component is Health or Performance related:

Flexibility | **Body Composition** | **Strength** | **Power** | **Ballistic**

Plyometric

Speed

Agility

Aerobic Endurance

Anaerobic Endurance

Balance

Coordination

Involves quick, powerful movements using a rapid stretch followed by a strong contraction. It trains the body to react fast by using stored elastic energy. Movements typically include a short ground contact time and aim to improve reactive strength.

The ability to produce force and apply it over a distance to move or resist a load. It involves voluntary muscle contractions and can be expressed in static, dynamic, or explosive forms.
Formula: Force × Distance
This ability supports other components like speed and power.

The ability to cover a certain distance in the shortest possible time
Formula: Distance ÷ Time
This quality is often confused with related terms like velocity (includes direction) and acceleration (rate of speed change), but each has a specific meaning in performance. It is closely linked to power and influenced by mechanics and sport-specific demands.

The ability to produce force and apply it over a distance to move or resist a load. It involves voluntary muscle contractions and can be expressed in static, dynamic, or explosive forms.
Formula: Force × Distance / Time
This ability supports other components like speed and strength.

	Involves explosive exercises where the body or object is accelerated through the full movement without slowing down. These focus on high-speed force during the concentric phase, often using resistance.
	The ability to perform continuous, moderate-intensity exercise over long periods without fatigue. It relies on the body's use of oxygen to produce energy and is important for both long-duration sports and recovery in high-intensity games. This quality supports sustained effort, quicker recovery between actions, and overall cardiovascular and muscular efficiency.
	The ability to stay stable and in control whether the body is still or moving. It includes both holding steady positions and staying balanced during actions like running or jumping. This skill depends on input from the eyes, inner ear, and muscles, and helps improve coordination and reduce injury risk.
	The ability to sustain or repeat high-intensity efforts over time without a major drop in performance. It relies on energy systems that don't require oxygen and is important in activities like sprinting, jumping, or repeated explosive actions with short rest. This quality helps athletes resist fatigue during intense efforts and is developed through interval and circuit-style training.
	The ability to perform smooth, controlled, and efficient movements by combining brain and body signals. It allows athletes to learn and execute complex skills with precision and improves through repeated, specific practice.
	The ability to quickly change direction, speed, or body position in response to a stimulus, while maintaining control and balance. It includes both physical skills—like coordination, strength, and balance—and mental skills such as reaction time and decision-making.
	The ability of a joint or group of joints to move through a full range of motion.
	Involves the proportion of fat, muscle, bone, and water in the body.

Consolidation:

By the end of this lesson, I am aware of:

-The difference between fitness and athleticism.

-The components that make up health-related fitness.

-The components that make up performance-related fitness.

LESSON 2: INTRODUCTION to STRENGTH & CONDITIONING – Drills Practice

Objectives:

-Practice all components of fitness through various drills.

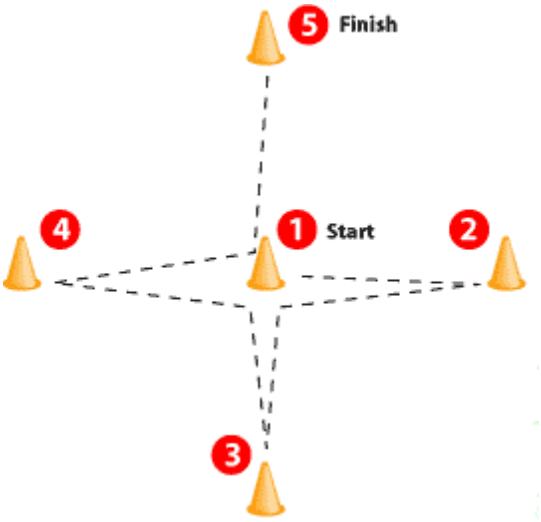
-Identify how to target different fitness components by making small adjustments to each drill.

-Design a circuit training session using the practiced drills.

Let's Compete, Play and Practice the following drills in teams at the assigned stations, and **identify** which component of fitness each drill targets:

- **Competition Time – Box Drill:**

Let's see who can crush this drill the fastest!



Which components of fitness does this drill mainly work on?

→

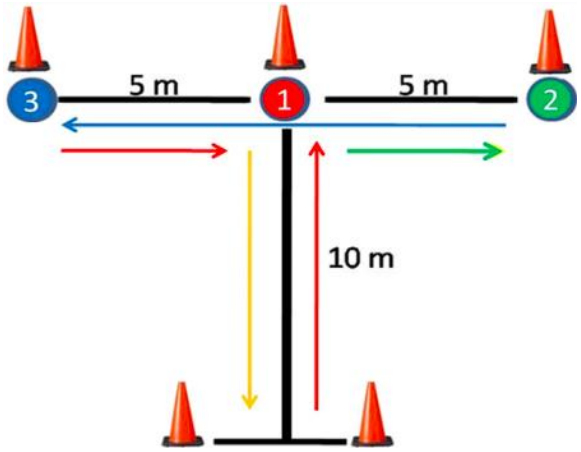
How could you adjust or modify this drill to target additional fitness components?

→

LET'S PUT THESE COMPONENTS INTO PRACTICE!

- **Competition Time - T-Drill:**

Let's see who's going to dominate this drill and finish first?

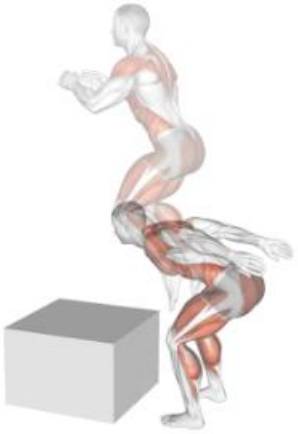


Which components of fitness does this drill mainly work on?

→

- **Box Jumps:**

Can you jump on all boxes of all heights?



Which components of fitness does this drill mainly work on?

→

How can we improve our landing?

→

- **Tuck Jumps:**

Let's see who can do 10 jumps in a row!



Which components of fitness does this drill mainly work on?

→

- **Ice-Skaters:**

Let's glide from side to side—who can land the smoothest?

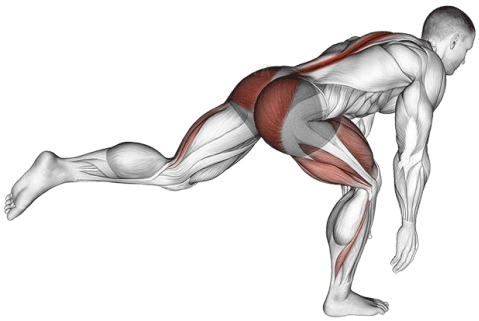


Which components of fitness does this drill mainly work on?

→

- **Single-leg Reach:**

Can you keep your foot off the ground the whole time?

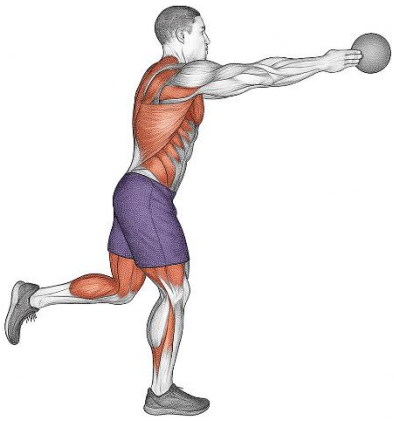


Which components of fitness does this drill mainly work on?

→

- **Single-leg Throw and Catch:**

Let's see who can last till the end!



Which components of fitness does this drill mainly work on?

→

GROUP ACTIVITY

Design a 30-minute circuit training session incorporating various drills for an intermediate-level client while trying to tackle all or most of the fitness components discussed (feel free to include any exercises or drills of your choice):

Exercise / Drill	Timing	Comments / Notes

Instructor's Feedback

How would you modify it if your client is a beginner or advanced?

Can circuit training include resistance training exercises?

What is the main goal of circuit training?

Did you use lots of gym equipment? Is it practical to be performed in a crowded gym?

Consolidation:

By the end of this lesson, I am aware of:

-How to apply each fitness component through practical drills.

-How small changes in exercises can shift the focus from one component to another.
-How to create a circuit targeting multiple components of fitness.

🎉 YOU MASTERED THE BASICS! 🤸💪

You've sprinted, coordinated, and powered through agility drills like a pro!
Your foundation in strength and conditioning is set, your skills are sharper, and

you’re ready to take on every component of fitness with confidence.

🏅 **Certified Strength & Conditioning Undergrad** 🏋️



CONGRATULATIONS!

LESSON 3: INTRODUCTION to PROGRAM DESIGN

Objectives:

-Lead a full training session in a realistic client scenario.
-Apply feedback to improve coaching skills.



“Real Case Scenario Simulation”:

This activity is designed to help learners step into the role of a real Personal Coach. Each participant will create a complete 30-minute training session, including a warm-up, full-body workout, cooldown, and feedback.

Once prepared, learners will pair up: one will coach their partner through the session while the other plays the role of the client, then they switch roles.

Objective: To practice leading a full training session and build confidence in handling real case scenario.

Instructor’s Feedback



Consolidation:

By the end of this lesson, I am aware of:

-How to design and coach a simple full-body session for a beginner client, as preparation for the next chapter on program design.



Post-Session 34:

Go over the material presented to practice it with a friend or relative and prepare to present it in the upcoming session’s recap

Pre-Session 35 Preparation:



Watch the below pre-recorded videos:

-Week 16: *Introduction to Strength and Conditioning + Program Design: Training Principles* (the videos related to the program design chapter only)

-Week 17: *Program Design: Training Principles + Components of Program Design* (training principles videos only)

Session 35: Program Design – Resistance Training

Session 34 Recap + Q&A:

LESSON 1: TRAINING PRINCIPLES

Objectives:
-Recognize the key principles for an effective training program design.
-Explain how each principal influence training outcomes.

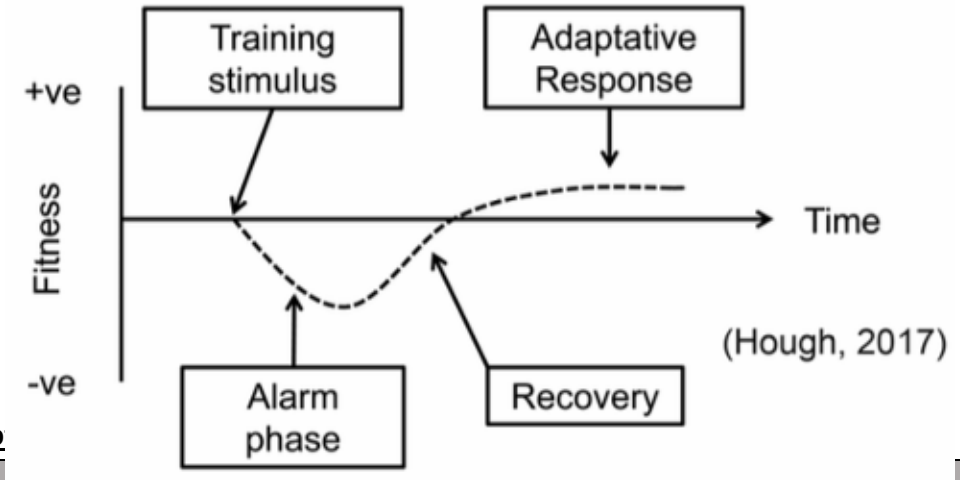
1. THE PRINCIPLE OF ADAPTATION:

- **Definition:**

Adaptation = The body's ability to adjust its structure, functions, and behaviors to handle the demands of training.

- **How it Works (General Adaptation Syndrome – GAS):**

Phase	What Happens	Training Example
Alarm	Temporary drop in performance as body reacts to new stress.	First week of resistance training – muscles feel sore and weak.
Resistance	Body recovers and improves beyond previous level.	After several weeks – strength and endurance improve.
Exhaustion	Stress is too much or too frequent for body to adapt.	Training too hard without recovery – risk of injury or burnout.



(Hough, 2017)

- **Types o**

Type	Duration	Example

Acute	Short-term (during/soon after exercise)	Increased heart rate during running.
Chronic	Long-term (after repeated training)	Stronger muscles after months of lifting.

- **Key points to Remember:**

✓ Body systems need to be stressed to adapt.

✓ Adequate overload → repeated cycle of stress + recovery → improvement.

✓ Too much stress without recovery = exhaustion phase.

2. Stimulus and Fatigue:



Define the following terms:

Stimulus	
Fatigue	
Performance	



Briefly explain the link between Stimulus, Fatigue, Recovery, Rest between sets / Exercises and Training Volume:

3. RECOVERY:

- **Definition:**

Recovery is the process that allows the body to meet or exceed previous performance levels by reducing fatigue after training stress.

- **Types of Recovery**:

Type	Description
Immediate Recovery	Between exertions (reps)
Short-term Recovery	Between sets or interval bouts
Training Recovery	Between workouts

- **Recovery Management:**



Brainstorm all the factors that could affect an athlete's recovery. **Write** your answers in the space below:

- **Managing the Training Program (Rest for Recovery):**

Rest days = No or very little activity
Active rest days = Light aerobic activity + stretching → helps blood flow, removes metabolites, restores muscle length/function



Which option leads to better recovery?



What is Deload for? Do we have to include it in our training programs?

- **Nutrition for Recovery**:

-Quantity: Calorie deficit / maintenance / surplus — more resources = better recovery

-Quality: Macronutrients (Protein → Carbs → Fat) + micronutrients + fiber

-Fluids: More than non-trainees; monitor urine color (pale yellow) & pre/post weight (~1–2% loss optimal)

	2–4 hrs before Exercise	During Exercise	After Exercise
Fluid intake recommendations	5 ml/kg	0.3–2.4 L/h of water or an electrolyte drink	1.25–1.5 L of fluid for every kg of bodyweight lost



How can we monitor our fluid levels in practical ways?

- **Sleep for Recovery**:

-Athletes need more sleep than non-athletes

-Aim: **8–9.5 hours/night** (top end for heavy training)

-Essential for physiological growth and repair

4. INDIVIDUALIZATION & RECOVERY:



Think of two people who trained together but saw different results. **List** possible genetic and physiological factors that could explain the difference:

Genetic Factors	Physiological Factors

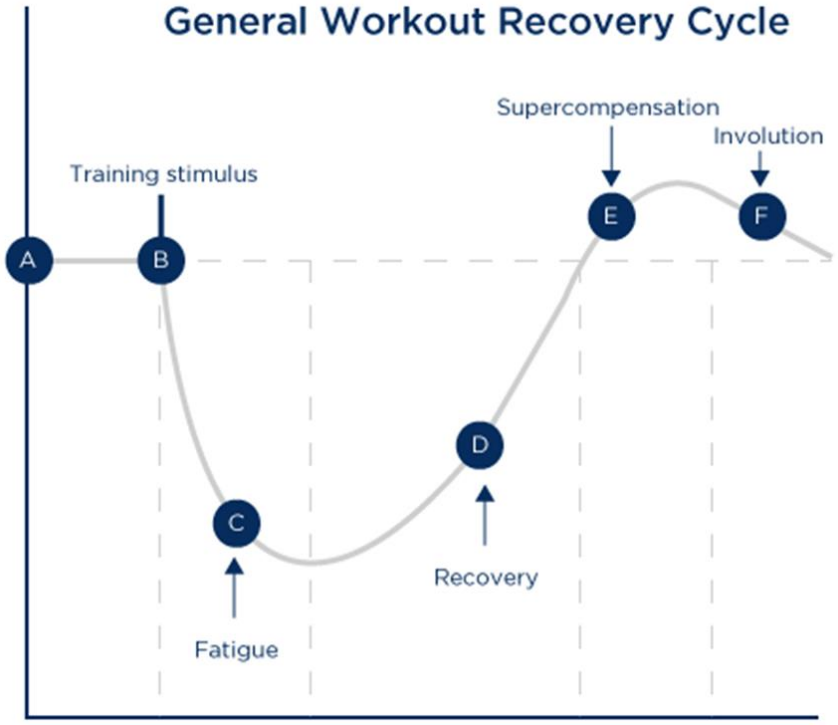
5. SPECIFICITY - (SAID) & RECOVERY:

SAID = Specific Adaptation to Imposed Demand

Briefly explain the specificity training principle and list the five components that should be considered:

6. STIMULUS RECOVERY ADAPTATION CURVE (SRA): *(Platform: Chapter Titled: Week 17: Program Design: Training Principles + Components of Program Design)*

Identify what each point on the SRA curve stands for:



- **Point A – Baseline :**

- **Point B – Stimulus :**

What practical signs indicate a good training stimulus?

- **Point C – Fatigue :**

What is the difference between CNS fatigue and PNS fatigue ?

How does immediate fatigue differ from delayed fatigue ?

How can we address these different types of fatigue when designing a training program ?

- **Point D – Recovery :**

Pair up and **summarize** the factors needed for good recovery :

-

-

-

-

- **Point E – Supercompensation :**

How do we reach supercompensation?

- **Point F – Involution :**

  How does it happen practically?

- **Point G – Exhaustion :**

7. GRADUAL PROGRESSIVE OVERLOAD (GPO):

GROUP ACTIVITY

Briefly explain what GPO is, and list examples of how you can practically apply this concept during training sessions:

Do you think training more is the best way to improve? Why or why not?



Consolidation:
By the end of this session, I am aware of:
- The key principles that guide effective training program design.
- How each principal influence training outcomes.



Post-Session 35 Tasks:
- Review the material covered during the session.
- Prepare a brief recap of the key training principles to be presented in the next session.



Pre-Session 36 Preparation:
Watch the below pre-recorded videos:
-Week 17: Program Design: Training Principles + Components of Program Design: Just the Intensity and Volume components of program design videos

Session 36: Program Design – Resistance Training

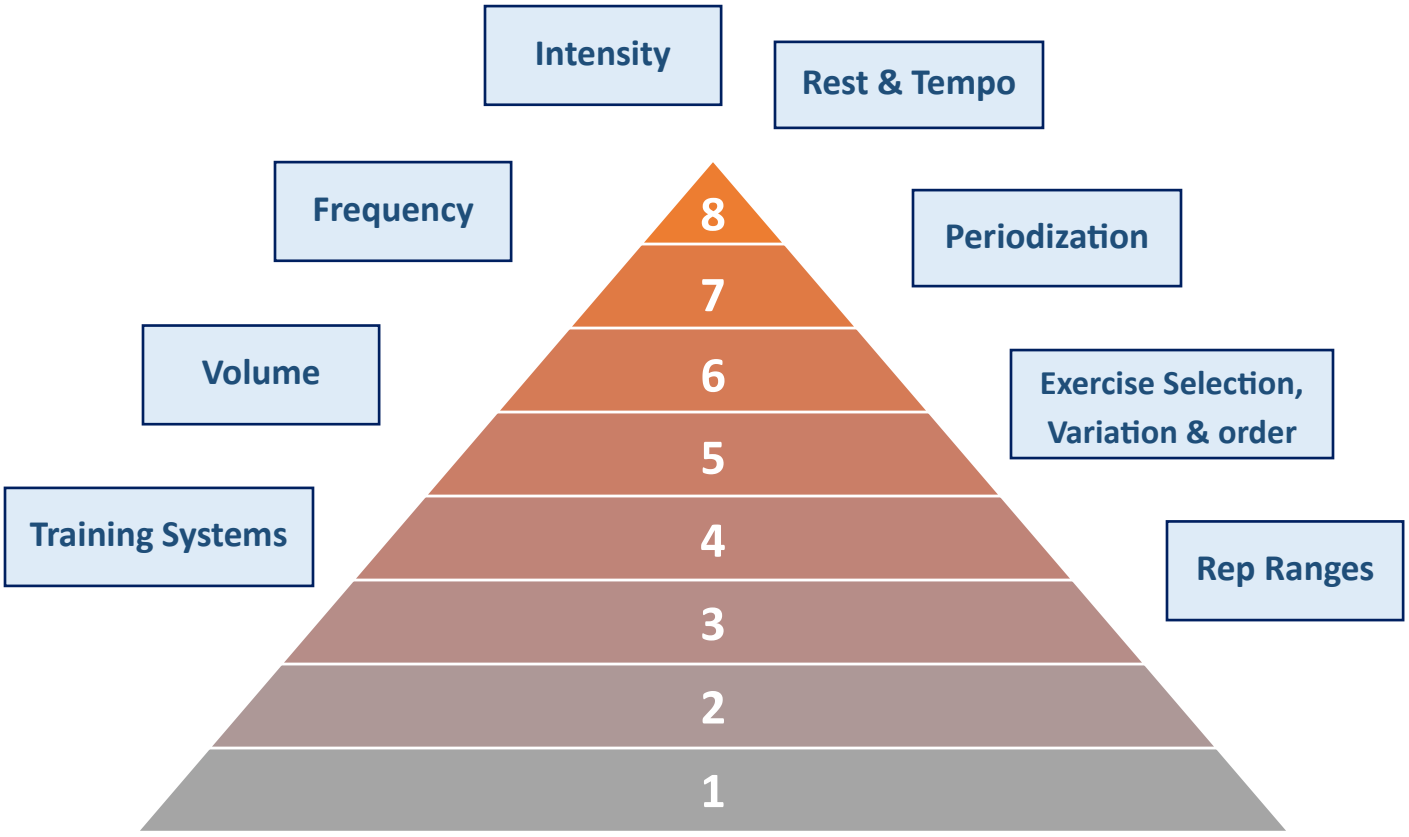
Session 35 Recap + Q&A:

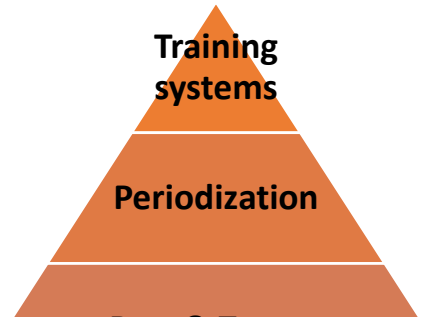
LESSON 1: COMPONENTS OF PROGRAM DESIGN – Order of Importance

Objectives:
- Understand the order of importance of the components of program design.

GROUP ACTIVITY

Match each component of program design (in blue) to its corresponding pyramid level, with level 1 being the most important





Consolidation:

By the end of this lesson, I am aware of:

-The different components of program design.

-The order of importance of these components.

LESSON 2: COMPONENTS OF PROGRAM DESIGN – Intensity

Intensity

Objectives:
-Understand what training intensity is and how to apply it in program design.
(Platform: Chapter Titled: Week 17: Program Design: Training Principles + Components of Program Design)

What is INTENSITY?

- **How can we measure Intensity?**

1. Repetition Maximum (RM):

• % of 1RM determines load category:
- Low: <60% of 1RM
- Moderate: 60–80% of 1RM
- High: >80% of 1RM

Repetitions (RM)	% of 1RM
2 RM	95%
3 RM	92.3%
4 RM	90%
5 RM	87.5%
6 RM	85%
7 RM	82.5%
8 RM	80%
9 RM	77.5%
10 RM	75%

Remember how can we calculate the estimated 1-RM?

2. Rate of Perceived Exertion (RPE)

Category Ratio Scale (CRS) Value	Description
0	Nothing at all
0.5	Very, very weak
1	Very weak
2	Weak
3	Moderate
4	Somewhat strong
5	Strong
6	
7	Very strong
8	
9	Very, very strong
10	Maximal

3. Reps in Reserve (RIR):

What are Reps in Reserve (RIR), and how can we practically use them during training?

RPE Score	RIR/Description
10	Maximal effort
9.5	No RIR, but could increase load
9	1 RIR
8.5	Definitely 1, maybe 2 RIR
8	2 RIR
7.5	Definitely 2, maybe 3 RIR
7	3 RIR
5-6	4–6 RIR
3-4	Light effort
1-2	Little to no effort

Practical Application:



RPE and RIR scales help you adjust training based on how you feel. To use them well, keep training logs with your workout details (RPE or RIR for each exercise, session, or set)

Load Adjustment for Next Set if RPE Target is Missed

Actual RPE	RIR	Load Adjustment
1	Little to no effort	Increase load by 20%
2	Little to no effort	Increase load by 16%
3	Light effort	Increase load by 12%
4	Light effort	Increase load by 8%
5	4-6 RIR	Increase load by 4%
6	4-6 RIR	Lifter/Trainer's choice
7	3 RIR	Lifter/Trainer's choice
7.5	Definitely 2, maybe 3 RIR	Lifter/Trainer's choice
8	2 RIR	Lifter/Trainer's choice
8.5	Definitely 1, maybe 2 RIR	Decrease load by 2%
9	1 RIR	Decrease load by 4%
9.5	No RIR, but could increase load	Decrease load by 6%
10	Maximal effort	Decrease load by 8%

Note: This table uses a target range of 6–8 RPE for a set. If you land within the RPE range, it is your choice how to adjust the weight for the next set. If you land outside the RPE range at the end of a set, adjust the load by 2% for every 0.5 RPE points outside of the target range.

How can I practically improve the client's RPE-RIR estimations?

Which type of training is better: Autoregulate or Fixed loading? And why?

How can we use the RPE-RIR to Autoregulate?

- **How hard should we train?**

1. Stimulating reps:

How can we practically identify a stimulating or challenging repetition?

Do all sets need to include stimulating reps when the goal is hypertrophy?

When training for hypertrophy, what is the minimal number of reps to be performed in one set?

2. Training to Failure:

Define the following terms:

Momentary Muscular Failure:	
Volitionary Muscular Failure:	



What is the number of stimulating reps in a set taken to failure for trained lifters?

3. Practical Applications:

Training for Strength	Training for Hypertrophy
Stop a little before failure for best results. It allows lifters to use higher movement velocity.	The last few reps in a set are the most effective for muscle growth.
At medium weights, keep 3 or more reps in reserve (RIR ≥ 3).	If training to failure (0 RIR), think about total training volume, fatigue levels, where the exercise is in your workout, exercise technique, and your weekly plan.
At heavy weights (≥ 85% of 1RM), you can train closer to failure (0–2 RIR) but keep total sets moderate.	Make sure to rest enough before the next session to ensure proper recovery
Too much training to failure with many sets can slow progress.	

4. Challenging Set:

Theoretically: Any set that includes 1 challenging rep. Can be anywhere between 5-30 reps

Optimal challenging set for hypertrophy: 3 challenging reps (2 RIR). Better to be between 6-15 reps

Consolidation:

By the end of this lesson, I am aware of:

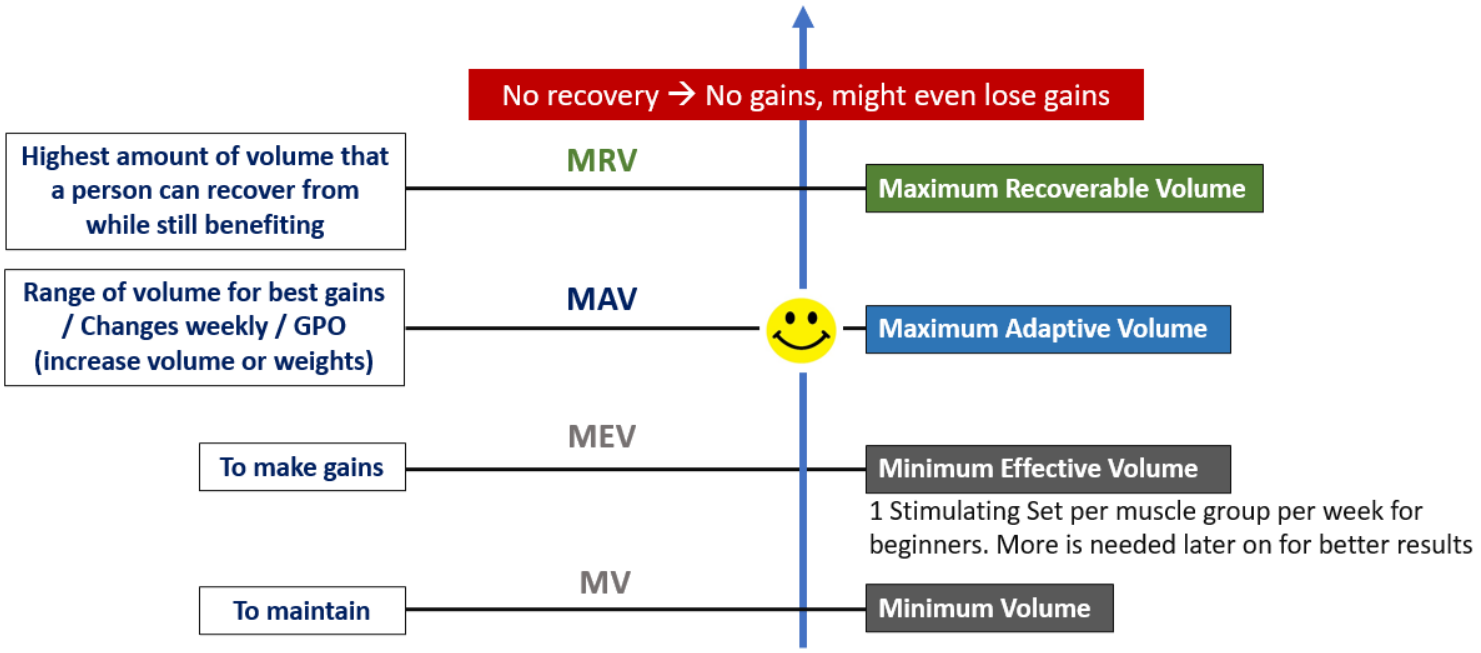
- The definition of training intensity.
- The different methods for measuring intensity (RM, RPE, RIR).
- How to apply intensity to strength and hypertrophy training.

LESSON 3: COMPONENTS OF PROGRAM DESIGN – Volume

Volume

Objectives:
-Understand what Volume is and how to apply it in program design.

1. Defining Volume:



2. Volume Recommendations:

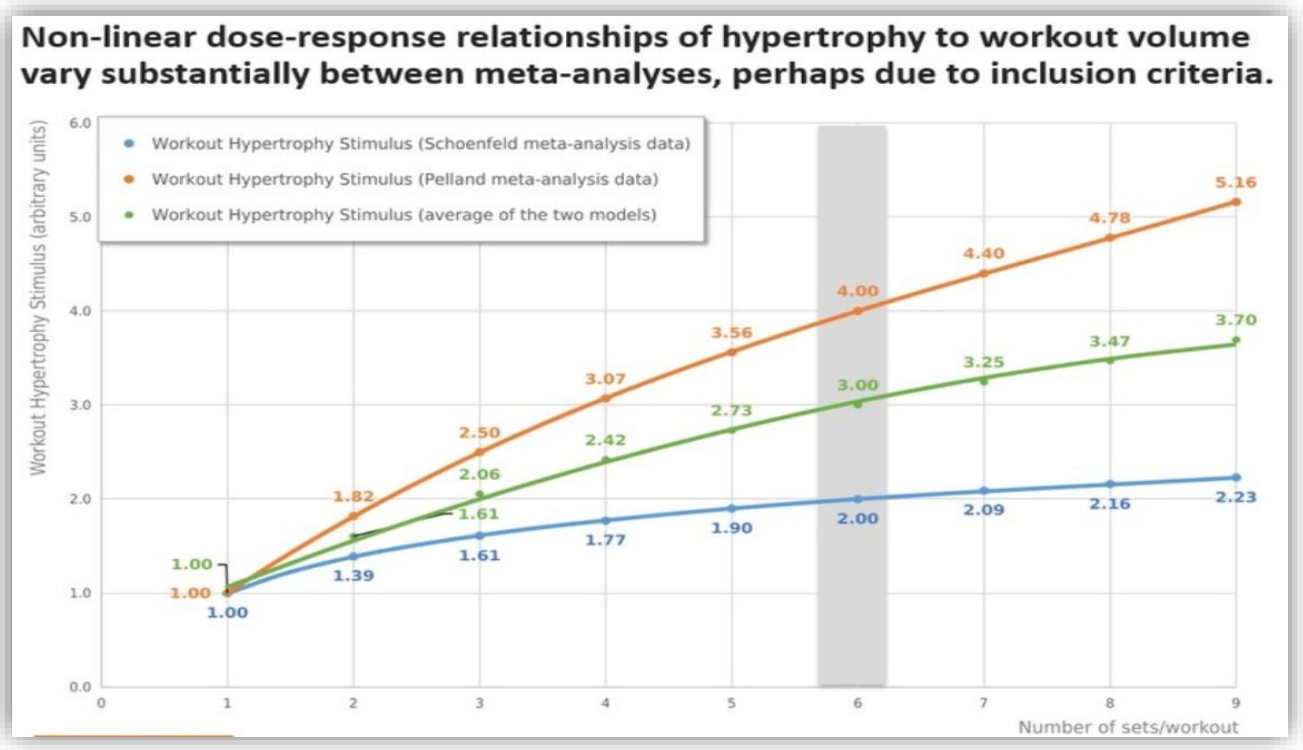
- **Dose–Response Relationship:**

More sets → more growth	As training volume (sets) increases, muscle growth also increases (+nuances)
Minimal Volume for gains?	Even 3 sets/week can give significant muscle growth (1 set per workout × 3 workouts/week).
Diminishing returns	Higher volumes don't lead to proportionally higher growth. Example: to double gains from 3 sets/week, we need ~18 sets/week. This is due to a non-linear dose-response relationship of hypertrophy to workout volume
Not all sets are equal	Later sets in a workout contribute less to growth than earlier sets.



Does increasing Volume always increase Hypertrophy?

Define the non-linear dose-response relationship of hypertrophy to workout volume presented in the graph below:



What is the physiological explanation to the above graph?

How does lifting experience affect the benefits of adding more training volume?

What is the best approach for each lifter type? (Beginner/Intermediate/Advanced)

- **Practical Applications – Strength & Hypertrophy:**

Recommendation	Notes
10–20 sets per muscle group/week	Works for most people.
High volume at high intensity is not needed	More isn't always better.
Advanced lifters	Focus on motor unit recruitment strategies, not just adding sets.



Think of a muscle group you're training, and **answer** the following questions:

1. How many sets per week are you currently doing? _____
2. Could you adjust this to be within the 10–20 set range? _____
3. If you're advanced, what non-volume strategies could you try? _____



Consolidation:

By the end of this session, I am aware of:

-What training volume means.
-The main volume landmarks (MV, MEV, MAV, MRV).
-The recommended set range for best results.



Post-Session 36 Tasks:

- Review the material covered during the session.
- Prepare a brief recap of the key training principles to be presented in the next session.



Pre-Session 37 Preparation:

Watch the below pre-recorded videos:
-Week 17: Program Design: Training Principles + Components of Program Design: Just the Frequency and the Exercise Selection components of program design videos.

Session 37: Program Design – Resistance Training

Session 36 Recap + Q&A:



Case Study:

Samir wants you to help him design a training program that allows him to engage his Glutes, Hamstrings, Quadriceps, Lats, Pec Major, Trapezius, Anterior Delts, Middle Delts, Posterior Delts, Biceps and Triceps in 6 challenging sets per week.

Suggest a list of exercises allowing Samir to reach his goals:
(Note: No need to mention the frequency, tempo nor resting time)

LESSON 1: COMPONENTS OF PROGRAM DESIGN – Frequency

Frequency

Objectives:
-Understand what Frequency is and how to apply it in program design.

Evidence in favor of frequency:

		1-Total Body Group	2-Split Group
Similarities	Mesocycle	8 weeks training block	8 weeks training block
	Volume	9 sets/body part/ week	9 sets/body part/ week
	Exercises	Same Exercises	Same Exercises
	Training Frequency	Trained 3 times/ week	Trained 3 times/ week
Differences	Volume	Full body 3 times/week	Day 1: Chest/Back Day 2: Legs Day 3: Shoulders and arms
	Body Part Frequency	Each muscle 3 times/week	Each muscle 1 time/ week

What conclusion can we draw from this table?

Evidence against frequency:

In another systematic review and meta-analysis published in 2019, they showed that as long as volume was equated, frequency did not matter.

Why do we have this contradiction in the frequency literature? (use the table below as a hint):

STUDY (blue = high frequency better)	FREQUENCY (HIGH GROUP)	VOLUME (HIGH GROUP)	CAN RECOVER?
Brigatto et al. (2018)	2/week	8 sets/workout	No
Gentil et al. (2015)	2/week	6 sets/workout	No
Gentil et al. (2018)	2/week	6 sets/workout	No
Tavares et al. (2017)	2/week	4 sets/workout	Yes
Lasevicius et al. (2019)	3/week	4-12 sets/workout	No
Yue et al. (2017)	2-4/week	2-10 sets/workout	Yes / No
Schoenfeld et al. (2015)	3/week	3 – 7 sets/workout	Yes / No
Neves et al. (2022)	3/week	3 sets/workout	Yes
Ochi et al. (2018)	3/week	2 sets/workout	Yes
Bartolomei et al. (2020)	4/week	5 sets/workout	No
Evangelista et al. (2021)	4/week	4-8 sets/workout	No
Zaroni et al. (2018)	5/week	3-6 sets/workout	No
Saric et al. (2018)	6/week	2-4 sets/workout	No

- **Did You Know – Sets to Failure & Recovery Time**

# of sets to failure	# of days to recover	Approximate recovery time (days)
8 sets	> 4 days	~ 5 to 6
6 sets	> 3 days	~ 4 to 5
4 sets	> 2 days	~ 2 to 3
2 sets	Within 2 days	

State example training programs (Split training, Push-Pull-Legs, Full body…) for clients with different schedules and availabilities as outlined in the table below:

Training Days per week	Training Program Examples
2 Days	
3 Days	
4 Days	
5 Days	
6 Days	

Frequency - Practical Applications

- Higher frequency (2-3x/week) yields more muscle growth than lower frequency (1x/week)
- If the trainee can only go to the gym 2-3 times/week, then opt for a full body workout on each of those days to make sure you train each muscle group 2+ times per week
- If the trainee is still a beginner (untrained individual), training frequency matters less.
- The more advanced you are as a lifter, the more you need to pay attention to your frequency (it is better to increase it to 2-3x/week).
- More is not always better; no evidence suggests that training a muscle group more than 3 times per week gives better results.



Case Study:

Re-design the training program for our old friend Samir, knowing that he is available to train 7 days a week if needed, in order to achieve the best hypertrophy results:

Consolidation:

By the end of this session, I am aware of:

-Practical application of training volume.

-How training Frequency is adjusted based on clients goals, recovery and schedule

-How training intensity, volume and frequency interact all together to influence results

Post-Session 37 Tasks:

- Review the material covered during the session.
- Prepare a brief recap of the key training principles to be presented in the next session.

Pre-Session 38 Preparation:

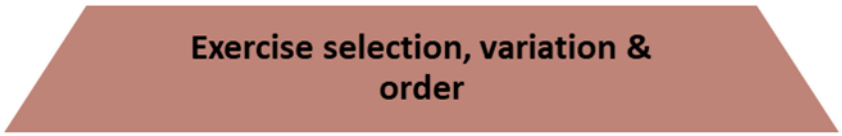
Watch the below pre-recorded videos:

-Week 17: Program Design: Training Principles + Components of Program Design: Just the Frequency and the Exercise Selection components of program design videos.

Session 38: Program Design – Resistance Training

Session 37 Recap + Q&A:

LESSON 1: COMPONENTS OF PROGRAM DESIGN – Exercise Selection, Variation and Order



Objectives:
-Understand how exercise selection, variation, and order influence training outcomes.
-Recognize the importance of exercise order for different training goals.

How can exercise selection, variation, and order affect a training program? Give 3 brief reasons.

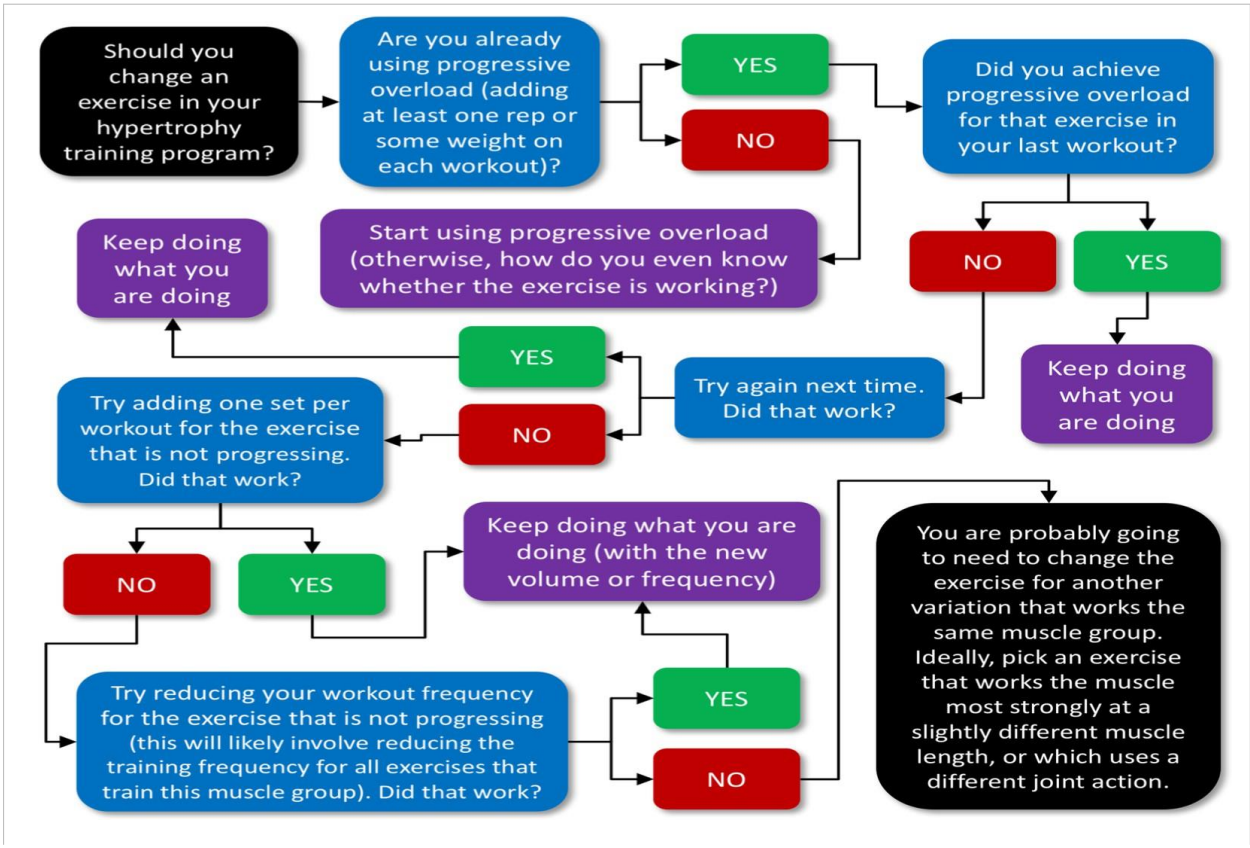
Why is it better to start with compound lifts when the goal is strength?

For hypertrophy, why is it important to include a variety of exercises for the same muscle group? Is it the same for all muscles?

GROUP ACTIVITY

Pair up and **answer** the following question:

When should you replace an exercise in a training program? Read the below picture and explain it to your classmates while answering the question:



- **Compound vs. Isolation Exercises:**

State 4 differences between compound exercises and isolation exercises while playing Basketball!!!:

Compound Exercises	Isolation Exercises

- **Practical Applications:**

Training for Strength	Training for Hypertrophy
- Do the exercise you want to improve on first in your session. - Start the week with this exercise so you are less fatigued.	- Use different exercises for the same muscle group to train all areas/heads of that muscle. - Train the muscle group at the start of the session/week. - Place single-joint or smaller muscle exercises later in the workout to reduce CNS fatigue.

Consolidation:

By the end of this lesson, I am aware of:

-How exercise selection, variation, and order influence training results.

-The importance of exercise order for achieving different training goals.

LESSON 3: COMPONENTS OF PROGRAM DESIGN – Rep Ranges

Rep Ranges

Objectives:
- Understand the different repetition ranges, their load categories, and how each impact training outcomes.

Suggest a rep range and the %1RM for each of the below categories, and **mention** one benefit of each category:

- **Heavy Loads:**
- **Moderate Loads:**
- **Light Loads:**
- **Very Light Loads:**

Why are moderate loads considered both practical and effective for training? Can't we just say training close to failure leads to hypertrophy regardless of the total number of reps per set?

Consolidation:
By the end of this lesson, I am aware of:
-The main repetition ranges and their corresponding load categories.
-How each repetition range affects fatigue, stimulating reps, and overall training results

LESSON 4: COMPONENTS OF PROGRAM DESIGN – Rest & Tempo



Objectives:

-Understand how rest periods and tempo affect training outcomes and how to apply them effectively in a program

-Understand what a deload is and how to apply it effectively in a training program.

(Platform: Chapter Titled: Week 18: Program Design: Components of Program Design for Resistance and Cardiovascular Training)

<u>Let's see how good you are in estimating time/tempo!</u>

Rest Periods	Tempo
Strength: Longer rests (for energy & nervous system recovery).	Tempo = speed of eccentric, pause, concentric, pause phases: ecc/pause/conc/pause ex: 2/0/1/0
Hypertrophy: Shorter rests increase stress but too short can reduce growth.	Lifting speed early in a set has little impact on hypertrophy if training to failure.
CNS fatigue can stop sets early (away from real failure) if rests are too short.	Slightly slower eccentric (< 4 sec) may increase growth without excess fatigue.

How do rest period lengths differ for strength vs hypertrophy training?

Guess the tempo of each of the following:

- **Fast phase:**
- **Regular speed:**
- **Slow speed:**

Which phase is best for power and strength training? What about hypertrophy?

Would a slow concentric phase during non-stimulating reps increase hypertrophy through increased time under tension?

What about slow eccentric phase and hypertrophy?

What is a deload? When and how can it be implemented in a training program?

Assignment time!

GROUP ACTIVITY

Case Study:

You are working with a male client who has already developed some muscle size in his lower body but has a noticeably smaller upper body. His main goal is to achieve overall hypertrophy, with greater emphasis on his arms, chest, and shoulders, while maintaining and slightly improving his lower body. The client is available to train up to 7 days per week if needed.

Design a complete training program for this client. Be sure to include all necessary details (frequency, exercises, sets, reps, tempo, and rest periods). You have all the knowledge required to create a well-structured and goal-oriented program!

Consolidation:

By the end of this lesson, I am aware of:

-How rest periods and tempo influence training results and ways to apply them in a program.

-What a deload is and how to use it effectively in a training plan.

Post-Session 37 Tasks:

- Review the material covered during the session.
- Prepare a brief recap of the key training principles to be presented in the next session.

Pre-Session 38 Preparation:

- Complete the case study assignment

Watch the below pre-recorded videos:

-Week 18: Components of Program Design for Resistance and Cardiovascular Training: Just the Periodization and Training Systems components of program design videos along with the Training Session video

Session 39: Program Design – Resistance Training

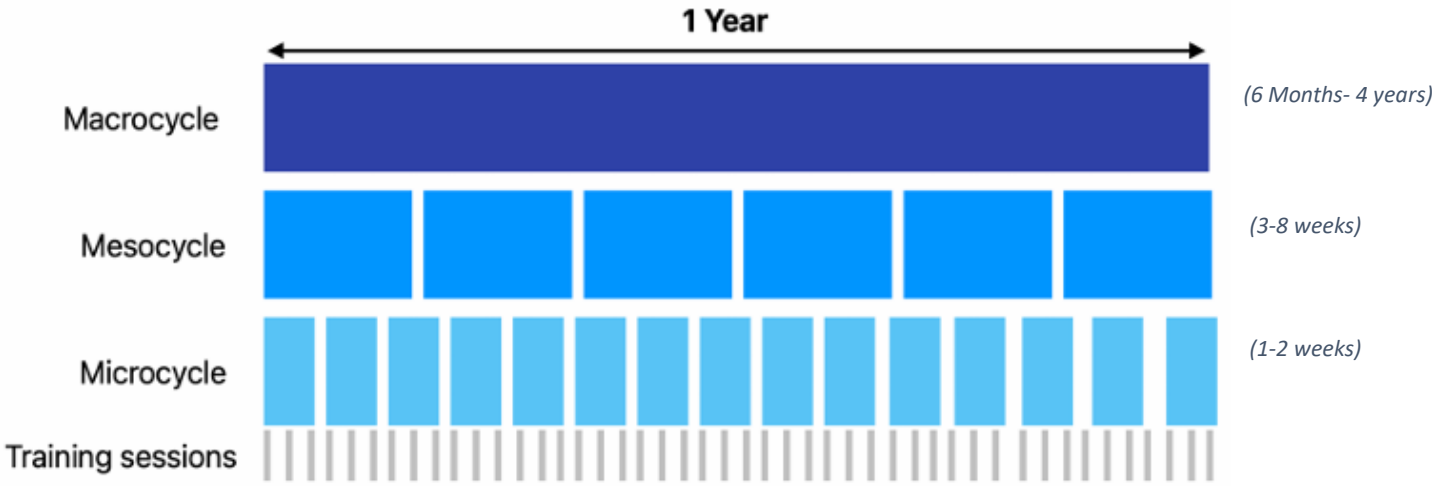
Session 38 Recap + Q&A:

LESSON 1: COMPONENTS OF PROGRAM DESIGN – Periodization



Objectives:
-Understand what Periodization is and how to apply it in program design.

- **Periodization Breakdown:**



- **Different types of periodization:**

Read the examples of periodization provided below and work with your partner to **answer** the questions that follow:

- **Linear Periodization (LP):**

	Rep range	Intensity (%1RM)
Week 1-4	12-15	60-70%
Week 5-8	8-12	70-80%
Week 9-12	4-8	80-85%

What about Reverse Linear Periodization? Which one is more effective for strength and power?

- **Undulating Periodization (UP):**

  - Daily Undulating Periodization (DUP):

	Rep scheme	Intensity (%1RM)
Day 1- Hypertrophy	4 sets of 10	60-75% RM
Day 2- Strength	7 sets of 4	> 80% RM
Day 3- Power	5 sets of 3	80-90% RM

  - Weekly Undulating Periodization (WUP):

	Rep scheme	Intensity (%1RM)
Week 1	4 sets of 10	60-70% RM
Week 2	7 sets of 4	> 80% RM
Week 3	4 sets of 10	65-75% RM
Week 4	7 sets of 4	80-90% RM

What is the benefit of Undulating Periodization vs Linear Periodization?

- **Block Periodization:**



What is Block periodization mainly used for?

What is the difference between Periodization and Programming?

Consolidation:

By the end of this lesson, I am aware of:

-What periodization is, the different types, and how to apply them effectively in a training program.

LESSON 2: COMPONENTS OF PROGRAM DESIGN – Training Systems



Objectives:
-Understand what training systems are and their purpose in program design.
-Recognize the benefits and drawbacks of different training systems.
(Platform: Chapter Titled: Week 18: Program Design: Components of Program Design for Resistance and Cardiovascular Training)

GROUP ACTIVITY

Each group will **prepare** one training system and **explain** it to their classmates, both theoretically and practically:

Training System	Description	Hypertrophy	Strength	Endurance
Drop Sets	Initial set to failure → reduce load → repeat to failure (no rest between). Final set with light or very light load.	Less effective long-term than straight sets due to CNS and post-workout fatigue (caused by calcium-ion related fatigue within workout).	No greater gains than traditional sets.	Greater improvements than traditional sets due to higher reps and longer sets.
Cluster Sets	Divide a set into smaller clusters with short rests (20–30 sec) between.	No greater adaptations than traditional sets. Minimizes fatigue.	Helps maintain velocity and rate of force development; viable under time constraints.	Lack of evidence.
Rest-Pause Sets	Perform to failure → rest 15–30 sec → continue at same weight to failure, repeat.	No greater adaptations than traditional sets; CNS fatigue from failure limits gains.	Similar to cluster in benefits; not superior to traditional sets.	Lack of evidence.

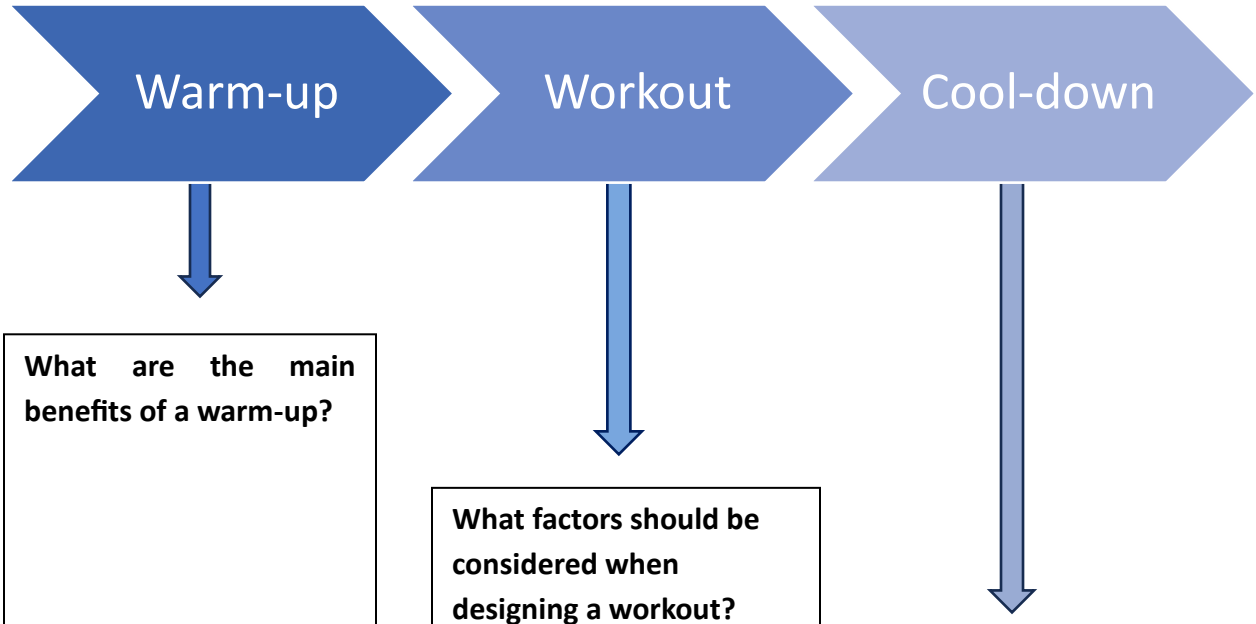
Training System	Description	Hypertrophy	Strength	Endurance
Supersets	Two exercises back-to-back without rest (agonist, antagonist, or unrelated).	Agonist: CNS fatigue limits second exercise. Antagonist/unrelated: supraspinal CNS (reduced drive from the brain to the spinal cord) fatigue limits gains.	Fatigue may impair neural adaptations for strength.	Increases training density (do more in less time); can help endurance (especially agonist type).
Giant Sets	Three or more exercises back-to-back without rest between, rest only at the end.	Similar effects to supersets.	Similar effects to supersets.	Similar effects to supersets.
Pre-Exhaustion	Single-joint exercise → multi-joint exercise for same muscle without rest.	No extra benefit; first set stimulates most growth.	No significant advantage over traditional training.	Fatigues target muscle in the first exercise and forces it to do the second exercise; increasing endurance demand.

What do you think about training systems in general? Do you have to use them in your training program? When would you incorporate them into a training program?

Consolidation:
By the end of this lesson, I am aware of:
-What training systems are, their advantages and disadvantages, and how to apply them effectively for different clients and goals.

LESSON 3: TRAINING SESSION – Resistance Training

Objectives:
-Understand the key components and flow of a training session and how to structure them effectively to meet client needs.



Consolidation:
By the end of this lesson, I am aware of:
-The main parts of a training session and how to apply them in the correct flow.

LESSON 4: CASE STUDIES

Objectives:
-Apply knowledge of training principles to design and adapt programs based on client goals, background, and progress.

GROUP ACTIVITY

Case Study: Sarah

Client Information:
General Goal: Weight maintenance (following up with a dietitian)

Specific Goals:
- Maintain upper body
- Hypertrophy in lower body and back
- Maintain calves

Training Background:
3 months of training experience
Current Intensity: 4 RIR
Volume per week:

Muscle Group	Sets per Week
Deltoids	4
Chest	6
Biceps/Triceps	0
Upper Back	7
Lats	7
Quads	12
Hamstrings	11
Glutes	14
Calves	0

Previous frequency: 3 sessions/week
Current availability: 7 days/week

Design a 2-week training program for Sarah that aligns with her goals.
(Consider: training days, splits, exercise selection, sets, reps, etc.)



Case Study (Continued)

📈 **Progress Update**

Sarah has now trained with you for 3 months.

She successfully reached her previous goals – great job!

🎯 **New Goal: Total body hypertrophy**

Design a new program for Sarah targeting total body hypertrophy.
What are the major changes compared to her previous program?



Case Study: Toufic

👤**Client Information:**

Toufic is a beginner male client coming for his very first training session.

🎯**Goal:**

Gain muscle and improve overall body shape.

Design an 8-week resistance training program for this client.



Consolidation:
By the end of this lesson, I am aware of:
-How to design and adjust a training program according to a client's goals and training history.



Post-Session 38:
Continue developing Sarah's case study
Apply what you learned by designing your own program



Pre-session 39:
Watch the below pre-recorded videos:
-Week 18: Components of Program Design for Resistance and Cardiovascular Training: Just the videos related to cardiovascular training.

Session 40: Program Design – Cardiovascular Training

Session 39 Recap + Q&A:

LESSON 1: AEROBIC TRAINING TYPE/MODE:

Objectives:
- Understand the differences between Steady-State Cardio (SSC) and various interval training methods (HIIT, VIIT, HVIT).

There are two main types of cardiovascular training: Steady-State Cardio and Interval Training

1- STEADY-STATE CARDIO (SSC):

Define SSC and list when it is considered safest.

When is it appropriate to incorporate this method of cardiovascular training?

After building up to ~30 minutes continuous work, what is the next programming focus?

Which energy systems predominate during SSC as described?

Frequency	Intensity	Time/Duration	Type
3 days/week	60-70% of MHR RPE of 3-4 (CRS) or 12-14 (Borg Scale)	25-30 min	- Running on a treadmill - Cycling of a recumbent bike

Table 1- Sample FIIT program for a Sedentary Individual Using Continuous Training

2- INTERVAL TRAINING:

GROUP ACTIVITY

Use the below tables to **teach** your fellow classmates about the Interval Training in question:

Method	Key Feature	Sample Duration	Intensity Focus	Best For
HIIT	Short intense bouts with longer recovery (≈1:2 work: rest)	~22 min (4 min speed)	RPE 7–9 during speed	Time efficiency, fitness gains
VIIT	Mix of high, medium, low intensities in one workout	~22 min (6 min high/max)	Cycles 5–6, 8–9, 9–10	Variety, training multiple systems
HVIT	Longer intervals at moderately high intensity	~33 min (6 min fast run)	70–80% MHR; RPE 6–8	Aerobic endurance, endurance athletes

- **<u>Practical Applications – Examples:</u>**

Time	Interval	Exertion Level (0-10)
5 min	Warm-up	3-4
1 min	Speed	7-9
2 min	Recovery	5-6
1 min	Speed	7-9
2 min	Recovery	5-6
1 min	Speed	7-9
2 min	Recovery	5-6
1 min	Speed	7-9
2 min	Recovery	5-6
5 min	Cool-down	3-4
22 min Total Time		
(4 min total speed)		

Table 2- Program Design Sample for HIIT using the treadmill

Time	Interval	Exertion Level (0-10)
5 min	Warm-up	3-4
2 min	High Speed	8-9
2 min	Medium Speed	5-6
1 min	Max Speed	9-10
2 min	Recovery	3-4
2 min	High Speed	8-9
2 min	Medium Speed	5-6
1 min	Max Speed	9-10
2 min	Recovery	3-4
5 min	Cool-down	3-4
22 min Total Time		
(6 min high speed & max speed)		

Table 3- Program Design Sample for VIIT using the treadmill

Time	Interval	Exertion Level (0-10)
5 min	Warm-up	3-4
3 min	Steady run	6-7
1 min	Fast Run	7-8
3 min	Steady run	6-7
1 min	Fast Run	7-8
3 min	Steady run	6-7
1 min	Fast Run	7-8
3 min	Steady run	6-7
1 min	Fast Run	7-8
3 min	Steady run	6-7
1 min	Fast Run	7-8
3 min	Steady run	6-7
1 min	Fast Run	7-8
5 min	Cool-down	3-4
33 min Total Time		
(6 min fast run)		

Table 4- HVIT Program Design Sample for Treadmill

How does the choice between SSC and Interval Training depend on an individual's goals (e.g., health, performance, fat loss)?

Which method would you recommend for someone short on time but seeking fast fitness results, and why?

Why is it important to first build a base with SSC before progressing to Interval Training?

What risks might an elderly client face with HIIT compared to SSC?



Discuss the following: Which client burns more fat: client A performing 30 minutes of steady-state cardio on a treadmill or client B performing 30 minutes of Interval Training on a treadmill as well?



Consolidation:
By the end of this section, I am aware of:
-The differences between Steady-State Cardio (SSC) and interval training methods (HIIT, VIIT, HVIT).

LESSON 2: TRAINING VARIABLES

Objectives:
- Understand different methods to measure and monitor cardiovascular training intensity.
-Recognize the importance of movement quantity, quality, specificity, and variation (cross-training).

1- TRAINING INTENSITY:

Briefly **explain** the main methods used to measure and monitor exercise intensity

Heart rate (% Maximum Heart Rate (MHR))	
Talk Test	
Rate of Perceived Exertion (RPE)	

Goals	RPE	Estimated Straight line % HR
Aerobic Endurance training	3–4	60–70
Anaerobic Endurance training	4–6	70–75
Muscular Endurance/Strength	6–7	75–80
Fast Race/Peak Performance	9–10	90–100

2- DURATION & FREQUENCY:

- **Recommended Weekly Physical Activity (WHO)**

✓ Adults should aim for 150 minutes/week of moderate-intensity activity

✓ Reminder: Some activity is better than none — even if you can't yet reach 150 min.

- **Classification of Weekly Aerobic Activity**

Level of Activity	Range (min/week)	Health Benefits	Comments
Inactive	No activity beyond baseline	None	Being inactive is unhealthy
Low	<150	Some	Better than being inactive
Medium	150–300	Substantial	High end = more extensive benefits
High	>300	Additional	No clear upper limit known

What would you recommend for a client who cannot reach 150 min/week at first?

3- QUANTITY & QUALITY:

How does specificity apply to cardiovascular training? Give an example.

Why is resistance training important alongside cardiovascular training?

What are the risks of performing only one activity for a long period at high stress/intensity?

How can cross-training (e.g., mixing running and cycling) reduce injury risk and improve fitness?



Consolidation

By the end of this section, I am aware of:

-How to monitor cardiovascular training intensity, apply duration and frequency guidelines, and use specificity and variation to improve safety and results.

LESSON 3: TRAINING SESSION – Cardiovascular Training

Objectives:
-Understand the key components and flow of a training session and how to structure them effectively to meet client needs.



Name some warm-up routines you can do before cardio:

What cool-down activities can you do after a cardio session:

Consolidation:
By the end of this lesson, I am aware of:
-The main parts of a training session and how to apply them in the correct flow.

LESSON 4: THE INTERFERENCE EFFECT & CONCURRENT TRAINING



Objectives:
-Recognize how aerobic and resistance training can interfere with each other's adaptations.
-Learn practical strategies to minimize the interference effect when combining cardio and strength training.

- **Practical Guidelines for Concurrent Training:**

Factor	Recommendation
Order	Perform resistance training before cardio (to reduce interference)
Separation	Ideally separate sessions when possible
Duration	Avoid very long cardio before lifting
Mode	Prefer cycling or swimming over running (less joint stress/interference)
Benefit	Better recovery between sets & sessions due to aerobic support

Why can aerobic and strength training interfere with each other's adaptations?

According to the practical guidelines, why is resistance training usually recommended before cardio?

Which modes of cardio are less likely to interfere with hypertrophy goals, and why?

How does a stronger aerobic system support recovery during anaerobic efforts?

Consolidation

By the end of this section, I am aware of:

-The adaptations of aerobic vs anaerobic training.

-The interference effect that may occur

-How to apply strategies to combine them effectively.

LESSON 5: CASE STUDY

Objectives:
-Apply principles of frequency, intensity, time, and type (FITT) in cardio programming.

GROUP ACTIVITY

Client Information

- Name: Samer
- Age: 38
- Goal: Improve cardiovascular endurance
- Current activity: Walks 2–3 times per week (~15 min)
- Availability: 4 days/week
- Fitness level: Beginner

Design a 1-week cardiovascular training plan for Samer using the FITT principle (Frequency, Intensity, Time, Type). Keep it simple and suitable for a beginner.

FITT Table – Weekly Plan

Day	Frequency	Intensity	Time	Type
Day 1				
Day 2				
Day 3				
Day 4				
Day 5				
Day 6				
Day 7				



Client Information

- Age: 29
- Height: 170 cm
- Weight: 100 kg
- Goal: Lose weight and gain muscle
- Current activity: Started training on his own 3 months ago
- Fitness level: Beginner

Design a 4-week training program for this client. In addition, provide nutritional guidance on how many calories, protein, carbohydrates, and fats he should eat to support fat loss while gaining muscle.

🎉 YOU MASTERED PROGRAM DESIGN! 📝🔥

You've decoded training principles, balanced variables, and built programs that *work*!

You now know how to structure both resistance and cardiovascular training with purpose, precision, and progression.

🏅 Certified Program Design Architect 🏋️‍♂️🏃



CONGRATULATIONS!